MATHEMATICS

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DEFINITE INTEGRATION

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JEE EXERCISES

Exercise J-I

INTEGRATION BASED ON STANDARD RESULTS

1. Which one of the following is TRUE :

निम्न में से कौनसा एक सत्य है :

(A) $x.\int \frac{dx}{x} = x\,ln|x| + C$ (B) $x.\int \frac{dx}{x} = x\,ln|x| + Cx$

(C) $\frac{1}{\cos x}.\int \cos x\ dx = \tan x + C$ (D) $\frac{1}{\cos x}.\int \cos x\ dx = x + C$

Ans. B

Sol. $x.\int \frac{dx}{x} = x(ln|x| + C) = x\,ln|x| + Cx$

$$\frac{1}{\cos x}\int \cos x dx = \frac{1}{\cos x}(\sin x + C)$$

$$= \tan x + \frac{C}{\cos x}$$

2. If $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + C$, then value of (A,B) is :

यदि $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + C$, तो (A,B) बराबर है :

(A) $(\sin\alpha, \cos\alpha)$ (B) $(\cos\alpha, \sin\alpha)$ (C) $(-\sin\alpha, \cos\alpha)$ (D) $(-\cos\alpha, \sin\alpha)$

Ans. B

Sol. $$\int \frac{\sin x}{\sin(x-\alpha)} dx = \int \frac{\sin(x-\alpha+\alpha)}{\sin(x-\alpha)} dx$$

$$= \int \frac{\sin(x-\alpha)\cos\alpha + \cos(x-\alpha)\sin\alpha}{\sin(x-\alpha)} dx$$

$$= \cos\alpha \int dx + \sin\alpha \int \frac{\cos(x-\alpha)}{\sin(x-\alpha)} dx$$

$$= \cos\alpha\, x + \sin\alpha\, \ell n\left|\sin(x-\alpha)\right| + C$$

$\Rightarrow A = \cos\alpha, B = \sin\alpha$

(A, B) = (cos α, sin α)

3. $\int \frac{dx}{\cos x+\sqrt{3}\sin x} =$

(A) $\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+C$ (B) $\frac{1}{2}\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+C$

(C) $\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+C$ (D) $\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+C$

Ans. A

Sol. $\int\frac{dx}{\cos x+\sqrt{3}\sin x}=\frac{1}{2}\int\frac{dx}{\frac{1}{2}\cos x+\frac{\sqrt{3}}{2}\sin x}$

$=\frac{1}{2}\int\sec\left(x-\frac{\pi}{3}\right)dx=\frac{1}{2}\ell n\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}-\frac{\pi}{6}\right)\right|+C$

$=\frac{1}{2}\ell n\left|\tan\left(\frac{\pi}{12}+\frac{x}{2}\right)\right|+C$ $\left\{\int\sec\theta d\theta=\ell n\left|\tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right|+C\right\}$

INTEGRATION BY SUBSTITUTION

4. If $\int\frac{(2x+3)dx}{x(x+1)(x+2)(x+3)+1} = C - \frac{1}{f(x)}$ where $f(x)$ is of the form of $ax^2 + bx + c$ then $(a + b + c)$ equals :

यदि $\int\frac{(2x+3)dx}{x(x+1)(x+2)(x+3)+1}=C-\frac{1}{f(x)}$ जहाँ $f(x)$, ax^2+bx+c रूप का है तब $(a+b+c)$ बराबर :

(A) 4 (B) 5 (C) 6 (D) none

Ans. B

Sol. $\int\frac{2x+3}{(x^2+3x)(x^2+3x+2)+1}dx$

put $x^2 + 3x = t$ $\Rightarrow$ $(2x + 3)\,dx = dt$

$\int\frac{dt}{t(t+2)+1}$; $\int\frac{dt}{(t+1)^2}=C-\frac{1}{t+1}=C-\frac{1}{x^2+3x+1}$

$\Rightarrow$ $a = 1, b = 3, c = 1$ $\Rightarrow$ $a + b + c = 5$

5. Consider $f(x)=\frac{x^2}{1+x^3}$; $g(t)=\int f(t)\,dt$. If $g(1)=0$ then $g(x)$ equals :

माना $f(x)=\frac{x^2}{1+x^3}$; $g(t)=\int f(t)\,dt$. If $g(1)=0$ तब $g(x)$ बराबर :

(A) $\frac{1}{3}\ell n(1+x^3)$ (B) $\frac{1}{3}\ell n\left(\frac{1+x^3}{2}\right)$ (C) $\frac{1}{2}\ell n\left(\frac{1+x^3}{3}\right)$ (D) $\frac{1}{3}\ell n\left(\frac{1+x^3}{3}\right)$

Ans. B

Sol. $g(t)=\int f(t)dt=\int\frac{t^2}{1+t^3}dt$

let $1+t^3=v$

$\Rightarrow 3t^2\,dt=dv$

$=\int\frac{dv}{3v}=\frac{1}{3}\ell n|v|+C$

$\Rightarrow g(t)=\frac{1}{3}\ell n|1+t^3|+C$

Given $g(1)=0$

$\Rightarrow\frac{1}{3}\ell n2+C=0\Rightarrow C=\frac{-1}{3}\ell n2$

$\Rightarrow g(t)=\frac{1}{3}\ell n\left|\frac{t^3+1}{2}\right|$

$\Rightarrow g(x)=\frac{1}{3}\ell n\left|\frac{1+x^3}{2}\right|$

INTEGRATION BY PARTS

6. $\int\frac{\cot^{-1}(e^x)}{e^x}\,dx=$

(A) $\frac{1}{2}\,ln\,(e^{2x}+1)-\frac{\cot^{-1}(e^x)}{e^x}+x+c$ (B) $\frac{1}{2}\,ln\,(e^{2x}+1)+\frac{\cot^{-1}(e^x)}{e^x}+x+c$

(C) $\frac{1}{2}\,ln\,(e^{2x}+1)-\frac{\cot^{-1}(e^x)}{e^x}-x+c$ (D) $\frac{1}{2}\,ln\,(e^{2x}+1)+\frac{\cot^{-1}(e^x)}{e^x}-x+c$

Ans. C

Sol. $\int\frac{\cot^{-1}\left(e^x\right)}{e^x}dx=\int\underbrace{\cot^{-1}\left(e^x\right)}_{I}\cdot\underbrace{e^{-x}}_{II}dx$

$=\cot^{-1}\left(e^x\right)\cdot(-1)e^{-x}+\int\frac{-1}{1+e^{2x}}e^{-x}\cdot e^x dx$

$=-\cot^{-1}\left(e^x\right)\cdot e^x-\int\frac{dx}{1+\left(e^x\right)^2}$

$=\frac{-\cot^{-1}\left(e^x\right)}{e^x}-\frac{1}{2}\int\frac{2dx}{\left(1+\left(e^x\right)^2\right)}$

$=\frac{-\cot^{-1}\left(e^x\right)}{e^x}-\frac{1}{2}\int\frac{2e^{-2x}}{e^{-2x}+1}dx$

$=\frac{-\cot^{-1}\left(e^x\right)}{e^x}+\frac{1}{2}\ell n\left(e^{-2x}+1\right)+C$

$=\frac{-\cot^{-1}\left(e^x\right)}{e^x}+\frac{1}{2}\ell n\left(1+e^{2x}\right)-x+C$

7. $\int x\,.\,\frac{ln\left(x+\sqrt{1+x^2}\right)}{\sqrt{1+x^2}}\,dx=$

(A) $\sqrt{1+x^2}\;ln\left(x+\sqrt{1+x^2}\right)-x+c$ (B) $\frac{x}{2}\,.\,ln^2\left(x+\sqrt{1+x^2}\right)-\frac{x}{\sqrt{1+x^2}}+c$

(C) $\frac{x}{2}\,.\,ln^2\left(x+\sqrt{1+x^2}\right)+\frac{x}{\sqrt{1+x^2}}+c$ (D) $\sqrt{1+x^2}\;ln\left(x+\sqrt{1+x^2}\right)+x+c$

Ans. A

Sol. $\int \underbrace{\ell n\left(x+\sqrt{1+x^2}\right)}_{I}\cdot\underbrace{\frac{x}{\sqrt{1+x^2}}}_{II}dx$

$=\ell n\left(x+\sqrt{1+x^2}\right)\sqrt{1+x^2}-\int\sqrt{x^2+1}\cdot\frac{1}{\sqrt{x^2+1}}dx$

$=\ell n\left(x+\sqrt{1+x^2}\right)\sqrt{1+x^2}-x+C$

8. $\int\frac{(\log x)}{x^2}dx=$

(A) $\frac{1}{2}(\log x+1)+c$ (B) $-\frac{1}{x}(\log x+1)+c$ (C) $\frac{1}{x}(\log x-1)+c$ (D) $\log(x+1)+c$

Ans. B

Sol. $\int\frac{\log x}{x^2}dx$

let $\ell n\, x=t\Rightarrow x=e^t$

$\frac{1}{x}dx=dt$

$=\int t\cdot e^{-t}dt=-te^{-t}-e^{-t}+C$

$=-e^{-t}(t+1)+C$

$=-\frac{1}{x}(1+\ell nx)+C$

9. If $\int f(x)dx=\Psi(x)$, then $\int x^5f(x^3)dx$ is equal to : **[JEE(Main)2013]**

यदि $\int f(x)dx=\Psi(x)$, तो $\int x^5f(x^3)dx$ बराबर है : **[JEE(Main)2013]**

(A) $\frac{1}{3}\left[x^3\Psi(x^3)-\int x^2\Psi(x^3)dx\right]+C$ (B) $\frac{1}{3}x^3\Psi(x^3)-3\int x^3\Psi(x^3)dx+C$

(C) $\frac{1}{3}x^3\Psi(x^3)-\int x^2\Psi(x^3)dx+C$ (D) $\frac{1}{3}\left[x^3\Psi(x^3)-\int x^3\Psi(x^3)dx\right]+C$

Ans. C

Sol. put $x^3=t$ $\Rightarrow$ $3x^2dx=dt$

$$I = \frac{1}{3}\int f(t)tdt$$

$$= \frac{1}{3}\left[t\int f(t)dt - \int (1.\int f(t)dt)dt\right]$$

$$= \frac{1}{3}x^3\Psi(x^3) - \int \Psi(x^3)x^2dx$$

TWO CLASSIC INTEGRALS

10. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\left(x+\sqrt{x}\right)dx$ =

(A) $2e^{\sqrt{x}}\left[x-\sqrt{x}+1\right]+C$ (B) $e^{\sqrt{x}}\left[x-2\sqrt{x}+1\right]$

(C) $e^{\sqrt{x}}\left(x+\sqrt{x}\right)+C$ (D) $e^{\sqrt{x}}\left(x+\sqrt{x}+1\right)+C$

Ans. A

Sol. $I = \frac{e^{\sqrt{x}}}{\sqrt{x}}\left(x+\sqrt{x}\right)dx$

let $\sqrt{x} = t$

$$\Rightarrow \frac{1}{2\sqrt{x}}dx = dt$$

$$I = 2\int \underset{II}{e^t}\underbrace{\left(t^2+t\right)}_{I}dt$$

$$= 2\left[e^t\left(t^2+t\right) - \int (2t+1)e^t dt\right]$$

$$= 2e^t\left(t^2+t\right) - 2(2t+1)e^t + 2\int 2.e^t dt$$

$$= 2\left[e^t\left(t^2+t\right) - (2t+1)e^t + 2e^t\right] + C$$

$$= 2\left[e^{\sqrt{x}}\left(x+\sqrt{x}-2\sqrt{x}-1+2\right)\right] + C$$

$$= 2e^{\sqrt{x}}\left(x+1-\sqrt{x}\right) + C$$

11. $\int\left\{\frac{(\log x-1)}{1+(\log x)^2}\right\}^2 dx =$

(A) $\frac{\log x}{(\log x)^2+1}+C$ (B) $\frac{x}{x^2+1}+C$ (C) $\frac{xe^x}{1+x^2}+C$ (D) $\frac{x}{(\log x)^2+1}+C$

Ans. D

Sol. $\int\left\{\frac{\ell nx-1}{1+(\ell nx)^2}\right\}^2 dx$

let $\ell nx = t \Rightarrow \frac{1}{x}dx = dt$

$$= \int\left[\frac{(t-1)}{1+t^2}\right]^2 \cdot e^t dt$$

$$= \int e^t\left[\frac{t^2-2t+1}{(1+t^2)^2}\right]dt$$

$$= \int e^t\left[\underbrace{\frac{1}{1+t^2}}_{f(t)} - \underbrace{\frac{2t}{(1+t^2)^2}}_{f'(t)}\right]dt$$

$$= e^t\cdot\left(\frac{1}{1+t^2}\right)+C = \frac{x}{1+(\ell nx)^2}+C$$

12. $\int\left(\int e^x\left(l\,n\,x+\frac{2}{x}-\frac{1}{x^2}\right)dx\right)dx =$

(A) $e^x lnx + C_1x + C_2$ (B) $e^x lnx + \frac{1}{x} + C_1x + C_2$

(C) $\frac{l\,n\,x}{x}+C_1x+C_2$ (D) None of these

Ans. A

Sol. $\int\left[\int e^x\left(\ell nx+\frac{2}{x}-\frac{1}{x^2}\right)dx\right]dx$

$\Rightarrow \int\left(\int\left(e^x\left(\ell nx+\frac{1}{x}\right)+e^x\left(\frac{1}{x}-\frac{1}{x^2}\right)\right)dx\right)dx$

$\Rightarrow \int\left(e^x\ell nx+\frac{e^x}{x}+C_1\right)dx$

$=e^x\ell nx+C_1x+C_2$

13. The integral $\int\left(1+x-\frac{1}{x}\right)e^{x+\frac{1}{x}}dx$ is equal to : **[JEE(Main)2014]**

समाकलन $\int\left(1+x-\frac{1}{x}\right)e^{x+\frac{1}{x}}dx$ बराबर है : **[JEE(Main)2014]**

(A) $(x+1)e^{x+\frac{1}{x}}+c$ (B) $-xe^{x+\frac{1}{x}}+c$ (C) $(x-1)e^{x+\frac{1}{x}}+c$ (D) $xe^{x+\frac{1}{x}}+c$

Ans. D

Sol. $\int\left(1+x-\frac{1}{x}\right)\cdot e^{x+\frac{1}{x}}dx$

$=\int e^{x+\frac{1}{x}}\left(1+x\left(1-\frac{1}{x^2}\right)\right)dx$

$=\int \underset{II}{1}\cdot\underbrace{e^{x+\frac{1}{x}}}_{I}dx+\int x\cdot e^{x+\frac{1}{x}}\left(1-\frac{1}{x^2}\right)dx$

$=x\cdot e^{x+\frac{1}{x}}-\int x\cdot e^{x+\frac{1}{x}}\left(1-\frac{1}{x^2}\right)dx+\int xe^{x+\frac{1}{x}}\left(1-\frac{1}{x^2}\right)dx$

$=xe^{x+\frac{1}{x}}+C$

INTEGRATION OF TRGONOMETRIC FUNCTION

14. $\int\frac{\sin 2x+\sin 4x-\sin 6x}{1+\cos 2x+\cos 4x+\cos 6x}dx$ is equal to : (where C is indefinite integration constant)

$\int \frac{\sin 2x + \sin 4x - \sin 6x}{1 + \cos 2x + \cos 4x + \cos 6x}\, dx$ बराबर होगा : (जहां C समाकलन अचर है)

(A) $\frac{1}{3}\, ln|\sec 3x| + \frac{1}{2}\, ln|\sec 2x| + ln|\sec x| + C$

(B) $\frac{1}{3}\, ln|\sec 3x| - \frac{1}{2}\, ln|\sec 2x| - ln|\sec x| + C$

(C) $C - \frac{1}{3}\, ln|\sec 3x| + \frac{1}{2}\, ln|\sec 2x| + ln|\sec x|$

(D) $C - \frac{1}{3}\, ln|\sec 3x| - \frac{1}{2}\, ln|\sec 2x| - ln|\sec x|$

Ans. B

Sol. $I = \int \frac{\sin 2x + \sin 4x - \sin 6x}{1 + \cos 2x + \cos 4x + \cos 6x}\, dx = \int \frac{2\sin 3x \cos x - 2\sin 3x \cos 3x}{2\cos^2 x + 2\cos x \cos 5x}\, dx$

$= \int \frac{\sin 3x\,(\cos x - \sin 3x)}{\cos x\,(\cos x + \cos 5x)}\, dx = \int \frac{2\sin 3x \sin 2x \sin x}{2\cos x \cos 2x \cos 3x}\, dx = \int \tan x \tan 2x \tan 3x\, dx$

$\because\ 3x = 2x + x \Rightarrow \tan 3x = \frac{\tan 2x + \tan x}{1 - \tan 2x \tan x} \Rightarrow \tan 3x - \tan 3x \tan 2x \tan x = \tan 2x + \tan x$

$\Rightarrow \tan 3x \tan 2x \tan x = \tan 3x - \tan 2x - \tan x$

$\therefore I = \int (\tan 3x - \tan 2x - \tan x)\, dx = \frac{1}{3} ln|\sec 3x| - \frac{1}{2} ln|\sec 2x| - ln|\sec x| + C$

15. $\int \sec^2\theta\,(\sec\theta + \tan\theta)^2\, d\theta$:

(A) $\frac{(\sec\theta + \tan\theta)}{2}[2 + \tan\theta(\sec\theta + \tan\theta)] + C$

(B) $\frac{(\sec\theta + \tan\theta)}{3}[2 + 4\tan\theta(\sec\theta + \tan\theta)] + C$

(C) $\frac{(\sec\theta + \tan\theta)}{3}[2 + \tan\theta(\sec\theta + \tan\theta)] + C$

(D) $\frac{3(\sec\theta + \tan\theta)}{2}[2 + \tan\theta(\sec\theta + \tan\theta)] + C$

Ans. C

Sol. $\sec\theta + \tan\theta = y$ ....(1)

$\sec\theta\,(\tan\theta + \sec\theta)\, d\theta = dy$

now, $\sec\theta - \tan\theta = \frac{1}{y}$ ....(2)

from (1) and (2)

$$2\sec\theta = y + \frac{1}{y}$$

$$\therefore \quad I = \frac{1}{2}\int y\left(y + \frac{1}{y}\right)dy = \frac{1}{2}\left[\frac{y^3}{3} + y\right] + C$$

$$\frac{1}{2}\left[\frac{(\sec\theta + \tan\theta)^3}{3} + (\sec\theta + \tan\theta)\right] = \frac{(\sec\theta + \tan\theta)}{6}[(\sec\theta + \tan\theta)^2 + 3]$$

16. $I = \int \frac{(\sin 2x)^{1/3}\, d\left(\tan^{1/3} x\right)}{\sin^{2/3} x + \cos^{2/3} x} =$

(A) $\frac{1}{2^{2/3}} l\,n\left(1 + \tan^{1/3} x\right) + C$ (B) $l\,n\left(1 + \tan^{2/3} x\right) + C$

(C) $2^{1/3} l\,n\left(1 + \tan^{2/3} x\right) + C$ (D) $\frac{1}{2^{2/3}} l\,n\left(1 + \tan^{2/3} x\right) + C$

Ans. D

Sol. $\int \frac{(\sin 2x)^{\frac{1}{3}}}{\sin^{\frac{2}{3}} x + \cos^{\frac{2}{3}} x} d\left(\tan^{\frac{1}{3}} x\right)$

$$= 2^{\frac{1}{3}} \int \frac{\sin^{\frac{1}{3}} x \cos^{\frac{1}{3}} x}{\left(\tan^{\frac{2}{3}} x + 1\right)\cos^{\frac{2}{3}} x} d\left(\tan^{\frac{1}{3}} x\right)$$

$$= \frac{2^{\frac{1}{3}}}{2} \int \frac{2\tan^{\frac{1}{3}} x}{\left(1 + \left(\tan^{\frac{1}{3}} x\right)^2\right)} d\left(\tan^{\frac{1}{3}} x\right)$$

$$= \frac{1}{2^{\frac{2}{3}}} \ell n\left(1 + \tan^{\frac{2}{3}} x\right) + C$$

17. If the integral $\int \frac{5\tan x}{\tan x - 2}dx = x + a\ln|\sin x - 2\cos x| + k$ then a is equal to : **[AIEEE - 2012]**

यदि समाकल $\int \frac{5\tan x}{\tan x - 2}dx = x + a\ln|\sin x - 2\cos x| + k$ है, तो a बराबर है : **[AIEEE - 2012]**

(A) 2 (B) –1 (C) –2 (D) 1

Ans. A

Sol. $\int \frac{5\tan x}{\tan x - 2}dx = \int \frac{5\sin x}{\sin x - 2\cos x}dx$

$5\sin x = A(\sin x - 2\cos x) + B\frac{d}{dx}(\sin x - 2\cos x)$

$= A(\sin x - 2\cos x) + B(\cos x + 2\sin x)$

$\left.\begin{matrix} A + 2B = 5 \\ -2A + B = 0 \end{matrix}\right\}$ $A = 1,\ B = 2$

Now $\int \frac{5\sin x}{\sin x - 2\cos x}dx$

$= \int\left(\frac{(\sin x - 2\cos x)}{\sin x - 2\cos x} + \frac{2(\cos x + 2\sin x)}{\sin x - 2\cos x}\right)dx$

$= x + 2\log|\sin x - 2\cos x| + K$

$a = 2$

KUTUR PUTUR

18. $\int \frac{1 - x^7}{x(1 + x^7)}\ dx =$

(A) $\ln x + \frac{2}{7}\ln(1 + x^7) + c$ (B) $\ln x - \frac{2}{7}\ln(1 - x^7) + c$

(C) $\ln x - \frac{2}{7}\ln(1 + x^7) + c$ (D) $\ln x + \frac{2}{7}\ln(1 - x^7) + c$

Ans. C

Sol. $I = \int \frac{dx}{x} - 2\int \frac{x^7}{x\left(1 + x^7\right)}dx$

$= \ell n\, x \frac{-2}{7}\ell n\left(1 + x^7\right) + C$

19. Primitive of $\frac{3x^4-1}{(x^4+x+1)^2}$ w.r.t. x is :

$\frac{3x^4-1}{(x^4+x+1)^2}$ का x के सापेक्ष समाकलन है :

(A) $\frac{x}{x^4+x+1}+c$ (B) $-\frac{x}{x^4+x+1}+c$ (C) $\frac{x+1}{x^4+x+1}+c$ (D) $-\frac{x+1}{x^4+x+1}+c$

Ans. B

Sol. $\int\frac{(3x^4-1)}{x^2(x^3+1+x^{-1})^2}dx=\int\frac{(3x^2-x^{-2})}{(x^3+1+x^{-1})^2}dx=-\frac{x}{x^4+x+1}+c$

20. The evaluation of $\int\frac{p\,x^{p+2q-1}-q\,x^{q-1}}{x^{2p+2q}+2x^{p+q}+1}\,dx$ is :

$\int\frac{p\,x^{p+2q-1}-q\,x^{q-1}}{x^{2p+2q}+2x^{p+q}+1}\,dx$ बराबर :

(A) $-\frac{x^p}{x^{p+q}+1}+C$ (B) $\frac{x^q}{x^{p+q}+1}+C$ (C) $-\frac{x^q}{x^{p+q}+1}+C$ (D) $\frac{x^p}{x^{p+q}+1}+C$

Ans. C

Sol. $\int\frac{px^{p+2q-1}-qx^{q-1}}{(x^{p+q}+1)^2}dx=\int\frac{px^{p+2q-1}-qx^{q-1}}{x^{2q}(x^p+x^{-q})^2}dx=\int\frac{px^{p-1}-qx^{-q-1}}{(x^p+x^{-q})^2}dx$

taking x^q as x^{2q} common from Denominator and take it in N^r

21. $\int\frac{dx}{\sqrt[3]{x^{5/2}\,(x+1)^{7/2}}}$

(A) $-\left(\frac{x+1}{x}\right)^{1/6}+C$ (B) $6\left(\frac{x+1}{x}\right)^{-1/6}+C$ (C) $\left(\frac{x}{x+1}\right)^{5/6}+C$ (D) $-\left(\frac{x}{x+1}\right)^{5/6}+C$

Ans. B

Sol. $\int\frac{dx}{\left[x^6\left(1+\frac{1}{x}\right)^{7/2}\right]^{1/3}}=\int\frac{dx}{x^2\left(1+\frac{1}{x}\right)^{7/6}}$

now put $1+\frac{1}{x}=t^6$

$=6\left(\frac{x+1}{x}\right)^{-1/6}+C$

22. The integral $\int \frac{2x^{12}+5x^9}{\left(x^5+x^3+1\right)^3}dx$ is equal to : (where c is an arbitary constant) **[JEE(Main)2016]**

समाकल $\int \frac{2x^{12}+5x^9}{\left(x^5+x^3+1\right)^3}dx$ बराबर है : (जहाँ c एक स्वेच्छ अचर है) **[JEE(Main)2016]**

(A) $\frac{x^5}{2\left(x^5+x^3+1\right)^2}+C$ (B) $\frac{-x^{10}}{2\left(x^5+x^3+1\right)^2}+C$ (C) $\frac{-x^5}{\left(x^5+x^3+1\right)^2}+C$ (D) $\frac{x^{10}}{2\left(x^5+x^3+1\right)^2}+C$

Ans. D

Sol. $\int \frac{\left(2x^{12}+5x^9\right)dx}{x^{15}\left(1+x^{-2}+x^{-5}\right)^3}$

$1+x^{-2}+x^{-5}=t$

$I=-\int \frac{dt}{t^3}$

$=\frac{1}{2t^2}+C$

$=\frac{x^{10}}{2(x^5+x^3+1)^2}+C$

23. The integral $\int \frac{\sin^2 x\cos^2 x}{\left(\sin^5 x+\cos^3 x\sin^2 x+\sin^3 x\cos^2 x+\cos^5 x\right)^2}dx$ is equal to : (Where C is a constant of integration) **[JEE(Main)2018]**

समाकलन $\int \frac{\sin^2 x\cos^2 x}{\left(\sin^5 x+\cos^3 x\sin^2 x+\sin^3 x\cos^2 x+\cos^5 x\right)^2}dx$ बराबर है : (जहाँ c एक स्वेच्छ अचर है)

[JEE(Main)2018]

(A) $\frac{1}{1+\cot^3 x}+C$ (B) $\frac{-1}{1+\cot^3 x}+C$ (C) $\frac{1}{3(1+\tan^3 x)}+C$ (D) $\frac{-1}{3(1+\tan^3 x)}+C$

Ans. D

Sol. Divide numberator & denominator by $\cos^{10}x$

$$\int \frac{\tan^2 x \sec^6 x\, dx}{(\tan^5 x + \tan^2 x + \tan^3 x + 1)^2}$$

$$\int \frac{\tan^2 x \sec^6 x\, dx}{\left(1+\tan^3 x\right)^2 \left(1+\tan^2 x\right)^2}$$

$$\int \frac{\tan^2 x \sec^6 x\, dx}{\left(1+\tan^3 x\right)^2}$$

Put $1 + \tan^3 x = t$

$$I = \frac{1}{3}\int \frac{dt}{t^2}$$

$$= -\frac{1}{3t} + C$$

$$-\frac{1}{3(1+\tan^3 x)} + C$$

MISCELLANEOUS

24. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ .....\circ f)}_{f \text{ occurs n times}}(x)$. Then $\int x^{n-2} g(x)\, dx$ equals :

माना $f(x) = \frac{x}{(1+x^n)^{1/n}}$ तो $n \geq 2$ के लिये $g(x) = \underbrace{(f \circ f \circ .....\circ f)}_{f \text{ occurs n times}}(x)$. तो $\int x^{n-2} g(x)\, dx$ का मान होगा :

(A) $\frac{1}{n(n-1)}(1+nx^n)^{1-\frac{1}{n}} + K$

(B) $\frac{1}{(n-1)}(1+nx^n)^{1-\frac{1}{n}} + K$

(C) $\frac{1}{n(n+1)}(1+nx^n)^{1+\frac{1}{n}} + K$

(D) $\frac{1}{(n+1)}(1+nx^n)^{1+\frac{1}{n}} + K$

Ans. A

Sol. Here $f(x) = \frac{f(x)}{[1+f(x)^n]^{1/n}} = \frac{x}{(1+2x^n)^{1/n}}$

and $f(x) = \frac{x}{(1+3x^n)^{1/n}}$

$\therefore\ g(x) = (fofo.....of)(x) = \frac{x}{(1+nx^n)^{1/n}}$

Let $I = \int x^{n-2} g(x)dx = \int \frac{x^{n-1}}{(1+nx^n)^{1/n}} dx$

$= \frac{1}{n^2} \int \frac{n^2 x^{n-1}}{(1+nx^n)^{1/n}} dx = \frac{1}{n^2} \int \frac{\frac{d}{dx}(1+nx^n)}{(1+nx^n)^{1/n}} dx$

$= \frac{1}{n(n-1)} (1+nx^n)^{1-\frac{1}{n}} + K$

25. $\int \frac{x^2(1-\ell n\, x)}{\ell n^4 x - x^4} dx$ equals :

$\int \frac{x^2(1-\ell n\, x)}{\ell n^4 x - x^4} dx$ बराबर :

(A) $\frac{1}{2}\ell n\left(\frac{x}{\ell n\, x}\right) - \frac{1}{4}\ell n\left(\ell n^2 x - x^2\right) + C$

(B) $\frac{1}{4}\ell n\left(\frac{\ell n\, x - x}{\ell n\, x + x}\right) - \frac{1}{2}\tan^{-1}\left(\frac{\ell n\, x}{x}\right) + C$

(C) $\frac{1}{4}\ell n\left(\frac{\ell n\, x + x}{\ell n\, x - x}\right) + \frac{1}{2}\tan^{-1}\left(\frac{\ell n\, x}{x}\right) + C$

(D) $\frac{1}{4}\left(\ell n\left(\frac{\ell n\, x - x}{\ell n\, x + x}\right) + \tan^{-1}\left(\frac{\ell n\, x}{x}\right)\right) + C$

Ans. B

Sol. $I = \int \frac{x^2(1-\ell n\, x)}{x^4\left(\left(\frac{\ell n\, x}{x}\right)^4 - 1\right)} dx = \int \frac{1-\ell n\, x}{x^2\left(\left(\frac{\ell n\, x}{x}\right)^4 - 1\right)} dx$

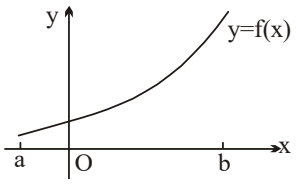
put $\frac{\ell n\, x}{x} = t \qquad \Rightarrow \quad \frac{1-\ell n\, x}{x^2} = dt$

$I = \int \frac{dt}{(t^4-1)} = \int \frac{dt}{(t^2+1)(t^2-1)} = \frac{1}{2}\int \frac{(t^2+1)-(t^2-1)}{(t^2+1)(t^2-1)} dt$

$I = \frac{1}{2}\left(\int \frac{dt}{t^2-1} - \int \frac{dt}{t^2+1}\right) = 7\frac{1}{2}\left(\frac{1}{2}\ell n\, \frac{t-1}{t+1} - \tan^{-1} t\right)$

$= \frac{1}{4}\ell n\left(\frac{\ell n\, x - x}{\ell n\, x + x}\right) - \frac{1}{2}\tan^{-1}\left(\frac{\ell n\, x}{x}\right) + C$

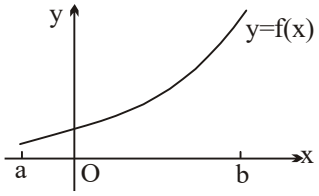
26. The graph of the function $y = f(x)$, where $f : [a, b] \to R$ and a is a negative constant and b a positive constant, is shown. The function $F : [a, b] \to R$ is defined by $F(t) = \int_0^t f(x)\, dx$. Then $F(t) > 0$ for :



(A) $t \in [a, b]\backslash\{0\}$ (B) $t \in [a, 0)$ only

(C) $t \in [0, b]$ only (D) $t \in (0, b]$ only

वक्र $y = f(x)$, जहाँ $f : [a, b] \to R$ एवं a एक ऋणात्मक नियतांक है तथा b एक धनात्मक नियतांक है। फलन $F : [a, b] \to R$ परिभाषित है $F(t) = \int_0^t f(x)\, dx$ हो तो $F(t) > 0$ के लिये :



(A) $t \in [a, b]\backslash\{0\}$ (B) केवल $t \in [a, 0)$

(C) केवल $t \in [0, b]$ (D) केवल $t \in (0, b]$

Ans. D

Sol. for negative values of t

(A) will not hold good for (0, a) and will be – ve

(B) will not be correct as $F(t) < 0$.

(C) at $t = 0$, $F(t) = 0 \Rightarrow$ (D) is correct

27. If $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} = 0$, where C_0, C_1, C_2 are all real, the equation $C_2x^2 + C_1x + C_0 = 0$ has :

(A) atleast one root in $(0, 1)$

(B) one root in $(1, 2)$ and other in $(3, 4)$

(C) one root in $(-1, 1)$ & the other in $(-5, -2)$

(D) both roots imaginary

यदि $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} = 0$, जहाँ C_0, C_1, C_2 वास्तविक संख्याऐं हो, तो समीकरण $C_2x^2 + C_1x + C_0 = 0$ का मान :

(A) कम से कम एकमूल $(0, 1)$ में होगा

(B) एक मूल $(1, 2)$ में तथा दूसरा $(3, 4)$ में होगा

(C) एक मूल $(-1, 1)$ में तथा दूसरा $(-5, -2)$ में होगा

(D) दोनों मूल काल्पनिक है

Ans. A

Sol. $\frac{C_0}{1}+\frac{C_1}{2}+\frac{C_2}{3}=0; C_1, C_2, C_3 \in R$

$C_2x^2+C_1x+C_0=0$

$\Rightarrow \int C_2x^2dx+\int C_1xdx+\int C_0dx=C$

$\Rightarrow \frac{C_2x^3}{3}+\frac{C_1x^2}{2}+C_0x=0$

let $f(x)=C_0x+\frac{C_1x^2}{2}+\frac{C_2x^3}{3}$

$\Rightarrow f(0)=0$

$f(1)=0$

& f(x) is continuous in [0,1]

& diff. in (0,1)

$\Rightarrow$ By Rolle's Theorem $\exists$ at least one real in (0,1) such that $f'(c)=0$

$\Rightarrow C_2x^2+C_1x+C_0=0$ have at least one real root in (0,1).

28. Let $I=\int \frac{e^x}{e^{4x}+e^{2x}+1}dx$, $J=\int \frac{e^{-x}}{e^{-4x}+e^{-2x}+1}dx$. Then, for an arbitrary constant C, the value of J – I equals: **[JEE-2008]**

माना $I=\int \frac{e^x}{e^{4x}+e^{2x}+1}dx$, $J=\int \frac{e^{-x}}{e^{-4x}+e^{-2x}+1}dx$ तो C अचर है तो J – I का मान होगा : **[JEE-2008]**

(A) $\frac{1}{2}ln\left(\frac{e^{4x}-e^{2x}+1}{e^{4x}+e^{2x}+1}\right)+C$

(B) $\frac{1}{2}ln\left(\frac{e^{2x}+e^{x}+1}{e^{2x}-e^{x}+1}\right)+C$

(C) $\frac{1}{2}ln\left(\frac{e^{2x}-e^{x}+1}{e^{2x}+e^{x}+1}\right)+C$

(D) $\frac{1}{2}ln\left(\frac{e^{4x}+e^{2x}+1}{e^{4x}-e^{2x}+1}\right)+C$

Ans. C

Sol. $J - I = \int\left(\frac{e^{-x}}{e^{-4x}+e^{-2x}+1}-\frac{e^{x}}{e^{4x}+e^{2x}+1}\right)dx$

$= \int\frac{e^{3x}-e^{x}}{e^{4x}+e^{2x}+1}dx = \int\frac{e^{x}(e^{2x}-1)}{e^{4x}+e^{2x}+1}dx$

Let $e^x = t$ $\qquad$ $e^x dx = dt$

$= \int\frac{t^2-1}{t^4+t^2+1}dt = \int\frac{1-1/t^2}{(t+1/t)^2-1}dt$

$= \frac{1}{2}\ell n\left|\frac{t+\frac{1}{t}-1}{t+\frac{1}{t}+1}\right| + C = \frac{1}{2}\ell n\left(\frac{e^{2x}-e^{x}+1}{e^{2x}+e^{x}+1}\right)+C$

DEFINITE INTEGRATION WITH THE HELP OF INDEFINITE INTEGRATION

29. If $g(x) = \int_0^x \cos^4 t\,dt$, then $g(x+\pi)$ equals :

यदि $g(x) = \int_0^x \cos^4 t\,dt$, तो $g(x+\pi)$ का मान है :

(A) $g(x)+g(\pi)$ (B) $g(x)-g(\pi)$ (C) $g(x)\,g(\pi)$ (D) $[g(x)/g(\pi)]$

Ans. A

Sol. $g(x+\pi) = \int_0^{x+\pi}\cos^4 t\,dt = \int_0^x \cos^4 t\,dt + \int_x^{x+\pi}\cos^4 t\,dt = g(x) + \int_0^{\pi}\cos^4 t\,dt = g(x)+g(\pi)$ **Ans.**

30. If $g(x) = \int_1^x e^{t^2}\,dt$ then the value of $\int_3^{x^3} e^{t^2}\,dt$ equals :

यदि $g(x) = \int_1^x e^{t^2}\,dt$ तो $\int_3^{x^3} e^{t^2}\,dt$ का मान है :

(A) $g(x^3)-g(3)$ (B) $g(x^3)+g(3)$ (C) $g(x^3)-3$ (D) $g(x^3)-3g(x)$

Ans. A

Sol. $\int_3^{x^3} e^{t^2}\,dt = \int_3^1 e^{t^2}\,dt + \int_1^{x^3} e^{t^2}\,dt = \int_1^{x^3} e^{t^2}\,dt - \int_1^3 e^{t^2}\,dt = g(x^3)-g(3)$

31. Given $\int_0^{\pi/2} \frac{dx}{1+\sin x+\cos x} = l\text{n } 2$, then the value of the def. integral. $\int_0^{\pi/2} \frac{\sin x}{1+\sin x+\cos x}\,dx$ is equal to :

दिया गया है $\int_0^{\pi/2} \frac{dx}{1+\sin x+\cos x} = l\text{n } 2$, तो $\int_0^{\pi/2} \frac{\sin x}{1+\sin x+\cos x}\,dx$ का मान है :

(A) $\frac{1}{2} l\text{n } 2$ (B) $\frac{\pi}{2} - l\text{n } 2$ (C) $\frac{\pi}{4} - \frac{1}{2} l\text{n } 2$ (D) $\frac{\pi}{2} + l\text{n } 2$

Ans. C

Sol. Let $I = \int_0^{\pi/2} \frac{\sin x\ dx}{1+\sin x+\cos x}$

$I = \int_0^{\pi/2} \frac{\cos x}{1+\sin x+\cos x} \Rightarrow 2I = \int_0^{\pi/2} \frac{\sin x+\cos x+1-1}{\sin x+\cos x+1}\,dx$

$\Rightarrow 2I = \frac{\pi}{2} - \ln 2 \Rightarrow I = \frac{\pi}{4} - \frac{1}{2} \ln 2$

32. Let $\frac{d}{dx} F(x) = \left(\frac{e^{\sin x}}{x}\right)$, $x > 0$. If $\int_1^4 \frac{3}{x} e^{\sin x^3}\,dx = F(k) - F(1)$, then one of the possible values of k, is :

मानाकि $\frac{d}{dx} F(x) = \left(\frac{e^{\sin x}}{x}\right)$, $x > 0$ है यदि $\int_1^4 \frac{3}{x} e^{\sin x^3}\,dx = F(k) - F(1)$, तो k का एक संभव मान होगा :

(A) 64 (B) 15 (C) 16 (D) 63

Ans. A

Sol. $\frac{d}{dx} F(x) = \frac{e^{\sin x}}{x}$ $x > 0$

$\Rightarrow \int_1^4 \frac{3}{x} e^{\sin x^3} dx$

let $x^3 = t$

$3x^2 dx = dt$

$= \int_1^{64} \frac{1}{t} e^{\sin t} dt = \left[F(t)\right]_1^{64} = F(64) - F(1)$

$= \int_1^{64} \frac{1}{t} e^{\sin t} dt = \left[F(t)\right]_1^{64} = F(64) - F(1)$

33. Let $F(x) = f(x) + f\left(\frac{1}{x}\right)$, where $f(x) = \int_1^x \frac{\log t}{1+t}dt$. Then F(e) equals :

माना $F(x) = f(x) + f\left(\frac{1}{x}\right)$, जहाँ $f(x) = \int_1^x \frac{\log t}{1+t}dt$, तब F(e) बराबर है :

(A) ½ (B) 0 (C) 1 (D) 2

Ans. A

Sol. $F(x) = f(x)+f(1/x)$ put $x = e$

$$F(e) = \int_1^e \frac{\log t}{1+t}dt + \int_1^{1/e} \frac{\log t}{1+t}dt$$

$$\text{let } t = \frac{1}{z} \Rightarrow \frac{dt}{dz} = \left(\frac{-1}{z^2}\right)$$

$$= \int_1^e \frac{\ln t}{(1+t)}dt + \int_1^e \frac{\ln 1/z}{(1+1/z)}\left(\frac{-1}{z^2}\right)dz$$

$$\text{by property } \int_a^b f(x)dx = \int_a^b f(t)dt$$

$$\int_1^e \frac{\ln t}{(1+t)}dt + \int_1^e \frac{\ln t}{t(1+t)}dt = \int_1^e \frac{\ln t}{t}dt = \frac{1}{2}$$

34. The value of $\int_0^1 \left(\prod_{r=1}^{n}(x+r)\right)\left(\sum_{k=1}^{n}\frac{1}{x+k}\right)dx$ equals :

$\int_0^1 \left(\prod_{r=1}^{n}(x+r)\right)\left(\sum_{k=1}^{n}\frac{1}{x+k}\right)dx$ का मान होगा :

(A) n (B) n ! (C) (n + 1) ! (D) n · n !

Ans. D

Sol. The given integrand is perfect differential coeff. of $\prod_{r=1}^{n}(x+r)$

$$\Rightarrow I = \left[\prod_{r=1}^{n}(x+r)\right]_0^1 = (n+1)! - n! = n \cdot n!$$

Alternatively: $(x + 1)(x + 2)(x + 3).....(x + n) = t$

so that when $x = 1,\ t = (n+1)!$

if $x = 0$ then $t = n!$

$$[\ell n\,(x+1) + \ell n\,(x+2) + \ldots\ldots + \ell n\,(x+n)] = \ell n\, t$$

$$\therefore \quad \left(\frac{1}{(x+1)} + \frac{1}{(x+2)} + \ldots\ldots + \frac{1}{(x+n)}\right) dx = \frac{1}{t}\,dt \Rightarrow \quad I = \int_{n!}^{(n+1)!} \frac{1}{t}\,t\,dt$$

Hence $I = \int_{\ell n\, n!}^{\ell n\,(n+1)!} e^t dt = e^t \Big|_{\ell n\, n!}^{\ell n\,(n+1)!} = e^{\ell n\,(n+1)!} - e^{\ell n\, n!} = (n+1)! - n! = n \cdot n!$ Ans.

35. The value(s) of $\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx$ is(are) : **[JEE-2010]**

$\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx$ का मान होगा : **[JEE-2010]**

(A) $\frac{22}{7} - \pi$ (B) $\frac{2}{105}$ (C) 0 (D) $\frac{71}{15} - \frac{3\pi}{2}$

Ans. A

Sol. $$I = \int_0^1 \frac{x^4\left(1-2x+x^2\right)^2}{1+x^2}\,dx$$

$$I = \int_0^1 \frac{x^4\left\{\left(1+x^2\right)^2 - 4x\left(1+x^2\right) + 4x^2\right\}}{1+x^2}\,dx$$

$$= \int_0^1 \left(1+x^2\right)x^4\,dx - \int_0^1 4x^5\,dx + 4\int_0^1 \frac{\left(x^6+1\right)-1}{1+x^2}\,dx$$

$$= \frac{1}{5} + \frac{1}{7} - 4.\frac{1}{6} + 4\int_0^1 \frac{\left(x^2+1\right)^3 - 3x^2\left(1+x^2\right)}{1+x^2}\,dx - 4\int_0^1 \frac{dx}{1+x^2}$$

$$= \frac{12}{35} - \frac{2}{3} + 4\int_0^1 \left(x^4 + 2x^2 + 1\right)dx - 12\int_0^1 x^2\,dx - \pi$$

$$= \frac{12}{35} - \frac{2}{3} + 4\left(\frac{1}{5} + \frac{2}{3} + 1\right) - 4 - \pi$$

$$= \frac{12}{35} - \frac{2}{3} + \frac{52}{15} - \pi = \frac{22}{7} - \pi$$

PROPERTY FOR DISCONTINUOUS NON UNIFORM FUNCTION

36. $\int_{-1/2}^{1/2}\left[\left(\frac{x+1}{x-1}\right)^2+\left(\frac{x-1}{x+1}\right)^2-2\right]^{1/2}dx$ has the value equal to :

$\int_{-1/2}^{1/2}\left[\left(\frac{x+1}{x-1}\right)^2+\left(\frac{x-1}{x+1}\right)^2-2\right]^{1/2}dx$ का मान होगा :

(A) 4 (B) $4\ell n\frac{3}{4}$ (C) $3\ell n\frac{3}{4}$ (D) $4\ell n\frac{4}{3}$

Ans. D

Sol. $\int_{-1/2}^{1/2}\left|\frac{x+1}{x-1}-\frac{x-1}{x+1}\right|dx$

$$=\int_{-\frac{1}{2}}^{\frac{1}{2}}\left|\frac{(x+)-(x-)^2}{x^2-1}\right|dx$$

$$=\int_{-\frac{1}{2}}^{\frac{1}{2}}\left|\frac{4x}{x^2-1}\right|dx$$

$$=-\int_{-\frac{1}{2}}^{0}\frac{4x}{x^2-1}+\int_{0}^{\frac{1}{2}}\frac{4x}{x^2-1}dx$$

$$=4\ell n\frac{4}{3}$$

37. Let [.] denote the greatest integet function then the value of $\int_0^{1.5}x\left[x^2\right]dx$ is : **[AIEEE-2011]**

माना [.] एक महत्तम पूर्णांक फलन दर्शाता है, तो $\int_0^{1.5}x\left[x^2\right]dx$ का मान है : **[AIEEE-2011]**

(A) $\frac{5}{4}$ (B) 0 (C) $\frac{3}{2}$ (D) $\frac{3}{4}$

Ans. D

Sol. $\int_0^{1.5}x[x^2]dx$

$$\int_0^1 0\,dx+\int_1^{\sqrt{2}}x\,dx+\int_{\sqrt{2}}^{1.5}2x\,dx$$

$$\left[\frac{x^2}{2}\right]_1^{\sqrt{2}} + \left[x^2\right]_{\sqrt{2}}^{1.5}$$

$$\left(\frac{2}{2} - \frac{1}{2}\right) + (2.25 - 2)$$

$$\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

38. The value of $\int_1^a$ [x] f′ (x) dx, a > 1, where [x] denotes the greatest integer not exceeding x is :

$\int_1^a$ [x] f′ (x) dx, a > 1, जहाँ [x] महत्तम पूर्णांक संख्या है जो x से बड़ी नही है, का मान है :

(A) [a] f(a) – {f(1) + f(2) + ... + f([a])} (B) [a] f([a]) – {f(1) + f(2) + ... + f(a)}
(C) a f([a]) – {f(1) + f(2) + ... + f(a)} (D) a f(a) – {f(1) + f(2) + ... + f([a])}

Ans. A

Sol. $\int_1^2 1.f'(x)dx + \int_2^3 2.f'(x)dx + ... + \int_{[a]}^a [a]f'(x)dx$

= [f(2) – f(1)] + 2[f(3) – f(2)] + ... + [a] [f(a)]
= [a] f(a) – {f(1) + f(2) + ... + f[a]}

39. The intergal $\int_0^{\pi} \sqrt{1+4\sin^2\frac{x}{2}-4\sin\frac{x}{2}}dx$ equals : **[JEE(Main)2014]**

समाकलन $\int_0^{\pi} \sqrt{1+4\sin^2\frac{x}{2}-4\sin\frac{x}{2}}dx$ बराबर है : **[JEE(Main)2014]**

(A) $4\sqrt{3}-4$ (B) $4\sqrt{3}-4-\frac{\pi}{3}$ (C) $\pi-4$ (D) $\frac{2\pi}{3}-4-4\sqrt{3}$

Ans. B

Sol. $I = \int_0^{\pi} \sqrt{1+4\sin^2\frac{\pi}{2}-4\sin\frac{x}{2}}dx$

$$\Rightarrow I = \int_0^{\pi} \sqrt{\left(2\sin\frac{x}{2}-1\right)^2}\,dx$$

$$\Rightarrow I = \int_0^{\pi} \left|2\sin\frac{x}{2}-1\right|dx$$

$$\Rightarrow I = \int_0^{\pi/3}\left(1-2\sin\frac{x}{2}\right)dx + \int_{\pi/3}^{\pi}\left(2\sin\frac{x}{2}-1\right)dx$$

$$= x + 4\cos\frac{x}{2}\Big|_0^{\pi} + \left[-\left(4\cos\frac{x}{2}+x\right)\right]_{\pi/3}^{\pi}$$

$$= \left[\left(\frac{\pi}{3}+4\frac{\sqrt{3}}{2}\right)-(0+4)\right]-\left[(0+\pi)-\left(4\frac{\sqrt{3}}{2}+\frac{\pi}{3}\right)\right]$$

$$= 4\sqrt{3}-4-\frac{\pi}{3}$$

PROPERTY FOR ODD AND EVEN FUNCTIONS

40. The integral $\int_{-\frac{1}{2}}^{\frac{1}{2}}\left([x]+\ell n\left(\frac{1+x}{1-x}\right)\right)dx$ equals, where [ ] denotes greatest integer function :

समाकलन $\int_{-\frac{1}{2}}^{\frac{1}{2}}\left([x]+\ell n\left(\frac{1+x}{1-x}\right)\right)dx$ का मान होगा, जहाँ [ ] महत्तम पूर्णांक फलन है।

(A) $-\frac{1}{2}$ (B) 0 (C) 1 (D) $2\ell n\left(\frac{1}{2}\right)$

Ans. A

Sol. $\int_{-\frac{1}{2}}^{\frac{1}{2}}\left([x]+\ell n\left(\frac{1+x}{1-x}\right)\right)dx$

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}}[x]dx + \underbrace{\int_{-\frac{1}{2}}^{\frac{1}{2}}\ell n\left(\frac{1+x}{1-x}\right)dx}_{\text{Odd Function}}$$

$$= \int_{-\frac{1}{2}}^{0}[x]dx + \int_{0}^{\frac{1}{2}}[x]dx + 0$$

$$= -[x]_{-\frac{1}{2}}^{0} + 0 = -\frac{1}{2}$$

41. $\int_{-1}^{1} \frac{x^3+|x|+1}{x^2+2|x|+1}dx = a\,ln\,2 + b$ then :

$\int_{-1}^{1} \frac{x^3+|x|+1}{x^2+2|x|+1}dx = a\,ln\,2 + b$ तो :

(A) a = 2 ; b = 1 (B) a = 2 ; b = 0 (C) a = 3 ; b = –2 (D) a = 4 ; b = – 1

Ans. B

Sol. $I = \int_{-1}^{1} \frac{x^3}{x^2+2|x|+1}\,dx + \int_{-1}^{1} \frac{|x|+1}{(|x|+1)^2}\,dx = 2\int_{0}^{1} \frac{dx}{1+x} = 2\,ln\,2$

42. The value of $\int_{-2}^{3} |1-x^2|\,dx$ is :

$\int_{-2}^{3} |1-x^2|\,dx$ का मान है :

(A) 28/3 (B) 14/3 (C) 7/3 (D) 1/3

Ans. A

Sol. $\int_{-2}^{3} |1-x^2|dx$ $\qquad |1-x^2| = \begin{cases} x^2-1 & x<-1 \\ 1-x^2 & -1\le x\le 1 \\ x^2-1 & x>-1 \end{cases}$

$= \int_{-2}^{-1} (x^2-1)dx + \int_{-1}^{1} (1-x^2)dx + \int_{1}^{3} (x^2-1)dx$

$= \left[\frac{x^3}{3}-x\right]_{-2}^{-1} + \left[x-\frac{x^3}{3}\right]_{-1}^{1} + \left[\frac{x^3}{3}-x\right]_{1}^{3}$

$= -\frac{1}{3}+3+\frac{8}{3}-2+\frac{2}{3}+\frac{2}{3}+9-3-\frac{1}{3}+1$

$= \frac{10}{3}+6 = \frac{28}{3}$

43. $\int_{-3\pi/2}^{-\pi/2} [(x+\pi)^3 + \cos^2(x+3\pi)]\,dx$ is equal to :

$\int_{-3\pi/2}^{-\pi/2} [(x+\pi)^3 + \cos^2(x+3\pi)]\,dx$ बराबर है :

(A) $(\pi^4/32) + (\pi/2)$ (B) $\pi/2$ (C) $(\pi/4) - 1$ (D) $\pi^4/32$

Ans. B

Sol. $\int_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}} \left((x+\pi)^3 + \cos^2(x+3\pi)\right)dx$

$$= \int_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}} (x+\pi)^3\,dx + \int_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}} \cos^2 x\,dx$$

$$= \left[\frac{(x+\pi)^4}{4}\right]_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}} + \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}}$$

$$= \frac{1}{4}\left[\left(\frac{\pi}{2}\right)^4 - \left(\frac{\pi}{2}\right)^4\right] + \frac{1}{2}\left[\pi + \frac{1}{2}(0-0)\right] = \frac{\pi}{2}$$

KING & QUEEN PROPERTY

44. If $I = \int_0^{\pi/2} \ell n(\sin x)\,dx$ then $\int_{-\pi/4}^{\pi/4} \ell n(\sin x + \cos x)\,dx =$

यदि $I = \int_0^{\pi/2} \ell n(\sin x)\,dx$, तो $\int_{-\pi/4}^{\pi/4} \ell n(\sin x + \cos x)\,dx =$

(A) $\frac{I}{2}$ (B) $\frac{I}{4}$ (C) $\frac{I}{\sqrt{2}}$ (D) I

Ans. A

Sol. $I_1 = \int_{-\pi/4}^{\pi/4} \ln(\sin x + \cos x)\,dx = \int_{-\pi/4}^{\pi/4} \ln(\cos x - \sin x)\,dx$ (using king)

$\Rightarrow 2I_1 = \int_{-\pi/4}^{\pi/4} \ln\cos 2x\,dx = 2\int_0^{\pi/4} \ln(\cos 2x) = \int_0^{\pi/2} \ln(\cos t)\,dt$ where $2x = t$

$\int_0^{\pi/2} \ln(\sin t)\,dt = I \Rightarrow I_1 = I/2$

45. The value of the definite integral $\int_0^{\pi/2} \sin x \sin 2x \sin 3x \, dx$ is equal to :

निश्चित समाकलन $\int_0^{\pi/2} \sin x \sin 2x \sin 3x \, dx$ का मान है :

(A) $\frac{1}{3}$ (B) $-\frac{2}{3}$ (C) $-\frac{1}{3}$ (D) $\frac{1}{6}$

Ans. D

Sol. $I = \int_0^{\pi/2} \sin x \sin 2x \sin 3x \, dx$

Apply king property

$I = -\int_0^{\pi/2} \cos x \sin 2x \cos 3x \, dx$

$\Rightarrow 2I = \int_0^{\pi/2} \sin 2x \cos 4x \, dx$

$I = \frac{1}{4}\int_0^{\pi/2} (\sin 6x - \sin 2x) \, dx$

$I = \frac{1}{6}$

46. The value of $\int_{\pi}^{2\pi} [2\cos x] dx$ where $[\cdot]$ represents the greatest integer function is :

(A) $-\frac{5\pi}{6}$ (B) $-\frac{\pi}{2}$ (C) □ π (D) none

$\int_{\pi}^{2\pi} [2\cos x] dx$ का मान होगा जहाँ $[\cdot]$ महत्तम पूर्णांक फलन है :

(A) $-\frac{5\pi}{6}$ (B) $-\frac{\pi}{2}$ (C) □ π (D) इनमें से कोई नहीं

Ans. B

Sol. $\int_{\pi}^{2\pi} [2\cos x] dx$

If $x \in (\pi, 2\pi)$ then $\cos x \in [-1,1]$

$\Rightarrow 2\cos x \in [-2,2]$.

$$= \int_{\pi}^{\pi+\frac{\pi}{3}} -2dx + \int_{\pi+\frac{\pi}{3}}^{\frac{3\pi}{2}} -1dx + \int_{\frac{3\pi}{2}}^{\frac{5\pi}{3}} 0dx + \int_{\frac{5\pi}{3}}^{2\pi} 1\cdot dx$$

$$= -\frac{2\pi}{3} - \frac{\pi}{6} + \frac{\pi}{3}$$

$$= -\frac{2\pi}{3} + \frac{\pi}{6} = \frac{-4\pi+\pi}{6} = -\frac{-\pi}{2}$$

47. If $f(a+b-x) = f(x)$, then $\int_a^b x\, f(x)\, dx$ is equal to :

यदि $f(a+b-x) = f(x)$, तो $\int_a^b x\, f(x)\, dx$ बराबर है :

(A) $\frac{a+b}{2}\int_b^a f(a+b-x)\, dx$ (B) $\frac{a+b}{2}\int_a^b f(b-x)\, dx$

(C) $\frac{a+b}{2}\int_a^b f(x)\, dx$ (D) $\frac{b-a}{2}\int_a^b f(x)\, dx$

Ans. C

Sol. $\int_a^b xf(x)dx = I$

Apply King Property

$$I = \int_a^b (a+b-x) f(x) dx$$

$$\Rightarrow 2I = (a+b)\int_a^b f(x)dx$$

$$\Rightarrow I = \frac{a+b}{2}\int_a^b f(x)dx = \frac{a+b}{2}\int_a^b f(a+b-x)dx$$

48. If $f(y) = e^y$, $g(y) = y$; $y > 0$ and $F(t) = \int_0^t f(t-y)\, g(y)\, dy$, then :

यदि $f(y) = e^y$, $g(y) = y$; $y > 0$ एवं $F(t) = \int_0^t f(t-y)\, g(y)\, dy$, तो :

(A) $F(t) = te^{-t}$ (B) $F(t) = 1 - e^{-1}(1+t)$

(C) $F(t) = e^t - (1+t)$ (D) $F(t) = te^t$

Ans. C

So. $f(y) = e^y, g(y) = y\ ; y > 0$

and $F(t) = \int_0^t f(t-y)g(y)dy$.

$= \int_0^t e^{t-y}ydy = e^t\int_0^t e^{-y}ydy = e^t[-ye^{-y} - e^{-y}]_0^t$

$= -e^t[te^{-t} + e^{-t} - 0 - 1] = e^t - (1+t)$

49. If $f(x) = \frac{e^x}{1+e^x}$, $I_1 = \int_{f(-a)}^{f(a)} xg\{x(1-x)\}\,dx$ and $I_2 = \int_{f(-a)}^{f(a)} g\{x(1-x)\}\,dx$, then the value of $\frac{I_2}{I_1}$ is :

यदि $f(x) = \frac{e^x}{1+e^x}$, $I_1 = \int_{f(-a)}^{f(a)} xg\{x(1-x)\}\,dx$ एवं $I_2 = \int_{f(-a)}^{f(a)} g\{x(1-x)\}\,dx$, तो $\frac{I_2}{I_1}$ का मान है :

(A) 2 (B) – 3 (C) – 1 (D) 1

Ans. A

Sol. $f(x) = \frac{e^x}{1+e^x}$ $I_1 = \int_{f(-a)}^{f(a)} xg[x(1-x)]dx$

$I_2 = \int_{f(-a)}^{f(a)} g[x(1-x)]dx$

$f(a) = \frac{e^a}{1+e^a}$, $f(-a) = \frac{e^{-a}}{1+e^{-a}}$

$\therefore\ f(a) + f(-a) = 1$

$2I_1 = \int_{f(-a)}^{f(a)} xg\{x(1-x)\}dx + \int_{f(-a)}^{f(a)} \{f(a)+f(-a)-x\}g(1-x)(x)dx$

$2I_1 = \int_{f(-a)}^{f(a)} g\{x(1-x)\}dx = I_2$

$\because\ f(a) + f(-a) = 1$

$2I_1 = I_2$

$\frac{I_2}{I_1} = 2$

50. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x}\,dx$, $a>0$, is :

$\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x}\,dx$, $a > 0$, का मान है :

(A) $\alpha\pi$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{a}$ (D) 2π

Ans. B

Sol. Putting $-x$ for x

$$I = \int_{\pi}^{-\pi} \frac{\cos^2 x}{1+a^{-x}}(-dx) = \int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^{-x}}dx$$

$$I + I = \int_{-\pi}^{\pi} \cos^2 x\left(\frac{1}{1+a^x}+\frac{1}{1+a^{-x}}\right)dx$$

$$= \int_{-\pi}^{\pi} \cos^2 x dx \Rightarrow 2I = 2\int_{0}^{\pi} \cos^2 x dx$$

$$= \int_{0}^{\pi} (1+\cos 2x)dx$$

$$2I = \left[x+\frac{\sin 2x}{2}\right]_0^{\pi}$$

$$2I = \pi \Rightarrow I = \frac{\pi}{2}$$

51. The value of the integral, $\int_{3}^{6} \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}}dx$ is :

समाकल $\int_{3}^{6} \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}}\,dx$ का मान है :

(A) $\frac{3}{2}$ (B) 2 (C) 1 (D) $\frac{1}{2}$

Ans. A

Sol. $\int_{3}^{6} \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}} = I$

Apply King Property

$$I=\int_3^6 \frac{\sqrt{9-x}}{\sqrt{x}+\sqrt{9-x}}dx$$

$$\Rightarrow 2I=\int_3^6 dx \Rightarrow I=\frac{3}{2}$$

52. The value of $I=\int_0^{\pi/2}\frac{(\sin x+\cos x)^2}{\sqrt{1+\sin 2x}}dx$ is :

$I=\int_0^{\pi/2}\frac{(\sin x+\cos x)^2}{\sqrt{1+\sin 2x}}dx$ का मान है :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $$I=\int_0^{\frac{\pi}{2}}\frac{(\sin x+\cos x)^2}{\sqrt{1+\sin 2x}}dx$$

$$\Rightarrow I=\int_0^{\frac{\pi}{2}}\frac{(\sin x+\cos x)^2}{|\sin x+\cos x|}dx$$

$$\Rightarrow I=\int_0^{\frac{\pi}{2}}(\sin x+\cos x)dx$$

$$\Rightarrow I=\left[-\cos x+\sin x\right]_0^{\frac{\pi}{2}}$$

$$\Rightarrow I=2$$

53. If $\int_0^{\pi} x\, f(\sin x)\,dx=A\int_0^{\pi/2} f(\sin x)\,dx$, then A is :

यदि $\int_0^{\pi} x\, f(\sin x)\,dx=A\int_0^{\pi/2} f(\sin x)\,dx$, तो A का मान है :

(A) 0 (B) π (C) $\pi/4$ (D) 2π

Ans. B

Sol. $$\int_0^{\pi} xf(\sin x)=I(\text{let})$$

Apply King Property

$$I=\int_0^{\pi}(\pi-x)f\left(\sin(\pi-x)\right)dx$$

$$=\int_0^{\pi}(\pi-x)f\left(\sin x\right)dx$$

$$\Rightarrow 2I=\pi\int_0^{\pi}f\left(\sin x\right)dx$$

$$\Rightarrow I=\frac{\pi}{2}\int_0^{\pi}f\left(\sin x\right)dx$$

Apply queen property

$$\Rightarrow I=\frac{\pi}{2}\times 2\int_0^{\frac{\pi}{2}}f\left(\sin x\right)dx$$

$$\Rightarrow A=\pi$$

54. $\int_{-\pi}^{\pi}\frac{2x(1+\sin x)}{1+\cos^2 x}=$

(A) π^2 (B) $\pi^2/4$ (C) $\pi/8$ (D) $\pi^2/8$

Ans. A

Sol. $$I=\int_{-\pi}^{\pi}\frac{2x\left(1+\sin x\right)}{1+\cos^2 x}dx$$

$$=\int_{-\pi}^{\pi}\underbrace{\frac{2x}{1+\cos^2 x}}_{\text{Odd Function}}dx+\int_{-\pi}^{\pi}\frac{2x\sin x}{1+\cos^2 x}dx$$

$$=\int_{-\pi}^{\pi}\frac{2x\sin x}{1+\cos^2 x}dx$$

Apply King Property

$$I=\int_{-\pi}^{\pi}\frac{2(\pi-x)}{1+\cos^2 x}\sin x dx$$

$$\Rightarrow 2I=2\pi\int_{-\pi}^{\pi}\frac{\sin x}{1+\cos^2 x}dx$$

Apply Queen Property

$$\Rightarrow 2I = 4\pi \int_{-\pi}^{\pi} \frac{\sin x}{1+\cos^2 x} dx$$

$$= -4\pi\left[\tan^{-1}(\cos x)\right]_0^{\pi} = -4\pi\left[-\frac{\pi}{2}\right] = 2\pi^2$$

$$\Rightarrow I = \pi^2$$

55. The integral $\int_2^4 \frac{\log x^2}{\log x^2 + \log(36-12x+x^2)} dx$ is equal to : **[JEE(Main)2015]**

समाकल $\int_2^4 \frac{\log x^2}{\log x^2 + \log(36-12x+x^2)} dx$ बराबर है : **[JEE(Main)2015]**

(A) 2 (B) 4 (C) 1 (D) 6

Ans. C

Sol. $$I = \int_2^4 \frac{\ln(x^2)}{\ln(x^2) + \ln(6-x)^2} dx$$

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

$$I = \int_2^4 \frac{\ln(6-x)^2}{\ln(6-x)^2 + \ln(x)^2} dx$$

$$2I = \int_2^4 dx \Rightarrow 2I = 2 \Rightarrow I = 1$$

56. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+2^x} dx$ is : **[JEE(Main)2018]**

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+2^x} dx$ का मान है : **[JEE(Main)2018]**

(A) 4π (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. $$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+2^x} dx$$

Use king & Add

$$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x\,dx$$

$$= 2\int_{0}^{\frac{\pi}{2}} \sin^2 x\,dx$$

$$I = \frac{\pi}{4}$$

57. The value of $\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ln 6 - x^2)}\,dx$ is : **[JEE-2011]**

$\int_{\sqrt{\ln 2}}^{\sqrt{\ln 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ln 6 - x^2)}\,dx$ का मान होगा : **[JEE-2011]**

(A) $\frac{1}{4}\ln\frac{3}{2}$ (B) $\frac{1}{2}\ln\frac{3}{2}$ (C) $\ln\frac{3}{2}$ (D) $\frac{1}{6}\ln\frac{3}{2}$

Ans. A

Sol. $I = \int_{\sqrt{\ell n 2}}^{\sqrt{\ell n 3}} \frac{x \sin x^2}{\sin x^2 + \sin(\ell n 6 - x^2)}\,dx$; put $x^2 = t$

$\Rightarrow 2xdx = dt$

$$\Rightarrow \quad I = \frac{1}{2}\int_{\ell n 2}^{\ell n 3} \frac{\sin t}{\sin t + \sin(\ell n 6 - t)}\,dt \qquad \text{.....(i)}$$

$$\Rightarrow \quad I = \frac{1}{2}\int_{\ell n 2}^{\ell n 3} \frac{\sin(\ell n 6 - t)}{\sin(\ell n 6 - t) + \sin t}\,dt \qquad \text{.....(ii)}$$

Adding equation (i) & (ii)

$$\Rightarrow \quad 2I = \frac{1}{2}\int_{\ell n 2}^{\ell n 3} dt \qquad \Rightarrow \quad I = \frac{1}{4}\ell n\left(\frac{3}{2}\right)$$

PROPERTY FOR PERIODIC FUCTIONS

58. Let T > 0 be a fixed real number. Suppose f is a continuous function such that for all $x \in R$ $f(x + T) = f(x)$. If $I = \int_0^T f(x)\,dx$ then the value of $\int_3^{3+3T} f(2x)\,dx$ is :

माना T > 0 एक निश्चित वास्तविक संख्या है तथा x के प्रत्येक मान के लिये f एक सतत फलन है तथा $f(x+T)=f(x)$. यदि $I=\int_0^T f(x)\,dx$ तो $\int_3^{3+3T} f(2x)\,dx$ का मान होगा :

(A) $\frac{3}{2}I$ (B) 2 I (C) 3 I (D) 6 I

Ans. C

Sol. $I=\int_0^T f(x)\,dx \qquad \left\{\because \int_a^{a+nT} f(x)\,dx = n\int_0^{nT} f(x)\,dx; n\in Z, a\in R\right\}$

$$\Rightarrow \int_3^{3+3T} f(2x)\,dx = \int_0^{3T} f(2x)\,dx = \frac{1}{2}\int_0^{6T} f(t)\,dt$$

$$= \frac{6}{2}\int_0^T f(t)\,dt \qquad \{\because f(x+t)=f(x);\forall x\in R\}$$

$$= 3I$$

59. The value of the definite integral $\int_{19}^{37}\left(\{x\}^2+3(\sin 2\pi x)\right)dx$ where $\{x\}$ denotes the fractional part function :

$\int_{19}^{37}\left(\{x\}^2+3(\sin 2\pi x)\right)dx$ का मान होगा जहाँ $\{x\}$ भिन्नात्मक फलन है :

(A) 0 (B) 6 (C) 9 (D) 18

Ans. B

Sol. put $x = 19 + y \quad \Rightarrow \quad dx = dy,$

$$I=\int_0^{18}\left(\{19+y\}^2+\sin 2\pi y\right)dy = \int_0^{18}(\{y\}^2+\underbrace{\sin 2\pi y}_{\text{zero}})\,dy = 18\int_0^1 y^2\,dy = 6$$

60. $\int_\pi^{10\pi} |\sin x|\,dx =$

(A) 9 (B) 10 (C) 18 (D) 20

Ans. C

Sol. $\int_\pi^{10\pi} |\sin x|\,dx \qquad \left[\because \int_{aT}^{bT} f(x)\,dx = (b-a)\int_0^T f(x)\,dx\right]$

$$= 9\int_0^\pi |\sin x|\,dx = -9\left[\cos x\right]_0^\pi$$

$$= 18$$

DIFFERENTIATION OF INTEGRATION BY LEIBNITZ

61. Let $f(x) = \int_1^x \sqrt{2-t^2}\ dt$. Then the real roots of the equation $x^2 - f'(x) = 0$ are :

(A) ± 1 (B) $\pm \frac{1}{\sqrt{2}}$ (C) $\pm \frac{1}{2}$ (D) 0 and 1

यदि $f(x) = \int_1^x \sqrt{2-t^2}\ dt$ तो समीकरण $x^2 - f'(x) = 0$ के वास्तविक मूल हैं :

(A) ± 1 (B) $\pm \frac{1}{\sqrt{2}}$ (C) $\pm \frac{1}{2}$ (D) 0 तथा 1

Ans. A

Sol. $f(x) = \int_1^x \sqrt{2-t^2}dt$

Apply Leibnitz rule

$f'(x) = \sqrt{2-x^2}\cdot 1 - 0 = \sqrt{2-x^2}$

$\Rightarrow f'(x) = x^2$

$\Rightarrow \sqrt{2-x^2} = x^2$

let $x^2 = v$

$\Rightarrow \sqrt{2-v} = v$

$\Rightarrow v^2 + v - 2 = 0$

$\Rightarrow (v-1)(v+2) = 0$

$\Rightarrow v = 1$ or $v = -2$

$\Rightarrow x^2 = 1$ or $x^2 = -2$ (not possible)

$\Rightarrow x = \pm 1$

62. Variable x and y are related by equation $x = \int_0^y \frac{dt}{\sqrt{1+t^2}}$. The value of $\frac{d^2y}{dx^2}$ is equal to :

चर x तथा y सम्बन्ध $x = \int_0^y \frac{dt}{\sqrt{1+t^2}}$ से जुड़ा है, तो $\frac{d^2y}{dx^2}$ का मान है :

(A) $\frac{y}{\sqrt{1+y^2}}$ (B) y (C) $\frac{2y}{\sqrt{1+y^2}}$ (D) 4y

Ans. B

Sol. $\frac{dx}{dy} = \frac{1}{\sqrt{y^2+1}}$

$\frac{dy}{dx} = \sqrt{y^2+1}\,; \quad \frac{d^2y}{dx^2} = \frac{y}{\sqrt{y^2+1}}\sqrt{y^2+1} = y$

63. Let $f(x) = \int_{-1}^{x} e^{t^2}dt$ and $h(x) = f\left(1+g(x)\right)$, where g (x) is defined for all x, g'(x) exists for all x, and g (x) < 0 for x > 0. If h'(1) = e and g'(1) = 1, then the possible values which g(1) can take :

माना $f(x) = \int_{-1}^{x} e^{t^2}dt$ तथा $h(x) = f\left(1+g(x)\right)$, जहाँ g (x), x के सभी मानों के लिए परिभाषित तथा g'(x), x के सभी मानों के fy, vfLr Ro esa gS rFkk g (x) < 0, x > 0 के लिए। यदि h'(1) = e तथा g'(1) = 1 हो, तो g(1) के सम्भव मान होगें :

(A) 0 (B) – 1 (C) – 2 (D) – 4

Ans. C

Sol. Given $f(x) = \int_{-1}^{x} e^{t^2}dt$; $h(x) = f\left(1+g(x)\right)$; $g(x) < 0$ for $x > 0$

now $h(x) = \int_{-1}^{1+g(x)} e^{t^2}dt = f\left(1+g(x)\right)$ (given)

differentiating $h'(x) = e^{(1+g(x))^2}\cdot g'(x)$

$h'(1) = e$ (given)

$e^{(1+g(1))^2}\cdot g'(1) = e$

$\therefore \quad (1+g(1))^2 = 1$

$1 + g(1) = \pm 1 \quad \Rightarrow g(1) = 0$ (not possible) or $g(1) = -2 \Rightarrow$ (C)

64. Let $f: [0, 1] \to R$ be a differentiable function with f (0) = 0, then $\lim_{n\to\infty} n^2 \int_0^{\frac{1}{n}} f(t)\,dt$ equals :

माना $f: [0, 1] \to R$ एक अवकलनीय फलन है जहाँ f (0) = 0, तो $\lim_{n\to\infty} n^2 \int_0^{\frac{1}{n}} f(t)\,dt$ का मान है :

(A) $\frac{1}{2}f'(0)$ (B) f' (0) (C) 2 f' (0) (D) 0

Ans. A

Sol. Let $I = \lim\limits_{n\to\infty} n^2 \int\limits_0^{\frac{1}{n}} f(t)\,dt = \lim\limits_{n\to\infty} \dfrac{\int\limits_0^{1/n} f(t)\,dt}{\frac{1}{n^2}}$ $\left(\dfrac{0}{0} \text{ form, as } n \to \infty\right)$

Using L'Hospital's rule and Using Leibnitz rule, we get

$$I = \lim_{n\to\infty} \frac{n^3 f\left(\frac{1}{n}\right)\left(\frac{-1}{n^2}\right)}{-2} = \lim_{n\to\infty} \frac{n}{2} f\left(\frac{1}{n}\right)$$

Put $n = \dfrac{1}{h}$, hence

$$I = \frac{1}{2} \lim_{h\to 0} \left(\frac{f(0+h) - f(0)}{h}\right) = \frac{1}{2} f'(0)$$

65. If a, b and c are real numbers then the value of $\lim\limits_{t\to 0} \ln\left(\dfrac{1}{t}\int\limits_0^t (1 + a\sin bx)^{c/x}\,dx\right)$ equals :

यदि a, b तथा c वास्तविक संख्या है, तो $\lim\limits_{t\to 0} \ln\left(\dfrac{1}{t}\int\limits_0^t (1 + a\sin bx)^{c/x}\,dx\right)$ का मान होगा :

(A) abc (B) $\dfrac{ab}{c}$ (C) $\dfrac{bc}{a}$ (D) $\dfrac{ca}{b}$

Ans. A

Sol. $l = \ln \lim\limits_{t\to 0} \dfrac{\int\limits_0^t (1 + a\sin bx)^{c/x}\,dx}{t} = \ln \lim\limits_{t\to 0} (1 + a\sin bt)^{c/t}$ (using L'Hospital's rule)

$$= \ln e^{\lim\limits_{t\to 0} \frac{c}{t}(a\sin bt)} = \lim_{t\to 0} \frac{abc\sin bt}{bt} = abc$$

66. $\lim\limits_{x\to 0} \dfrac{\int\limits_0^x \sin t^2 dt}{x(1-\cos x)}$ equals :

$\lim\limits_{x\to 0} \dfrac{\int\limits_0^x \sin t^2 dt}{x(1-\cos x)}$ का मान होगा :

(A) $\frac{1}{3}$ (B) 2 (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

Ans. D

Sol. $\lim\limits_{x\to 0} \dfrac{\int_0^x \sin t^2 dt}{x^3 \dfrac{(1-\cos x)}{x^2}}$ (using $\lim\limits_{x\to 0} \dfrac{1-\cos x}{x^2} = \dfrac{1}{2}$)

$= 2 \lim\limits_{x\to 0} \dfrac{\int_0^x \sin t^2 dt}{x^3}$ (Using L'Hospital Rule)

$= 2 \lim\limits_{x\to 0} \dfrac{\sin x^2}{3x^2} = \dfrac{2}{3}$

67. The value of $\lim\limits_{x\to 0} \dfrac{\int_0^{x^2} \sec^2 t \, dt}{x \sin x}$ is :

$\lim\limits_{x\to 0} \dfrac{\int_0^{x^2} \sec^2 t \, dt}{x \sin x}$ का मान है :

(A) 0 (B) 3 (C) 2 (D) 1

Ans. D

Sol. $\lim\limits_{x\to 0} \dfrac{\int_0^{x^2} \sec^2 t dt}{x \sin x} \left(\dfrac{0}{0} \text{ form}\right) = \lim\limits_{x\to 0} \dfrac{\int_0^{x^2} \sec^2 t dt}{x^2} \left(\dfrac{0}{0} \text{ form}\right)$

Apply L' Hospital Rule & Leibnitz Rule

$= \lim\limits_{x\to 0} \dfrac{\sec^2\left(x^2\right)\cdot 2x}{2x} = 1$

68. Let $f : R \to R$ be a differentiable function having $f(2) = 6$, $f'(2) = \left(\frac{1}{48}\right)$. Then $\lim\limits_{x\to 2} \int_6^{f(x)} \dfrac{4t^3}{x-2} dt$ equals :

माना $f : R \to R$ एक अवकलनीय फलन हे, जिसके लिए $f(2) = 6$, $f'(2) = \left(\frac{1}{48}\right)$ है, तो $\lim\limits_{x\to 2} \int_6^{f(x)} \dfrac{4t^3}{x-2} dt$ बराबर है :

(A) 24 (B) 36 (C) 12 (D) 18

Ans. D

Sol. $f : R \xrightarrow{\text{diff.}} R$

$f(2) = 6,\ f'(2) = \frac{1}{48}$

$$\lim_{x\to 2} \int_6^{f(x)} \frac{4t^3}{x-2}dt = \lim_{x\to 2} \frac{\int_6^{f(x)} 4t^3 dt}{x-2} \quad \left(\frac{0}{0} \text{ form}\right)$$

Apply L'Hospital & Leibnitz Rule

$$= \lim_{x\to 0} \frac{4(f(x))^3 \cdot f'(x)}{1}$$

$= 4(f(2))^3 \cdot f'(2)$

$= 4 \cdot 216 \cdot \frac{1}{48} = 18$

69. $\underset{x\to \frac{\pi}{4}}{\text{Lim}} \frac{\int_2^{\sec^2 x} f(t)dt}{x^2 - \frac{\pi^2}{16}} =$

(A) $\frac{8}{\pi} f(2)$ (B) $\frac{2}{\pi} f(2)$ (C) $\frac{2}{\pi} f\left(\frac{1}{2}\right)$ (D) $4f(2)$

Ans. A

Sol. $$\lim_{x\to \frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(t)dt}{x^2 - \frac{\pi^2}{16}} \quad \left(\frac{0}{0} \text{ form}\right)$$

Apply Leibnitz & L'Hospital Rule

$$= \lim_{x\to \frac{\pi}{4}} \frac{f(\sec^2 x) \cdot 2\sec^2 x \tan x}{2x} = \frac{f(2)\cdot 4}{\frac{\pi}{2}} = \frac{8}{\pi} f(2)$$

70. The value of $\underset{x\to 0}{\text{Lim}} \frac{1}{x^3} \int_0^x \frac{t\, ln(1+t)}{t^4+4} dt$ is : **[JEE-2010]**

$\lim_{x\to0}\frac{1}{x^3}\int_0^x\frac{t\,\ell n(1+t)}{t^4+4}dt$ का मान होगा : **[JEE-2010]**

(A) 0 (B) $\frac{1}{12}$ (C) $\frac{1}{24}$ (D) $\frac{1}{64}$

Ans. B

Sol. Applying L-Hospital rule,

$$\lim_{x\to0}\frac{\int_0^x\frac{t\ell n(1+t)}{t^4+4}dt}{x^3}=\lim_{x\to0}\frac{\frac{x\ell n(1+x)}{x^4+4}}{3x^2}$$

$$=\lim_{x\to0}\frac{\ell n(1+x)}{3x(x^4+4)}=\frac{1}{12}$$

DETERMINATION OF FUNCTION

71. If $f'(x)=f(x)+\int_0^1 f(x)dx$ and given f(0) = 1, then $\int f(x)dx$ is equal to :

यदि $f'(x)=f(x)+\int_0^1 f(x)dx$ तथा दिया गया है कि f(0) = 1 है, तो $\int f(x)dx$ बराबर होगा :

(A) $\frac{2}{3-e}e^x+\left(\frac{3-e}{1-e}\right)x+C$ (B) $\frac{2}{3-e}e^x+\left(\frac{1-e}{3-e}\right)x+C$

(C) $\frac{3}{2-e}e^x+\left(\frac{1+e}{3+e}\right)x+C$ (D) $\frac{2}{2-e}e^x+\left(\frac{1-e}{3+e}\right)x+C$

Ans. B

Sol. $f'(x)=f(x)+\int_0^1 f(x)dx;\quad f(0)=1$

$\Rightarrow f''(x)=f'(x)$

$\Rightarrow \ell n|f'(x)|=x+\ell nC$

$\Rightarrow f'(x)=C\cdot e^x$

$\Rightarrow f(x)=Ce^x+d$

Given f(0) = 1

$\Rightarrow$ c + d = 1

$\Rightarrow$ f(x) = c(e^x – 1) + 1

$$\because f'(x) = f(x) + \int_0^1 f(x)\,dx$$

$$\Rightarrow ce^x = c\left(e^x - 1\right) + 1 + 1\left[ce^x - cx + x\right]_0^1$$

$$\Rightarrow C - 1 = \left[ce - c + 1 - c\right]$$

$$\Rightarrow C - 1 = ce - 2c + 1$$

$$\Rightarrow 3c = ce + 2$$

$$\Rightarrow c = \frac{2}{3-e}$$

$$\therefore f(x) = \frac{2}{3-e}\left(e^x - 1\right) + 1$$

$$\Rightarrow \int f(x)\,dx = \frac{2}{3-e}e^x + x\left(\frac{1-e}{3-e}\right) + \lambda; \lambda \in R$$

72. The set of values of 'a' which satisfy the equation $\int_0^2 (t - \log_2 a)\,dt = \log_2\left(\frac{4}{a^2}\right)$ is :

'a' के मानों का समुच्चय जो समीकरण $\int_0^2 (t - \log_2 a)\,dt = \log_2\left(\frac{4}{a^2}\right)$ को सन्तुष्ट करता है :

(A) $a \in R$ (B) $a \in R^+$ (C) $a < 2$ (D) $a > 2$

Ans. B

Sol. $\frac{t^2}{2} - \log_2 a \cdot t\Big|_0^2 = 2 - \log_2(a^2)$ (note $a > 0$)

$(2 - 2\log_2 a) = 2 - 2\log_2 a$ (an identity)

$2\log_2 a = 2\log_2 a \quad \Rightarrow \quad a \in R^+$

73. If $\int_{\sin x}^1 t^2(f(t))dt = (1 - \sin x)$, then $f\left(\frac{1}{\sqrt{3}}\right)$ is :

यदि $\int_{\sin x}^1 t^2(f(t))dt = (1 - \sin x)$, तो $f\left(\frac{1}{\sqrt{3}}\right)$ होगा :

(A) 1/3 (B) $1/\sqrt{3}$ (C) 3 (D) $\sqrt{3}$

Ans. C

Sol. $\int_{\sin x}^{1} t^2 f(t)\,dt = 1 - \sin x$

diff. w.r.t. x

$\Rightarrow -\sin^2 x\ f(\sin x)\cos x = -\cos x$

If $\sin x = \frac{1}{\sqrt{3}}$ $\sin x = \frac{1}{\sqrt{3}} \Rightarrow \cos x = \pm\sqrt{\frac{2}{3}}$

$\Rightarrow -\frac{1}{3} f\left(\frac{1}{\sqrt{3}}\right) = -1$

$\Rightarrow f\left(\frac{1}{\sqrt{3}}\right) = 3$

74. Let p(x) be a function defined on R such that $p'(x) = p'(1 - x)$, for all $x \in [0, 1]$, $p(0) = 1$ and $p(1) = 41$. Then $\int_0^1 p(x)\,dx$ equals : **[AIEEE-2010]**

माना p(x), R पर परिभाषित एक ऐसा फलन है कि $p'(x) = p'(1 - x)$, प्रत्येक $x \in [0, 1]$, $p(0) = 1$ तथा $p(1) = 41$ है। तो $\int_0^1 p(x)\,dx$: **[AIEEE-2010]**

(A) $\sqrt{41}$ (B) 21 (C) 41 (D) 42

Ans. B

Sol. $\int P'(x)dx = \int P'(1-x)dx$

$P(x) = -P(1-x) + C$

$C = P(x) + P(1 - x)$

Put $x = 1$ $\therefore$ $P(1) + P(0) = 42$ .....(i)

Let $I = \int_0^1 P(x)dx$ .....(ii)

$I = \int_0^1 P(1-x)dx$ .....(iii)

(ii) + (iii)

$2I = \int_0^1 P(x) + P(1-x)dx$

$= 42\int_0^1 dx = 42$

$\therefore$ I = 21

75. Let f be a non–negative function defined on the interval [0,1]. If $\int_0^x \sqrt{1-(f'(t))^2}\,dt = \int_0^x f(t)\,dt$, $0 \le x \le 1$, and f(0) = 0, then : **[JEE-2009]**

(A) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$ (B) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$

(C) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$ (D) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

यदि f अऋणात्मक फलन है जो कि [0,1] में परिभाषित है यदि $\int_0^x \sqrt{1-(f'(t))^2}\,dt = \int_0^x f(t)\,dt$, $0 \le x \le 1$, तथा f(0) = 0, तो : **[JEE-2009]**

(A) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ तथा $f\left(\frac{1}{3}\right) > \frac{1}{3}$ (B) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ तथा $f\left(\frac{1}{3}\right) > \frac{1}{3}$

(C) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ तथा $f\left(\frac{1}{3}\right) < \frac{1}{3}$ (D) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ तथा $f\left(\frac{1}{3}\right) < \frac{1}{3}$

Ans. C

Sol. $\int_0^x \sqrt{1-(f'(t))^2}\,dt = \int_0^x f(t)dt$, $0 \le x \le 1$

differentiating both the sides & squreing

$\Rightarrow 1-\left(f'(x)\right)^2 = f^2(x) \Rightarrow \frac{f'(x)}{\sqrt{1-f^2(x)}} = 1$

$\Rightarrow \sin^{-1} f(x) = x + c$

$f(0) = 0$

$\Rightarrow f(x) = \sin x \Rightarrow \because \sin x \le x$ for $x \in [0, 1]$

$\Rightarrow f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

76. Let f be a real-valued function defined on the interval (–1, 1) such that $e^{-x} f(x) = 2 + \int_0^x \sqrt{t^4+1}\,dt$, for all $x \in (-1,1)$, and let f^{-1} be the inverse function of f. Then $(f^{-1})'(2)$ is equal to : **[JEE-2010]**

यदि अन्तराल (–1, 1) में f एक वास्तविक फलन है जहाँ $e^{-x} f(x) = 2 + \int_0^x \sqrt{t^4+1}\,dt$, सभी $x \in (-1,1)$, एवं f^{-1} फलन f का प्रतिलोम है तो $(f^{-1})'(2)$ का मान होगा : **[JEE-2010]**

(A) 1 (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{e}$

Ans. B

Sol. $e^{-x}f(x) = 2 + \int_0^x \sqrt{t^4+1}dt$

$e^{-x}f'(x) - e^{-x}f(x) = \sqrt{x^4+1}$

$\Rightarrow \quad f'(x) - f(x) = e^x\sqrt{x^4+1}$

$\Rightarrow \quad \frac{dy}{dx} = y + e^x\sqrt{x^4+1}$ (say) .....(i)

considering $y = f(x)$. so that $x = f'(y)$

$f^{-1}'(2) = \left(\frac{dx}{dy}\right)_{y=2}$ ..... (ii)

for $x = 0$ $\Rightarrow$ $f(x) = 2$ i.e. $y = 2$

$\Rightarrow$ $f^{-1}(2) = 0$

$\frac{dy}{dx} = 2 + 1\sqrt{1} = 3$

from (2), $f^{-1}'(2) = \frac{1}{3}$

DEFINITE INTEGRALAS A LIMIT OF A SUM

77. $\lim\limits_{n\to\infty} \frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \ldots\ldots + \frac{1}{\sqrt{n}\sqrt{4n}}$ is equal to :

$\lim\limits_{n\to\infty} \frac{1}{\sqrt{n}\sqrt{n+1}} + \frac{1}{\sqrt{n}\sqrt{n+2}} + \ldots\ldots + \frac{1}{\sqrt{n}\sqrt{4n}}$ का मान है :

(A) 2 (B) 4 (C) $2\left(\sqrt{2}-1\right)$ (D) $2\sqrt{2}-1$

Ans. A

Sol. $T_n = \frac{1}{\sqrt{n}\sqrt{n+r}}$ where r varies from 1 to 3n

$\Rightarrow \quad \frac{1}{n} < \frac{r}{n} < 3 \qquad \Rightarrow \quad 0 < x < 3$ as $n \to \infty$

$\lim\limits_{n\to\infty} \sum_{r=1}^{3n} \frac{1}{n\sqrt{1+\frac{r}{n}}} = \int_0^3 \frac{dx}{\sqrt{1+x}} = \int_0^3 (1+x)^{-1/2}\,dx = 2\sqrt{1+x}\Big]_0^3 = 2[2-1] = 2$

78. $f(x) = \lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \frac{n}{n^2 + k^2x^2}$, $x > 0$ is equal to :

$\lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \frac{n}{n^2 + k^2x^2}$, $x > 0$ का मान है :

(A) $x\tan^{-1}(x)$ (B) $\tan^{-1}(x)$ (C) $\frac{\tan^{-1}(x)}{x}$ (D) $\frac{\tan^{-1}(x)}{x^2}$

Ans. C

Sol. $T_n = \frac{n}{n^2 + k^2x^2} = \frac{1}{n\left(1 + (k/n)^2x^2\right)}$;

$$S = \frac{1}{n}\sum_{k=1}^{n} \frac{1}{1 + \underbrace{(k/n)}_{t}^{2}x^2} = \int_0^1 \frac{dt}{1 + t^2x^2} = \frac{1}{x^2}\int_0^1 \frac{dt}{t^2 + (1/x^2)} = \frac{1}{x}\tan^{-1} tx\Big]_0^1 = \frac{\tan^{-1} x}{x}$$

79. $\lim\limits_{n\to\infty} \frac{1^P + 2^P + 3^P + \ldots\ldots + n^P}{n^{P+1}} =$

(A) 1 (B) $\frac{1}{P+1}$ (C) $\frac{1}{P+2}$ (D) P^2

Ans. B

Sol. $\lim\limits_{n\to\infty} \sum\limits_{r=1}^{n} \frac{r^p}{n^{p+1}} = \lim\limits_{n\to\infty} \sum\limits_{r=1}^{n} \left(\frac{r}{n}\right)^p \cdot \frac{1}{n}$

$$= \int_0^1 x^p \cdot dx = \left[\frac{x^{p+1}}{p+1}\right]_0^1 = \frac{1}{p+1}$$

80. $\lim\limits_{n\to\infty} \left[\frac{1}{n^2}\sec^2\frac{1}{n^2} + \frac{2}{n^2}\sec^2\frac{4}{n^2} + \ldots\ldots + \frac{1}{n}\sec^2 1\right]$ equals :

$\lim\limits_{n\to\infty} \left[\frac{1}{n^2}\sec^2\frac{1}{n^2} + \frac{2}{n^2}\sec^2\frac{4}{n^2} + \ldots\ldots + \frac{1}{n}\sec^2 1\right]$ का मान है :

(A) $\frac{1}{2}$ sec 1 (B) $\frac{1}{2}$ cosec 1 (C) tan 1 (D) $\frac{1}{2}$ tan 1

Ans. D

Sol. $\lim\limits_{n\to\infty} \sum\limits_{r=1}^{n} \frac{r}{n^2}\sec^2\frac{r^2}{n^2}$

$= \int_0^1 x \sec^2 x^2 dx$

Put $x^2 = t$; $2xdx = dt$; $xdx = \frac{dt}{2}$

$= \frac{1}{2}\int_0^1 \sec^2 t dt$

$= \frac{1}{2}(\tan t)_0^1 \qquad = \frac{1}{2}\tan 1$

81. $\lim_{n\to\infty}\left(\frac{(n+1)(n+2)....3n}{n^{2n}}\right)^{1/n}$ is equal to : **[JEE(Main)2016]**

$\lim_{n\to\infty}\left(\frac{(n+1)(n+2)....3n}{n^{2n}}\right)^{1/n}$ बराबर है : **[JEE(Main)2016]**

(A) $\frac{9}{e^2}$ (B) $3\log 3 - 2$ (C) $\frac{18}{e^4}$ (D) $\frac{27}{e^2}$

Ans. D

Sol. $L = \prod_{r=1}^{2n}\left(\frac{n+r}{n}\right)^{\frac{1}{n}}$

$\ell n\, L = \frac{1}{n}\sum_{r=1}^{2n}\ell n\left(1+\frac{r}{n}\right)$

$= \int_0^2 \ell n(1+x)dx$

$\ell n\, L = 3\,\ell n\, 3 - 2$

$L = \frac{27}{e^2}$

DEFINITE INTEGRATION BY ESTIMATION

82. If $I_1 = \int_0^1 2^{x^2} dx$, $I_2 = \int_0^1 2^{x^3} dx$, $I_3 = \int_1^2 2^{x^2} dx$ and $I_4 = \int_1^2 2^{x^3} dx$ then :

यदि $II_1 = \int_0^1 2^{x^2} dx$, $I_2 = \int_0^1 2^{x^3} dx$, $I_3 = \int_1^2 2^{x^2} dx$ तथा $I_4 = \int_1^2 2^{x^3} dx$ तो :

(A) $I_2 > I_1$ (B) $I_1 > I_2$ (C) $I_3 = I_4$ (D) $I_3 > I_4$

Ans. B

Sol. for $0 < x < 1$, $x^2 > x^3$ and

for $1 < x < 2$, $x^3 > x^2$

$\therefore$ for, $0 < x < 1$, $2^{x^2} > 2^{x^3}$ and

for $1 < x < 2$, $2^{x^2} < 2^{x^3}$

$\therefore \int_0^1 2^{x^2} dx > \int_0^1 2^{x^3} dx$ and $\int_1^2 2^{x^2} dx < \int_1^2 2^{x^3} dx$

$\therefore I_1 > I_2$ and $I_3 < I_4$

83. Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$. Then which one of the following is true? **[AIEEE-2008]**

(A) $1 > \frac{2}{3}$ and $J > 2$ (B) $I < \frac{2}{3}$ and $J < 2$ (C) $I < \frac{2}{3}$ and $J > 2$ (D) $I > \frac{2}{3}$ and $J < 2$

माना $I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$ तथा $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$ तब निम्न में से कौनसा कथन सत्य है? **[AIEEE-2008]**

(A) $1 > \frac{2}{3}$ तथा $J > 2$ (B) $I < \frac{2}{3}$ तथा $J < 2$ (C) $I < \frac{2}{3}$ तथा $J > 2$ (D) $I > \frac{2}{3}$ तथा $J < 2$

Ans. B

Sol. Now

$\sigma\nu\, \xi < \xi \Rightarrow$ $\frac{\sin x}{\sqrt{x}} < \sqrt{x}$

$\int_0^1 \frac{\sin x}{\sqrt{x}} dx < \int_0^1 \sqrt{x}\, dx$

$I < \left[\frac{2}{3} x^{3/2}\right]_0^1$

$I < \frac{2}{3}$

$\because \cos x < 1$

$\frac{\cos x}{\sqrt{x}} < \frac{1}{\sqrt{x}} \Rightarrow \int_0^1 \frac{\cos x}{\sqrt{x}} dx < \int_0^1 \frac{1}{\sqrt{x}} dx < \left[2\sqrt{x}\right]_0^1 < 2$

$J < 2$

INTEGRATION BY REDUCTION

84. If $I_n = \int (\sin x)^n \, dx \quad n \in N$, then $5I_4 - 6I_6$ is equal to :

यदि $I_n = \int (\sin x)^n \, dx \quad n \in N$, तब $5I_4 - 6I_6$ बराबर :

(A) $\sin x \cdot (\cos x)^5 + C$ (B) $\sin 2x \cdot \cos 2x + C$

(C) $\frac{\sin 2x}{8} [\cos^2 2x + 1 - 2\cos 2x] + C$ (D) $\frac{\sin 2x}{8} [\cos^2 2x + 1 + 2\cos 2x] + C$

Ans. C

Sol. reduction formula of $(\sin x)^n$

85. Let $I_n = \int \tan^n x \, dx, (n > 1)$. If $I_4 + I_6 = a \tan^5 x + bx^5 + C$, where C is a constant of integration, then the ordered pair (a, b) is equal to : **[JEE(Main)-2017]**

माना $I_n = \int \tan^n x \, dx, (n > 1)$ है। यदि $I_4 + I_6 = a \tan^5 x + bx^5 + C$ है, जहाँ C एक समाकलन अचर हैं, तो क्रमित युग्म (a, b) बराबर है : **[JEE(Main)-2017]**

(A) $\left(\frac{1}{5}, -1\right)$ (B) $\left(-\frac{1}{5}, 0\right)$ (C) $\left(-\frac{1}{5}, 1\right)$ (D) $\left(\frac{1}{5}, 0\right)$

Ans. D

Sol. $I_4 + I_6 = \int \tan^4(x)\left(1 + \tan^2 x\right) dx$

$= \frac{(\tan x)^5}{5} + C$

$a = \frac{1}{5}, b = 0$

SUBJECTIVE / INTEGER TYPE QUESTIONS

INTEGRATION BASED ON STANDARD RESULTS

1. $\int \frac{\sin 2x - \sin 2k}{\sin x - \sin k + \cos x - \cos k} dx$:

Ans. $(\sin x - \cos x) + (\sin k + \cos k)x + C$

Sol. $\int \frac{\sin 2x - \sin 2k}{\sin x - \sin k + \cos x - \cos k} dx$

$= \int \frac{(\sin x + \cos x)^2 - (\sin k + \cos k)^2}{\sin x - \sin k + \cos x - \cos k} dx$

$= \int (\sin x + \cos x + \sin k + \cos k) dx$

$= (\sin x - \cos x) + x(\sin k + \cos k) + C$

2. $\int \frac{x^2 + 3}{x^6(x^2 + 1)} dx$:

Ans. $C - \frac{2}{x} + \frac{2}{3}\frac{1}{x^3} - \frac{3}{5}\frac{1}{x^5} - 2\tan^{-1}x$

Sol. $\int \frac{x^2 + 3}{x^6(x^2 + 1)} dx = \int \frac{(3x^2 + 3 - 2x^2)}{x^6(x^2 + 1)} dx$

$= 3\int \frac{dx}{x^6} - 2\int \frac{dx}{x^4(x^2 + 1)}$

$= -\frac{3}{5}\cdot\frac{1}{x^5} - \int \frac{(2 + 2x^2 - 2x^2)}{x^4(x^2 + 1)} dx$

$= -\frac{3}{5x^5} - \int \frac{2dx}{x^4} + 2\int \frac{dx}{x^2(x^2 + 1)}$

$$= -\frac{3}{5x^5} + \frac{2}{3x^3} + 2\int\left(\frac{1}{x^2} - \frac{1}{x^2+1}\right)dx$$

$$= -\frac{3}{5x^5} + \frac{2}{3x^3} - \frac{2}{x} - 2\tan^{-1}x + C$$

3. $\int \sin x \cos x \cos 2x \cos 4x \, dx$:

Ans. $-\frac{1}{64}\cos 8x + C$

Sol. $\int \sin x \cos x \cos 2x \cos 4x dx$

$$= \frac{1}{8}\int \sin 8x dx$$

$$= -\frac{1}{64}\cos 8x + C$$

INTEGRATION BY SUBSTITUTION

4. $\int \frac{l\text{n}\left(l\text{n}\left(\frac{1+x}{1-x}\right)\right)}{1-x^2}dx$:

Ans. $\frac{1}{2}\left[l\text{n}\left(\frac{1+x}{1-x}\right)\cdot l\text{n}\left(l\text{n}\frac{1+x}{1-x}\right) - l\text{n}\left(\frac{1+x}{1-x}\right)\right] + C$

Sol. $\int \frac{\ell\text{n}\left(\ell\text{n}\left(\frac{1+x}{1-x}\right)\right)}{1-x^2}dx$

let $\ell\text{n}\left(\frac{1+x}{1-x}\right) = t$

$$\Rightarrow \frac{1-x}{1+x} \times \frac{(1-x)+(1+x)}{(1-x)^2}dx = dt$$

$$\Rightarrow \frac{2}{1-x^2}dx = dt$$

$$= \frac{1}{2}\int \ell\text{n}(t)\,dt = \frac{1}{2}\left[t(\ell\text{n}t) - t\right] + C$$

$$= \frac{1}{2}\left[\ell n\left(\frac{1+x}{1-x}\right)\cdot \ell n\left(\ell n\left(\frac{1+x}{1-x}\right)\right) - \ell n\left(\frac{1+x}{1-x}\right)\right] + C$$

5. $\int \left[\left(\frac{x}{e}\right)^x + \left(\frac{e}{x}\right)^x\right] \ell nx\, dx$:

Ans. $\left(\frac{x}{e}\right)^x - \left(\frac{e}{x}\right)^x + C$

Sol. $I = \int \left[\left(\frac{x}{e}\right)^x + \left(\frac{e}{x}\right)^x\right] \ell nx\ dx = \int \left(1 + \frac{1}{\left(\frac{x}{e}\right)^{2x}}\right) \ell nx \left(\frac{x}{e}\right)^x dx$

$$\left\{ Put\left(\frac{x}{e}\right)^x = t \Rightarrow \ell nx\left(\frac{x}{e}\right)^x dx = dt \right\}$$

$$I = \int \left(1 + \frac{1}{t^2}\right) dt = t - \frac{1}{t} + C = \left(\frac{x}{e}\right)^x - \left(\frac{e}{x}\right)^x + C$$

6. $\int \frac{\log_6 \sqrt[6]{\left((\sin x) 6^{\cos x}\right)} \cos x}{\sin x} dx$:

Ans. $\frac{1}{6}\left(\frac{ln^2(\sin x)}{ln\, 36} + ln\left(\tan\frac{x}{2}\right) + \cos x\right) + C$

Sol. $\int \frac{\log_6 \left((\sin x) 6^{\cos x}\right)^{\frac{1}{6}}}{\sin x} \cdot \cos x dx$

$$= \frac{1}{6}\int \frac{\left(\log_6 \sin x + \cos x\right)\cos x}{\sin x} dx$$

let $\sin x = t \Rightarrow \cos x\, dx = dt$

$$= \frac{1}{6}\int \left(\frac{\log_6 t}{t} + \frac{\sqrt{1-t^2}}{t}\right) dt$$

$$= \frac{1}{6}\int\left[\frac{\ell nt}{\ell n6t} + \frac{\sqrt{1-t^2}}{t}\right]dt$$

$$= \frac{1}{6\ell n6}\int \ell nt\cdot\frac{1}{t}dt + \frac{1}{6}\int\frac{\sqrt{1-t^2}}{t}dt$$

$$= \frac{1}{6\ell n6}\frac{\left[\ell nt\right]^2}{2} + \frac{1}{6}\int\frac{\cos^2 x}{\sin x}dx$$

$$= \frac{1}{6\ell n36}\left(\ell n\sin x\right)^2 + \frac{1}{6}\left[\int\left(\operatorname{cosec}x - \sin x\right)dx\right.$$

$$= \frac{1}{6\ell n36}\left(\ell n\sin x\right)^2 + \frac{1}{6}\left[\ell n\left|\tan\frac{x}{2}\right| + \cos x\right] + C$$

7. $\int\frac{x^2+x}{(e^x+x+1)^2}dx$:

Ans. $C - ln(1 + (x + 1)e^{-x}) - \frac{1}{1+(x+1)e^{-x}}$

Sol. $\int\frac{x^2+x}{(e^x+x+1)^2}dx = \int\frac{e^{-2x}x(x+1)}{\left((x+1)e^{-x}+1\right)^2}dx$

Let $(x + 1)\, e^{-x} + 1 = t$

$\left(e^{-x}[1] - e^{-x}(x+1)\right)dx = dt$

$= \int\frac{(1-t)}{t^2}dt = -\frac{1}{t} - \ln|t| + C$

$= \frac{-1}{(x+1)e^{-x}+1} - \ln\left|(x+1)e^{-x}+1\right| + C$

8. Let $\int\frac{f'(x)g(x) - g'(x)f(x)}{(f(x)+g(x))\sqrt{f(x)g(x)-g^2(x)}}dx = \sqrt{m}\tan^{-1}\left(\sqrt{\frac{f(x)-g(x)}{ng(x)}}\right) + C$, where m, n ∈ N and 'C' is constant of integration $(g(x) > 0)$. Find the value of $(m^2 + n^2)$.

माना $\int \frac{f'(x)g(x)-g'(x)f(x)}{(f(x)+g(x))\sqrt{f(x)g(x)-g^2(x)}}dx = \sqrt{m}\tan^{-1}\left(\sqrt{\frac{f(x)-g(x)}{ng(x)}}\right)+C$, जहाँ m, n ∈ N तथा 'C', समाकलन का अचर है $(g(x)>0)$ तो (m^2+n^2) का मान बताओं।

Ans. 8

Sol. $\int \frac{f'(x)g(x)-g'(x)f(x)}{(f(x)+g(x))\sqrt{f(x)g(x)-g^2(x)}}dx$

$$=\int \frac{1}{\left(\frac{f(x)}{g(x)}+1\right)\sqrt{\frac{f(x)}{g(x)}-1}}.\frac{f'(x)g(x)-g'(x)f(x)}{g'(x)}dx$$

Let $\frac{f(x)}{g(x)}-1=t^2$

$$\Rightarrow \left(\frac{f'(x)g(x)-g'(x)f(x)}{g'(x)}\right)dx = 2t\ dt$$

$$=\int \frac{2tdt}{t\left(t^2+2\right)} = 2\int \frac{dt}{t^2+2}$$

$$=\sqrt{2}\left(\tan^{-1}\left(\frac{\sqrt{\frac{f(x)}{g(x)}-1}}{\sqrt{2}}\right)\right)+C$$

$\Rightarrow m=2,\ n=2$

$\Rightarrow m^2+n^2=4+4=8$

9. $\int \frac{\sin^p x}{\cos^{p+2} x}\,dx$:

Ans. $\frac{\tan^{p+1} x}{p+1}+c$

Sol. $\int \frac{\sin^p x}{\cos^{p+2} x}\,dx = \int \tan^p x \sec^2 x dx$

Let $\tan x = t$

$\sec^2 x\, dx = dt$

$$= \int t^p dt = \frac{1}{p+1}\tan^{p+1} x + C$$

10. $\int x^x ln(ex)dx$:

Ans. $x^x + C$

Sol. $$= \int x^x \left(\ln(ex)\right)dx = \int x^x \left(1+\ln x\right)dx$$

$= x^x + C$

INTEGRATION BY TMRIGONOMETERIC SUBSTITUTION

11. $\int \sin^{-1}\sqrt{\frac{x}{a+x}}\,dx$:

Ans. $(a + x)\arctan\sqrt{\frac{x}{a}} - \sqrt{ax} + C$

Sol. $$\int \sin^{-1}\sqrt{\frac{x}{a+x}}\,dx = \int \sin^{-1}\left(\frac{\sqrt{x}}{\sqrt{a+x}}\right)dx$$

Let $x = a\tan^2\theta$

$dx = 2a\tan\theta\sec^2\theta\, d\theta$

$$= \int \sin^{-1}\left(\frac{\sqrt{a}+\tan\theta}{\sqrt{a}\sec\theta}\right).2a\tan\theta\sec^2\theta d\theta$$

$$= 2a\int \underset{I}{\theta}\underbrace{\left(\tan\theta.\sec^2\theta\right)}_{II}d\theta$$

$$= 2a\theta\frac{\tan^2\theta}{2} - a\int\tan^2\theta\, d\theta$$

$$= a\theta\tan^2\theta - a\tan\theta + a\theta + C$$

$$= \left(x+a\right)\tan^{-1}\sqrt{\frac{x}{a}} - \sqrt{ax} + C$$

12. $\int \sin^{-1}\left(\frac{2x+2}{\sqrt{4x^2+8x+13}}\right)dx$:

Ans. $(x+1)\tan^{-1}\frac{2(x+1)}{3}-\frac{3}{4}\ell n(4x^2+8x+13)+C$

Sol. $\int \sin^{-1}\left(\frac{2x+2}{\sqrt{4x^2+8x+13}}\right)dx$

$$=\int \sin^{-1}\left(\frac{2x+2}{\sqrt{(2x+2)^2+9}}\right)$$

Let $(2x+2)=3\tan\theta \Rightarrow 2dx=3\sec^2\theta\ d\theta$

$$=\int \sin^{-1}\left(\frac{3\tan\theta}{3\sec\theta}\right).\frac{3}{2}.\sec^2\theta\ d\theta$$

$$=\frac{3}{2}\int \underset{\text{I}}{\theta}.\underbrace{\sec^2\theta d\theta}_{\text{II}}$$

$$=\frac{3}{2}\left[\theta\tan\theta-\int\tan\theta\ d\theta\right]$$

$$=\frac{3}{2}\frac{(2x+2)}{3}\tan^{-1}\left(\frac{2x+2}{3}\right)+\frac{3}{2}\ln\cos\theta+C$$

$$=(2x+1)\tan^{-1}\left(\frac{2x+2}{3}\right)-\frac{3}{2}\ln\left(\sqrt{1+\tan^2\theta}\right)+C$$

$$=(x+1)\tan^{-1}\left(\frac{2x+2}{3}\right)-\frac{3}{4}\ln\left(1+\frac{(2x+2)^2}{9}\right)+C$$

$$=(x+1)\tan^{-1}\left(\frac{2x+2}{3}\right)-\frac{3}{4}\ln(4x^2+8x+13)+C$$

INTEGRATION BY PARTS

13. $\int \frac{x \ln x}{\left(x^2-1\right)^{3/2}}\, dx$

Ans. $\operatorname{arc\,sec} x - \frac{\ln x}{\sqrt{x^2-1}} + C$

Sol. $\int \frac{x \ln x}{\left(x^2-1\right)^{3/2}}\, dx = \int \frac{x}{\left(x^2-1\right)^{3/2}} . \ln x\, dx$

Let $x = \sec\theta$

$dx = \sec\theta \tan\theta\, d\theta$

$$= \int \frac{\sec\theta \ln(\sec\theta) . \sec\theta \tan\theta\, d\theta}{\tan^3\theta}$$

$$= \int \frac{\sec^2\theta \ln(\sec\theta)}{\tan^2\theta}\, d\theta$$

$$= \int \underbrace{\frac{\sec^2\theta}{\tan^2\theta}}_{II} . \underbrace{\ln(\sec\theta)}_{I}\, d\theta$$

$$= \ln(\sec\theta) . \frac{-1}{\tan\theta} + \int \frac{d\theta}{\sec\theta\tan\theta} \times \sec\theta\tan\theta$$

$$= -\frac{\ln(\sec\theta)}{\tan\theta} + \int d\theta$$

$$= -\frac{\ln(x)}{\sqrt{x^2-1}} + \sec^{-1} x + C$$

14. $\int x^3 e^{x^2}\, dx$

Ans. $\frac{1}{2}(x^2-1)\, e^{x^2} + c$

Sol. $\int x^3 e^{x^2}\, dx$

Let $x^2 = t \Rightarrow 2x dx = dt$

$$= \frac{1}{2}\int \underset{\text{I}}{t}\,\underset{\text{II}}{e^t}\,dt = \frac{1}{2}\left[te^t - e^t\right] + C$$

$$= \frac{1}{2}\left[x^2 - 1\right]e^{x^2} + C$$

INTEGRATION OF RATIONAL FUNCTIONS

15. $\int \frac{x^5 + 3x^4 - x^3 + 8x^2 - x + 8}{x^2 + 1}dx$

Ans. $\frac{x^4}{4} + x^3 - x^2 + 5x + \frac{1}{2}ln\,(x^2 + 1) + 3\tan^{-1}x + C$

Sol. $\int \frac{x^5 + 3x^4 - x^3 + 8x^2 - x + 8}{x^2 + 1}dx$

$$= \int \frac{\left(x^3 + 3x^2 - 2x + 5\right)\left(x^2 + 1\right) + \left(x + 3\right)dx}{\left(x^2 + 1\right)}$$

$$= \int \left(x^3 + 3x^2 - 2x + 5\right)dx + \int \frac{\left(x + 3\right)}{x^2 + 1}dx$$

$$= \frac{x^4}{4} + x^3 - x^2 + 5x + \frac{1}{2}ln\,(x^2 + 1) + 3\tan^{-1}x + C$$

16. $\int \frac{(\sqrt{x} + 1)dx}{\sqrt{x}\,(\sqrt[3]{x} + 1)}$

Ans. $6\left[\frac{t^4}{4} - \frac{t^2}{2} + t + \frac{1}{2}ln(1 + t^2) - \tan^{-1}t\right] + C$ where $t = x^{1/6}$

Sol. $\int \frac{\left(\sqrt{x} + 1\right)}{\sqrt{x}\left(x^{1/3} + 1\right)}dx = \int \frac{\left(\sqrt{x} + 1\right)}{\sqrt{x}\left(x^{1/3} + 1\right)}dx$

Let $x^{1/6} = t$

$$\Rightarrow \frac{1}{6}x^{-\frac{5}{6}}dx = dt$$

$$= \int \frac{\left(t^3 + 1\right)}{t^3\left(t^2 + 1\right)} \cdot \frac{6t^5 dt}{5}$$

$$= 6\int \frac{t^2\left(t^3+1\right)}{t^2+1}dt = 6\int \frac{\left(t^5+t^2\right)}{t^2+1}dt$$

$$= 6\int \frac{\left(t^2+1+t^5-1\right)}{t^2+1}dt$$

$$= 6\left[t+\int \frac{t^5-1}{t^2+1}dt \right]$$

$$= 6\left[t-\tan^{-1}t+\int \frac{t^4.t}{\underbrace{t^2+1}_{\text{let } t^2+1=\theta}}\,dt \right]$$

$$= 6\left[t-\tan^{-1}t+\frac{1}{2}\int \frac{(\theta-1)^2}{\theta}d\theta \right]$$

$$= 6\left[t-\tan^{-1}t+\frac{\theta^2}{4}-\theta+\frac{1}{2}\ln|\theta| \right]+C$$

$$= 6\left[t-\tan^{-1}t+\frac{\left(t^2+1\right)^2}{4}-t^2-1+\frac{1}{2}\ln\left(1+t^2\right) \right]+C$$

$$= 6\left[\frac{t^4}{4}-\frac{t^2}{2}+t+\frac{1}{2}\ln\left(1+t^2\right)-\tan^{-1}t \right]+C;\ t = x^{1/6}$$

17. $\int \sqrt{x^2-8x+7}\,dx$

Ans. $-\frac{9}{2}\log\left[x-4+\sqrt{x^2-8x+7}\right]+\frac{1}{2}(x-4)\sqrt{x^2-8x+7}+c$

Sol. $\int \sqrt{x^2-8x+7}\,dx$

$$= \int \sqrt{(x-4)^2-9}$$

$$= \frac{x-4}{2}\sqrt{x^2-8x+7}-\frac{9}{2}\log\left|x-4+\sqrt{x^2-8x+7}\right|+C$$

TWO CLASSIC INTEGRALS

18. $\int \frac{e^x\left(2-x^2\right)}{(1-x)\sqrt{1-x^2}}\,dx$

Ans. $e^x\sqrt{\frac{1+x}{1-x}}+c$

Sol. $\int \frac{e^x\left(2-x^2\right)}{(1-x)\sqrt{1-x^2}}\,dx$

$$=\int e^x\left(\frac{1+(1+x)(1-x)}{(1-x)\sqrt{1-x^2}}\right)dx$$

$$=\int e^x\left(\underbrace{\sqrt{\frac{1+x}{1-x}}}_{f(x)}+\underbrace{\frac{1}{(1-x)\sqrt{1-x^2}}}_{f'(x)}\right)dx$$

$$=e^x\sqrt{\frac{1+x}{1-x}}+C$$

19. $\int \frac{e^{\cos x}(x\sin^3 x+\cos x)}{\sin^2 x}dx$

Ans. $C-e^{\cos x}(x+\operatorname{cosec} x)$

Sol. $\int e^{\cos x}\left(\frac{x\sin^3 x+\cos x}{\sin^2 x}\right)dx$

$$=\int e^{\cos x}\left(x\sin x+\cot x.\operatorname{cosec} x\right)dx$$

$$=\int xe^{\cos x}\sin x\,dx+\int e^{\cos x}\operatorname{cosec} x\cot x\,dx$$

$$=x\int e^{\cos x}\sin x\,dx-\int\left(\int e^{\cos x}\sin x\,dx\right)dx+e^{\cos x}\int \operatorname{cosec} x\cot x\,dx$$

$$+\int e^{\cos x}\sin x\left[\int \operatorname{cosec} x\cot x\,dx\right]dx+C$$

$$=-xe^{\cos x}+\int e^{\cos x}dx-e^{\cos x}\operatorname{cosec} x+\int e^{\cos x}dx+C$$

$= -(x + \operatorname{cosec} x)e^{\cos x} + C$

INTEGRATION OF TRGONOMETRIC FUNCTION

20. **(a)** $\int \frac{dx}{\cot\frac{x}{2}\cdot\cot\frac{x}{3}\cdot\cot\frac{x}{6}}$; **(b)** $\int \frac{\tan(\ln x)\tan\left(\ln\frac{x}{2}\right)\tan(\ln 2)}{x}dx$

Ans. (a) $2\ln\left(\sec\frac{x}{2}\right)-3\ln\left(\sec\frac{x}{3}\right)-6\ln\left(\sec\frac{x}{6}\right)+C$; (b) $\ln\left(\frac{\sec(\ln x)}{\sec(\ln(x/2))x^{\tan(\ln 2)}}\right)+C$

Sol. (a) $\int \frac{dx}{\cot\left(\frac{x}{2}\right)\cot\left(\frac{x}{3}\right)\cot\left(\frac{x}{6}\right)}$

$= \int \tan\frac{x}{2}.\tan\frac{x}{3}.\tan\frac{x}{6}dx$

$\because \frac{x}{2}-\frac{x}{3}=\frac{x}{6}$

$\Rightarrow \tan\left(\frac{x}{2}-\frac{x}{3}-\frac{x}{6}\right)=0$

$\Rightarrow \tan\frac{x}{2}+\tan\left(\frac{-x}{3}\right)+\tan\left(\frac{-x}{6}\right)=\tan\frac{x}{2}.\tan\left(\frac{-x}{3}\right).\tan\left(\frac{-x}{6}\right)$

$= \int\left(\tan\frac{x}{2}-\tan\frac{x}{3}-\tan\frac{x}{6}\right)dx$

$= 2\ln\left|\sec\frac{x}{2}\right|-3\ln\left|\sec\frac{x}{3}\right|-6\ln\left|\sec\frac{x}{6}\right|+C$

(b) $\int \frac{\tan(\ln x)\tan\left(\ln\frac{x}{2}\right)\tan(\ln 2)}{x}dx$

Let $\ln x = t$

$= \int \tan t.\tan(t-\ln 2).\tan(\ln 2)dx$

$= \int(\tan t-\tan(\ln 2)-\tan(t-\ln 2))dt$

$= \ln|\sec t| - t\tan(\ln 2) - \ln|\sec t - \ln 2| + C$

$$= \ln\left|\frac{\sec(\ln x)}{\sec\left(\ln\left(\frac{x}{2}\right)\right).x^{\tan(\ln 2)}}\right| + C$$

21. $\int\sqrt{\frac{\text{cosec}\,x - \cot x}{\text{cosec}\,x + \cot x}}.\frac{\sec x}{\sqrt{1+2\sec x}}dx$

Ans. $\sin^{-1}\left(\frac{1}{2}\sec^2\frac{x}{2}\right) + C$

Sol. $\int\sqrt{\frac{\text{cosec}\,x - \cot x}{\text{cosec}\,x + \cot x}}.\frac{\sec x}{\sqrt{1+2\sec x}}dx$

$$= \int\frac{1}{\text{cosec}\,x + \cot x}.\frac{1}{\sqrt{\cos^2 x + 2\cos x + 1 - 1}}dx$$

$$= \int\frac{\sin x}{1+\cos x}.\frac{1}{\sqrt{\cos^2 x + 2\cos x + 1 - 1}}dx$$

Let $1 + \cos x = t$

$-\sin x\,dx = dt$

$$= -\int\frac{dt}{t\sqrt{t^2-1}}$$

$= +\,\text{cosec}^{-1}(t) + C$

$$= \sin^{-1}\left(\frac{1}{2\cos^2\frac{x}{2}}\right) + C$$

$$= \sin^{-1}\left(\frac{1}{2}\sec^2\frac{x}{2}\right) + C$$

22. $\int\frac{\tan 2\theta}{\sqrt{\cos^6\theta + \sin^6\theta}}d\theta$

Ans. $l\text{n}\left(\dfrac{1+\sqrt{1+3\cos^2 2\theta}}{\cos 2\theta}\right)+C$

Sol. $\displaystyle\int \frac{\tan 2\theta}{\sqrt{\cos^6\theta+\sin^6\theta}}\,d\theta$

$$=\int \frac{\sin 2\theta}{\cos 2\theta\sqrt{1-3\sin^2\theta\cos^2\theta}}\,d\theta$$

$$=\int \frac{\sin 2\theta}{\cos 2\theta\sqrt{\sin^4\theta+\cos^4\theta-\sin^2\theta\cos^2\theta}}\,d\theta$$

$$=\int \frac{\sin 2\theta}{\cos 2\theta}.\frac{1}{\sqrt{\left(\cos^2\theta-\sin^2\theta\right)^2+\sin^2\theta\cos^2\theta}}\,d\theta$$

$$=\int \frac{2\sin 2\theta}{\cos 2\theta}.\frac{1}{\sqrt{\cos^2 2\theta+\frac{1}{4}\sin^2 2\theta}}\,d\theta$$

$$=\int \frac{2\sin 2\theta}{\cos 2\theta}.\frac{1}{\sqrt{3\cos^2 2\theta+1}}\,d\theta$$

Let $\cos 2\theta = t$

$\Rightarrow -2\sin 2\theta\ d\theta = dt$

$$=-\int \frac{dt}{t\sqrt{3t^2+1}}=-\int \frac{t\,dt}{t^2\sqrt{3t^2+1}}$$

Let $3t^2 + 1 = \theta^2$

$\Rightarrow 6t\ dt = 2\theta d\theta$

$$=-\int \frac{2\theta}{6\left(\frac{\theta^2-1}{3}\right).\theta}d\theta=-\int \frac{d\theta}{\theta^2-1}$$

$$=\frac{-1}{2}\ln\left|\frac{\theta-1}{\theta+1}\right|+C$$

$$=\frac{-1}{2}\ln\left|\frac{\theta+1}{\theta-1}\right|+C$$

$$= \frac{1}{2}\ln\left|\frac{1+\sqrt{1+3\cos^2 2\theta}}{\sqrt{1+3\cos^2 2\theta}-1}\right| + C$$

$$= \ln\left|\frac{1+\sqrt{1+3\cos^2 2\theta}}{\sqrt{3}\cos 2\theta}\right| + C$$

$$= \ln\left|\frac{1+\sqrt{1+3\cos^2 2\theta}}{\cos 2\theta}\right| + C$$

23. $\int \frac{a^2\sin^2 x + b^2\cos^2 x}{a^4\sin^2 x + b^4\cos^2 x}dx$

Ans. $\frac{1}{a^2+b^2}\left(x + \tan^{-1}\left(\frac{a^2\tan x}{b^2}\right)\right) + C$

Sol. $\int \frac{a^2\sin^2 x + b^2\cos^2 x}{a^4\sin^2 x + b^4\cos^2 x}dx$

$$= \int \frac{\left(a^2 + \tan^2 x + b^2\right)}{\left(a^4\tan^2 x + b^4\right)}dx$$

$$= \frac{1}{a^2+b^2} = \int \frac{\left(a^4 + \tan^2 x + b^4 + b^2a^2 + a^2\tan^2 x\right)}{a^4\tan^2 x + b^4}dx$$

$$= \frac{1}{a^2+b^2}\left[\int dx + \int \frac{a^2b^2 + a^2\tan^2 x}{a^4\tan^2 x + b^4}dx\right]$$

$$= \frac{1}{a^2+b^2}\left[x + \int \left(\frac{\frac{a^2}{b^2} + \frac{a^2}{b^2}\tan^2 x}{\left(\frac{a^2}{b^2}\tan x\right)^2 + 1}\right)dx\right]$$

$$= \frac{1}{a^2+b^2}\left[x + \tan^{-1}\left(\frac{a^2\tan x}{b^2}\right)\right] + C$$

24. $\int \frac{\cos^2 x}{1+\tan x} dx$

Ans. $\frac{1}{4} ln(\cos x + \sin x) + \frac{x}{2} + \frac{1}{8} (\sin 2x + \cos 2x) + C$

Sol. $\int \frac{\cos^2 x}{1+\tan x} dx = \int \frac{\sec^2 x dx}{\left(1+\tan^2 x\right)^2 (1+\tan x)}$

Let $\tan x = t$

$\sec^2 x \, dx = dt$

$= \int \frac{dt}{(1+t)\left(1+t^2\right)^2}$

$\because \frac{1}{(1+t)\left(1+t^2\right)^2} = \frac{A}{1+t} + \frac{B}{\left(1+t^2\right)} + \frac{Ct+D}{\left(t^2+1\right)^2}$

$\Rightarrow A = \frac{1}{4}, A + B = 0$

$\Rightarrow A = \frac{1}{4}, B = -\frac{1}{4}, E = \frac{1}{4}, C = -\frac{1}{2}, D = \frac{1}{2}$

$\Rightarrow \int \frac{dt}{(1+t)\left(1+t^2\right)^2} = \frac{1}{4}\int\left(\frac{1}{1+t}\right) + \frac{1-t}{4\left(1+t^2\right)} + \frac{1-t}{2\left(t^2+1\right)^2} dt$

$= \frac{1}{4}\int \frac{dt}{1+t} + \frac{1}{4}\int \frac{dt}{1+t^2} - \frac{1}{4}\int \frac{t}{1+t^2} dt$

$+ \frac{1}{4}\int \frac{(1-t)}{\left(t^2+1\right)^2} dt$

$= \frac{-1}{8} \log\left(\sec^2 x\right) + \frac{1}{4} \ln|1+\tan x| + \frac{1}{4} x$

$+ \frac{1}{2} I_1 \quad ; \; I_1 = \int \frac{1-t}{\left(t^2+1\right)^2} dt$

Now, $I_1 = \int \frac{(1-t)}{\left(1+t^2\right)^2} dt$

Let $\tan\theta = t$

$\sec^2\theta\, d\theta = dt$

$$= \int \frac{1-\tan\theta}{\sec^4\theta}.\sec^2\theta d\theta$$

$$= \int \cos^2\theta\left(1-\tan\theta\right)d\theta$$

$$= \int \left(1-\sin^2\theta+\sin^2\theta+\tan\theta-\tan\theta\right)d\theta$$

$$= \frac{\theta}{2}+\frac{1}{4}\sin 2\theta+\frac{\cos^2\theta}{2}$$

$$\Rightarrow \int \frac{\cos^2 x}{1+\tan x}dx$$

$$= \frac{x}{2}+\frac{1}{8}\left(\sin 2x+\cos 2x\right)$$

$$+\frac{1}{4}\log\left(\sin x+\cos x\right)+C$$

25. $\int (\sin x)^{-11/3}(\cos x)^{-1/3}dx$

Ans. $-\frac{3\left(1+4\tan^2 x\right)}{8(\tan x)^{8/3}} + C$

Sol. $\int (\sin x)^{-11/3}(\cos x)^{-1/3}dx$

$$= \int \frac{dx}{\left(\sin^{11} x\cos x\right)^{1/3}} = \int \frac{1}{\left(\sin^{12} x\cot x\right)^{1/3}}dx$$

Let $\cot x = t$

$\text{cosec}^2 x\, dx = -dt$

$$= -\int\left(\frac{1+t^2}{t^{1/3}}\right)dt - \int t^{-1/3}dt - \int t^{5/3}dt$$

$$= -\left[\frac{t^{2/3}}{2/3}+\frac{t^{8/3}}{8/3}\right]+C$$

$$= \frac{-3}{8}\left[4\cos^{2/3} x + \cot^{8/3} x\right] + C$$

26. $\int \frac{dx}{\sec x + \cos ec x}\, dx$

Ans. $\frac{1}{2}\left[\sin x - \cos x - \frac{1}{\sqrt{2}} \ln \tan\left(\frac{x}{2} + \frac{\pi}{8}\right)\right] + C$

Sol. $\int \frac{dx}{\sec x + \cos ecx} = \frac{1}{2}\int \frac{\sin 2x}{\sin x + \cos x} dx$

$$= \frac{1}{2}\int\left(\frac{(\sin x + \cos x)^2 - 1}{\sin x \cos x}\right) dx$$

$$\frac{1}{2}\left[\int (\sin x + \cos x)\, dx - \frac{1}{\sqrt{2}}\int \frac{dx}{\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x}\right]$$

$$= \frac{1}{2}\left[\sin x - \cos x - \frac{1}{\sqrt{2}}\int \frac{dx}{\sin\left(x + \frac{\pi}{4}\right)}\right]$$

$$= \frac{1}{2}\left[\sin x - \cos x - \frac{1}{\sqrt{2}} \ln\left|\tan\left(\frac{x}{2} + \frac{\pi}{8}\right)\right|\right] + C$$

27. $\int \frac{dx}{\sin x + \sec x}$

Ans. $\frac{1}{2\sqrt{3}} \ln \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} + \arctan(\sin x + \cos x) + C$

Sol. $\int \frac{dx}{\sin x + \sec x} = \int \frac{\cos x\, dx}{\sin x \cos x + 1}$

$$= \int \frac{2\cos x\, dx}{\sin 2x + 2} = \int \frac{(\cos x + \sin x) + (\cos x - \sin x)}{\sin 2x + 1 + 1} dx$$

$$= \int \frac{(\cos x + \sin x) + (\cos x - \sin x)}{(\sin x + \cos x)^2 + 1} dx$$

$$= \int \frac{\cos x - \sin x}{1 + (\sin x + \cos x)^2} dx + \int \frac{(\cos x + \sin x)}{1 + (\cos x + \sin x)^2} dx$$

$$= \tan^{-1}(\sin x + \cos x) + \int \frac{(\cos x + \sin x)}{3 - (\sin x - \cos x)^2} dx$$

$$= \tan^{-1}(\sin x + \cos x) + \frac{1}{2\sqrt{3}} \ln \left| \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} \right| + C$$

28. $\int \frac{\cos x - \sin x}{7 - 9\sin 2x} dx$

Ans. $\frac{1}{24} ln \frac{(4 + 3\sin x + 3\cos x)}{(4 - 3\sin x - 3\cos x)} + C$

Sol. $\int \frac{\cos x - \sin x}{7 - 9\sin 2x} dx$

$$= \int \frac{\cos x - \sin x}{16 - 9(\sin x + \cos x)^2} dx$$

$$= \frac{1}{24} \ln \left| \frac{4 + 3\sin x + 3\cos x}{4 - 3\sin x - 3\cos x} \right| + C$$

29. $\int \frac{\sqrt{\cot x} - \sqrt{\tan x}}{1 + 3\sin 2x} dx$

Ans. $\tan^{-1} \left(\frac{\sqrt{2\sin 2x}}{\sin x + \cos x} \right) + C$

Sol. $\int \frac{\sqrt{\cot x} - \sqrt{\tan x}}{1 + 3\sin 2x} dx$

$$= \int \frac{\cos x - \sin x}{\sqrt{\sin x \cos x} \left(3(\sin x + \cos x)^2 - 2 \right)} dx$$

$$= \sqrt{2}\int \frac{\cos x - \sin x}{\sqrt{\sin 2x}\left(3\left(\sin x + \cos x\right)^2 - 2\right)} dx$$

Let $\sin x + \cos x = t \Rightarrow 1 + \sin 2x = t^2$

$\Rightarrow (\cos x - \sin x)\, dx = dt$

$$= \sqrt{2}\int \frac{dt}{\sqrt{t^2 - 1}\left(3t^2 - 2\right)}$$

Let $t = \frac{1}{\theta}$

$$\Rightarrow dt = \frac{-1}{\theta^2} d\theta$$

$$= \sqrt{2}\int \frac{\frac{-1}{\theta^2} d\theta}{\frac{\sqrt{1-\theta^2}}{\theta}.\left(\frac{3-2\theta^2}{\theta^2}\right)} = \sqrt{2}\int \frac{-\theta}{\sqrt{1-\theta^2}\left(3-2\theta^2\right)} d\theta$$

Let $\sqrt{1-\theta^2} = \alpha$

$$\Rightarrow \frac{\theta d\theta}{\sqrt{1-\theta^2}} = d\alpha$$

$$= \sqrt{2}\int \frac{d\alpha}{1+2\alpha^2} = \frac{\sqrt{2}}{\sqrt{2}} \tan^{-1}\left(\sqrt{2}\alpha\right) + C$$

$$= \tan^{-1}\left(\sqrt{2}.\sqrt{1-\theta^2}\right) + C$$

$$= \tan^{-1}\left(\frac{\sqrt{2}\sqrt{t^2-1}}{t}\right) + C$$

$$= \tan^{-1}\left(\frac{\sqrt{2\sin 2x}}{\sin x + \cos x}\right) + C$$

INTEGRATION BY PARTIAL FRACTIONS

30. $\int \frac{3x^2+1}{(x^2-1)^3}dx$

Ans. $C - \frac{x}{(x^2-1)^2}$

Sol. $\int \frac{3x^2+1}{(x^2-1)^3}dx = \int \frac{3x^2+1}{(x-1)^3(x+1)^3}dx$

$= \int -\frac{1}{2}\left[\frac{1}{(x+1)^3} - \frac{1}{(x-1)^3}\right]dx$

$= \frac{1}{2}\left[\frac{(x-1)^{-2}}{-2} + \frac{(x+1)^{-2}}{-2}\right] = C - \frac{x}{(x^2-1)^2}$

31. Let f(x) is a quadratic function such that f(0) = 1 and $\int \frac{f(x)\,dx}{x^2(x+1)^3}$ is a rational function, find the value of $f'(0)$.

माना f(x) एक द्विघातीय फलन है जहाँ f(0) = 1 तथा $\int \frac{f(x)\,dx}{x^2(x+1)^3}$ एक परिमेय फलन है तो $f'(0)$ का मान बताइये।

Ans. 3

Sol. $\int \frac{f(x)dx}{x^2(x+1)^3} = \int\left(\frac{1}{x^2} - \frac{x}{(x+1)^3}\right)dx$

$\int \frac{f(x)dx}{x^2(x+1)^3} = \int \frac{3x^2+3x+1}{x^2(x+1)^3}dx$

$\Rightarrow f(x) = 3x^2 + 3x + 1$

$f'(0) = 3$

32. $\int \frac{4x^5 - 7x^4 + 8x^3 - 2x^2 + 4x - 7}{x^2(x^2+1)^2}dx$

Ans. $4\,\ln x + \frac{7}{x} + 6\tan^{-1}(x) + \frac{6x}{1+x^2} + C$

Sol. $I = \int \frac{4x^5 - 7x^4 + 8x^3 - 2x^2 + 4x - 7}{x^2(x^2+1)^2} dx$

We can split into partial fraction as $\frac{A}{x^2} + \frac{B}{x} + \frac{Cx+D}{x^2+1} + \frac{Ex+F}{(x^2+1)^2}$ _____(1)

Multiplying both sides by x^2 and putting $x = 0$ we get $A = -7$

Similaly multiplying by $(x^2 + 1)$ and substituting $x = i$ we get $Ei + F = 12$

$\therefore$ $E = 0$, $F = 12$

to solve it substituting different and convenient values of x.

put $x = 1$ we get $2B + C + D = 8$ _____(2)

$x = -1$ we get $-2B - C + D = -8$ _____(3)

$x = 2$ we get $5B + 4C = 20$ _____(4)

From (2), (3) and (4) we get $D = 0$, $C = 0$ and $B = 4$

$$I = \int \left(\frac{-7}{x^2} + \frac{4}{x} + \frac{12}{(x^2+1)^2} \right) dx$$

$$I = \frac{7}{x} + 4\log|x| + \int \frac{12}{(x^2+1)^2} dx$$

Put $x = \tan\theta$

$$\frac{7}{x} + 4\log|x| + 12\int \cos^2\theta \, d\theta$$

$$I = 4 \, ln \, x + \frac{7}{x} + 6\tan^{-1}(x) + \frac{6x}{1+x^2} + C$$

KUTUR PUTUR

33. $\int \frac{(ax^2 - b)\ dx}{x\sqrt{c^2x^2 - (ax^2+b)^2}}$

Ans. $\sin^{-1}\left(\frac{ax^2+b}{cx}\right) + k$

Sol. $$\int \frac{\left(ax^2-b\right)}{cx^2\sqrt{1-\left(\frac{ax^2+b}{cx}\right)^2}}dx$$

$$\int \frac{\left(\frac{a}{c}-\frac{b}{cx^2}\right)}{\sqrt{1-\left(\frac{a}{c}x+\frac{b}{cx}\right)^2}}dx$$

$$=\sin^{-1}\left(\frac{ax^2+b}{cx}\right)+K$$

34. $$\int \frac{dx}{\left(x+\sqrt{x(1+x)}\right)^2}$$

Ans. $2ln\ \frac{t}{2t+1}+\frac{1}{2t+1}+C$ when $t=x+\sqrt{x^2+x}$

Sol. $$\int \frac{dx}{\left(x+\sqrt{x(1+x)}\right)^2}=\frac{\int \frac{dx}{x^2\left(1+\sqrt{1+\frac{1}{x}}\right)^2}}{}$$

Let $1+\frac{1}{x}=t^2$

$$=\int \frac{-2t\,dt}{(1+t)^2}=-2\int \frac{tdt}{(1+t)^2}$$

$$=-2\left[\int \frac{dt}{t+1}-\int \frac{1}{(1+t)^2}dt\right]$$

$$=-2\ln|t+1|+\frac{2}{t+1}+C$$

$$=-2\ln\left|\frac{\sqrt{x}+\sqrt{x+1}}{\sqrt{x}}\right|+\frac{2\sqrt{x}}{\sqrt{x+1}+\sqrt{x}}+C$$

35. $\int \frac{5x^4 + 4x^5}{\left(x^5 + x + 1\right)^2}\, dx$

Ans. $C - \frac{x+1}{x^5+x+1}$ or $C + \frac{x^5}{x^5+x+1}$

Sol. $\int \frac{5x^4 + 4x^5}{\left(x^5 + x + 1\right)^2}\, dx = \int \frac{\left(\frac{5}{x^6} + \frac{4}{x^5}\right)}{\left(1 + \frac{1}{x^4} + \frac{1}{x^5}\right)^2} dx$

Let $\frac{1}{x^4} + \frac{1}{x^5} + 1 = t$

$-\left(\frac{4}{x^5} + \frac{5}{x^6}\right) dx = dt$

$= -\int \frac{dt}{t^2} = \frac{1}{t} + C = \frac{x^5}{x^5+x+1} + C$

36. For any natural number m, evaluate $\int \left(x^{3m} + x^{2m} + x^m\right) \left(2x^{2m} + 3x^m + 6\right)^{\frac{1}{m}} dx$, where $x > 0$

यदि $m \in N$ तो $\int \left(x^{3m} + x^{2m} + x^m\right) \left(2x^{2m} + 3x^m + 6\right)^{\frac{1}{m}} dx$, का मान ज्ञात करो जहाँ $x > 0$

Ans. $\frac{1}{6(m+1)}\left(2x^{3m} + 3x^{2m} + 6x^m\right)^{\frac{m+1}{m}} + C$

Sol. $I = \int (x^{3m} + x^{2m} + x^m)(2x^{2m} + 3x^m + 6)^{1/m} dx$

$= \int (x^{3m} + x^{2m} + x^m)\left[\frac{2x^{3m} + 3x^{2m} + 6x^m}{x^m}\right]^{1/m} dx$

$= \int \left(\frac{x^{3m} + x^{2m} + x^m}{x}\right)(2x^{3m} + 3x^{2m} + 6x^m)^{1/m} dx$

$= \int (x^{3m-1} + x^{2m-1} + x^{m-1})(2x^{3m} + 3x^{2m} + 6x^m)^{1/m} dx$

Put $2x^{3m} + 3x^{2m} + 6x^m = y$

$\therefore I = \frac{1}{6m}\int y^{1/m} dy = \frac{1}{6m} \frac{y^{(1/m)+1}}{\left(\frac{m+1}{m}\right)} + C$

$$= \frac{1}{6}\frac{y^{\frac{m+1}{m}}}{(m+1)} + C$$

$$= \frac{1}{6}\frac{(2x^{3m} + 3x^{2m} + 6x^{m})^{\frac{m+1}{m}}}{m+1} + C$$

INTEGRATION OF IRRATIONAL ALGEBRAIC FUNCTIONS

37. $\int \frac{dx}{(x-\alpha)\sqrt{(x-\alpha)(x-\beta)}}$

Ans. $\frac{-2}{\alpha-\beta}\sqrt{\frac{x-\beta}{x-\alpha}} + C$

Sol. $\int \frac{dx}{(x-\alpha)\sqrt{(x-\alpha)(x-\beta)}}$

let $x - \alpha = \frac{1}{t} \quad \Rightarrow x = \frac{\alpha+1}{t}$

$\Rightarrow dx = \frac{-1}{t^2}dt$

$$= -\int \frac{t}{\sqrt{\frac{1}{t}\cdot\left(\alpha-\beta+\frac{1}{t}\right)}}\cdot\frac{1}{t^2}dt$$

$$= -\int \frac{1}{t\frac{\sqrt{(\alpha-\beta)t+1}}{t}}dt$$

$$= -\int \frac{dt}{\sqrt{(\alpha-\beta)t+1}}$$

$$= \frac{-2}{(\alpha-\beta)}\sqrt{(\alpha-\beta)t+1} + C$$

$$= \frac{-2}{(\alpha-\beta)}\sqrt{\frac{(\alpha-\beta)}{x-\alpha}+1} + C$$

$$= \frac{-2}{\alpha-\beta}\sqrt{\frac{x-\beta}{x-\alpha}} + C$$

MISCELLANEOUS

38. $\int \frac{x^3+x+1}{x^4+x^2+1}dx$

Ans. $\frac{1}{2}\ln(x^2+x+1)-\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right)-\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right)+C$

Sol. $\int \frac{x^3+x+1}{x^4+x^2+1}dx$

$$=\int \frac{\left(x^3+1\right)}{x^4+x^2+1}dx+\int \frac{x}{x^4+x^2+1}dx$$

$$=\frac{1}{2}\int \frac{2\left(x+1\right)\left(x^2-x+1\right)}{\left(x^2+x+1\right)\left(x^2-x+1\right)}+\frac{1}{2}\int \frac{2x}{\underbrace{x^4+x^2+1}_{\text{Let } x^2=t}}dx$$

$$2x\,dx = dt$$

$$=\frac{1}{2}\int \frac{2x+1}{x^2+x+1}+\frac{1}{2}\int \frac{dx}{x^2+x+1}\int \frac{dx}{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}}+\frac{1}{2}\int \frac{dt}{t^2+t+1}$$

$$=\frac{1}{2}\ln|x^2+x+1|+\frac{1}{2}\times\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)+\frac{1}{2}\times\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{t+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)+C$$

$$=\frac{1}{2}\ln|x^2+x+1|+\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right)+C$$

39. $\int \frac{(1+x^2)dx}{1-2x^2\cos\alpha+x^4}\ \alpha\in(0,\ \pi)$

Ans. $\frac{1}{2}\left(\operatorname{cosec}\frac{\alpha}{2}\right)\cdot\tan^{-1}\left(\left(\frac{x^2-1}{2x}\right)\operatorname{cosec}\frac{\alpha}{2}\right)$

Sol. $\int \frac{(1+x^2)dx}{1-2x^2\cos\alpha+x^4}\ \alpha\in(0,\ \pi)$

$$= \int \frac{\left(1+x^2\right)}{x^2\left(x^2-2\cos\alpha+\frac{1}{x^2}\right)}dx$$

$$= \int \frac{\left(1+\frac{1}{x^2}\right)dx}{\left(x-\frac{1}{x}\right)^2+2-2\cos\alpha}$$

Let $x-\frac{1}{x}=t$

$$\left(1+\frac{1}{x^2}\right)dx = dt$$

$$= \int \frac{dt}{t^2+2-2\cos\alpha} = \int \frac{dt}{t^2+\left(2\sin\frac{\alpha}{2}\right)^2}$$

$$= \frac{1}{2.\sin\frac{\alpha}{2}}\left[\tan^{-1}\left(\frac{t}{2\sin\frac{\alpha}{2}}\right)\right]+C$$

$$= \frac{1}{2}\cos ec\frac{\alpha}{2}\tan^{-1}\left(\left(\frac{x^2-1}{2x}\right)\cos ec\frac{\alpha}{2}\right)+C$$

DEFINITE INTEGRATION WITH THE HELP OF INDEFINITE INTEGRATION

40. Evaluate: $\int_0^1 e^{l\text{n}\tan^{-1}x}\cdot\sin^{-1}(\cos x)\,dx$.

ज्ञात करो : $\int_0^1 e^{l\text{n}\tan^{-1}x}\cdot\sin^{-1}(\cos x)\,dx$.

Ans. $\frac{\pi^2}{8}-\frac{\pi}{4}(1+l\text{n}2)+\frac{1}{2}$

Sol. $\int_0^1 e^{ln \tan^{-1} x} \cdot \sin^{-1}(\cos x)\, dx$

$= \int_0^1 \tan^{-1} x . \sin^{-1}(\cos x)\, dx = \int_0^1 \tan^{-1} x \left(\frac{\pi}{2} - x\right) dx$

$= \frac{\pi}{2}\int_0^1 \tan^{-1} x\ dx - \int_0^1 x \tan^{-1} x\, dx$

$= \frac{\pi}{2}\int_0^1 \tan^{-1} x dx - \int_0^1 x \tan^{-1} x\, dx$

Let $I = \int_0^1 \tan^{-1} x dx = \left[x \tan^{-1} x\right]_0^1 - \frac{1}{2}\int_0^1 \frac{2x}{1+x^2} dx$

$= \frac{\pi}{4} - \frac{1}{2}\left[\ln|1+x^2|\right]_0^1$

$= \frac{\pi}{4} - \frac{\ln 2}{2}$

Now, $I' = \int_0^1 x \tan^{-1} x = \left[\frac{x^2}{2}\tan^{-1} x\right]_0^1 - \frac{1}{2}\int_0^1 \frac{x^2}{1+x^2} dx$

$= \frac{\pi}{8} - \frac{1}{2}\left[I - \frac{\pi}{4}\right] = \frac{\pi}{4} - \frac{1}{2}$

$\Rightarrow \int_0^1 e^{\ln \tan^{-1} x} . \sin^{-1}(\cos x)\, dx = \frac{\pi^2}{8} - \frac{\pi}{4}\ln 2 - \frac{\pi}{4} + \frac{1}{2} + C$

41. Prove that :

(a) $\int_\alpha^\beta \sqrt{(x-\alpha)(\beta-x)}\ dx = \frac{(\beta-\alpha)^2\pi}{8}$

(b) $\int_\alpha^\beta \sqrt{\frac{x-\alpha}{\beta-x}}\ dx = (\beta-\alpha)\frac{\pi}{2}$

(c) $\int_\alpha^\beta \frac{dx}{x\sqrt{(x-\alpha)(\beta-x)}} = \frac{\pi}{\sqrt{\alpha\beta}}$ where $\alpha, \beta > 0$

(d) $\int_\alpha^\beta \frac{x.dx}{\sqrt{(x-\alpha)(\beta-x)}} = (\alpha+\beta)\frac{\pi}{2}$ where $\alpha < \beta$

सिद्ध कीजिए :

(a) $\int_{\alpha}^{\beta} \sqrt{(x-\alpha)(\beta-x)}\, dx = \frac{(\beta-\alpha)^2\pi}{8}$ **(b)** $\int_{\alpha}^{\beta} \sqrt{\frac{x-\alpha}{\beta-x}}\, dx = (\beta-\alpha)\frac{\pi}{2}$

(c) $\int_{\alpha}^{\beta} \frac{dx}{x\sqrt{(x-\alpha)(\beta-x)}} = \frac{\pi}{\sqrt{\alpha\beta}}$ जहाँ $\alpha, \beta > 0$ **(d)** $\int_{\alpha}^{\beta} \frac{x.dx}{\sqrt{(x-\alpha)(\beta-x)}} = (\alpha+\beta)\frac{\pi}{2}$ जहाँ $\alpha < \beta$

Ans.

Sol. (a) $\int_{\alpha}^{\beta} \sqrt{(x-\alpha)(\beta-x)}\, dx = \frac{(\beta-\alpha)^2\pi}{8}$

Let $x = \alpha\cos^2\theta + \beta\sin^2\theta \Rightarrow (x-\alpha) = (\beta-\alpha)\sin^2(\theta)$ and $(\beta-\alpha)\cos^2(\theta) = \beta - x$

$\delta\xi = (\beta-\alpha)\sin 2\theta d\theta$

$$= \frac{1}{2}\int (\beta-\alpha)\sin 2\theta.(\beta-\alpha)\sin 2\theta d\theta$$

$$= \frac{(\beta-\alpha)^2}{2}\int_0^{\pi/2} \sin^2 2\theta.d\theta = \frac{(\beta-\alpha)^2}{2}.\frac{\pi}{4} = \frac{(\beta-\alpha)^2\pi}{8}$$

(b) $\int_{\alpha}^{\beta} \sqrt{\frac{x-\alpha}{\beta-x}}\, dx = (\beta-\alpha)\frac{\pi}{2}$

Let $x = \alpha\cos^2\theta + \beta\sin^2\theta$

$$= (\beta-\alpha)\int \sqrt{\tan^2\theta}.\sin^2\theta d\theta$$

$$= 2(\beta-\alpha)\sin^2\theta\, d\theta = 2(\beta-\alpha).\frac{\pi}{4} = \frac{\pi(\beta-\alpha)}{2}$$

(c) $\int_{\alpha}^{\beta} \frac{dx}{x\sqrt{(x-\alpha)(\beta-x)}}$

Let $x = \alpha\cos^2\theta + \beta\sin^2\theta$

$$I = \int_0^{\pi/2} \frac{(\beta-\alpha)\sin 2\theta}{(\alpha\cos^2\theta + \beta\sin^2\theta)\sqrt{(\beta-\alpha)^2\sin^2\theta\cos^2\theta}} d\theta$$

$$I = 2\int_0^{\pi/2} \frac{1}{(\alpha\cos^2\theta + \beta\sin^2\theta)} d\theta$$

$$I = 2\int_0^{\pi/2} \frac{\sec^2\theta}{\alpha+\beta\tan^2\theta}d\theta$$

Let $\tan\theta = t$

$\sec^2\theta\, d\theta = dt$

$$I = \frac{2}{\alpha}\int_0^{\infty} \frac{d}{1+\frac{\beta t^2}{\alpha}}$$

If $\alpha, \beta > 0$, then

$$I = \frac{2}{\alpha}.\frac{\sqrt{\alpha}}{\sqrt{\beta}} = \left[\tan^{-1}\frac{t\sqrt{\alpha}}{\sqrt{\beta}}\right]_0^{\infty}$$

$$I = \frac{2}{\sqrt{\alpha\beta}}.\frac{\pi}{2} = \frac{\pi}{\sqrt{\alpha\beta}}$$

(d) $\int_{\alpha}^{\beta} \frac{x.dx}{\sqrt{(x-\alpha)(\beta-x)}}$

If $\beta > \alpha$ then

$\Lambda\varepsilon\tau\ \xi = \alpha\cos^2\theta + \beta\sin^2\theta$

$$= \int_0^{\pi/2} \frac{\left(\alpha\cos^2\theta+\beta\sin^2\theta\right)}{(\beta-\alpha)\sin\theta\cos\theta}.(\beta-\alpha)\sin 2\theta d\theta$$

$$= 2\int_0^{\pi/2} \left(\alpha\cos^2\theta+\beta\sin^2\theta\right)d\theta$$

$$= (\alpha+\beta)\frac{\pi}{2}$$

42. Evaluate: **(a)** $\int_0^{\pi/2} e^x\left\{\cos(\sin x)\cos^2\frac{x}{2}+\sin(\sin x)\sin^2\frac{x}{2}\right\}dx$ **(b)** $\int_1^e \left\{(1+x)e^x+(1-x)e^{-x}\right\}.\ln x\, dx$

मान बताइये : **(a)** $\int_0^{\pi/2} e^x\left\{\cos(\sin x)\cos^2\frac{x}{2}+\sin(\sin x)\sin^2\frac{x}{2}\right\}dx$ **(b)** $\int_1^e \left\{(1+x)e^x+(1-x)e^{-x}\right\}.\ln x\, dx$

Ans. (a) $\frac{1}{2}\left[e^{\pi/2}\left(\cos 1+\sin 1\right)-1\right]$; (b) $e^{1+e}+e^{1-e}+e^{-e}-e^{e}+e-e^{-1}$

Sol. (a) $\int_0^{\pi/2} e^x\left\{\cos(\sin x)\cos^2\frac{x}{2}+\sin(\sin x)\sin^2\frac{x}{2}\right\}dx$

$\because \cos^2\frac{x}{2}=\frac{1+\cos x}{2}$ & $\sin^2\frac{x}{2}=\frac{1-\cos x}{2}$

$$=\frac{1}{2}\int_0^{\pi/2} e^x\left(\underbrace{\cos\left(\sin x\right)+\sin\left(\sin x\right)}_{f(x)}+\underbrace{-\sin\left(\sin x\right).\cos x+\cos\left(\sin x\right)\cos x}_{f'(x)}\right)dx$$

$$=\frac{1}{2}\left[e^x\left(\cos\left(\sin x\right)+\left(\sin x\right)\right)\right]_0^{\pi/2}$$

$$=\frac{1}{2}\left[e^{\pi/2}\left(\cos 1+\sin 1\right)-1\right]$$

(b) $I=\int_1^e \underbrace{e^x\left(1+x\right)}_{II}\underbrace{\ell n\, x}_{I}\,dx+\int_1^e \underbrace{e^{-x}\left(1-x\right)}_{II}\underbrace{\ell n\, x}_{I}\,dx$

$$=\left[\ell n\, x\, e^x.x\right]_1^e-\int_1^e e^x dx+\left[e^{-x}\left(x\right).\ell n\, x\right]_1^e-\int_1^e e^{-x}dx$$

$= e^{1+e}+e^{1-e}+e^{-e}-e^{e}+e-e^{-1}$

43. If $P=\int_0^\infty \frac{x^2}{1+x^4}dx$; $Q=\int_0^\infty \frac{x\,dx}{1+x^4}$ and $R=\int_0^\infty \frac{dx}{1+x^4}$ then prove that

(a) $P=\frac{\pi}{4}$, **(b)** $P=R$, **(c)** $P-\sqrt{2}\,Q+R=\frac{\pi}{2\sqrt{2}}$

यदि $P=\int_0^\infty \frac{x^2}{1+x^4}dx$; $Q=\int_0^\infty \frac{x\,dx}{1+x^4}$ तथा $R=\int_0^\infty \frac{dx}{1+x^4}$ तो सिद्ध कीजिए।

(a) $P=\frac{\pi}{4}$, **(b)** $P=R$, **(c)** $P-\sqrt{2}\,Q+R=\frac{\pi}{2\sqrt{2}}$

Ans.

Sol. $P=\int_0^\infty \frac{x^2}{1+x^4}dx$

Let $x^2 = \tan\theta \Rightarrow 2x\,dx = \sec^2\theta\,d\theta$

$$\Rightarrow P = \int_0^{\pi/2} \frac{\tan\theta}{1+\tan^2\theta}\cdot\frac{\sec^2\theta\,d\theta}{2\sqrt{\tan\theta}}$$

$$P = \frac{1}{2}\int_0^{\pi/2} \sqrt{\tan\theta}\;d\theta$$

Apply king prop.

$$P = \frac{1}{2}\int_0^{\pi/2} \sqrt{\cot\theta}\,d\theta$$

$$\Rightarrow 2P = \frac{1}{2}\int_0^{\pi/2} \left(\sqrt{\tan\theta}+\sqrt{\cot\theta}\right)d\theta$$

$$\Rightarrow 2P = \frac{1}{4}\int_0^{\pi/2} \frac{(1+\tan\theta)}{\sqrt{\tan\theta}}d\theta$$

$$\Rightarrow P = \frac{\pi}{4}$$

$$R = \int_0^{\infty} \frac{dx}{1+x^4}$$

Let $x^2 = \tan\theta$

$2xdx = \sec^2\theta\,d\theta$

$$R = \int_0^{\pi/2} \frac{1}{\sec^2\theta}\frac{\sec^2\theta}{2\sqrt{\tan\theta}}d\theta$$

$$\Rightarrow \frac{1}{2}\int_0^{\pi/2} \sqrt{\cot\theta}\;d\theta$$

Apply king prop.

$$\Rightarrow R = \frac{1}{2}\int_0^{\pi/2} \sqrt{\tan\theta}\,d\theta$$

$$\Rightarrow R = \frac{\pi}{4}$$

$\Rightarrow P = R$

$$P - \sqrt{2}Q + R = \int_0^\infty \frac{\left(x^2 - \sqrt{2}x + 1\right)}{x^4 + 1} dx$$

$$= \int_0^\infty \frac{\left(x^2 - \sqrt{2}x + 1\right)dx}{\left(x^2 - \sqrt{2}x + 1\right)\left(x^2 + \sqrt{2}x + 1\right)} = \int_0^\infty \frac{dx}{x^2 + \sqrt{2}x + 1}$$

$$= \int_0^\infty \frac{dx}{\left(x + \frac{1}{\sqrt{2}}\right)^2 + \frac{1}{2}} = \sqrt{2}\left[\tan^{-1}\left(\frac{x + \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}\right)\right]_0^\infty = \frac{\pi}{2\sqrt{2}}$$

44. For $a \geq 2$, if the value of the definite integral $\int_0^\infty \frac{dx}{a^2 + (x - (1/x))^2} = \frac{\pi}{5050}$. Find the value of a.

$a \geq 2$ के लिये यदि $\int_0^\infty \frac{dx}{a^2 + (x - (1/x))^2}$ का मान $\frac{\pi}{5050}$ है तो a का मान बताओं।

Ans. 2525

Sol. $\int_0^\infty \frac{dx}{a^2 + (x - (1/x))^2} = \frac{\pi}{5050}$

Let $x = \frac{1}{t}$

$$= \int_\infty^0 \frac{-1}{t^2} \cdot \frac{1}{\left(a^2 + \left(\frac{1}{t} - t\right)^2\right)} dt$$

$$I = \int_0^\infty \frac{1}{x^2\left(a^2 + \left(x - \frac{x}{x}\right)^2\right)} dx$$

$$\Rightarrow 2I = \int_0^\infty \frac{\left(1 + \frac{1}{x^2}\right)}{a^2 + \left(x - \frac{1}{x}\right)^2} \text{ Let } x - \frac{1}{x} = \theta$$

$$2I = \int_0^{\infty} \frac{dx}{a^2+\theta^2} = \frac{1}{a}\left[\tan^{-1}\left(x-\frac{1}{x}\right)\right]_0^{\infty}$$

$$\Rightarrow I = \frac{1}{2a}\left[\frac{\pi}{2}+\frac{\pi}{2}\right] = \frac{\pi}{2a}$$

$$\Rightarrow a = 2525$$

45. If $\int_0^{\pi} \sqrt{(\cos x+\cos 2x+\cos 3x)^2+(\sin x+\sin 2x+\sin 3x)^2}\,dx$ has the value equal to $\left(\frac{\pi}{k}+\sqrt{w}\right)$ where k and w are positive integers find the value of (k^2+w^2).

यदि $\int_0^{\pi} \sqrt{(\cos x+\cos 2x+\cos 3x)^2+(\sin x+\sin 2x+\sin 3x)^2}\,dx$ का मान $\left(\frac{\pi}{k}+\sqrt{w}\right)$ जहाँ k एवं w धनात्मक पूर्णांक है तो $(k^2 + w^2)$ का मान होगा।

Ans. 153

Sol. $\int_0^{\pi} \sqrt{(\cos x+\cos 2x+\cos 3x)^2+(\sin x+\sin 2x+\sin 3x)^2}\,dx$

$$= \int_0^{\pi} \sqrt{\cos^2 2x\left(1+2\cos x\right)^2+\sin^2 2x\left(1+2\cos x\right)^2}\,dx$$

$$= \int_0^{\pi} \left|1+2\cos x\right|dx$$

$$= \int_0^{\pi/3} \left(1+2\cos x\right)dx - \int_{2\pi/3}^{\pi} \left(1+2\cos x\right)dx$$

$$= \frac{2\pi}{3}+2.\frac{\sqrt{3}}{2}-\frac{\pi}{3}+2.\frac{\sqrt{3}}{2}$$

$$= \frac{\pi}{3}+2\sqrt{3}$$

$\Rightarrow k = 3$ & $w = 12$

$\Rightarrow k^2 + w^2 = 144 + 9 = 153$

46. $\int_0^{\frac{l\ln 3}{2}} \frac{e^x+1}{e^{2x}+1}dx$

Ans. $\frac{1}{2}\left[\frac{\pi}{6}+l\ln 3-l\ln 2\right]$

Sol. $\int_0^{\frac{l\ln 3}{2}} \frac{e^x+1}{e^{2x}+1}dx =$

$$= \underbrace{\int_0^{\ln 3/2} \frac{e^x}{e^{2x}+1}dx}_{\text{let } e^x=t} + \int_0^{\ln 3/2} \frac{dx}{1+e^{2x}}$$

$$= \int_1^{\sqrt{3}} \frac{dt}{1+t^2} + \underbrace{\int_0^{\ln 3/2} \frac{e^{-2x}}{e^{-2x}+1}dx}_{\substack{e^{-2x}=t \\ -2e^{-2x}dx=dt}}$$

$$= \frac{\pi}{2}-\frac{\pi}{4}-\frac{1}{2}\left[\ln\left(1+e^{-2x}\right)\right]_0^{\ln\sqrt{3}}$$

$$= \frac{1}{2}\left[\frac{\pi}{6}+\ln 3-\ln 2\right]$$

47. Let $I = \int_0^1 \frac{2+3x+4x^2}{2\sqrt{1+x+x^2}}dx$. Find the value of I^2.

माना कि $I = \int_0^1 \frac{2+3x+4x^2}{2\sqrt{1+x+x^2}}dx$. तो I^2 का मान बताइये।

Ans. 3

Sol. $I = \int_0^1 \frac{2+3x+4x^2}{2\sqrt{1+x+x^2}}dx$

$$\Rightarrow I = \int_0^1 \left(\sqrt{1+x+x^2}+\frac{x(2x+1)}{x\sqrt{1+x+x^2}}\right)dx$$

Let $f(x)=\sqrt{1+x+x^2}$

$$f'(x)=\frac{2x+1}{2\sqrt{1+x+x^2}}$$

$$\Rightarrow \int_0^1 \left(f(x)+xf'(x)\right)dx$$

$$=\left[xf(x)\right]_0^1$$

$$=f(1)=\sqrt{3}$$

$$\Rightarrow I^2=3$$

48. Show that the sum of the two integrals $\int_{-4}^{-5} e^{(x+5)^2}dx+3\int_{1/3}^{2/3} e^{9(x-2/3)^2}dx$ is zero.

दर्शाइये कि दोनों समाकलन का योग $\int_{-4}^{-5} e^{(x+5)^2}dx+3\int_{1/3}^{2/3} e^{9(x-2/3)^2}dx$ शून्य है।

Ans.

Sol. $\int_{-4}^{-5} e^{(x+5)^2}dx+3\int_{1/3}^{2/3} e^{9(x-2/3)^2}dx$

$$\int_{-4}^{-5} e^{(x+5)^2}dx+3\underbrace{\int_{1/3}^{2/3} e^{(3x-2)^2}dx}_{\text{let } 2-3x=t}$$

$$=\int_{-4}^{-5} e^{(x+5)^2}dx-\int_{-1}^{0} e^{t^2}dt$$

Let $x+5=\theta$

$$I=\int_1^0 e^{t^2}d\theta-\int_{+1}^0 e^{t^2}dt=0$$

49. $\int_0^{\pi/2} \tan^{-1}\left[\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right]dx$

Ans. $\frac{3\pi^2}{16}$

Sol. $\int_0^{\pi/2} \tan^{-1}\left[\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right]dx$

$$= \int_0^{\pi/2} \tan^{-1}\left[\frac{\left|1\sin\frac{x}{2}+\cos\frac{x}{2}\right|+\left|\sin\frac{x}{2}-\cos\frac{x}{2}\right|}{\left|\sin\frac{x}{2}+\cos\frac{x}{2}\right|-\left|\sin\frac{x}{2}-\cos\frac{x}{2}\right|}\right]dx$$

$$= \int_0^{\pi/2} \tan^{-1}\left(\cot\frac{x}{2}\right)dx$$

$$= \left[\frac{\pi}{2}x-\frac{x^2}{4}\right]_0^{\pi/2}$$

$$= \frac{\pi^2}{4}-\frac{\pi^2}{16}=\frac{3\pi^2}{16}$$

50. $\int_{\frac{\sqrt{3a^2+b^2}}{2}}^{\sqrt{\frac{a^2+b^2}{2}}} \frac{x.dx}{\sqrt{(x^2-a^2)(b^2-x^2)}}$

Ans. $\frac{\pi}{12}$

Sol. $\int_{\frac{\sqrt{3a^2+b^2}}{2}}^{\sqrt{\frac{a^2+b^2}{2}}} \frac{x.dx}{\sqrt{(x^2-a^2)(b^2-x^2)}}$

Let $x^2 - a^2 = t^2$

$\Rightarrow xdx = tdt$

$$I = \int \frac{tdt}{t\sqrt{\left(b^2-a^2\right)-t^2}} = \left(\sin^{-1}\left(\frac{t}{\sqrt{b^2-a^2}}\right)\right)$$

$$= \left[\sin^{-1}\left(\sqrt{\frac{x^2 - a^2}{b^2 - a^2}} \right) \right]_{\frac{\sqrt{3a^2+b^2}}{2}}^{\frac{\sqrt{a^2+b^2}}{2}}$$

$$= \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) - \sin^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12}$$

51. (a) Show that $\int_0^{\infty} \frac{dx}{x^2 + 2x\cos\theta + 1} = 2\int_0^{1} \frac{dx}{x^2 + 2x\cos\theta + 1}$

(b) Evaluate: $f(\theta) = \int_0^{\infty} \frac{\tan^{-1} x}{x^2 + 2x\cos\theta + 1} dx$, $\theta \in (0, \pi)$

(a) सिद्ध करो कि $\int_0^{\infty} \frac{dx}{x^2 + 2x\cos\theta + 1} = 2\int_0^{1} \frac{dx}{x^2 + 2x\cos\theta + 1}$

(b) ज्ञात करो : $f(\theta) = \int_0^{\infty} \frac{\tan^{-1} x}{x^2 + 2x\cos\theta + 1} dx$, $\theta \in (0, \pi)$

Ans. (b) $\frac{\pi\theta}{4\sin\theta}$

Sol. (a) $\int_0^{\infty} \frac{dx}{x^2 + 2x\cos\theta + 1} = \int_0^{\infty} \frac{dx}{(x + \cos\theta)^2 + 1 - \cos^2\theta}$

Let $x + \cos\theta = t$

$$= \int \frac{dt}{t^2 + \sin^2\theta} = \frac{1}{\sin\theta} \tan^{-1}\left[\frac{t}{\sin\theta}\right]$$

$$= \frac{1}{\sin\theta} \tan^{-1}\left[\frac{x + \cos\theta}{\sin\theta}\right]_0^{\infty}$$

$$= \frac{1}{\sin\theta}\left(\frac{\pi}{2} - \left(\frac{\pi}{2} - \theta\right)\right) = \frac{\theta}{\sin\theta}$$

Now, $2\int_0^{1} \frac{dx}{(x + \cos\theta)^2 + \sin^2\theta}$ Let $x + \cos\theta = t$

$$= 2\int_{\cos\theta}^{1+\cos\theta} \frac{dt}{t^2+\sin^2\theta} = \frac{2}{\sin\theta}\tan^{-1}\left[\frac{t}{\sin\theta}\right]_{\cos\theta}^{1+\cos\theta}$$

$$= \frac{2}{\sin\theta}\left[\tan^{-1}\left[\cos ec\theta + \cot\theta\right] - \tan^{-1}\left(\cot\theta\right)\right]$$

$$= \frac{2}{\sin\theta}\left[\frac{\pi}{2} - \frac{\theta}{2} - \left(\frac{\pi}{2} - \theta\right)\right] = \frac{\theta}{\sin\theta}$$

$$\Rightarrow \int_0^{\infty} \frac{dx}{x^2+2x\cos\theta+1} = 2\int_0^{1} \frac{dx}{x^2+2x\cos\theta+1}$$

(b) $f(\theta) = \int_0^{\infty} \frac{\tan^{-1} x}{x^2+2x\cos\theta+1} dx$, $\theta \in (0, \pi)$

Let $x = \frac{1}{t}$

$$dx = \frac{-1}{t^2}dt$$

$$f(\theta) = \int_0^{\infty} \frac{\cot^{-1} t}{t^2+2t\cos\theta+1} dt$$

$$\Rightarrow 2f(\theta) = \int_0^{\infty} \frac{\left(\tan^{-1} t + \cot^{-1} t\right)}{t^2+2t\cos\theta+1} dt$$

$$= \frac{\pi}{2} \times \frac{\theta}{\sin\theta}$$

$$\Rightarrow f(\theta) = \frac{\pi\theta}{4\sin\theta}$$

52. Let $I = \int_0^{\pi/2} \frac{\cos x + 4}{3\sin x + 4\cos x + 25} dx$ and $J = \int_0^{\pi/2} \frac{\sin x + 3}{3\sin x + 4\cos x + 25} dx$. If $25I = a\pi + b\,ln\frac{c}{d}$ where a, b, c and $d \in N$ and $\frac{c}{d}$ is not a perfect square of a rational then find the value of (a + b + c + d).

माना कि $I = \int_0^{\pi/2} \frac{\cos x + 4}{3\sin x + 4\cos x + 25} dx$ और $J = \int_0^{\pi/2} \frac{\sin x + 3}{3\sin x + 4\cos x + 25} dx$ यदि $25I = a\pi + b\ln\frac{c}{d}$ जहाँ a, b, c तथा $d \in N$ तथा $\frac{c}{d}$ किसी परिमेय संख्या का पूर्णवर्ग नही है तो (a + b + c + d) का मान बताइये।

Ans. 62

Sol. $I = \int_0^{\pi/2} \frac{\cos x + 4}{3\sin x + 4\cos x + 25} dx$

$J = \int_0^{\pi/2} \frac{\sin x + 3}{3\sin x + 4\cos x + 25} dx$

$\Rightarrow 3I - 4J = \int_0^{\pi/2} \frac{(3\cos x - 4\sin x)}{(3\sin x + 4\cos x + 25)} dx$

$\Rightarrow 3I - 4J = \left[\ln|3\sin x + 4\cos x + 25|\right]_0^{\pi/2}$

$3I - 4J = \ln\left(\frac{28}{29}\right)$ .....(i)

$4I + 3J = \int_0^{\pi/2} \frac{4\cos x + 3\sin x + 25}{3\sin x + 4\cos x + 25} dx$

$4I + 3J = \frac{\pi}{2}$ .....(ii)

$\Rightarrow$ From (i) & (ii)

$25I = 3\ln\left(\frac{28}{29}\right) + 2II$

$\Rightarrow a = 2, b = 3, c = 28, d = 29$

$\Rightarrow a + b + c + d = 62$

PROPERTY FOR DISCONTINUOUS/NON UNIFORM FUNCTION

53. Let $\beta(n) = \int_0^{n\pi} \sqrt{1 - \sin t}\, dt$. Find the value of $\beta(2) - \beta(1)$.

माना $\beta(n) = \int_0^{n\pi} \sqrt{1-\sin t}\ dt$ हो तो $\beta(2) - \beta(1)$ का मान ज्ञात कीजिये।

Ans. 4

Sol. $\beta(n) = \int_0^{n\pi} \sqrt{1-\sin t}\ dt$

$$\beta(n) = \int_0^{n\pi} \left|\sin\frac{t}{2} - \cos\frac{t}{2}\right| dt$$

$$\beta(2) - \beta(1) = \int_0^{2\pi} \left|\sin\frac{t}{2} - \cos\frac{t}{2}\right| dt - \int_0^{\pi} \left|\sin\frac{t}{2} - \cos\frac{t}{2}\right| dt$$

$$= \int_{\pi}^{2\pi} \left|\sin\frac{t}{2} - \cos\frac{t}{2}\right| dt = 2\int_{\pi/2}^{\pi} |\sin\theta - \cos\theta|\, d\theta \quad \text{If } \frac{t}{2} = \theta$$

$$= 2\int_{\pi/2}^{\pi} (\sin\theta - \cos\theta)\, d\theta$$

$$= 2\left[-\cos\theta - \sin\theta\right]_{\pi/3}^{\pi} = 2[1+1] = 4$$

54. $\int_0^{\pi/2} \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}\, dx$

Ans. $l\text{n}\, 2$

Sol. $\int_0^{\pi/2} \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}\, dx$

$$= \int_0^{\pi/2} \frac{|\sin x - \cos x|}{|\sin x + \cos x|} dx$$

$$= -\int_0^{\pi/4} \frac{(\sin x - \cos x)}{\sin x + \cos x} dx + \int_{\pi/4}^{\pi/2} \left(\frac{\sin x - \cos x}{\sin x + \cos x}\right) dx$$

$$= \left[\ln|\sin x + \cos x|\right]_0^{\pi/4} - \left[\ln(\sin x + \cos x)\right]_{\pi/4}^{\pi/2}$$

$$= \ln\sqrt{2} + \ln\sqrt{2}$$

$$= \ln 2$$

55. Find the value of the definite integral $\int_0^{\pi} \left|\sqrt{2}\sin x + 2\cos x\right| dx$.

निश्चित समाकलन $\int_0^{\pi} \left|\sqrt{2}\sin x + 2\cos x\right| dx$ का मान ज्ञात कीजिये।

Ans. $2\sqrt{6}$

Sol. $I = \int_0^{\pi} \left|\sqrt{2}\sin x + 2\cos x\right| dx$

$I = \sqrt{2}\int_0^{\pi} \left|\sin x + \sqrt{2}\cos x\right| dx$

let $\cos\theta = \frac{1}{\sqrt{3}}$

$= \sqrt{6}\int_0^{\pi} \left|\sin(x+\theta)\right| dx$

let $x + \theta = t$

$= \sqrt{6}\int_{\theta}^{\pi+\theta} \left|\sin t\right| dt$

$= \sqrt{6}\int_{\theta}^{\pi+\theta} \left|\sin t\right| dt$

$= \sqrt{6}\int_{\theta}^{\pi+\theta} \left|\sin t\right| dt$

$= \sqrt{6}\left[\int_{\theta}^{\pi} \sin t\, dt + \int_{\pi}^{\pi+\theta} -\sin t\, dt\right]$

$= \sqrt{6}\left[(-\cos t)_{\theta}^{\pi} + (\cos t)_{n}^{n+\theta}\right]$

$= \sqrt{6}\left[1 + \cos\theta - \cos\theta + 1\right]$

$= 2\sqrt{6}$

PROPERTY FOR ODD AND EVEN FUNCTIONS

56. $\int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7+3x^6-10x^5-7x^3-12x^2+x+1}{x^2+2}\,dx$

Ans. $\frac{\pi}{2\sqrt{2}}-\frac{16\sqrt{2}}{5}$

Sol. $I=\int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7+3x^6-10x^5-7x^3-12x^2+x+1}{x^2+2}\,dx$

Apply king

$$I=\int_{-\sqrt{2}}^{\sqrt{2}} \frac{-2x^7+3x^6+10x^5+7x^3-12x^2-x+1}{x^2+2}\,dx$$

$$\Rightarrow 2I=\int_{-\sqrt{2}}^{\sqrt{2}} \frac{6x^6-24x^2+2}{x^2+2}dx$$

$$\Rightarrow I=\int_{-\sqrt{2}}^{\sqrt{2}} \frac{3x^6-12x^2+1}{x^2+2}dx$$

$$\Rightarrow I=2\int_{0}^{\sqrt{2}} \frac{3x^6-12x^2+1}{x^2+2}dx$$

$$=2\left[\int_{0}^{\sqrt{2}} \frac{3\left(x^4-4\right)x^2}{x^2+2}dx+\int_{0}^{\sqrt{2}} \frac{dx}{x^2+2}\right]$$

$$=2\left[3\int_{0}^{\sqrt{2}} x^2\left(x^2-2\right)dx+\frac{1}{\sqrt{2}}\left[\tan^{-1}\left(\frac{x}{\sqrt{2}}\right)\right]_0^{\sqrt{2}}\right]$$

$$=2\left[3\left[\frac{x^5}{5}-\frac{2x^3}{3}\right]_0^{\sqrt{2}}+\frac{1}{\sqrt{2}}\frac{\pi}{4}\right]$$

$$=\frac{\pi}{2\sqrt{2}}+\left[\frac{2\sqrt{2}}{5}-4\sqrt{2}\right]2=\frac{\pi}{2\sqrt{2}}-\frac{16\sqrt{2}}{5}$$

57. $\int_{-2}^{2} \frac{x^2 - x}{\sqrt{x^2+4}}\,dx$

Ans. $4\sqrt{2} - 4\ln(\sqrt{2}+1)$

Sol. $\int_{-2}^{2} \frac{x^2 - x}{\sqrt{x^2+4}}\,dx = \underbrace{\int_{-2}^{2} \frac{x^2}{\sqrt{x^2+4}}\,dx}_{\text{even}} - \underbrace{\int_{-2}^{2} \frac{x}{\sqrt{x^2+4}}\,dx}_{\text{odd}}$

$= 2\int_{0}^{2} \frac{x^2}{\sqrt{x^2+4}}\,dx$

$= 2\int_{0}^{2} \left(\sqrt{x^2+4} - \frac{4}{\sqrt{x^2+4}} \right) dx$

$= 2\left[\frac{x}{2}\sqrt{x^2+4} + 2\ln\left| x + \sqrt{x^2+4} \right| - 4\ln\left| x + \sqrt{x^2+4} \right| \right]_0^2$

$= 4\sqrt{2} - 4\ln\left(\sqrt{2}+1\right)$

58. Evaluate : $\int_{-1}^{1} \frac{(2x^{332} + x^{998} + 4x^{1668} \cdot \sin x^{691})}{1+x^{666}}\,dx$

ज्ञात बताइये : $\int_{-1}^{1} \frac{(2x^{332} + x^{998} + 4x^{1668} \cdot \sin x^{691})}{1+x^{666}}\,dx$

Ans. $\frac{\pi+4}{666}$

Sol. $\int_{-1}^{1} \frac{(2x^{332} + x^{998} + 4x^{1668} \cdot \sin x^{691})}{1+x^{666}}\,dx$

$= \underbrace{\int_{-1}^{1} \left(\frac{2x^{332} + x^{998}}{1+x^{666}} \right) dx}_{\text{even}} + \underbrace{\int_{-1}^{1} \frac{4x^{1668}.\sin x^{6941}}{1+x^{666}}\,dx}_{\text{odd}}$

$= 2\int_{0}^{1} \frac{2x^{332} + x^{998}}{1+x^{666}} dx + 0$

$$= 2\int_0^1 \frac{x^{332} + x^{332}\left(1+x^{666}\right)}{1+x^{666}}dx$$

$$= 2\left[\int_0^1 \frac{x^{332}}{1+\left(x^{333}\right)^2}dx + \int_0^1 x^{332}dx\right]$$

$$= 2\left[\frac{\tan^{-1}\left(x^{33}\right)}{333}\right]_0^1 + \frac{2}{333}$$

$$= \frac{\pi}{666} + \frac{2}{333} = \frac{\pi+4}{666}$$

59. Evaluate $\int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2-\cos\left(|x| + \frac{\pi}{3}\right)}dx$.

ज्ञात बताइये $\int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2-\cos\left(|x| + \frac{\pi}{3}\right)}dx$.

Ans. $\frac{4\pi}{\sqrt{3}}\tan^{-1}\left(\frac{1}{2}\right)$

Sol. $\int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2-\cos\left(|x| + \frac{\pi}{3}\right)}dx = 2\int_0^{\frac{\pi}{3}} \frac{\pi}{2-\cos\left(x + \frac{\pi}{3}\right)}dx$

Let $x + \frac{\pi}{3} = t$

$$= 2\pi \int_{\pi/3}^{2\pi/3} \frac{\sec^2\frac{t}{2}}{\sec^2\frac{t}{2} + 2\tan^2\frac{t}{2}}dt$$

$$= \frac{4\pi}{\sqrt{3}}\tan^{-1}\left(\frac{1}{2}\right)$$

KING & QUEEN PROPERTY

60. Let $u=\int_0^{\pi/4}\left(\frac{\cos x}{\sin x+\cos x}\right)^2 dx$ and $v=\int_0^{\pi/4}\left(\frac{\sin x+\cos x}{\cos x}\right)^2 dx$. Find the value of $\frac{v}{u}$.

माना कि $u=\int_0^{\pi/4}\left(\frac{\cos x}{\sin x+\cos x}\right)^2 dx$ तथा $v=\int_0^{\pi/4}\left(\frac{\sin x+\cos x}{\cos x}\right)^2 dx$. तो $\frac{v}{u}$ का मान बताओं।

Ans. 4

Sol. $u=\int_0^{\pi/4}\left(\frac{\cos x}{\sin x+\cos x}\right)^2 dx$

$v=\int_0^{\pi/4}\left(\frac{\sin x+\cos x}{\cos x}\right)^2 dx$

$\Rightarrow u=\int_0^{\pi/4}\left[\frac{\cos\left(\frac{\pi}{4}-x\right)}{\sin\left(\frac{\pi}{4}-x\right)+\cos\left(\frac{\pi}{4}-x\right)}\right]^2 dx$

$=\int_0^{\pi/4}\left(\frac{\cos x+\sin x}{\sqrt{2}\left(\sqrt{2}\cos x\right)}\right)^2 dx$

$=\frac{1}{4}\int_0^{\pi/4}\left(\frac{\cos x+\sin x}{\cos x}\right)^2 dx$

$\Rightarrow 4u=v$

$\Rightarrow \frac{v}{u}=4$

61. $\int_0^{\pi/4}\frac{x\,dx}{\cos x\,(\cos x+\sin x)}$

Ans. $\frac{\pi}{8}ln\,2$

Sol. $\int_0^{\pi/4}\frac{x\,dx}{\cos x\,(\cos x+\sin x)}dx$ .....(i)

Apply king

$$I=\int_0^{\pi/4}\frac{\left(\frac{\pi}{4}-x\right)}{\cos\left(\frac{\pi}{4}-x\right)\left(\cos\left(\frac{\pi}{4}-x\right)+\sin\left(\frac{\pi}{4}-x\right)\right)}dx$$

$$\Rightarrow I=\int_0^{\pi/4}\frac{\sqrt{2}\left(\frac{\pi}{4}-x\right)}{(\cos x+\sin x)\sqrt{2}\cos x}dx \qquad \text{.....(ii)}$$

By (i) & (ii)

$$2I=\frac{\pi}{4}\int_0^{\pi/4}\frac{1}{\cos x(\cos x+\sin x)}dx$$

$$=\frac{\pi}{4}\int_0^{\pi/4}\frac{\sec^2 x dx}{(1+\tan x)}x$$

$$=\frac{\pi}{4}\left[\ln|1+\tan x|\right]_0^{\pi/4}$$

$$2I=\frac{\pi}{4}\ln 2$$

$$\Rightarrow I=\frac{\pi}{8}\ln 2$$

62. $\int_0^1\frac{\sin^{-1}\sqrt{x}}{x^2-x+1}dx$

Ans. $\frac{\pi^2}{6\sqrt{3}}$

Sol. $I=\int_0^1\frac{\sin^{-1}\sqrt{x}}{x^2-x+1}dx$ .....(i)

Apply king property

$$=\int_0^1\frac{\sin^{-1}\sqrt{1-x}}{(1-x)^2-(1-x)+1}dx=\int_0^1\frac{\sin^{-1}\sqrt{1-x}}{x^2-x+1}dx \qquad \text{.....(ii)}$$

Add. (i) and (ii)

$$2I = \int_0^1 \frac{\sin^{-1}\sqrt{x} + \sin^{-1}\sqrt{1-x}}{x^2 - x + 1}\,dx$$

$$\Rightarrow 2I = \frac{\pi}{2}\int_0^1 \frac{dx}{(x-1/2)^2 + (\sqrt{3}/2)^2}$$

$$\Rightarrow I = \frac{\pi}{4}\frac{2}{\sqrt{3}}\left[\tan^{-1}\frac{2x-1}{\sqrt{3}}\right]_0^1 = \frac{\pi^2}{6\sqrt{3}}$$

63. $\int_0^{\pi/2} \frac{a\sin x + b\cos x}{\sin\left(\frac{\pi}{4}+x\right)}\,dx$

Ans. $\frac{\pi(a+b)}{2\sqrt{2}}$

Sol. $I = \int_0^{\pi/2} \frac{a\sin x + b\cos x}{\sin\left(\frac{\pi}{4}+x\right)}\,dx$

$$I = \sqrt{2}\int_0^{\pi/2} \frac{a\sin x + b\cos x}{\sin x + \cos x}\,dx \qquad .....(1)$$

Apply king property

$$I = \sqrt{2}\int_0^{\pi/2} \frac{a\cos x + b\sin x}{\cos x + \sin x} \qquad .....(2)$$

add.(1) and (2)

$$2I = \sqrt{2}(a+b)\int_0^{\pi/2} dx \quad \Rightarrow \quad I = \frac{(a+b)\pi}{2\sqrt{2}}$$

64. $\int_0^{\pi} \frac{(ax+b)\sec x\tan x}{4+\tan^2 x}\,dx \;\; (a,b>0)$

Ans. $\frac{(a\pi+2b)\pi}{3\sqrt{3}}$

Sol. $\int_0^{\pi} \frac{(ax+b)\sec x\tan x}{4+\tan^2 x}\,dx \;\; (a,b>0)$

Apply king property

$$I=\int_0^{\pi}\frac{(a\pi+b-ax)\sec x\tan x}{4+\tan^2 x}dx$$

$$\Rightarrow 2I=\int_0^{\pi}\frac{(a\pi+2b)\sec x\tan x}{4+\tan^2 x}dx$$

$$\Rightarrow 2I=(a\pi+2b)\int_0^{\pi}\frac{\sec x\tan x}{3+\sec^2 x}dx$$

Let sec x = t

sec x tan x dx = dt

$$\Rightarrow 2I=(a\pi+2b)=\int_1^{-1}\frac{dt}{3+t^2}$$

$$\Rightarrow 2I=-(a\pi+2b)\left[\frac{1}{\sqrt{3}}\tan^{-1}\frac{t}{\sqrt{3}}\right]_{-1}^{1}$$

$$\Rightarrow 2I=\frac{-(a\pi+2b)}{\sqrt{3}}\left[\frac{2\pi}{6}-\pi\right]\Rightarrow I=\frac{(a\pi+2b)\pi}{3\sqrt{3}}$$

65. $\int_0^{\pi}\frac{(2x+3)\sin x}{(1+\cos^2 x)}dx$

Ans. $\frac{\pi(\pi+3)}{2}$

Sol. $\int_0^{\pi}\frac{(2x+3)\sin x}{(1+\cos^2 x)}dx$

Apply king property

$$I=\int_0^{\pi}\frac{(2\pi-2x+3)\sin x}{1+\cos^2 x}dx$$

$$\Rightarrow 2I=\int_0^{\pi}\frac{(2\pi+6)\sin x}{1+\cos^2 x}dx$$

$$\Rightarrow I=(\pi+3)\int_0^{\pi}\frac{\sin x}{1+\cos^2 x}dx$$

Let $\cos x = t$

$\sin x\,dx = dt$

$$\Rightarrow I=-(\pi+3)\int_1^{-1}\frac{dt}{1+t^2}$$

$$\Rightarrow I=(\pi+3)\int_{-1}^{1}\frac{dt}{1+t^2}$$

$$\Rightarrow I=(\pi+3)\left[\tan^{-1}t\right]_{-1}^{1}$$

$$I=\frac{\pi(\pi+3)}{2}$$

66. Let $I_n=\int_{-n}^{n}\left(\{x+1\}\cdot\{x^2+2\}+\{x^2+2\}\{x^3+4\}\right)dx$ where { } denotes the fractional part of x. Find I_1.

माना $I_n=\int_{-n}^{n}\left(\{x+1\}\cdot\{x^2+2\}+\{x^2+2\}\{x^3+4\}\right)dx$ जहाँ { } भिन्नात्मक फलन है तो I_1 का मान बताइये।

Ans. 2/3

Sol. $I_n=\int_{-n}^{n}\left(\{x+1\}\cdot\{x^2+2\}+\{x^2+2\}\{x^3+4\}\right)dx$

$$\Rightarrow I_n=\int_{-n}^{n}\left(\{x\}\{x^2\}+\{x^2\}\{x^3\}\right)dx$$

$$\Rightarrow I_1=\int_{-1}^{n}\left(\{x\}+\{x^3\}\right)\{x^2\}dx$$

Apply king prop.

$$\Rightarrow I_1=\int_{-1}^{1}\left(\{-x\}+\{-x^3\}\right)\{x^2\}dx$$

$$\Rightarrow 2I_1=\int_{-1}^{1}\left(\{x\}+\{-x\}+\{x^3\}+\{-x^3\}\right)\{x^2\}dx$$

$$\Rightarrow 2I_1 = \int_{-1}^{1} x^2 (1+1)\,dx = 2\left[\frac{x^3}{3}\right]_{-1}^{1}$$

$$= \frac{4}{3}$$

$$\Rightarrow I_1 = \frac{2}{3}$$

67. $\int_0^{\pi} \frac{x^2 \sin 2x . \sin\left(\frac{\pi}{2}.\cos x\right)}{2x-\pi}\,dx.$

Ans. $\frac{8}{\pi}$

Sol. $I = \int_0^{\pi} \frac{x \sin 2x \sin\left(\frac{\pi}{2}\cos x\right)}{2x-\pi}\,dx$ _____(1)

Apply king property

$$I = \int_0^{\pi} \frac{(\pi - x)\sin 2(\pi - x)\sin\left(\frac{\pi}{2}\cos(\pi - x)\right)}{2(\pi - x) - \pi}\,dx$$

$$= \int_0^{\pi} \frac{(\pi - x)\sin 2x \sin\left(\frac{\pi}{2}\cos x\right)}{\pi - 2x}\,dx$$

$$= \int_0^{\pi} \frac{(x - \pi)\sin 2x \sin\left(\frac{\pi}{2}\cos x\right)}{2x - \pi}\,dx$$ _____(2)

add equation (1) & (2)

$$2I = \int_0^{\pi} \sin 2x \sin\left(\frac{\pi}{2}\cos x\right)dx$$

$\therefore \quad I = \int_0^{\pi} \sin x \cos x \sin\left(\frac{\pi}{2}\cos x\right)dx$

Put $\frac{\pi}{2}\cos x = t \Rightarrow \sin x\,dx = -\frac{2}{\pi}dt$

$$\therefore \quad I = -\frac{2}{\pi}\int_{\pi/2}^{-\pi/2}\frac{2t}{\pi}\sin t\,dt \qquad = \frac{4}{\pi^2}\int_{-\pi/2}^{\pi/2} t\sin t\,dt$$

$$\Rightarrow \quad I = \frac{4}{\pi^2}\int_{-\pi/2}^{\pi/2} t\sin t\,dt = \frac{4}{\pi^2}\left[-t\cos t + \sin t\right]_{-\pi/2}^{\pi/2}$$

$$= \frac{4}{\pi^2}\times 2 = \frac{8}{\pi^2}$$

68. $\int_0^{2\pi}\frac{dx}{2+\sin 2x}$.

Ans. $\frac{2\pi}{\sqrt{3}}$

Sol. $$\int_0^{2\pi}\frac{dx}{2+\sin 2x} = \int_0^{2\pi}\frac{dx}{2+\frac{2\tan x}{1+\tan^2 x}}$$

$$= \frac{1}{2}\int_0^{2\pi}\frac{1+\tan^2 x}{1+\tan^2 x+\tan x}dx$$

Apply queen property

$$= \frac{4}{2}\int_0^{\pi/2}\frac{\sec^2 x dx}{\left(\tan x+\frac{1}{2}\right)+\frac{3}{4}}$$

Let $\tan x = t$

$\Rightarrow \sec^2 x\,dx = dt$

$$= 2\int_0^{\infty}\frac{dt}{\left(t+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} = \frac{4}{\sqrt{3}}\left[\tan^{-1}\left(\frac{2t+1}{\sqrt{3}}\right)\right]_0^{\infty}$$

$$= \frac{4}{\sqrt{3}}\left[\frac{\pi}{2}\right]$$

69. If f is an even function then prove that $\int_0^{\pi/2} f(\cos 2x)\cos x\,dx = \sqrt{2}\int_0^{\pi/4} f(\sin 2x)\cos x\,dx$.

यदि f एक सम फलन है तो सिद्ध कीजिए $\int_0^{\pi/2} f(\cos 2x)\cos x\,dx = \sqrt{2}\int_0^{\pi/4} f(\sin 2x)\cos x\,dx$

Ans.

Sol. $I = \int_0^{\frac{\pi}{2}} f\left(\cos 2x\right)\cos x\,dx$ _____(1)

Given f is an even function.

$$I = \int_0^{\frac{\pi}{2}} f\left(\cos\left(\pi - 2x\right)\right)\cos\left(\frac{\pi}{2} - x\right)dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} f\left(\cos 2x\right)\sin x\,dx$$

By (1) + (2)

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} f\left(\cos 2x\right)\left(\sin x + \cos x\right)dx$$

$$2I = \sqrt{2}\int_0^{\frac{\pi}{2}} \cos\left(x - \frac{\pi}{4}\right) f\left(\cos 2x\right)dx$$

let $x - \frac{\pi}{4} = t$

$$I = \frac{1}{\sqrt{2}}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos t\, f\left(\sin 2t\right)dt$$

$$I = \sqrt{2}\int_0^{\frac{\pi}{4}} \cos t\, f\left(\sin 2t\right)dt$$

$$I = \sqrt{2}\int_0^{\frac{\pi}{4}} \cos x\, f\left(\sin 2x\right)dx$$

PROPERTY FOR PERIODIC FUCTIONS

70. Show that $\int_0^{p+q\pi} |\cos x|\, dx = 2q + \sin p$ where $q \in N$ & $-\frac{\pi}{2} < p < \frac{\pi}{2}$.

दर्शाइये कि $\int_0^{p+q\pi} |\cos x|\, dx = 2q + \sin p$ जहाँ $q \in N$ तथा $-\frac{\pi}{2} < p < \frac{\pi}{2}$

Ans.

Sol. $\int_0^{p+q\pi} |\cos x|\, dx = \int_0^{p} |\cos x|\, dx + \int_p^{p+q\pi} |\cos x|\, dx$

$= \int_0^{p} |\cos x|\, dx + \int_0^{q\pi} |\cos x|\, dx$

$= \int_0^{p} \cos x\, dx + q\int_0^{\pi} |\cos x|\, dx$

$\{\because q \in N\}$ $\{\&\ (|\cos x|) \text{ is periodic function with period } \pi\}$

$= \sin p + q\left(\int_0^{\pi/2} \cos x\, dx - \int_{\pi/2}^{\pi} \cos x\, dx\right)$

$= \sin p + 2q$

71. For any real number x, let [x] denote the largest integer less than or equal to x. Let f be a real valued function defined on the interval [–10, 10] by $f(x) = \begin{cases} x-[x] & \text{if } [x] \text{ is odd} \\ 1+[x]-x & \text{if } [x] \text{ is even} \end{cases}$. Then the value of $\frac{\pi^2}{10}\int_{-10}^{10} f(x)\cos \pi x\, dx$ is. **[JEE-2010]**

यदि x एक वास्तविक संख्या है, माना [x], x के महत्तम पूर्णांक मान जो x से छोटा अथवा बराबर है को दर्शाता है। यदि f एक वास्तविक मान का फलन अन्तराल [–10, 10] में $f(x) = \begin{cases} x-[x] & \text{यदि } [x] \text{ विषम है} \\ 1+[x]-x & \text{यदि } [x] \text{ सम है} \end{cases}$ द्वारा परिभाषित है। तो $\frac{\pi^2}{10}\int_{-10}^{10} f(x)\cos \pi x\, dx$ का मान है। **[JEE-2010]**

Ans. 4

Sol. $f(x) = \begin{cases} \{x\} & \text{when } -9 \le x < -8;\ -7 \le x < -6, \ldots\ldots \\ 1-\{x\} & \text{when } -10 \le x \le -9;\ -8 \le x < -7, \ldots\ldots \end{cases}$

Since f(x) & $\cos\pi x$ both are periodic functions having period 2.

$$I = \frac{10\times\pi^2}{10}\left(\int_0^1 (1-\{x\})\cos\pi x dx + \int_1^2 \{x\}\cos\pi x dx\right)$$

$$= \pi^2\left(\int_0^1 (1-x)\cos\pi x dx + \int_1^2 (x-1)\cos\pi x dx\right)$$

$$= \pi^2\left(\int_0^1 \cos\pi x dx - \int_1^2 \cos\pi x dx + \int_1^2 x\cos\pi x dx - \int_0^1 x\cos\pi x dx\right)$$

$$\Rightarrow \quad I = 4$$

DIFFERENTIATION OF INTEGRATION BY LEIBNITZ

72. $\lim\limits_{n\to\infty} n^2 \int\limits_{-1/n}^{1/n} (2010\sin x + 2012\cos x)|x|dx$.

Ans. 2012

Sol. $\lim\limits_{n\to\infty} n^2 \int\limits_{-1/n}^{1/n} (2010\sin x + 2012\cos x)|x|dx$

$$\text{Let } g(x) = \int_{-\frac{1}{n}}^{1/n} \underbrace{(2010\sin x + 2012\cos x)|x|dx}_{\text{let } f(x)}$$

$$= \int_0^{1/n} \left(f(x) + f(-x)\right)dx$$

$$= \int_0^{1/n} 1024\cos x|x|dx$$

$$= 1024\int_0^{1/n} \cos x.|x|dx$$

$$\Rightarrow \lim_{n\to\infty} \frac{1024\int_0^{1/n}\cos x.xdx}{\frac{1}{\pi^2}} \quad \left(\frac{0}{0}\text{form}\right)$$

$$= \lim_{n\to\infty} \frac{1024.\cos\left(\frac{1}{n}\right).\frac{1}{n}.-\frac{1}{n^2}}{\frac{-2}{n^3}} = 2012$$

73. If $y(x)=\int\limits_{\pi^2/16}^{x^2}\frac{\cos x.\cos\sqrt{\theta}}{1+\sin^2\sqrt{\theta}}.d\theta$ then find $\frac{dy}{dx}$ at $x=\pi$.

यदि $y(x)=\int\limits_{\pi^2/16}^{x^2}\frac{\cos x.\cos\sqrt{\theta}}{1+\sin^2\sqrt{\theta}}.d\theta$ तो $\frac{dy}{dx}$ का $x=\pi$ पर मान बताइये।

Ans. 2π

Sol. $y(x)=\int\limits_{\pi^2/16}^{x^2}\frac{\cos x\cos\sqrt{\theta}}{1+\sin^2\sqrt{\theta}}d\theta$

$\Rightarrow\frac{dy}{dx}=\frac{\cos^2 x}{1+\sin^2 x}.2x$

$\Rightarrow\left.\frac{dy}{dx}\right|_{atx=\pi}=\frac{2\pi}{1}=2\pi$

DETERMINATION OF FUNCTION

74. A continuous real function f satisfies $f(2x)=3f(x)\ \forall\ x\in R$ If $\int\limits_0^1 f(x)\,dx=1$, then compute the value of definite integral $\int\limits_1^2 f(x)\,dx$.

एक सतत् वास्तविक फलन f सम्बन्ध $f(2x)=3f(x)\ \forall\ x\in R$ को संतुष्ट कर रहा है। यदि $\int\limits_0^1 f(x)\,dx=1$, तो $\int\limits_1^2 f(x)\,dx$ का मान बताइये।

Ans. 5

Sol. $f(2x)=3f(x)\forall x\in R$

$\Rightarrow\int\limits_0^1\frac{1}{3}f(2x)dx=\int\limits_0^1 f(x)dx$

$=$ let $2x=t$

$\Rightarrow dx=\frac{dt}{2}$

$\Rightarrow\frac{1}{6}\int\limits_0^2 f(t)dt=7$

$\Rightarrow\int\limits_0^1 f(t)dt+\int\limits_1^2 f(t)dt=6$

$\Rightarrow \int_1^2 f(x)dx = 5$

75. Let f (x) be a function defined on R such that $f'(x) = f'(3-x)\ \forall\, x \in [0,3]$ with $f(0) = -32$ and $f(3) = 46$. Then find the value of $\int_0^3 f(x)\,dx$.

माना f(x), R पर परिभाषित फलन निम्न प्रकार से है $f'(x) = f'(3-x)\ \forall\, x \in [0,3]$ तथा $f(0) = -32$, $f(3) = 46$ तो $\int_0^3 f(x)\,dx$ का मान ज्ञात कीजिये।

Ans. 21

Sol. $f'(x)f'(3-x) \forall x \in [0,3],\ f(0) = -32$

$f(3) = 46$

let $I = \int_0^3 f(x)dx$

$\Rightarrow I = [xf(x)]_0^3 - \int_0^3 xf'(x)dx$

$\Rightarrow I = 3f(3) - \int_0^3 xf'(x)dx$

let $I' = \int_0^3 xf'(x)dx$

Apply king prop.

$I' = \int_0^3 (3-x)f'(3-x)dx = \int_0^3 f'(x).(3-x)dx$

$\Rightarrow I' = 3I[f(x)]_0^3 - I'$

$\Rightarrow 2I' = 3[f(3) - f(0)] = 3.[46+32]$

$= 3.[78] = 234$

$\Rightarrow I' = 117$

$\Rightarrow I = 138 - 117 = 21$

76. Let $f: R \to R$ be a continuous function which satisfies $f(x) = \int_0^x f(t)dt$. Then the value of $f(\ln 5)$ is.

[JEE-2009]

यदि $f: R \to R$ एक सतत् फलन है जो कि $f(x) = \int_0^x f(t)dt$ को संतुष्ट करता है तो $f(\ln 5)$ का मान बताइये। **[JEE 2009]**

Ans. 0

Sol. $f(x)=\int_0^x f(t)dt \Rightarrow f(0)=0$

$\Rightarrow f'(x)=f(x)$

$\Rightarrow \int \frac{f'(x)}{f(x)}dx=\int dx$

$\Rightarrow \ln|f(x)|=(x)+\ln c$

$\Rightarrow f(x)=ce^x$

$\Rightarrow \because f(0)=0$

$\Rightarrow c=0$

$\Rightarrow f(x)=0$

$\Rightarrow f(\ln 5)=0$

DEFINITE INTEGRALAS A LIMIT OFA SUM

77. Evaluate: $\underset{n\to\infty}{\text{Lim}} \frac{1}{n^2}\sum_{k=0}^{n-1}\left[k\int_k^{k+1}\sqrt{(x-k)(k+1-x)}\,dx\right]$

ज्ञात कीजिए : $\underset{n\to\infty}{\text{Lim}} \frac{1}{n^2}\sum_{k=0}^{n-1}\left[k\int_k^{k+1}\sqrt{(x-k)(k+1-x)}\,dx\right]$

Ans. $\frac{\pi}{16}$

Sol. $\underset{n\to\infty}{\text{Lim}} \frac{1}{n^2}\sum_{k=0}^{n-1}\left[k\int_k^{k+1}\sqrt{(x-k)(k+1-x)}\,dx\right]$

for $\int_k^{k+1}\sqrt{(x-k)(k+1-x)}dx$

let x = $k\cos^2\theta+(k+1)\sin^2\theta$

$\Rightarrow dx=\sin^2\theta d\theta$

$\Rightarrow x-k=\sin^2\theta$

$(k+1-x)=\cos^2\theta$

$=\frac{1}{2}\int_0^{\pi/2}\sin^2 2\theta d\theta=\frac{\pi}{8}$

$\Rightarrow \lim_{n\to\infty}\frac{1}{n^2}\left[\sum_{k=0}^{n-1}\frac{\pi k}{8}\right]=\frac{\pi}{8}\left(\lim_{n\to\infty}\frac{n(n-1)}{2n^2}\right)=\frac{\pi}{16}$

78. Evaluate: (a) $\lim_{n\to\infty}\left[\left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)\left(1+\frac{3^2}{n^2}\right).....\left(1+\frac{n^2}{n^2}\right)\right]^{1/n}$; (b) $\lim_{n\to\infty}\frac{1}{n}\left[\frac{1}{n+1}+\frac{2}{n+2}+.....+\frac{3n}{4n}\right]$

ज्ञात कीजिए : (a) $\lim_{n\to\infty}\left[\left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)\left(1+\frac{3^2}{n^2}\right).....\left(1+\frac{n^2}{n^2}\right)\right]^{1/n}$; (b) $\lim_{n\to\infty}\frac{1}{n}\left[\frac{1}{n+1}+\frac{2}{n+2}+.....+\frac{3n}{4n}\right]$

Ans. (a) $2e^{\frac{\pi-4}{2}}$; (b) $3-\ln 4$

Sol. $(A)\left(\sum_{r=1}^{n}\left(1+\frac{r^2}{n^2}\right)\right)^{1/n}=\{A(let)$

$$\Rightarrow\sum_{r=1}^{n}\frac{1}{n}\left(\ell n\left(1+\frac{r}{n}\right)^2\right)=\ell n(A)$$

$$\Rightarrow\lim_{n\to\infty}\ell nA=\lim_{n\to\infty}\sum_{r=1}^{n}\ell n\left(1+\left(\frac{r}{n}\right)^2\right).\frac{1}{n}$$

$$=\int_0^1\ell n(1+x^2)dx$$

$$\Rightarrow\ell nA=\left[x\ell n(1+x^2)\right]_0^1-\int_0^1\frac{2x^2}{1+x^2}dx$$

$$=\ell n2-2\int_0^1\frac{(x^2+1-1)}{x^2+1}dx$$

$$=\ell n2-2+2\left[\tan^{-1}x\right]_0^1$$

$$=\ell n2-2+2.\frac{\pi}{4}$$

$$\ell nA=\frac{\pi-4}{2}+\ell n2$$

$$\Rightarrow A=2.e\frac{\pi-4}{2}$$

$$(B)\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{3n}\frac{r}{n+r}$$

$$=\lim_{n\to\infty}\sum_{r=1}^{3n}\left(\frac{\frac{r}{n}}{1+\frac{r}{n}}\right).\frac{1}{n}$$

$$=\int_0^3\frac{x}{1+x}dx=\int_0^3\left(1-\frac{1}{1+x}\right)dx$$

$= 3 - \ell n\ 4$

79. Consider a function $f(n) = \dfrac{1}{1+n^2}$. Let $\alpha_n = \dfrac{1}{n}\sum_{r=1}^{n} f\left(\dfrac{r}{n}\right)$ and $\beta_n = \dfrac{1}{n}\sum_{r=0}^{n-1} f\left(\dfrac{r}{n}\right)$ for n = 1, 2, 3, ...

Also $\alpha = \underset{n\to\infty}{\text{Lim}}\,\alpha_n$ & $\beta = \underset{n\to\infty}{\text{Lim}}\,\beta_n$. Then prove **(a)** $\alpha_n < \beta_n$ **(b)** $\alpha = \beta$ **(c)** $\alpha_n < \dfrac{\pi}{4} < \beta_n$

दिया है $f(n) = \dfrac{1}{1+n^2}$. माना कि $\alpha_n = \dfrac{1}{n}\sum_{r=1}^{n} f\left(\dfrac{r}{n}\right)$ एवं $\beta_n = \dfrac{1}{n}\sum_{r=0}^{n-1} f\left(\dfrac{r}{n}\right)$ जहाँ n = 1, 2, 3, ... तथा $\alpha = \underset{n\to\infty}{\text{Lim}}\,\alpha_n$ और $\beta = \underset{n\to\infty}{\text{Lim}}\,\beta_n$. तो सिद्ध कीजिए कि **(a)** $\alpha_n < \beta_n$ **(b)** $\alpha = \beta$ **(c)** $\alpha_n < \dfrac{\pi}{4} < \beta_n$

Ans.

Sol. (a) $\alpha_n = \dfrac{1}{n}\left[\sum_{r=1}^{n} f\left(\dfrac{r}{n}\right)\right]$

$= \dfrac{1}{n}\left[f\left(\dfrac{1}{n}\right) + f\left(\dfrac{2}{n}\right) + - - - + f\left(\dfrac{n-1}{n}\right) + f(1)\right]$

$\beta_n = \dfrac{1}{n}\left[\sum_{r=0}^{n-1} f\left(\dfrac{r}{n}\right)\right]$

$= \dfrac{1}{n}\left[f\left(\dfrac{1}{n}\right) + f\left(\dfrac{2}{n}\right) + - - - + f\left(\dfrac{n-1}{n}\right)\right]$

$\Rightarrow \alpha_n = \beta_n + \dfrac{1}{2n} - \dfrac{f(0)}{n}$

$\Rightarrow \alpha_n = \beta_n - \dfrac{1}{2n}$

$\therefore \beta_n > \alpha_n$

(b) $f(n) = \dfrac{1}{1+n^2}$

$\lim\limits_{n\to\infty} \alpha_n = \lim\limits_{n\to\infty}\left(\sum_{r=1}^{n} f\left(\dfrac{r}{n}\right)\right)\dfrac{1}{n} = \int_0^1 f(x)\,dx$

$= \int_0^1 \dfrac{1}{1+x^2}dx = \left[\tan^{-1} x\right]_0^1 = \dfrac{\pi}{4}$

$\Rightarrow \alpha = \dfrac{\pi}{4}$

$\lim\limits_{n\to\infty} \beta_n = \left(\sum_{r=1}^{n-1} f\left(\dfrac{r}{n}\right)\right).\dfrac{1}{n} = \int_0^1 f(x)dx = \dfrac{\pi}{4}$

$\Rightarrow \beta = \frac{\pi}{4}$

$\Rightarrow \alpha = \beta$

(c) $\alpha_n < \frac{\pi}{4} < \beta_n$

DEFINITE INTEGRATION BY ESTIMATION

80. Prove the inequalities :

सिद्ध कीजिए :

(a) $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{2}}{8}$ **(b)** $2\,e^{-1/4} < \int_0^2 e^{x^2-x}\,dx < 2e^2.$

(c) $\frac{1}{3} < \int_0^1 x^{(\sin x+\cos x)^2} dx < \frac{1}{2}$ **(d)** $\frac{1}{2} \le \int_0^2 \frac{dx}{2+x^2} \le \frac{5}{6}$

Ans.

Sol. (a) $\int_0^1 \frac{dx}{\sqrt{4-x^2}} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^2}}$

$\Rightarrow \frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{2}}{8}$

(b) If $x \in [0,2]$

then $y = x^2 - x \in \left[\frac{-1}{4}, 2\right]$

$2\,e^{-1/4} < \int_0^2 e^{x^2-x}\,dx < 2e^2.$

(c) If $x \in [0,1]$

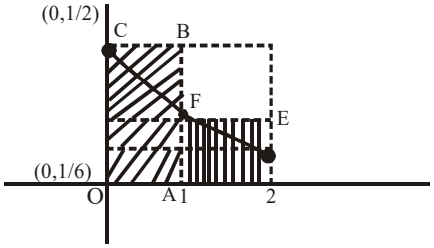
then $1+\sin 2x \in [1,2]$

$\Rightarrow \int_0^1 x^2 dx < \int_0^1 x^{1+\sin 2x}\,dx < \int_0^1 x\,dx$

$\Rightarrow \frac{1}{3} < \int_0^1 x^{1+\sin 2x} dx < \frac{1}{2}$

(d) If $x \in [0,2]$

then $x^2 + 2 \in [2,6]$



$$\Rightarrow \frac{1}{2} \le \int_0^2 \frac{dx}{2+x^2} \le \frac{1}{2}+\frac{1}{3}=\frac{5}{6}$$

↑ ↑

Area of OABC — Area of OABC + Area of ADEF

WALLI'S THEOREM

81. If a_1, a_2 and a_3 are the three values of a which satisfy the equation $\int_0^{\pi/2}(\sin x + a\cos x)^3 dx - \frac{4a}{\pi-2}\int_0^{\pi/2} x\cos x\,dx = 2$ then find the value of $1000(a_1^2+a_2^2+a_3^2)$

यदि a के तीन मान a_1, a_2 एवं a_3 है तथा समीकरण $\int_0^{\pi/2}(\sin x + a\cos x)^3 dx - \frac{4a}{\pi-2}\int_0^{\pi/2} x\cos x\,dx = 2$ को संतुष्ट करते है तो $1000(a_1^2+a_2^2+a_3^2)$ का मान बताइये।

Ans. 5250

Sol. $\int_0^{\pi/2}(\sin x + a\cos x)^3 dx - \frac{4a}{\pi-2}\int_0^{\pi/2} x\cos x dx = 2$

$$\int_0^{\frac{\pi}{2}} \sin^3 x\,dx + a^3\int_0^{\frac{\pi}{2}} \cos^3 x\,dx + 3a\int_0^{\frac{\pi}{2}} \sin^2 x\cos x + 3a^2\int_0^{\frac{\pi}{2}} \cos^2 x\sin x\,dx + \frac{4a}{\pi-2}\left[x\sin x+\cos x\right]_0^{\frac{\pi}{2}} = 2$$

$$\Rightarrow \frac{2}{3}+\frac{2}{3}a^3+a+a^2-\frac{4a}{\pi-2}\left[\frac{\pi-2}{2}\right]=2$$

$$\Rightarrow \frac{2a^3}{3}+a^2+a+\frac{2}{3}-2a=2$$

$$\Rightarrow 2a^3+3a^2-3a-4=0 \Leftarrow \begin{matrix} a_1 \\ a_2 \\ a_3 \end{matrix}$$

$$\left(a_1+a_2+a_3\right)^2=a_1^2+a_2^2+a_3^2+\left(\frac{-3}{2}\right).2$$

$$\Rightarrow a_1^2+a_2^2+a_3^2=\frac{9}{4}+3=\frac{21}{4}$$

$$\Rightarrow 1000\left(a_1^2+a_2^2+a_3^2\right)=5250$$

INTEGRATION BY REDUCTION

82. Determine a positive integer $n \le 5$, such that $\int_0^1 e^x(x-1)^n\ dx = 16-6e$.

धनात्मक पूर्णांक $n \le 5$ ज्ञात कीजिए जहाँ $\int_0^1 e^x(x-1)^n\ dx = 16-6e$.

Ans. $n = 3$

Sol. $I_n=\int_0^1 e^x.(x-1)^n\,dx=16-6e;\ n\le 5$

$$I_n=e^x.(x-1)^n\Big|_0^1-n\int_0^1 e^x(x-1)^{n-1}dx$$

$$I_n=-(-1)^n-n\,I_{n-1}=(-1)^{n+1}-n\,I_{n-1}$$

$$n=1,\ I=\int_0^1 e^x(x-1)dx$$

$$=(x-1)e^x\Big|_0^1-\int_0^1 e^x dx=2-e$$

$I_2 = -1-2(2-e) = 2e-5$

$I_3 = 1-3.(2e-5) = 16-6e$

so $n = 3$

83. Let $U_{10}=\int_0^{\frac{\pi}{2}} x\ \sin^{10}x\ dx$, then find the value of $\left(\frac{100U_{10}-1}{U_8}\right)$

माना $U_{10}=\int_0^{\frac{\pi}{2}} x\ \sin^{10}x\ dx$, हो तो $\left(\frac{100U_{10}-1}{U_8}\right)$ का मान ज्ञात कीजिये।

Ans. 90

Sol. Use reduction formula

$$10U_{10} = \frac{1}{10} + 9U_8$$

$$100U_{10} - 1 = 90U_8$$

84. Find the value of $\dfrac{5050\int_0^1 \left(1-x^{50}\right)^{100} dx}{\int_0^1 \left(1-x^{50}\right)^{101} dx}$

$\dfrac{5050\int_0^1 \left(1-x^{50}\right)^{100} dx}{\int_0^1 \left(1-x^{50}\right)^{101} dx}$ ज्ञात कीजिए।

Ans. 5051

Sol. Let $I = \int_0^1 (1-x^{50})^{100} dx$ and $I' = \int_0^1 (1-x^{50})^{101} dx$

Then, $I' = \int_0^1 1.(1-x^{50})^{101} dx = (x(1-x^{50})^{101})_0^1$

$$+101\int_0^1 50x^{50}(1-x^{50})^{100} dx$$

$$= 5050\int_0^1 x^{50}(1-x^{50})^{100} dx$$

$$-I' = 5050\int_0^1 -x^{50}(1-x^{50})^{100} dx$$

$$\Rightarrow 5050I - I' = 5050\int_0^1 (1-x^{50})^{100} dx$$

$$+5050\int_0^1 -x^{50}(1-x^{50})^{100} dx$$

$$\Rightarrow 5050\int_0^1 (1-x^{50})^{101}\,dx = 5050\ I'$$

$$\Rightarrow 5050\ I = 5051\ I' \Rightarrow 5050\frac{I}{I'} = 5051$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 85 to 87

Let the function f satisfies $f(x)\cdot f'(-x) = f(-x)\cdot f'(x)$ for all x and $f(0) = 3$.

माना फलन f प्रतिबन्ध $f(x)\cdot f'(-x) = f(-x)\cdot f'(x)$ को x के सभी मानों के लिए सन्तुष्ट करता है तथा $f(0) = 3$ है।

85. The value of $f(x)\cdot f(-x)$ for all x, is :

$f(x)\cdot f(-x)$ का x के सभी मानों के लिए मान होगा :

(A) 4 (B) 9 (C) 12 (D) 16

Ans. B

Sol. $f(x).f'(-x) = f(-x)f'(x)\ \forall x$

$f(0) = 3$

$$\Rightarrow \int \frac{f'(x)}{f(x)}dx = \int \frac{f'(-x)}{f(-x)}dx$$

$$= \log|f(x)| = -\log|f(-x)| + \log c$$

$$\Rightarrow f(x).f(-x) = c$$

at $x = 0$

$$(f(0))^2 = C \Rightarrow C = 9$$

$$\Rightarrow f(x).f(-x) = 9$$

86. $\int_{-51}^{51} \frac{dx}{3+f(x)}$ has the value equal to :

$\int_{-51}^{51} \frac{dx}{3+f(x)}$ का मान है :

(A) 17 (B) 34 (C) 102 (D) 0

Ans. A

Sol. $I = \int_{-51}^{51} \frac{dx}{3+f(x)}$ _____(1)

Apply king prop.

$$\Rightarrow I = \int_{-51}^{51} \frac{1}{3+f(-x)}dx = \int_{-51}^{51} \frac{f(x)}{3f(x)+9}dx$$ _____(2)

by (1) + (2)

$$\Rightarrow 2I = \int_{-51}^{51} \left(1+\frac{f(x)}{3}\right).\frac{1}{3+f(x)}dx = \frac{1}{3}.102$$

$$\Rightarrow I = 17$$

87. Number of roots of $f(x) = 0$ in $[-2, 2]$ is :

समीकरण $f(x) = 0$ का अन्तराल $[-2, 2]$ में हलों की संख्या होगी :

(A) 0 (B) 1 (C) 2 (D) 4

Ans. A

Sol. Let $x = \alpha$ be the root of $f(x) = 0$

$\therefore \quad f(\alpha) = 0$

$f(x) \cdot f(-x) = 9$

put $\xi = \alpha \Rightarrow 0 = 9$ impossible

hence $f(x)$ has no root

but $f(0) = 3$

hence $f(x) > 0 \ \forall \ x \in R$ as f is continuous

possible function $f(x) = 3e^{-x}$

Paragraph for Question nos. 88 to 90

Consider the integral $I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1+e^{2\sin^3 4x}} dx$

माना समाकलन $I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1+e^{2\sin^3 4x}} dx$ है।

88. If $I = k\int_0^{\pi/2} \cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x \, dx$ then 'k' is equal to :

यदि $I = k\int_0^{\pi/2} \cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x \, dx$ हो, तो k का मान है :

(A) 1 (B) 5 (C) 10 (D) 20

Ans. C

89. If $I = c \cdot \int_0^{\pi/4} \cos 6x \cdot \cos 8x \cdot \cos 2x \, dx$ then 'c' equals :

यदि $I = c \cdot \int_0^{\pi/4} \cos 6x \cdot \cos 8x \cdot \cos 2x \, dx$ हो, तो c का मान होगा :

(A) 5 (B) 10 (C) 20 (D) $\frac{5}{2}$

Ans. B

90. The value of 'I' equals :

ट का मान होगा :

(A) $\frac{5\pi}{4}$ (B) $\frac{5\pi}{16}$ (C) $\frac{5\pi}{32}$ (D) $\frac{5\pi}{8}$

Ans. D

Sol. (Que. 88, 89, 90)

$$I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1+e^{2\sin^3 4x}} dx \quad ....(1)$$

$$I = \int_0^{10\pi} \frac{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}{1+e^{-2\sin^3 4x}} dx \quad \text{(using King)}$$

$$I = \int_0^{10\pi} \frac{e^{2\sin^3 4x}(\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x)}{1+e^{2\sin^3 4x}} dx \quad \text{(using King)} \quad ....(2)$$

add (1) + (2)

$$2I = \int_0^{10\pi} \underbrace{\cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x}_{f(x)} dx$$

$f(x)$ repeats after 2π

$$2I = 5\int_0^{2\pi} f(x)\,dx \; ; \quad \text{again} \quad f(2a-x) = f(x)$$

$$2I = 10\int_0^{\pi} f(x)\,dx \; ; \quad \text{again} \; f(\pi - x) = f(x)$$

$$2I = 20\int_0^{\pi/2} f(x)\,dx \quad \Rightarrow \quad I = 10\int_0^{\pi/2} f(x)\,dx$$

$$\therefore \quad I = 10\int_0^{\pi/2} \cos 6x \cdot \cos 7x \cdot \cos 8x \cdot \cos 9x\,dx \quad ....(3)$$

$$I = 10\int_0^{\pi/2} \cos(3\pi - 6x)\cos\left(\frac{7\pi}{2} - 7x\right)\cos(4\pi - 8x)\cos\left(\frac{9\pi}{2} - 9x\right)dx$$

$$I = 10\int_0^{\pi/2} \cos 6x \cdot \sin 7x \cdot \cos 8x \cdot \sin 9x\, dx \qquad ....(4)$$

(3) + (4)

$$2I = 10\int_0^{\pi/2} \cos 6x \cdot \cos 8x[\cos 7x \cdot \cos 9x + \sin 7x \cdot \sin 9x]\, dx$$

$$2I = 10\int_0^{\pi/2} \cos 6x \cdot \cos 8x \cdot \cos 2x\, dx$$

$$I = 5\int_0^{\pi/2} \cos 6x \cdot \cos 8x \cdot \cos 2x\, dx = 10\int_0^{\pi/4} \cos 6x \cdot \cos 8x \cdot \cos 2x\, dx \Rightarrow (B)$$

$$I = 5\int_0^{\pi/2} \frac{\cos 8x}{2}(\cos 8x + \cos 4x)\, dx = \frac{5}{2}\int_0^{\pi/2} \cos 8x(\cos 8x + \cos 4x)dx$$

$$= \frac{5}{2}\int_0^{\pi/2} \cos^2 8x\, dx + \frac{5}{2}\underbrace{\int_0^{\pi/2} \cos 8x \cos 4x\, dx}_{\text{zero (using Queen)}}$$

$$\left(\int_0^{\pi/2}(\cos 12x + \cos 4x) = \frac{\sin 12x}{12} + \frac{1}{4}\sin 4x\Big]_0^{\frac{\pi}{2}} \to 0\right)$$

$$= \frac{1}{8}\cdot\frac{5}{2}\int_0^{4\pi} \cos^2 t\, dt \qquad (8x = t)$$

$$= \frac{5}{4}\int_0^{\pi} \cos^2 t\, dt \qquad (\text{using Queen})$$

$$I = \frac{5}{2}\int_0^{\pi/2} \cos^2 t\, dt = \frac{5}{2}\int_0^{\pi/2} \sin^2 t\, dt$$

$$2I = \frac{5}{2}\int_0^{\pi/2} dt \quad \Rightarrow \quad I = \frac{5}{4}\cdot\frac{\pi}{2} \quad \Rightarrow \quad I = \frac{5\pi}{8}$$

Paragraph for Question nos. 91 to 93

Let $y = f(x)$ be a twice differentiable, non-negative function defined on [a, b]. The area $\int_a^b f(x)dx, b > a$ bounded by $y = f(x)$, the x-axis and the ordinates at $x = a$ and $x = b$ can be approximated as $\int_a^b f(x)dx \cong \frac{(b-a)}{2}\{f(a)+f(b)\}$.

Since $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$, $c \in (a, b)$, a better approximation to $\int_a^b f(x)dx$ can be written as $\int_a^b f(x)dx \cong \frac{(c-a)}{2}\{f(a)+f(c)\} + \frac{(b-c)}{2}\{f(c)+f(b)\} \equiv F(c)$. If $c = \frac{a+b}{2}$, then this gives : $\int_a^b f(x)dx \cong \frac{b-a}{4}\{f(a)+2f(c)+f(b)\}$, ... (1) **[JEE-2006]**

यदि $y = f(x)$ अन्तराल [a, b] में ऋणात्मक नही है तथा दो बार अवकलनीय है तो क्षेत्रफल $\int_a^b f(x)dx, b > a$ जो कि $y = f(x)$, x-अक्ष तथा रेखा x = a एवं x = b से परिबद्ध है को $\int_a^b f(x)dx \cong \frac{(b-a)}{2}\{f(a)+f(b)\}$ के लगभग लिखा जा सकता है। क्योंकि $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$, $c \in (a, b)$, तो उपरोक्त का एक और लगभग मान $\int_a^b f(x)dx$ यह भी होगा $\int_a^b f(x)dx \cong \frac{(c-a)}{2}\{f(a)+f(c)\} + \frac{(b-c)}{2}\{f(c)+f(b)\} \equiv F(c)$. यदि $c = \frac{a+b}{2}$, तो दिया गया है : $\int_a^b f(x)dx \cong \frac{b-a}{4}\{f(a)+2f(c)+f(b)\}$, ... (1) **[JEE-2006]**

91. The approximate value of $\int_0^{\pi/2} \sin x\,dx$ using rule (1) given above is :

नियम 1 से $\int_0^{\pi/2} \sin x\,dx$ का लगभग मान होगा :

(A) $\frac{\pi}{8\sqrt{2}}(1+\sqrt{2})$ (B) $\frac{\pi}{4\sqrt{2}}(1+\sqrt{2})$ (C) $\frac{\pi}{8}(1+\sqrt{2})$ (D) $\frac{\pi}{4}(1+\sqrt{2})$

Ans. C

Sol. $\int_0^{\pi/2} \sin x dx = \frac{\left(\frac{\pi}{2}-0\right)}{4}\left(\sin 0 + \sin\frac{\pi}{2} + 2\sin\frac{\pi}{4}\right)$

$= \frac{\pi}{8}(1+\sqrt{2})$

92. If $\lim\limits_{t\to a}\left\{\dfrac{\int_a^t f(x)dx - \frac{(t-a)}{2}(f(t)+f(a))}{(t-a)^3}\right\} = 0$, for each fixed a, then f(x) is a polynomial of degree utmost :

यदि $\lim\limits_{t\to a}\left\{\dfrac{\int_a^t f(x)dx - \frac{(t-a)}{2}(f(t)+f(a))}{(t-a)^3}\right\} = 0$, जहाँ a एक निश्चित संख्या है तो, f(x) किस अधिकतम घात का बहुपदीय फलन

gksxk:

(A) 4 (B) 3 (C) 2 (D) 1

Ans. D

Sol. $\lim\limits_{x\to a}\dfrac{\int_a^x f(x)dx - \left(\frac{x-a}{2}\right)(f(x)+f(a))}{(x-a)^3} = 0$

$$\lim_{h\to 0}\frac{\int_a^{a+h} f(x)dx - \frac{h}{2}(f(a+h)+f(a))}{h^3} = 0$$

$$\Rightarrow \lim_{h\to 0}\frac{f(a+h) - \frac{1}{2}[f(a)+f(a+h)] - \frac{h}{2}(f'(a+h))}{3h^2} = 0$$

[Using L'Hospital rule]

$$\Rightarrow \lim_{h\to 0}\frac{\frac{1}{2}f(a+h) - \frac{1}{2}f(a) - \frac{h}{2}f'(a+h)}{3h^2} = 0$$

$$\Rightarrow \lim_{h\to 0}\frac{\frac{1}{2}f'(a+h) - \frac{1}{2}f'(a+h) - \frac{h}{2}f''(a+h)}{6h} = 0$$

[Using L' Hospital rule]

$\Rightarrow \lim_{h\to 0}\frac{-f''(a+h)}{12} = 0 \Rightarrow f'(x) = 0, \ \forall \ a \in R$

$\Rightarrow$ f(x) must be of max. degree 1

93. If $f''(x) < 0, \ x \in (a, b)$, then at the point $C\ (c, f(c))$ on $y = f(x)$ for which F(c) is a maximum, f'(c) is given by :

यदि $f''(x) < 0,$ जहाँ $x \in (a, b)$ तो वक्र $y = f(x)$ का बिन्दु $C\ (c, f(c))$ पर वह मान जिस पर F(c) अधिकतम है तो f'(c) का मान ज्ञात है:

(A) $f'(c) = \frac{f(b)-f(a)}{b-a}$ (B) $f'(c) = \frac{f(b)-f(a)}{a-b}$

(C) $f'(c) = \frac{2(f(b)-f(a))}{b-a}$ (D) $f'(c) = 0$

Ans. A

Sol. $f''(x) < 0, \ \forall \ x \in (a, b)$, for $c \in (a, b)$

$$F(c) = \frac{c-a}{2}(f(a)+f(c)) + \frac{b-c}{2}(f(b)+f(c))$$

$$= \frac{b-a}{2}f(c) + \frac{c-a}{2}f(a) + \frac{b-c}{2}f(b)$$

$$\Rightarrow F'(c) = \frac{b-a}{2}f'(c) + \frac{1}{2}f(a) - \frac{1}{2}f(b)$$

$$= \frac{1}{2}[(b-a)f'(c) + f(a) - f(b)]$$

$$F''(c) = \frac{1}{2}(b-a)f''(c) < 0$$

$[\because f''(x) < 0, \forall x \in (a, b) \text{and } b > a]$

$\therefore$ F(c) is max. at the point (c, f(c)) where

$$F'(c) = 0 \Rightarrow f'(c) = \left(\frac{f(b)-f(a)}{b-a}\right)$$

Paragraph for Questions 94 to 96

Consider the function f(x) and g(x), both defined from $R \to R$, $f(x) = \frac{x^3}{2} + 1 - x\int_0^x g(t)dt$ and $g(x) = x - \int_0^1 f(t)dt$, then

फलन f(x) तथा g(x), दोनों $R \to R$ में परिभाषित है। $f(x) = \frac{x^3}{2} + 1 - x\int_0^x g(t)\,dt$ तथा $g(x) = x - \int_0^1 f(t)\,dt$ है, तो

94. Minimum value of f(x) is :

(A) 0 (B) 1 (C) $\frac{3}{2}$ (D) Does not exist

f(x) का न्यूनतम मान होगा :

(A) 0 (B) 1 (C) $\frac{3}{2}$ (D) विद्यमान नहीं है

Ans. B

95. The number of points of intersection of f(x) and g(x) is/are :

f(x) तथा g(x) के प्रतिच्छेद बिन्दुओं की संख्या है :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. A

96. The area bounded by g(x) with co-ordinate axes is (in square units) :

(A) $\frac{9}{4}$ (B) $\frac{9}{2}$ (C) $\frac{9}{8}$ (D) None of these

g(x) तथा निर्देशांकी अक्षो के मध्य का क्षेत्रफल होगा (वर्ग इकाई में) :

(A) $\frac{9}{4}$ (B) $\frac{9}{2}$ (C) $\frac{9}{8}$ (D) इनमें से कोई नहीं

Ans. C

Sol. (Que. 94, 95, 96)

$$f(x) = \frac{x^3}{2} + 1 - x\int_0^x g(t)\,dt$$

and $g(x) = x - \int_0^1 f(t)dt = x - k$ where $k = \int_0^1 f(t)\,dt$

$$\Rightarrow f(x) = \frac{x^3}{2} + 1 - x\int_0^x (t-k)dt$$

$$\Rightarrow f'(x) = \frac{3x^2}{2} - \int_0^x (t-k)\,dt - x(x-k)$$

$$= \frac{3x^2}{2} - \left[\frac{t^2}{2} - kt\right]_0^x - x^2 + kx$$

$$= \frac{3x^2}{2} - \frac{x^2}{2} + kx - x^2 + kx$$

$$f'(x) = 2kx$$

$$\Rightarrow \int f'(x)\,dx = 2k\int x\,dx$$

$$\Rightarrow f(x) = kx^2 + C$$

$\therefore$ f(0) = 1

$$\Rightarrow C = 1$$

$$\Rightarrow f(x) = kx^2 + 1$$

Now $k = \int_0^1 f(t)\,dt$

$$\Rightarrow k = \int_0^1 \left(kt^2 + 1\right) dt = \frac{k}{3} + 1$$

$$\Rightarrow k = \frac{3}{2}$$

$$\Rightarrow f(x) = \frac{3}{2}x^2 + 1 \text{ and } g(x) = x - \frac{3}{2}$$

$\therefore$ minimum value of f(x) is 1

intersection point of f(x) and g(x) are

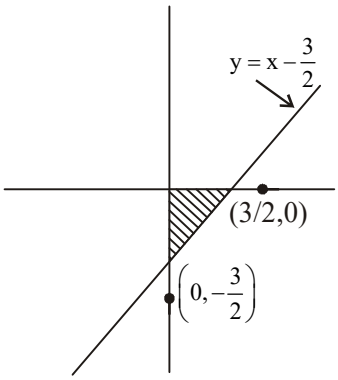
$$\frac{3}{2}x^2 + 1 = x - \frac{3}{2}$$

$$\Rightarrow 3x^2 + 2 = 2x - 3$$

$$\Rightarrow 3x^2 - 2x + 5 = 0$$

Here a = 3 and D < 0

$\Rightarrow$ Intersection point of f(x) and g(x) are zero.



Area = $\frac{9}{8}$

Paragraph for Questions 97 to 98

Let $f : R \to \left[\frac{3}{4}, \infty\right)$ be a surjective quadratic function with line of symmetry 2x – 1 = 0 and f(1) = 1.

माना $f : R \to \left[\frac{3}{4}, \infty\right)$ एक आच्छादक द्विघात फलन जो कि रेखा 2x – 1 = 0 के सापेक्ष सम्मित है तथा f(1) = 1.

97. If $g(x)=\frac{f(x)+f(-x)}{2}$ then $\int\frac{dx}{\sqrt{g(e^x)-2}}$ is equal to : (Where C is constant of integration)

(A) $\sec^{-1}(e^{-x})+C$ (B) $\sec^{-1}(e^{x})+C$ (C) $\sin^{-1}(e^{-x})+C$ (D) $\sin^{-1}(e^{x})+C$

यदि $g(x)=\frac{f(x)+f(-x)}{2}$ है, तो $\int\frac{dx}{\sqrt{g(e^x)-2}}$ बराबर होगा : (जहाँ C एक समाकलन नियतांक है)

(A) $\sec^{-1}(e^{-x})+C$ (B) $\sec^{-1}(e^{x})+C$ (C) $\sin^{-1}(e^{-x})+C$ (D) $\sin^{-1}(e^{x})+C$

Ans. B

98. $\int\frac{e^x}{f(e^x)}dx$:

(A) $\cot^{-1}\left(\frac{2e^x-1}{\sqrt{3}}\right)+C$ (B) $\frac{2}{\sqrt{3}}\cot^{-1}\left(\frac{2e^x+1}{\sqrt{3}}\right)+C$

(C) $\tan^{-1}\left(\frac{2e^x+1}{\sqrt{3}}\right)+C$ (D) $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2e^x-1}{\sqrt{3}}\right)+C$

Ans. D

Sol. (Que. 97, 98)

$f(x)=a\left(x-\frac{1}{2}\right)^2+\frac{3}{4}$

$\because f(1)=1$

$\Rightarrow a=1$

$\Rightarrow g(x)=\frac{f(x)+f(-x)}{2}=\frac{\left(x-\frac{1}{2}\right)^2+\left(x+\frac{1}{2}\right)^2+\frac{3}{2}}{2}$

$=x^2+1$

$\Rightarrow\int\frac{dx}{\sqrt{g(e^x)-2}}=\int\frac{dx}{\sqrt{e^{2x}+1-2}}=\int\frac{dx}{\sqrt{(e^x)^2-1}}$

$=\int\frac{e^x dx}{e^x\sqrt{e^{2x}-1}}=\sec^{-1}(e^x)+C$

98. $\int\frac{e^x}{f(e^x)}dx$

$$= \int \frac{e^x}{\left(e^x - \frac{1}{2}\right)^2 + \frac{3}{4}} dx$$

$$= \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2e^x - 1}{\sqrt{3}}\right) + C$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

99. Suppose $J = \int \frac{\sin^2 x + \sin x}{1 + \sin x + \cos x} dx$ and $K = \int \frac{\cos^2 x + \cos x}{1 + \sin x + \cos x} dx$. If C is an arbitrary constant of integration then which of the following is/are correct :

माना $J = \int \frac{\sin^2 x + \sin x}{1 + \sin x + \cos x} dx$ और $K = \int \frac{\cos^2 x + \cos x}{1 + \sin x + \cos x} dx$. यदि C स्वेच्छ समाकलन अचर है। तब निम्न में से कौन कौन से सत्य है :

(A) $J = \frac{1}{2}(x - (\sin x + \cos x)) + C$ (B) $J = K - (\sin x + \cos x) + C$

(C) $J = x - K + C$ (D) $K = \frac{1}{2}(x - \sin x + \cos x) + C$

Ans. ABC

Sol. $J + K = \int \frac{1 + \sin x + \cos x}{1 + \sin x + \cos x} dx$

$J + K = x + C$ ....(1) $\Rightarrow$ (C)

again $J - K = \int \frac{(\sin^2 x - \cos^2 x) + \sin x - \cos x}{1 + \sin x + \cos x} dx = \int \frac{(\sin x - \cos x)(\sin x + \cos x + 1)}{1 + \sin x + \cos x} dx$

$J - K = -\cos x - \sin x + C$ ....(2)

hence $J = K - (\sin x + \cos x) + C$ $\Rightarrow$ (B)

Also (1) + (2)

$2J = x - (\cos x + \sin x) + C$

$J = \frac{1}{2}[x - \sin x - \cos x] + C$

and (1) – (2)

$2K = x + (\sin x + \cos x) + C$

$K = \frac{1}{2}(x + \sin x + \cos x) + C$

from (1) $J = x - K + C \Rightarrow$ (C)

100. Let $u = \int_0^\infty \frac{dx}{x^4+7x^2+1}$ & $v = \int_0^\infty \frac{x^2\,dx}{x^4+7x^2+1}$ then :

माना $u = \int_0^\infty \frac{dx}{x^4+7x^2+1}$ और $v = \int_0^\infty \frac{x^2\,dx}{x^4+7x^2+1}$ तो :

(A) $v > u$ (B) $6\,\varpi = \pi$ (C) $3u + 2v = 5\pi/6$ (D) $u + v = \pi/3$

Ans. BCD

Sol. $u = \underbrace{\int_0^\infty \frac{dx}{x^4+7x^2+1}}_{\text{let } x=\frac{1}{t}}$ $v = \int_0^\infty \frac{x^2dx}{x^4+7x^2+1}$

$$\Rightarrow u = \int_\infty^0 \frac{t^4 - \frac{1}{t^2}dt}{t^4+7t^2+1}$$

$$= \int_0^\infty \frac{t^4}{t^2\left(t^4+7t^2+1\right)}dt = \int_0^\infty \frac{t^2}{t^4+7t^2+1}dt$$

$$\Rightarrow u = v$$

$$u = \int_0^\infty \frac{dx}{x^4+7x^2+1} = \int \frac{\frac{1}{x^2}dx}{\left(x^2+\frac{1}{x^2}+7\right)}$$

$$u = \int_0^\infty \frac{\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+9}dx \text{ and } v = \int_0^\infty \frac{1}{\left(x-\frac{1}{x}\right)^2+9}.dx$$

$$\Rightarrow u + v = \int_0^\infty \frac{\left(1+\frac{1}{x^2}\right)dx}{\left(x-\frac{1}{x}\right)^2+9}$$

$$= \int \frac{dt}{t^2+9} = \frac{1}{3}\tan^{-1}\left(\frac{t}{3}\right)$$

$$= \left[\frac{1}{3}\tan^{-1}\left[\frac{x-\frac{1}{x}}{3}\right]\right]_0^{\infty} = \frac{\pi}{6}+\frac{\pi}{6}$$

$$\Rightarrow u+v=\frac{\pi}{3}$$

$$\Rightarrow u=\frac{\pi}{6}=v$$

$$\Rightarrow 3u+2v=54=\frac{5\pi}{6}$$

101. Let $S_n=\sum_{k=1}^{n}\frac{k^2+n^2}{n^3}$ and $T_n=\sum_{k=0}^{n-1}\frac{k^2+n^2}{n^3}$ for n = 1,2,3,..... Then :

माना $S_n=\sum_{k=1}^{n}\frac{k^2+n^2}{n^3}$ और $T_n=\sum_{k=0}^{n-1}\frac{k^2+n^2}{n^3}$, जहाँ n = 1,2,3,..... तो :

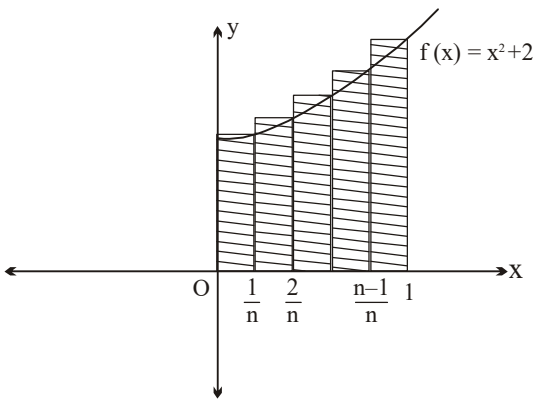
(A) $S_n<\frac{4}{3}$ (B) $T_n>\frac{4}{3}$ (C) $S_n>\frac{4}{3}$ (D) $T_n<\frac{4}{3}$

Ans. CD

Sol. We have $S_n=\frac{1}{n}\sum_{k=1}^{n}\left(\frac{k^2}{n^2}+1\right)=\frac{1}{n}\sum_{k=1}^{n}f\left(\frac{k}{n}\right)$

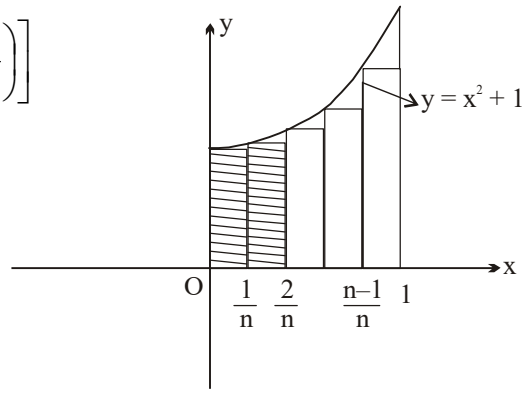
$$\Rightarrow S_n=\frac{1}{n}\left[f\left(\frac{1}{n}\right)+f\left(\frac{2}{n}\right)+.....+f\left(\frac{n}{n}\right)\right]$$

We have $f(x)=x^2+1$, which is increasing in [0,1]



Clearly $S_n>\int_0^1(x^2+1)\,dx \Rightarrow S_n>\frac{4}{3}$ .....(1)

Similarly $T_n=\frac{1}{n}\left[f(0)+f\left(\frac{1}{n}\right)+.....+f\left(\frac{n-1}{n}\right)\right]$



$T_n<\int_0^1(x^2+1)\,dx \Rightarrow T_n<\frac{4}{3}$ .....(2)

Clearly from graph, we conclude that

$T_n<\frac{4}{3}<S_n$.

102. If $f(\theta)=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n\theta}\frac{2r}{n\sqrt{(3\theta n-2r)(n\theta+2r)}}$ then :

(A) $f(1)=\frac{\pi}{6}$

(B) $f(\theta)=\frac{\theta}{2}\int_0^{\theta}\frac{dx}{\sqrt{\theta^2-\left(x-\frac{\theta}{2}\right)^2}}$

(C) f(θ) is a constant function

(D) y = f(θ) is invertible

यदि $f(\theta)=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n\theta}\frac{2r}{n\sqrt{(3\theta n-2r)(n\theta+2r)}}$ है, तो :

(A) $f(1)=\frac{\pi}{6}$

(B) $f(\theta)=\frac{\theta}{2}\int_0^{\theta}\frac{dx}{\sqrt{\theta^2-\left(x-\frac{\theta}{2}\right)^2}}$

(C) f(θ) एक अचर फलन है

(D) y = f(θ) एक व्युत्क्रमणीय फलन है

Ans. ABD

Sol. $f(\theta)=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n\theta}\frac{2r}{n\sqrt{(3\theta n-2r)(n\theta+2r)}}$

$=\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n\theta}\frac{2\left(\frac{r}{n}\right)}{\sqrt[n]{\left(3\theta-2\frac{r}{n}\right)\left(\theta+\frac{2r}{n}\right)}}$

$=\int_0^{\theta}\frac{2x}{\sqrt{(3\theta-2x)(\theta+2x)}}dx$

$=\int_0^{\theta}\frac{2x}{\sqrt{3\theta^2+4\theta x-4x^2}}=\int_0^{\theta}\frac{2x}{2\sqrt{\theta x-x^2+\frac{3}{2}\theta^2}}$

$I=\int_0^{\theta}\frac{x}{\sqrt{\theta^2-(x-\theta/2)^2}}dx$

$\Rightarrow$ Apply king prop.

$I=\int_0^{\theta}\frac{(\theta-x)dx}{\sqrt{\theta^2-(x-\theta/2)^2}}$

$$\Rightarrow 2I = \theta\int_0^{\theta} \frac{dx}{\sqrt{\theta^2 - (x-\theta/2)^2}} = \frac{\theta}{2}\left[\frac{\sin^{-1}(x-\theta/2)}{\theta}\right]_0^{\theta}$$

$$\Rightarrow I = \frac{\theta}{2}\left[\frac{\pi}{6}+\frac{\pi}{6}\right] = \frac{\pi\theta}{6}$$

$$\Rightarrow f(I) = \frac{\pi}{6}$$

$$\Rightarrow f(\theta) = \frac{\theta}{2}\int_0^{\theta} \frac{dx}{\sqrt{\theta^2 - (x-\theta/2)^2}} = \frac{\pi\theta}{6}$$

$y = f(\theta)$ is invertible

103. Let $J = \int_{-1}^{2}\left(\cot^{-1}\frac{1}{x} + \cot^{-1} x\right)dx$, $K = \int_{-2\pi}^{7\pi}\frac{\sin x}{|\sin x|}dx$. Then which of the following alternative(s) is/are correct?

माना $J = \int_{-1}^{2}\left(\cot^{-1}\frac{1}{x} + \cot^{-1} x\right)dx$, $K = \int_{-2\pi}^{7\pi}\frac{\sin x}{|\sin x|}dx$ है, तो निम्न में से कौनसा/कौनसे विकल्प सत्य है?

(A) $2J + 3K = 8\pi$ (B) $4J^2 + K^2 = 26\pi^2$ (C) $2J - K = 3\pi$ (D) $\frac{J}{K} = \frac{2}{5}$

Ans. AB

Sol. $J = \int_{-1}^{2}\left(\cot^{-1}\frac{1}{x} + \cot^{-1} x\right)dx$

$$= \int_{-1}^{0}\left(\pi + \tan^{-1} x + \cot^{-1} x\right)dx + \int_0^2\left(\tan^{-1} x + \cot^{-1} x\right)dx$$

$$= \pi + \frac{\pi}{2} + \pi = \frac{5\pi}{2}$$

$$K = \int_{-2\pi}^{7\pi}\frac{\sin x}{|\sin x|}dx$$

$$= \int_{-2\pi}^{-\pi}dx - \int_{-\pi}^{0}dx + \int_0^{\pi}dx - \int_{\pi}^{2\pi}dx + \int_{2\pi}^{3\pi}dx - \int_{3\pi}^{4\pi}dx + \int_{4\pi}^{5\pi}dx - \int_{5\pi}^{6\pi}dx + \int_{6\pi}^{7\pi}dx$$

$$= \pi - \pi + \pi - \pi + \pi - \pi + \pi - \pi + \pi = \pi$$

$2J + 3K = 8\pi$ and $4J^2 + K^2 = 26\pi^2$

$2J - 3K = 2\pi$ and $\frac{J}{K} = \frac{5}{2}$

104. Which of the following function(s) is/are even?

निम्न में से कौनसा/कौनसे सम फलन है?

(A) $f(x)=\int_0^x \ln\left(t+\sqrt{1+t^2}\right)dt$ (B) $g(x)=\int_0^x \frac{\left(2^t+1\right)t}{2^t-1}dt$

(C) $h(x)=\int_0^x \left(\sqrt{1+t+t^2}-\sqrt{1-t+t^2}\right)dt$ (D) $l(x)=\int_0^x \ln\left(\frac{1-t}{1+t}\right)dt$

Ans. ACD

Sol. (A) $f(x)=\int_0^x \ln\left(t+\sqrt{1+t^2}\right)dt$

$=\int_0^x \ln\left(\frac{1}{-t+\sqrt{1+t^2}}\right)dt$

$f(-x)=\int_0^{-x} \ln\left(\frac{1}{\sqrt{1+t^2}-t}\right)dt$

let t = - v

$=\int_0^x \ln\left(\frac{1}{\sqrt{1+v^2}+v}\right)dv$

$=\int_0^x \ln\left(v+\sqrt{1+v^2}\right)dv$

$\Rightarrow f(-x)=f(x)$

$\therefore$ f is an even function.

(B) $g(-x)=\int_0^{-x} \frac{\left(2^t+1\right)dt}{2^t-1}$

let –t = v

$\Rightarrow g(-x)=-1\int_0^x \frac{\left(1+2^v\right)-v}{\left(1-2^v\right)}dv$

$g(-x)=-\int_0^x \frac{\left(2^v+1\right)v}{2^v-1}dv$

$g(-x)=-g(x)$

$\therefore$ g is odd

(C) $h(x)=\int_0^x \left(\sqrt{1+t+t^2}-\sqrt{1-t+t^2}\right)dt$

$=\int_0^x \left(\frac{2t}{\sqrt{1+t+t^2}+\sqrt{1-t+t^2}}\right)dt$

$$h(-x)=\int_0^{-x}\frac{2t\,dt}{\left(\sqrt{1+t+t^2}+\sqrt{1-t+t^2}\right)}$$

$let\ t=-v$

$$h(-x)=+\int_0^{x}\frac{2v\,dv}{\left(\sqrt{1-v+v^2}+\sqrt{1+v+v^2}\right)}=h(x)$$

(D) $l(x)=\int_0^x \ln\left(\frac{1-t}{1+t}\right)dt$

$$l(-x)=\int_0^{-x}\ln\left(\frac{1-t}{1+t}\right)dt$$

let - t = v

$$l(-x)=\int_0^{-x}\ln\left(\frac{1+V}{1-V}\right)dv=l(x)$$

105. Let $S_n=\sum_{k=1}^{n}\frac{n}{n^2+kn+k^2}$ and $T_n=\sum_{k=0}^{n-1}\frac{n}{n^2+kn+k^2}$, for n = 1, 2, 3, ... Then : **[JEE-2008]**

माना $S_n=\sum_{k=1}^{n}\frac{n}{n^2+kn+k^2}$ तथा $T_n=\sum_{k=0}^{n-1}\frac{n}{n^2+kn+k^2}$, के लिए n = 1, 2, 3, ... तब : **[JEE-2008]**

(A) $S_n<\frac{\pi}{3\sqrt{3}}$ (B) $S_n>\frac{\pi}{3\sqrt{3}}$ (C) $T_n<\frac{\pi}{3\sqrt{3}}$ (D) $T_n>\frac{\pi}{3\sqrt{3}}$

Ans. AD

Sol. $S_n=\frac{1}{n}\sum_{k=1}^{n}\frac{1}{1+k/n+k^2/n^2}$

$$S_n<\int_0^1\frac{dx}{x^2+x+1}$$

(∵ the function is decreasing)

$$S_n<\int_0^1\frac{dx}{\left(x+1/2\right)^2+\left(\sqrt{3}/2\right)^2}$$

$$S_n<\frac{2}{\sqrt{3}}\left[\tan^{-1}\frac{2x+1}{\sqrt{3}}\right]_0^1$$

$$S_n<\frac{2}{\sqrt{3}}\left[\frac{\pi}{3}-\frac{\pi}{6}\right]$$

$S_n < \frac{\pi}{3\sqrt{3}}$

Now $T_n - S_n = 1 - \frac{1}{3n} \Rightarrow T_n - S_n > \frac{2}{3}$

106. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1+\pi^x)\sin x} dx$, $n = 0,1,2,.....$, then: **[JEE-2009]**

यदि $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1+\pi^x)\sin x} dx$, $n = 0,1,2,.....$, तो : **[JEE-2009]**

(A) $I_n = I_{n+2}$ (B) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$ (C) $\sum_{m=1}^{10} I_{2m} = 0$ (D) $I_n = I_{n+1}$

Ans. ABC

Sol. $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1+\pi^x)\sin x} dx$

Apply king property

$I_n = \int_{-\pi}^{\pi} \frac{\pi^x \sin nx}{(1+\pi^x)\sin x} dx$

$2I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{\sin x} dx$ ..(i)

$2\, I_{n+2} = \int_{-\pi}^{\pi} \frac{\sin(n+2)x}{\sin x} dx$ ...(i)

(ii) – (i)

$\Rightarrow 2(I_{n+2} - I_n) = \int_{-\pi}^{\pi} \cos(n+1)x = 0 \Rightarrow I_{n+2} = I_n$

$\sum_{m=1}^{10} I_{2m} = 10 \sum_{m=1}^{10} I_2 = \frac{10}{2} \int_{-\pi}^{\pi} \frac{\sin 2x}{\sin x} dx = 0$

Put n = 1 in equation (i)

$2I_1 = \int_{-\pi}^{\pi} \frac{\sin x dx}{\sin x} = 2\pi$

$I_1 = \pi$

$\sum_{m=1}^{10} I_{2m+1} = 10\pi$

MATCH THE COLUMN TYPE QUESTION(S)

107. Match the following columns :

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) $\int \frac{x^4-1}{x^2\sqrt{x^4+x^2+1}}dx$	(P) $l\text{n}\left(\frac{(x^2+1)+\sqrt{x^4+1}}{x}\right)+C$
(B) $\int \frac{x^2-1}{x\sqrt{1+x^4}}dx$	(Q) $C-\frac{1}{\sqrt{2}}l\text{n}\left(\frac{\sqrt{x^4+1}-\sqrt{2}x}{(x^2-1)}\right)$
(C) $\int \frac{1+x^2}{(1-x^2)\sqrt{1+x^4}}dx$	(R) $C-\tan^{-1}\left(\sqrt{\sqrt{1+\frac{1}{x^4}}-1}\right)$
(D) $\int \frac{1}{(1+x^4)\sqrt{\sqrt{1+x^4}-x^2}}dx$	(S) $\frac{\sqrt{x^4+x^2+1}}{x}+C$

Ans. (A) S; (B) P ; (C) Q ; (D) R

Sol. (A) $\int \frac{x^4-1}{x^2\sqrt{x^4+x^2+1}}dx = \int \frac{x^2-\frac{1}{x^2}}{\sqrt{x^4+x^2+1}}dx$

$$= \int \frac{\left(x-\frac{1}{x^3}\right)dx}{\sqrt{x^2+1+\frac{1}{x^2}}}$$

Let $x^2+\frac{1}{x^2}=t$

$$\Rightarrow 2\left(x-\frac{1}{x^3}\right)dx = dt$$

$$= \frac{1}{2}\int \frac{dt}{\sqrt{t+1}} = \sqrt{t+1}+C = \sqrt{\frac{x^4+x^2+1}{x^2}}+C$$

(B) $\int \frac{x^2-1}{x\sqrt{1+x^4}}dx = \int \frac{1-\frac{1}{x^2}}{\sqrt{\frac{1}{x^2}+x^2}}dx$

$$\int \frac{\left(1-\frac{1}{x^2}\right)}{\sqrt{\left(x+\frac{1}{x}\right)^2-2}}dx$$

Let $x+\frac{1}{x}=t \Rightarrow \left(1-\frac{1}{x^2}\right)dx=dt$

$$=\int \frac{dt}{\sqrt{t^2-2}}=\frac{1}{2\sqrt{2}}\ln\left|\frac{t-\sqrt{2}}{1+\sqrt{2}}\right|+C$$

$$=\frac{1}{2\sqrt{2}}\ln\left|\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right|+C$$

$$=\frac{1}{2\sqrt{2}}\ln\left|\frac{\left(x^2+1\right)^2-2\sqrt{2}\left(x^2+1\right)+2x^2}{\left(x^2+1\right)^2-2x^2}\right|+C$$

$$=\frac{1}{2\sqrt{2}}\ln\left|\frac{\left(x^2\sqrt{2}x+1\right)^2}{\left(x+1\right)^2\left(x-1\right)^2}\right|+C$$

$$=\frac{1}{2\sqrt{2}}\ln\left|\frac{x^2-\sqrt{2}+1}{x^2-1}\right|+C$$

(C) $\int \frac{1+x^2}{(1-x^2)\sqrt{1+x^4}}dx = \int \frac{\left(1+\frac{1}{x^2}\right)dx}{\left(\frac{1}{x}-x\right)\sqrt{x^2+\frac{1}{x^2}+2-2}}$

$$=-\int \frac{\left(1+\frac{1}{x^2}\right)dx}{\left(x-\frac{1}{x}\right)\sqrt{\left(x-\frac{1}{x}\right)^2+2}}$$

let $x-\frac{1}{x}=t$

$$=-\int \frac{dt}{t\sqrt{t^2+2}} \text{ let } \sqrt{t^2+2}=v$$

$$\Rightarrow \frac{t}{\sqrt{t^2+2}}dt = d\nu$$

$$= -\int \frac{1}{\upsilon^2-2}d\nu = \frac{-1}{2\sqrt{2}}\ln\left|\frac{\nu-\sqrt{2}}{\nu+\sqrt{2}}\right| + C$$

$$= \frac{-1}{2\sqrt{2}}\ln\left|\frac{\sqrt{t^2+2}-\sqrt{2}}{\sqrt{t^2+2}+\sqrt{2}}\right| + C$$

$$= \frac{-1}{2\sqrt{2}}\ln\left|\frac{\sqrt{x^4+1}-\sqrt{2}x}{\sqrt{x^4+1}+\sqrt{2}x}\right| + C$$

$$= \frac{+1}{2\sqrt{2}}\ln\left|\frac{\sqrt{x^4+1}+\sqrt{2}x}{\left(\sqrt{x^4+1}\right)^2-\left(\sqrt{2}x\right)^2}\right| + C$$

(D) $\int \frac{1}{(1+x^4)\sqrt{\sqrt{1+x^4}-x^2}}dx$

$$= \int \frac{dx}{x^4\left(1+\frac{1}{x^4}\right)x\sqrt{\sqrt{1+\frac{1}{x^4}}-1}}$$

Let $1+\frac{1}{x^4} = \nu^2$

$$\Rightarrow \frac{dx}{x^5} = -\frac{\nu d\nu}{2}$$

$$= \frac{-1}{2}\int \frac{\nu d\nu}{\nu^2\sqrt{\nu-1}}$$

$$= \frac{-1}{2}\int \frac{d\nu}{\nu\sqrt{\nu-1}}$$

Let $\nu - 1 = t^2$

$\Rightarrow d\nu = 2tdt$

$$= \frac{-1}{2}\int \frac{2tdt}{\left(t^2+1\right)\sqrt{t^2}} = -\int \frac{dt}{t^2+1}$$

$= -\tan^{-1}(t) + C$

$= -\tan^{-1}\left(\sqrt{\nu-1}\right) + C$

$$= -\tan^{-1}\left(\sqrt{1+\frac{1}{x^4}}-1\right)+C$$

108. Let $I_1 = \int \tan x \ \tan(ax+b)\,dx$ and $I_2 = \int \cot x \ \cot(ax+b)\,dx$:

Column-I	**Column-II**
(A) value of I_1 for $a = 1$ is	(P) $x - \cot b \ ln \dfrac{\cos(x-b)}{\cos x} + C$
(B) value of I_2 for $a = 1$ is	(Q) $\cot b \ ln \dfrac{\sin x}{\sin(x+b)} - x + C$
(C) value of I_1 for $a = -1$ is	(R) $\cot b \ ln \left(\dfrac{\cos x}{\cos(x+b)}\right) - x + C$
(D) value of I_2 for $a = -1$ is	(S) $x + \cot b \ ln \left(\dfrac{\sin x}{\sin(b-x)}\right) + C$

माना $I_1 = \int \tan x \ \tan(ax+b)\,dx$ तथा $I_2 = \int \cot x \ \cot(ax+b)\,dx$:

स्तम्भ–I	**स्तम्भ–II**
(A) I_1 का मान $a = 1$ के लिये है	(P) $x - \cot b \ ln \dfrac{\cos(x-b)}{\cos x} + C$
(B) I_2 का मान $a = 1$ के लिये है	(Q) $\cot b \ ln \dfrac{\sin x}{\sin(x+b)} - x + C$
(C) I_1 का मान $a = -1$ के लिये है	(R) $\cot b \ ln \left(\dfrac{\cos x}{\cos(x+b)}\right) - x + C$
(D) I_2 का मान $a = -1$ के लिये है	(S) $x + \cot b \ ln \left(\dfrac{\sin x}{\sin(b-x)}\right) + C$

Ans. (A) R ; (B) Q ; (C) P ; (D) S

Sol. $I_1 = \int \tan x \tan(ax+b)\,dx$

For $a = 1$

(A) $I_1 = \int \tan x \left(\dfrac{\tan x + \tan b}{1 - \tan x \tan b}\right) dx$

$$= \int \frac{\tan^2 x}{1-\tan x \tan b}\,dx + \tan \int \frac{\tan x}{1-\tan x \tan b}\,dx$$

$$= \int \frac{\sec^2 x - 1}{(1 - \tan x \tan b)} dx + \tan b \int \frac{\tan x}{1 - \tan x \tan b} dx$$

$$= \int \frac{\sec^2 x}{1 - \tan x \tan b} dx - \int dx$$

$$\cot b \int \frac{\sec^2 x dx}{\cot b - \tan x} - x + C$$

$$= -\cot b \ln |\cot b - \tan x| - x + C$$

$$= -\cot b \ln \left| \frac{\cos(b + x)}{\sin b \cos x} \right| - x + C$$

$$= \cot b \ln \left(\frac{\cos x}{\cos(b + x)} \right) - x + C$$

(B) For a = 1

$$I_2 = \int \cot x \cot(x + b) dx$$

$$= \int \cot x \left(\frac{\cot b \cot x - 1}{\cot b + \cot x} \right) dx$$

$$= \int \frac{\cot b (\cos ec^2 x - 1) - \cot x}{\cot b + \cot x} dx$$

$$= \cot b \int \frac{\cos ec^2 x}{\cot B + \cot x} dx - x + C$$

$$= -\cot b \ln \left| \frac{\cos b}{\sin b} + \frac{\cos x}{\sin x} \right| - x + C$$

$$= \cot b \ln \left| \frac{\sin x}{\sin(x + b)} \right| - x + C$$

(C) For a = –1

$$I_1 = \int \tan x \tan(b - x) dx = \int \tan x \frac{(\tan b - \tan c)}{1 + \tan b \tan c} dx$$

$$= \int \frac{\tan x \tan b - (\sec^2 x - 1)}{1 + \tan b \tan x} dx$$

$$= \int \left(1 - \frac{\sec^2 x}{1 + \tan x \tan b} \right) dx$$

$$= x - \cot b \ln |\tan x + \cot b| + C$$

$$= x - \cot b \ln\left|\frac{\cos(x-b)}{\cos x \sin b}\right| + C$$

$$= -\cot b \ln\left|\frac{\cos(x-b)}{\cos x}\right| + x + C$$

(D) For a = – 1

$$I_2 = \int \cot x \left(\frac{\cot x \cot b + 1}{\cot x - \cot b}\right) dx$$

$$= \int \frac{\left(\cos ec^2 x - 1\right)\cot b + \cot x}{\cot x - \cot b} dx$$

$$= -\cot b \ln|\cot x - \cot b| + x + C$$

$$= x + \cot b \ln\left|\frac{\sin x}{\sin(b-x)}\right| + C$$

109. Let $I = \int \frac{e^x}{e^{4x}+1} dx$ and $J = \int \frac{e^{-x}}{e^{-4x}+1} dx$. Then for any arbitrary constant C, match the following :

यदि $I = \int \frac{e^x}{e^{4x}+1} dx$ और $J = \int \frac{e^{-x}}{e^{-4x}+1} dx$ तब किसी स्वेच्छ अचर C के लिए :

	Column-I	**Column-II**
(A)	I	(P) $\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right) + C$
(B)	J + I	(Q) $\frac{1}{2\sqrt{2}} \ln\left(\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right) + C$
(C)	J – I	(R) $\frac{1}{2\sqrt{2}}\left(\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right) - \frac{1}{2}\ln\left(\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right)\right) + C$
		(S) $\frac{1}{2\sqrt{2}}\left(\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right) + \frac{1}{2}\ln\left(\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right)\right) + C$

Ans. (A) R ; (B) P; (C) Q

Sol. Given $I = \int \frac{e^x}{1+e^{4x}} dx$ and $J = \int \frac{e^{-x}}{1+e^{-4x}} dx$

$$J = \int \frac{e^{4x}\cdot e^{-x}}{1+e^{4x}}dx = \int \frac{e^{3x}\,dx}{1+e^{4x}}$$

$\therefore \quad I + J = \int \frac{e^x(1+e^{2x})}{1+e^{4x}}dx$

put $\quad e^x = t \quad \Rightarrow \; e^x dx = dt$

$$\therefore \quad I + J = \int \frac{1+t^2}{1+t^4}dt = \int \frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}}dt = \int \frac{1+\frac{1}{t^2}}{\left(t-\frac{1}{t}\right)^2+2}dt$$

put $\quad t-\frac{1}{t} = y \quad \Rightarrow \quad y = \left(t+\frac{1}{t^2}\right)dt = dy$

$$= \int \frac{dy}{y^2+2} = \frac{1}{\sqrt{2}}\tan^{-1}\frac{y}{\sqrt{2}} = \frac{1}{\sqrt{2}}\tan^{-1}\frac{t^2-1}{\sqrt{2}\,t}$$

$$J + I = \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right)+C \qquad ....(1) \quad \Rightarrow \quad (P)$$

|||ly $\quad J - I = \int \frac{e^x(-1+e^{2x})}{1+e^{4x}}dx = \int \frac{-1+t^2}{1+t^4}dt = + \int \frac{-1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}}dt = + \int \frac{-1+\frac{1}{t^2}}{\left(t+\frac{1}{t}\right)^2-2}dt \quad \left(t+\frac{1}{t}=y\right)$

$$\therefore \quad J - I = + \int \frac{dy}{y^2-2} = -\frac{1}{2\sqrt{2}}\ln\frac{y-\sqrt{2}}{y+\sqrt{2}}$$

$$J - I = -\frac{1}{2\sqrt{2}}\ln\frac{t+\frac{1}{t}-\sqrt{2}}{t+\frac{1}{t}+\sqrt{2}} = \frac{1}{2\sqrt{2}}\cdot\frac{t^2-\sqrt{2}\,t+1}{t^2+\sqrt{2}\,t+1} = \frac{1}{2\sqrt{2}}\cdot\frac{e^{2x}-\sqrt{2}\,e^x+1}{e^{2x}+\sqrt{2}\,e^x+1} \;....(2) \Rightarrow (Q)$$

(1) – (2) gives

$$2I = \frac{1}{\sqrt{2}}\tan^{-1}\frac{e^{2x}-1}{\sqrt{2}\,e^x} - \frac{1}{2\sqrt{2}}\frac{e^{2x}-\sqrt{2}\,e^x+1}{e^{2x}+\sqrt{2}\,e^x+1}$$

$$I = \frac{1}{2\sqrt{2}}\left[\tan^{-1}\frac{e^{2x}-1}{\sqrt{2}\,e^x} - \frac{1}{2}\cdot\frac{e^{2x}-\sqrt{2}\,e^x+1}{e^{2x}+\sqrt{2}\,e^x+1}\right] \quad \Rightarrow \quad (R)$$

110. Match the following column : (where C is the constant of integration.)

निम्न कॉलम को सुमेलित कीजिए : (जहाँ C समाकलन नियंताक है)

Column-I		**Column-II**	
(A)	$\int \frac{x^2(x^6+x^5-1)\,dx}{(2x^6+3x^5+2)^2}$	(P)	$\frac{1}{6}\left(\frac{1}{x^{-3}}+\frac{3}{x^{-2}}\right)^{\frac{1}{2}}+C$
(B)	$\int \frac{(x^5+x^4+x^2)\,dx}{\sqrt{4x^7+5x^6+10x^4}}$	(Q)	$\frac{1}{2}(1+x^{-2}+x^{-5})^{-2}+C$
(C)	$\int \frac{(2x^{12}+5x^9)dx}{(x^5+x^3+1)^3}$	(R)	$\frac{-1}{6}(2x^3+3x^2+2x^{-3})^{-1}+C$
(S)	$x\left(\frac{x^3}{25}+\frac{x^2}{20}+\frac{1}{10}\right)^{\frac{1}{2}}+C$		

Ans. (A) R, (B) S, (C) Q

Sol. **(A)** Let $I_1=\int \frac{x^2(x^6+x^5-1)dx}{(2x^6+3x^5+2)^2} = \int \frac{x^2(x^6+x^5-1)dx}{x^6\left(2x^3+3x^2+\frac{2}{x^3}\right)^2} = \int \frac{(x^2+x-x^{-4})\,dx}{(2x^3+3x^2+2x^{-3})^2}$

Putting $2x^3+3x^2+2x^{-3}=t \Rightarrow 6\left(x^2+x-\frac{1}{x^4}\right)dx=dt \Rightarrow \frac{(x^6+x^5-1)}{x^4}dx=\frac{1}{6}dt$

Hence $I_1=\frac{1}{6}\int\frac{dt}{t^2}=\frac{-1}{6t}+C=\frac{-1}{6(2x^3+3x^2+2x^{-3})}+C=\frac{-1}{6}(2x^3+3x^2+2x^{-3})^{-1}+C$.

(B) Let $I_2=\int \frac{(x^5+x^4+x^2)dx}{\sqrt{4x^7+5x^6+10x^4}}=\int \frac{(x^5+x^4+x^2)dx}{x\sqrt{4x^5+5x^4+10x^2}}=\int \frac{(x^4+x^3+x)\,dx}{\sqrt{4x^5+5x^4+10x^2}}$

Putting $4x^5+5x^4+10x^2=t \Rightarrow 20\,(x^4+x^3+x)\,dx=dt$

Hence $I_2=\frac{1}{20}\int\frac{dt}{\sqrt{t}}=\frac{1}{10}\sqrt{t}+C=\frac{1}{10}\sqrt{4x^5+5x^4+10x^2}+C=\frac{1}{10}x\sqrt{4x^3+5x^2+10}+C$

$=x\left(\frac{x^3}{25}+\frac{x^2}{20}+\frac{1}{10}\right)^{\frac{1}{2}}+C$

(C) Let $I_3=\int \frac{2x^{12}+5x^9}{(x^5+x^3+1)^3}dx=\int \frac{2x^{12}+5x^9}{x^{15}(1+x^{-2}+x^{-5})^3}dx=\int \frac{(2x^{-3}+5x^{-6})\,dx}{(1+x^{-2}+x^{-5})^3}dx$

Putting $1+x^{-2}+x^{-5}=t \Rightarrow (-2x^{-3}-5x^{-6})dx=dt$

$$\therefore I_3 = -\int \frac{dt}{t^3} = \frac{1}{2t^2} + C = \frac{1}{2}\left(1 + x^{-2} + x^{-5}\right)^{-2} + C$$

111. Match the integrals in **Column I** with the values in **Column II**.

स्तम्भ का मिलान कीजिए।

Column I		**Column II**	
(A)	$\int_{-1}^{1} \frac{dx}{1+x^2}$	(P)	$\frac{1}{2}\log\left(\frac{2}{3}\right)$
(B)	$\int_{0}^{1} \frac{dx}{\sqrt{1-x^2}}$	(Q)	$2\log\left(\frac{2}{3}\right)$
(C)	$\int_{2}^{3} \frac{dx}{1-x^2}$	(R)	$\frac{\pi}{3}$
(D)	$\int_{1}^{2} \frac{dx}{x\sqrt{x^2-1}}$	(S)	$\frac{\pi}{2}$

Ans. (A) S, (B) S, (C) P, (D) R

Sol. (A) $\int_{-1}^{1} \frac{dx}{1+x^2} dx = \left[\tan^{-1} x\right]_{-1}^{1} = \frac{\pi}{2}$

(B) (B) $\int_{0}^{1} \frac{dx}{\sqrt{1-x^2}} = \left[\sin^{-1} x\right]_{0}^{1} = \frac{\pi}{2}$

(C) $\int_{2}^{3} \frac{dx}{1-x^2} = \left[\frac{1}{2}\ln\left|\frac{1+x}{1-x}\right|\right]_{2}^{3}$

$= \frac{1}{2}\ln\left(\frac{2}{3}\right)$

(D) $\int_{1}^{2} \frac{dx}{x\sqrt{x^2-1}} = \left(\sec^{-1} x\right)_{1}^{2} = \frac{\pi}{3} - 0 = \frac{\pi}{3}$

Exercise J-M

1. Let $f(x)=\int \frac{2x}{(x^2+1)(x^2+3)}dx$. If $f(3)=\frac{1}{2}(\log_e 5-\log_e 6)$, then f(4) is equal to :

[JEE(Main)Jan.2023]

माना $f(x)=\int \frac{2x}{(x^2+1)(x^2+3)}dx$ है। यदि $f(3)=\frac{1}{2}(\log_e 5-\log_e 6)$ तो f(4) बराबर है :

[JEE(Main)Jan.2023]

(A) $\frac{1}{2}(\log_e 17-\log_e 19)$ (B) $\frac{1}{2}(\log_e 19-\log_e 17)$

(C) $\log_e 17-\log_e 18$ (D) $\log_e 19-\log_e 20$

Ans. A

Sol. $f(x)=\int \frac{2x}{(x^2+1)(x^2+3)}dx$

Let $x^2=t$

$2xdx=dt$

$\int \frac{dt}{(t+1)(t+3)}$

$=\frac{1}{2}\int \frac{(t+3)-(t+1)}{(t+1)(t+3)}dt$

$=\frac{1}{2}[\ln|t+1|-\ln|t+3|]+\frac{C}{2}$

$=\frac{1}{2}\left[\ln\left|x^2+1\right|-\ln\left|x^2+3\right|\right]+\frac{C}{2}$

$\because \quad f(3)=\frac{1}{2}[\ln 5-\ln 6]$

$\therefore \quad \frac{1}{2}[\ln 5-\ln 6]=\frac{1}{2}[\ln 10-\ln 12]+\frac{C}{2}$

$\Rightarrow \quad C=0$

$\therefore \quad f(x)=\frac{1}{2}\left[\ln\left|x^2+1\right|-\ln\left|x^2+3\right|\right]$

$f(4)=\frac{1}{2}[\ln 17-\ln 19]$

2. If $\int\sqrt{\sec 2x-1}dx=\alpha\log_e\left|\cos 2x+\beta+\sqrt{\cos 2x\left(1+\cos\frac{1}{\beta}x\right)}\right|+\text{constant}$, then $\beta-\alpha$ is equal to _____. **[JEE(Main)Jan.2023]**

यदि $\int\sqrt{\sec 2x-1}dx=\alpha\log_e\left|\cos 2x+\beta+\sqrt{\cos 2x\left(1+\cos\frac{1}{\beta}x\right)}\right|+\text{constant}$, तो $\beta-\alpha$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 1

Sol. $\int\sqrt{\sec 2x-1}dx=\int\sqrt{\frac{1-\cos 2x}{\cos 2x}}dx=\sqrt{2}\int\frac{\sin x}{\sqrt{2\cos^2 x-1}}$

put $\cos x=t\Rightarrow -\sin x\,dx=dt$

$$=-\sqrt{2}\int\frac{dt}{\sqrt{2t^2-1}}=-\ln\left|\sqrt{2}\cos x+\sqrt{\cos 2x}\right|+c=-\frac{1}{2}\ln\left|\cos 2x+\frac{1}{2}+\sqrt{\cos 2x}.\sqrt{1+\cos 2x}\right|+c$$

$\because \beta=\frac{1}{2},\alpha=-\frac{1}{2}\Rightarrow\beta-\alpha=1$

3. If $\int_0^1\left(x^{21}+x^{14}+x^7\right)\left(2x^{14}+3x^7+6\right)^{\frac{1}{7}}dx=\frac{1}{\ell}(11)^{\frac{m}{n}}$ where l, m, n $\in$ N, m and n are coprime then $l+m+n$ is equal to _____. **[JEE(Main)Jan.2023]**

यदि $\int_0^1\left(x^{21}+x^{14}+x^7\right)\left(2x^{14}+3x^7+6\right)^{\frac{1}{7}}dx=\frac{1}{\ell}(11)^{\frac{m}{n}}$ है, जहाँ l, m, n $\in$ N, m एवं n सहअभाज्य हैं, तब $l+m+n$ का मान है। **[JEE(Main)Jan.2023]**

Ans. 63

Sol. $\int\left(x^{20}+x^{13}+x^6\right)\left(2x^{21}+3x^{14}+6x^7\right)^{1/7}dx$

$2x^{21}+3x^{14}+6x^7=t$

$42(x^{20}+x^{13}+x^6)\,dx=dt$

$$\frac{1}{42}\int_0^{11}t^{\frac{1}{7}}dt=\left(\frac{t^{\frac{8}{7}}}{\frac{8}{7}}\times\frac{1}{42}\right)_0^{11}$$

$$\frac{1}{48}\left(t^{\frac{8}{7}}\right)_0^{11}=\frac{1}{48}(11)^{8/7}$$

$l=48,\ m=8,\ n=7$

$l + m + n = 63$

4. $\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}}dx$ is equal to : **[JEE(Main)Jan.2023]**

$\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}}dx$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{\pi}{3}$ (B) 2π (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

Ans. B

Sol. $\frac{48}{2}\left(\sin^{-1}\left(\frac{2x}{3}\right)\right)_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} = 24\left(\sin^{-1}\frac{\sqrt{3}}{2} - \sin^{-1}\frac{1}{\sqrt{2}}\right)$

$= 24\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = 24\left(\frac{\pi}{12}\right) = 2\pi$

5. The minimum value of the function $f(x) = \int_0^2 e^{|x-t|}\,dt$ is : **[JEE(Main)Jan.2023]**

फलन $f(x) = \int_0^2 e^{|x-t|}\,dt$ का निम्नतम मान है : **[JEE(Main)Jan.2023]**

(A) 2 (B) e (e – 1) (C) 2 (e – 1) (D) 2e – 1

Ans. C

Sol. For $x > 2$

$f(x) = \int_0^2 e^{x-t}dt$

$= e^x\left(-e^{-t}\right)\Big|_0^2$

$= e^x\left(1 - e^{-2}\right)$

For $x < 0$

$f(x) = \int_0^2 e^{t-x}dt = e^{-x}e^t\Big|_0^2 = e^{-x}\left(e^2 - 1\right)$

For $0 \le x \le 2$

$f(x) = \int_0^x e^{x-t}dt + \int_x^2 e^{t-x}dt$

$= -e^xe^{-t}\Big|_0^x + e^{-x}e^t\Big|_x^2$

$\Rightarrow -e^{x}\left(e^{-x}-1\right)+e^{-x}\left(e^{2}-e^{x}\right)$

$\Rightarrow -1+e^{x}+e^{2-x}-1$

$=e^{2-x}+e^{x}-2$

$$f(x)=\begin{cases} e^{x}\left(1-e^{-2}\right); & x>2 \\ e^{2-x}+e^{x}-2; & 0\le x\le 2 \\ e^{-x}\left(e^{2}-1\right); & x<0 \end{cases}$$

For $x>2$

$f(x)_{min}=e^{2}-1$

For $0\le x\le 2$

$f'(x)=-e^{2-x}+e^{x}=0\Rightarrow e^{x}=e^{2-x}\Rightarrow e^{2x}=e^{2}\Rightarrow x=1$

$f(x)=2e-2=2(e-1)$

For $x<0$

$f(x)_{min}=e^{2}-1$

6. The integral $16\int_{1}^{2}\frac{dx}{x^{3}\left(x^{2}+2\right)^{2}}$ is equal to : **[JEE(Main)Jan.2023]**

समाकलन $16\int_{1}^{2}\frac{dx}{x^{3}\left(x^{2}+2\right)^{2}}$ बराबर है **[JEE(Main)Jan.2023]**

(A) $\frac{11}{6}+\log_{e}4$ (B) $\frac{11}{12}+\log_{e}4$ (C) $\frac{11}{6}-\log_{e}4$ (D) $\frac{11}{12}-\log_{e}4$

Ans. C

Sol. $\int_{1}^{2}\frac{dx}{x^{3}x^{4}}\left(1+\frac{2}{x^{2}}\right)^{2}$

Let $1+\frac{2}{x^{2}}=t\Rightarrow\frac{-4}{x^{3}}dx=dt$

$I=\frac{-1}{4}\int_{3}^{\frac{3}{2}}\left(\frac{t-1}{2}\right)^{2}\frac{dt}{t^{2}}=-\frac{1}{16}\int_{3}^{\frac{3}{2}}\frac{t^{2}-2t+1)}{t^{2}}dt$

$=\frac{-1}{16}\int_{3}^{\frac{3}{2}}\left(1-\frac{2}{t}+\frac{1}{t^{2}}\right)dt$

$=\frac{-1}{16}\left(t-2\ln|t|-\frac{1}{t}\right)_{3}^{\frac{3}{2}}$

$$=\frac{-1}{16}\left(\frac{3}{2}-2\ln\frac{3}{2}-\frac{2}{3}\right)-(3-2)n2-\frac{1}{3}\Bigg)$$

$$=\frac{-1}{16}\left(2\left(\ln 2-\ln\frac{3}{2}\right)-\frac{3}{2}-\frac{1}{3}\right)$$

$$=\frac{-1}{16}\left(2\ln 3-\frac{11}{6}\right)=\frac{-1}{16}\left(\ln 4-\frac{11}{6}\right)=\frac{11-6\ln 4}{16\times 6}$$

$$\frac{11-6\ln 4}{96}$$

7. Let [x] denote the greatest integer $\le x$. Consider the function $f(x) = \max\{x^2, 1+[x]\}$. Then the value of the integral $\int_0^2 f(x)dx$ is : **[JEE(Main)Jan.2023]**

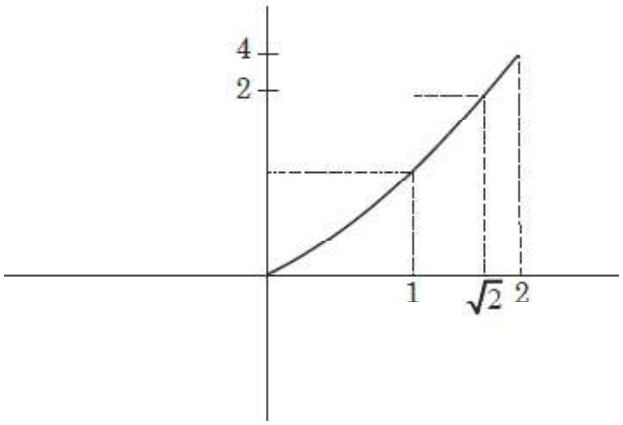
माना [x] महत्तम पूर्णांक $\le x$ है। माना फलन $f(x) = \max\{x^2, 1+[x]\}$ है। तब समाकलन $\int_0^2 f(x)dx$ का मान है :

[JEE(Main)Jan.2023]

(A) $\frac{8+4\sqrt{2}}{3}$ (B) $\frac{5+4\sqrt{2}}{3}$ (C) $\frac{1+5\sqrt{2}}{3}$ (D) $\frac{4+5\sqrt{2}}{3}$

Ans. B

Sol.



$$A=\int_0^1 1.dx+\int_1^{\sqrt{2}} 2dx+\int_{\sqrt{2}}^2 x^2dx$$

$$=1+2\sqrt{2}-2+\frac{8}{3}-\frac{2\sqrt{2}}{3}$$

$\frac{5}{3}+\frac{4\sqrt{2}}{3}$,

8. Let $f(x)=x+\frac{a}{\pi^2-4}\sin x+\frac{b}{\pi^2-4}\cos x, x\in IR$ be a function which satisfies

$f(x)=x+\int_0^{\pi/2}\sin(x+y)f(y)dy$. Then $(a+b)$ is equal to : **[JEE(Main)Jan.2023]**

माना एक फलन $f(x)=x+\frac{a}{\pi^2-4}\sin x+\frac{b}{\pi^2-4}\cos x, x\in IR$,

$f(x)=x+\int_0^{\pi/2}\sin(x+y)f(y)dy$ को संतुष्ट करता है। तब $(a+b)$ का मान है : **[JEE(Main)Jan.2023]**

(A) $-\pi(\pi+2)$ (B) $-2\pi(\pi+2)$ (C) $-2\pi(\pi-2)$ (D) $-\pi(\pi-2)$

Ans. B

Sol. $f(x)=x+\int_0^{\pi/2}(\sin x\cos y+\cos x\sin y)f(y)dy$

$f(x)=x+\int_0^{\pi/2}\left((\cos yf(y)dy)\sin x+(\sin yf(y)dy)\cos x\right)...(1)$

On comparing with

$f(x)=x+\frac{a}{\pi^2-4}\sin x+\frac{b}{\pi^2-4}\cos x, x\in R$, then

$\Rightarrow\frac{a}{\pi^2-4}=\int_0^{\pi/2}\cos yf(y)dy$ ...(2)

$\Rightarrow\frac{b}{\pi^2-4}=\int_0^{\pi/2}\sin y\ f(y)dy$ ...(3)

Add (2) and (3)

$\frac{a+b}{\pi^2-4}=\int_0^{\pi/2}(\sin y+\cos y)f(y)dy$ ....(4)

$\frac{a+b}{\pi^2-4}=\int_0^{\pi/2}(\sin y+\cos y)f\left(\frac{\pi}{2}-y\right)dy$ ....(5)

Add (4) and (5)

$\frac{2(a+b)}{\pi^2-4}=\int_0^{\pi/2}(\sin y+\cos y)\left(\frac{\pi}{2}+\frac{(a+b)}{\pi^2-4}(\sin y+\cos y)\right)dy$

$$= \pi + \frac{a+b}{\pi^2 - 4}\left(\frac{\pi}{2} + 1\right)$$

$(a + b) = -2\pi\,(\pi + 2)$

9. The value of the integral $\int_{1/2}^{2} \frac{\tan^{-1} x}{x} dx$ is equal to : **[JEE(Main)Jan.2023]**

समाकलन $\int_{1/2}^{2} \frac{\tan^{-1} x}{x} dx$ का मान बराबर है **[JEE(Main)Jan.2023]**

(A) $\pi \log_e 2$ (B) $\frac{1}{2}\log_e 2$ (C) $\frac{\pi}{4}\log_e 2$ (D) $\frac{\pi}{2}\log_e 2$

Ans. D

Sol. $I = \int_{1/2}^{2} \frac{\tan^{-1} x}{x} dx$ .....(i)

Put $x = \frac{1}{t}$ $dx = -\frac{1}{t^2} dt$

$$I = -\int_{2}^{1/2} \frac{\tan^{-1}\frac{1}{t}}{\frac{1}{t}} \cdot \frac{1}{t^2} dt = -\int_{2}^{1/2} \frac{\tan^{-1}\frac{1}{t}}{t} dt$$

$$I = \int_{2}^{1/2} \frac{\cot^{-1} t}{t} dt = \int_{2}^{1/2} \frac{\cot^{-1} x}{x} dx \qquad .....(ii)$$

Add both equation

$$2I = \int_{1/2}^{2} \frac{\tan^{-1} x + \cot^{-1} x}{x} dx = \frac{\pi}{2}\int_{1/2}^{2} \frac{dx}{x} = \frac{\pi}{2}(\ell n\, 2)_{1/2}^{2}$$

$$= \frac{\pi}{2}\left(\ell n 2 - \ell n \frac{1}{2}\right) = \pi \ell n 2$$

$$I = \frac{\pi}{2}\ell n 2$$

10. The value of the integral $\int_{1}^{2}\left(\frac{t^4+1}{t^6+1}\right) dt$ is : **[JEE(Main)Jan.2023]**

समाकलन $\int_{1}^{2}\left(\frac{t^4+1}{t^6+1}\right) dt$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\tan^{-1}\frac{1}{2} - \frac{1}{3}\tan^{-1} 8 + \frac{\pi}{3}$ (B) $\tan^{-1}\frac{1}{2} + \frac{1}{3}\tan^{-1} 8 - \frac{\pi}{3}$

(C) $\tan^{-1}2-\frac{1}{3}\tan^{-1}8+\frac{\pi}{3}$ (D) $\tan^{-1}2+\frac{1}{3}\tan^{-1}8-\frac{\pi}{3}$

Ans. D

Sol. $=\int_1^2\frac{\left(t^4+1\right)}{\left(t^6+1\right)}\frac{\left(t^2+1\right)}{\left(t^2+1\right)}dt$

$=\int_1^2\frac{t^6+t^4+t^2+1}{\left(t^6+1\right)\left(t^2+1\right)}dt$

$=\int_1^2\frac{t^6+1}{\left(t^6+1\right)\left(t^2+1\right)}dt+\int_1^2\frac{t^4+t^2}{\left(t^6+1\right)\left(t^2+1\right)}dt$

$=\int_1^2\frac{1}{t^2+1}dt+\int_1^2\frac{t^2}{t^6+1}dt,$

$I=\left[\tan^{-1}t\right]_1^2+I_1$

$\tan^{-1}-\frac{\pi}{4}+I_1\ \tan^{-1}-\frac{\pi}{4}+I_1$

$I_1=\int_{11}^2\frac{t^2}{t^6+1}dt$

put $t^3=\theta\cdot 3t^2\,dt=d\theta$

$t=1$ then $\theta=1$

$t=2$ then $\theta=8$

$=\frac{1}{3}\int_1^8\frac{d\theta}{\theta^2+1}$

$=\frac{1}{3}\left[\tan^{-1}(\theta)\right]_1^8$

$=\frac{1}{3}\left[\tan^{-1}8-\frac{\pi}{4}\right]$

$I=\tan^{-1}2-\tan^{-1}(1)+\frac{1}{3}\tan^{-1}8-\frac{\pi}{12}$

$\tan^{-1}2+\frac{1}{3}\tan^{-1}8-\frac{\pi}{3}$

11. If $[t]$ denotes the greatest integer $\le t$, then the value of $\frac{3(e-1)}{e}\int_1^2 x^2e^{[x]+\left[x^3\right]}dx$ is :

यदि [t] महत्तम पूर्णांक $\le t$, $\frac{3(e-1)}{e}\int_1^2 x^2 e^{[x]+[x^3]}dx$ का मान है : **[JEE(Main)Jan.2023]**

(A) $e^8 - 1$ (B) $e^7 - 1$ (C) $e^8 - e$ (D) $e^9 - 1$

Ans. C

Sol. $\frac{3(e-1)}{e}\int_1^2 x^2 e^{[x]+[x^3]}dx$

Let $k = \int_1^2 x^2 e^{1+[x^3]}dx$

$x^3 = t$

$3x^2\,dx = dt$

$k = \frac{1}{3}\int_1^8 e.e^{[t]}dt$

$\frac{e}{3}\left[e + e^2 + e^3 + ... + e^7\right] = \frac{e^2}{3}\left(\frac{e^7-1}{e-1}\right)$

$\frac{3(e-1)}{e}k = \frac{3(e-1)}{e}\frac{e^2}{3}\left(\frac{e^7-1}{e-1}\right) = e(e^7-1)$

$= e^8 - e$

12. $\lim_{n\to\infty}\frac{3}{n}\left\{4+\left(2+\frac{1}{n}\right)^2+\left(2+\frac{2}{n}\right)^2+.....+\left(3-\frac{1}{n}\right)^2\right\}$ is equal to : **[JEE(Main)Jan.2023]**

$\lim_{n\to\infty}\frac{3}{n}\left\{4+\left(2+\frac{1}{n}\right)^2+\left(2+\frac{2}{n}\right)^2+.....+\left(3-\frac{1}{n}\right)^2\right\}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 12 (B) 19 (C) $\frac{19}{3}$ (D) 0

Ans. B

Sol. $\lim_{n\to\infty}\frac{3}{n}\sum_{r=0}^{n-1}\left(2+\frac{r}{n}\right)^2$

$= 3\int_0^1 (2+x)^2 dx = 27 - 8 = 19$

13. The value of $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\frac{(2+3\sin x)}{\sin x(1+\cos x)}dx$ is equal to : **[JEE(Main)Jan.2023]**

$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{(2+3\sin x)}{\sin x(1+\cos x)}dx$ का मान बराबर है **[JEE(Main)Jan.2023]**

(A) $\frac{10}{3}-\sqrt{3}-\log_e\sqrt{3}$ (B) $-2+3\sqrt{3}+\log_e\sqrt{3}$

(C) $\frac{10}{3}-\sqrt{3}+\log_e\sqrt{3}$ (D) $\frac{7}{2}-\sqrt{3}-\log_e\sqrt{3}$

Ans. C

Sol. $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{2(1-\cos x)}{\sin x\left(1-\cos^2 x\right)}dx+\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{3}{1+\cos x}dx$

$=\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{2}{\sin^3 x}dx-2\int_{\frac{\pi}{2}}^{\frac{\pi}{2}} \cot x.\cos ec^2 x dx+\frac{3}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sec^2\frac{x}{2}dx$

$=2\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sqrt{1+\cot^2 x}.\cos ec^2 x dx-2\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot x.\cos ec^2 x ds+3\left[\tan\frac{x}{2}\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$

Let $\cot x = 1$

$=-2\int_{\frac{1}{\sqrt{3}}}^{0} \sqrt{1+t^2}dt+2\int_{\frac{1}{\sqrt{3}}}^{0} tdt+3\left[1-\frac{1}{\sqrt{3}}\right]$

$=2\left[\frac{t}{2}\sqrt{1+t^2}+\frac{1}{2}\log\left(t+\sqrt{1+t^2}\right)\right]_0^{\frac{1}{\sqrt{3}}}-\left[t^2\right]_0^{\frac{1}{\sqrt{3}}}+3-\sqrt{3}$

$=\frac{2}{3}+\log\sqrt{3}-\frac{1}{3}+3-\sqrt{3}=\frac{10}{3}-\sqrt{3}+\frac{1}{2}\log 3$

14. Let $\alpha\in(0,1)$ and $\beta=\log_e(1-\alpha)$. Let $P_n(x)=x+\frac{x^2}{2}+\frac{x^3}{3}+.....+\frac{x^n}{n}, x\in(0,1)$. Then the integral $\int_0^{\alpha}\frac{t^{50}}{1-t}dt$ is equal to : **[JEE(Main)Jan.2023]**

माना $\alpha\in(0,1)$ तथा $\beta=\log_e(1-\alpha)$ है। माना $P_n(x)=x+\frac{x^2}{2}+\frac{x^3}{3}+.....+\frac{x^n}{n}, x\in(0,1)$ है। तो समाकलन $\int_0^{\alpha}\frac{t^{50}}{1-t}dt$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $-\left(\beta+P_{50}(\alpha)\right)$ (B) $\beta+P_{50}(\alpha)$ (C) $\beta-P_{50}(\alpha)$ (D) $P_{50}(\alpha)-\beta$

Ans. A

Sol. $\int_0^{\alpha}\frac{t^{50}-1+1}{1-t}=-\int_0^{\alpha}\left(1+t+.....+t^{49}\right)+\int_0^{\alpha}\frac{1}{1-t}dt$

$$=-\left(\frac{\alpha^{50}}{50}+\frac{\alpha^{49}}{49}+.....+\frac{\alpha^{1}}{1}\right)+\left(\frac{\ln(1-f)}{-1}\right)_0^{\alpha}$$

$= -P_{50}(\alpha) - \ln(1-\alpha)$

$= \square\Pi_{50}(\alpha) - \beta$

15. Let $\alpha>0$. If $\int_0^{\alpha}\frac{x}{\sqrt{x+\alpha}-\sqrt{x}}dx=\frac{16+20\sqrt{2}}{15}$, then α is equal to : **[JEE(Main)Jan.2023]**

माना $\alpha>0$ है। यदि $\int_0^{\alpha}\frac{x}{\sqrt{x+\alpha}-\sqrt{x}}dx=\frac{16+20\sqrt{2}}{15}$ है, तो α बराबर है : **[JEE(Main)Jan.2023]**

(A) $\sqrt{2}$ (B) 4 (C) $2\sqrt{2}$ (D) 2

Ans. D

Sol. After rationalising

$\int_0^{\alpha}\frac{x}{\alpha}(\sqrt{x+\alpha}+\sqrt{x})$

$\int_0^{\alpha}\frac{1}{\alpha}\left[(x+\alpha)^{3/2}-\alpha(x+\alpha)^{1/2}+x^{3/2}\right]$

$\frac{1}{\alpha}\left[\frac{2}{5}(x+\alpha)^{5/2}-\alpha\frac{2}{3}(x+\alpha)^{3/2}+\frac{2}{5}x^{5/2}\right]\Bigg|_0^{\alpha}$

$=\frac{1}{\alpha}\left(\frac{2}{5}(2\alpha)^{5/2}-\frac{2\alpha}{3}(2\alpha)^{3/2}+\frac{2}{5}\alpha^{5/2}-\frac{2}{5}\alpha^{5/2}+\frac{2}{3}\alpha^{5/2}\right.$

$=\frac{1}{\alpha}\left(\frac{2^{7/2}\alpha^{5/2}}{5}-\frac{2^{5/2}\alpha^{5/2}}{3}+\frac{2}{3}\alpha^{5/2}\right)$

$=\alpha^{3/2}\left(\frac{2^{7/2}}{5}-\frac{2^{5/2}}{3}+\frac{2}{3}\right)$

$=\frac{\alpha^{3/2}}{15}(24\sqrt{2}-20\sqrt{2}+10)=\frac{\alpha^{3/2}}{15}(4\sqrt{2}+10)$

Now, $\frac{\alpha^{\frac{3}{2}}}{15}\left(4\sqrt{2}+10\right)=\frac{16+20\sqrt{2}}{15}$

$\Rightarrow \alpha = 2$

16. If $\phi(x)=\frac{1}{\sqrt{x}}\int_{\frac{\pi}{4}}^{x}\left(4\sqrt{2}\sin t-3\phi'(t)\right)dt$, $x>0$, then $\phi'\left(\frac{\pi}{4}\right)$ is equal to : **[JEE(Main)Jan.2023]**

यदि $\phi(x)=\frac{1}{\sqrt{x}}\int_{\frac{\pi}{4}}^{x}\left(4\sqrt{2}\sin t-3\phi'(t)\right)dt$, $x>0$ है, तो $\phi'\left(\frac{\pi}{4}\right)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{8}{6+\sqrt{\pi}}$ (B) $\frac{4}{6+\sqrt{\pi}}$ (C) $\frac{8}{\sqrt{\pi}}$ (D) $\frac{4}{6-\sqrt{\pi}}$

Ans. A

Sol. $\phi'(x)=\frac{1}{\sqrt{x}}\left[\left(4\sqrt{2}\sin x-3\phi'(x)\right)\cdot 1-0\right]-\frac{1}{2}x^{-3/2}$

$\int_{\frac{\pi}{4}}^{x}\left(4\sqrt{2}\sin t-3\phi'(t)\right)dt$

$\phi'\left(\frac{\pi}{4}\right)=\frac{2}{\sqrt{\pi}}\left[4-3\phi'\left(\frac{\pi}{4}\right)\right]+0$

$\left(1+\frac{6}{\sqrt{\pi}}\right)\phi'\left(\frac{\pi}{4}\right)=\frac{8}{\sqrt{\pi}}$

$\phi'\left(\frac{\pi}{4}\right)=\frac{8}{\sqrt{\pi}+6}$

17. $\lim_{n\to\infty}\left[\frac{1}{1+n}+\frac{1}{2+n}+\frac{1}{3+n}+.....+\frac{1}{2n}\right]$ is equal to : **[JEE(Main)Jan.2023]**

$\lim_{n\to\infty}\left[\frac{1}{1+n}+\frac{1}{2+n}+\frac{1}{3+n}+.....+\frac{1}{2n}\right]$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\log_e 2$ (B) $\log_e\left(\frac{2}{3}\right)$ (C) 0 (D) $\log_e\left(\frac{3}{2}\right)$

Ans. A

Sol. $\lim_{n\to\infty}\left(\frac{1}{1+n}+\frac{1}{2+n}+....+\frac{1}{2n}\right)=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{r+n}$

$=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{\left(\frac{r}{n}+1\right)}\cdot\frac{1}{n}$

$\int_0^1\frac{dx}{1+x}=\left[\ln(1+x)\right]_0^1$

$\ln 2 - \ln 1 = \ln 2$

18. The value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x+\frac{\pi}{4}}{2-\cos 2x}dx$ is : **[JEE(Main)Jan.2023]**

समाकलन $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x+\frac{\pi}{4}}{2-\cos 2x}dx$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{\pi^2}{6\sqrt{3}}$ (B) $\frac{\pi^2}{6}$ (C) $\frac{\pi^2}{3\sqrt{3}}$ (D) $\frac{\pi^2}{12\sqrt{3}}$

Ans. A

Sol. $I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x+\frac{\pi}{4}}{2-\cos 2x}dx$ (A)

$x \to -x$

$I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{-x+\frac{\pi}{4}}{2-\cos 2x}dx$ (B)

(A) + (2)

$2I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\frac{\pi}{2}}{2-\cos 2x}dx$

$I = \frac{\pi}{4}\cdot 2\int_{0}^{\frac{\pi}{4}} \frac{dx}{2-\cos 2x}dx$

$I = \frac{\pi}{4}\cdot 2\int_{0}^{\frac{\pi}{4}} \frac{\left(1+\tan^2 x\right)dx}{2\left(1+\tan^2 x\right)-\left(1-\tan^2 x\right)}$

$I = \frac{\pi}{2}\int_{0}^{1} \frac{dt}{3t^2+1}$

$$\Rightarrow I = \frac{\pi}{2\sqrt{3}} \tan^{-1}\sqrt{3}$$

$$I = \frac{\pi^2}{6\sqrt{3}}$$

19. Let f be a differentiable function defined on $\left[0, \frac{\pi}{2}\right]$. such that f(x) > 0 and $f(x) + \int_0^x f(t)\sqrt{1-\left(\log_e f(t)\right)^2}\,dt = e$, $\forall x \in \left[0, \frac{\pi}{2}\right]$. Then $\left(6\log_e f\left(\frac{\pi}{6}\right)\right)^2$ is equal to _____. **[JEE(Main)Jan.2023]**

माना $\left[0, \frac{\pi}{2}\right]$ पर परिभाषित एक अवकलनीय फलन f है जिसके लिए f(x) > 0 और $f(x) + \int_0^x f(t)\sqrt{1-\left(\log_e f(t)\right)^2}\,dt = e$, $\forall x \in \left[0, \frac{\pi}{2}\right]$ है। तब $\left(6\log_e f\left(\frac{\pi}{6}\right)\right)^2$ बराबर है _____ **[JEE(Main)Jan.2023]**

Ans. 27

Sol. $t(x) = \int_0^x f(t)\sqrt{1-\ell nt(x)^2}\,dt = e : \nabla x \in \left[0, \frac{\pi}{2}\right]$

Differentiate on both side

$$f\cdot(x) + f(x)\sqrt{1-knf\left(xm^2\right.} = 0$$

$$\int \frac{f(x)dx}{f(x)\sqrt{1-tnf(x)m^2}} = -\int dx$$

Let $(n+\alpha) = U$

$$\frac{1}{f(x)}\cdot f'(x)dx = du$$

$$\int \frac{du}{\sqrt{1-u^2}} = -\int dx$$

$$\sin^{-1}\left(\ell n\left(f\left(x\right)\right)\right) = -x + c$$

put $x = 0$

$$\sin^{-1}\left(\ell n\left(f\left(0\right)\right)\right) = 0 + c$$

$$\sin^{-1}(1) = c$$

$$c = \frac{\pi}{2}$$

$$\sin^{-1}\left(\ell n\left(f\left(x\right)\right)\right) = -x + \frac{\pi}{2}$$

$$\ell n\left(f\left(x\right)\right)=\sin\left(\frac{\pi}{2}-x\right)=\cos x$$

$$f\left(x\right)=e^{\cos x}$$

$$f\left(\frac{\pi}{6}\right)=e^{\frac{\sqrt{3}}{2}}$$

$$\text{Now}\left(6\ell ne^{\frac{\sqrt{3}}{2}}\right)^2=\left(6\times\frac{\sqrt{3}}{2}\right)^2=27$$

20. The value of $\frac{8}{\pi}\int_0^{\frac{\pi}{2}}\frac{\left(\cos x\right)^{2023}}{\left(\sin x\right)^{2023}+\left(\cos x\right)^{2023}}dx$ is _____. **[JEE(Main)Jan.2023]**

$\frac{8}{\pi}\int_0^{\frac{\pi}{2}}\frac{\left(\cos x\right)^{2023}}{\left(\sin x\right)^{2023}+\left(\cos x\right)^{2023}}dx$ का मान है _____. **[JEE(Main)Jan.2023]**

Ans. 2

Sol. $I=\frac{8}{\pi}\int_0^{\frac{\pi}{2}}\frac{(\cos x)^{2023}}{(\sin x)^{2023}+(\cos x)^{2023}}dx$ _____(1)

Using $\int_0^a f(x)dx=\int_0^a f(a-x)dx$

$I=\frac{8}{\pi}\int_0^{\frac{\pi}{2}}\frac{(\sin x)^{2023}}{(\sin x)^{2023}+(\cos x)^{2023}}dx$ _____(2)

Adding (1) and (2)

$2I=\frac{8}{\pi}\int_0^{\frac{\pi}{2}}1dx$

$I=2$

21. The value of $12\int_0^3\left|x^2-3x+2\right|dx$ is _____.

$12\int_0^3\left|x^2-3x+2\right|dx$ का मान है _____

Ans. 22

Sol. $12\int_0^3\left|x^2-3x+2\right|dx$

$=12\int_0^3\left|\left(x-\frac{3}{2}\right)^2-\frac{1}{4}\right|dx$

If $x-\frac{3}{2}=t$

$dx = dt$

$$= 24\int_0^{3/2}\left|t^2-\frac{1}{4}\right|dt$$

$$= 24\left[-\int_0^{1/2}\left(t^2-\frac{1}{4}\right)dt+\int_{1/2}^{3/2}\left(t^2-\frac{1}{4}\right)dt\right]=22$$

22. If $\int_{\frac{1}{3}}^{3}|\log_e x|\,dx=\frac{m}{n}\log_e\left(\frac{n^2}{e}\right)$, where m and n are coprime natural numbers, then m^2+n^2-5 is equal _____.

[JEE(Main)Jan.2023]

यदि $\int_{\frac{1}{3}}^{3}|\log_e x|\,dx=\frac{m}{n}\log_e\left(\frac{n^2}{e}\right)$ है, जहाँ m तथा n असहभाज्य धन पूर्णांक है, तो m^2+n^2-5 बराबर है।

[JEE(Main)Jan.2023]

Ans. 20

Sol. $\int_{\frac{1}{3}}^{3}|\ell nx|\,dx=\int_{\frac{1}{3}}^{1}(-\ell nx)dx+\int_1^3(\ell nx)dx$

$$=-[x\ell nx-x]_{1/3}^1+[x\ell nx-x]_1^3$$

$$=-\left[-1-\left(\frac{1}{3}\ln\frac{1}{3}-\frac{1}{3}\right)\right]+[3\ln 3-3-(-1)]$$

$$=\left[-\frac{2}{3}-\frac{1}{3}\ln\frac{1}{3}\right]+[3\ln 3-2]$$

$$=-\frac{4}{3}+\frac{8}{3}\ln 3$$

$$=\frac{4}{3}(2\ln 3-1)$$

$$=\frac{4}{3}\left(\ell n\frac{9}{2}\right)$$

$\therefore m=4, n=3$

Now, $m^2+n^2-5=16+9-5=20$

23. $\lim_{x\to 0}\frac{48}{x^4}\int_0^x\frac{t^3}{t^6+1}dt$ is equal to _____. **[JEE(Main)Jan.2023]**

$\lim\limits_{x\to 0}\dfrac{48}{x^4}\int\limits_0^x\dfrac{t^3}{t^6+1}dt$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 12

Sol. $\lim\limits_{x\to 0}\dfrac{48}{x^4}\int\limits_0^x\dfrac{t^3}{t^6+1}dt$

$$=\lim_{x\to 0}\frac{48\int\limits_0^x\frac{t^3}{t^6+1}\cdot dt}{x^4}$$

$$=\lim_{x\to 0}\frac{48\times\frac{x^3}{x^6+1}}{4x^3}=\lim_{x\to 0}\frac{12}{x^6+1}=12$$

24. If $\int\limits_0^{\pi}\dfrac{5^{\cos x}\left(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x\right)dx}{1+5^{\cos x}}=\dfrac{k\pi}{16}$, then k is equal to _____ .

[JEE(Main)Jan.2023]

यदि $\int\limits_0^{\pi}\dfrac{5^{\cos x}\left(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x\right)dx}{1+5^{\cos x}}=\dfrac{k\pi}{16}$ है, तो k बराबर है।

[JEE(Main)Jan.2023]

Ans. 26

Sol. $I=\int\limits_0^{\pi}\dfrac{5^{\cos x}\left(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x\right)dx}{1+5^{\cos x}}$

$$=2\int_0^{\pi/2}\left(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x\right)dx$$

$$=\int_0^{\pi/2}2+\cos 4x+\cos 2x+1+\cos 2x+2\cos 3x\left(\frac{\cos 3x+3\cos x}{4}\right)dx$$

$$=\int_0^{\pi/2}3dx+0+0+0+\int_0^{\pi/2}\frac{1}{2}\left(\cos^2 3x+3\cos x\cos 3x\right)dx$$

$$=\frac{3\pi}{2}+\frac{1}{4}\int_0^{\pi/2}\left(1+\cos 6x+3\left(\cos 4x+\cos 2x\right)\right)dx$$

$$= \frac{3\pi}{4} + \frac{1}{4}\cdot\frac{\pi}{2} = \frac{13\pi}{8} = \frac{k\pi}{16}$$

$\Rightarrow k = 26$

25. Let $I(x) = \int \frac{x^2\left(x\sec^2 x + \tan x\right)}{(x\tan x + 1)^2}dx$. If $I(0) = 0$, then $I\left(\frac{\pi}{4}\right)$ is equal to : **[JEE(Main)April,2023]**

माना $I(x) = \int \frac{x^2\left(x\sec^2 x + \tan x\right)}{(x\tan x + 1)^2}dx$ है। यदि $I(0) = 0$ है, तब $I\left(\frac{\pi}{4}\right)$ का मान है : **[JEE(Main)April,2023]**

(A) $\log_e \frac{(\pi+4)^2}{16} + \frac{\pi^2}{4(\pi+4)}$ (B) $\log_e \frac{(\pi+4)^2}{32} - \frac{\pi^2}{4(\pi+4)}$

(C) $\log_e \frac{(\pi+4)^2}{32} + \frac{\pi^2}{4(\pi+4)}$ (D) $\log_e \frac{(\pi+4)^2}{16} - \frac{\pi^2}{4(\pi+4)}$

Ans. B

Sol. $\int x^2\left(\frac{x\sec^2 x + \tan x}{(x\tan x + 1)^2}\right)dx$

$$= \frac{x^2}{(x\tan x + 1)} + \int \frac{2x}{x\tan x + 1}dx$$

$$I = 2\int \frac{x}{x\tan x + 1}dx$$

$$= 2\int \frac{x\cos x}{x\sin x + \cos x}dx$$

Let $x\sin x + \cos x = t$

$(x\cos x + \sin x - \sin x)dx = dt$

$$= 2\int \frac{dt}{t} = 2\log t + c$$

$$= 2\log|x\sin x + \cos x| + c$$

$$\therefore \int \frac{x^2\left(x\sec^2 x + \tan x\right)}{\left(x\tan + 1\right)^2}dx = \frac{-x^2}{x\tan x + 1} + 2\log\left|x\sin x + \cos x\right| + c$$

$I(0) = 0 \qquad \Rightarrow c = 0$

$$I\left(\frac{\pi}{4}\right) = \frac{-\pi}{\frac{\pi}{4}\times 1 + 1} + 2\log\left|\frac{1}{\sqrt{2}}\left(\frac{\pi}{4}\right) + 1\right|$$

$$= \log_e \frac{(\pi+4)^2}{32} - \frac{\pi^2}{4(\pi+4)}$$

26. Let $I(x)=\int\frac{(x+1)}{x\left(1+xe^x\right)^2}dx, x>0$. If $\lim\limits_{x\to\infty} I(x)=0$, then $I(1)$ is equal to : **[JEE(Main)April,2023]**

माना $I(x)=\int\frac{(x+1)}{x\left(1+xe^x\right)^2}dx, x>0$ है। यदि $\lim\limits_{x\to\infty} I(x)=0$ है, तो $I(1)$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{e+2}{e+1}-\log_e(e+1)$ (B) $\frac{e+1}{e+2}+\log_e(e+1)$

(C) $\frac{e+1}{e+2}-\log_e(e+1)$ (D) $\frac{e+2}{e+1}+\log_e(e+1)$

Ans. A

Sol. $I(x)=\int\frac{(x+1)}{x\left(1+xe^x\right)^2}dx$

$1+xe^x=t$

$\left(xe^x+e^x\right)dx=dt$

$(x+1)dx=\frac{1}{e^x}dt$

$\therefore\int\frac{dt}{xe^x\cdot t^2}=\int\frac{dt}{(t-1)t^2}=\int\frac{dt}{t(t-1)\cdot t}\Rightarrow\int\frac{t-(t-1)}{t(t-1)t}dt$

$=\int\frac{dt}{t(t-1)}-\int\frac{dt}{t^2}=\int\frac{t-(t-1)}{t(t-1)}dt+\frac{1}{t}+c$

$=\ln|t-1|-\ln|t|+\frac{1}{t}+c$

$\Rightarrow\ln\left|xe^x\right|-\ln\left|1+xe^x\right|+\frac{1}{1+xe^x}+c$

$I(x)=\ln\left|\frac{xe^x}{1+xe^x}\right|+\frac{1}{1+xe^x}+c$

$\lim\limits_{x\to\infty} I(x)=c=0$

$\therefore I(1)=\ln\left|\frac{e}{1+e}\right|+\frac{1}{1+e}$

$=\ln e-\ln(1+e)+\frac{1}{1+e}$

$=\frac{e+2}{e+1}-\ln(1+e)$

27. The integral $\int\left(\left(\frac{x}{2}\right)^x+\left(\frac{2}{x}\right)^x\right)\log_2 x dx$ is equal to : **[JEE(Main)April,2023]**

समाकलन $\int\left(\left(\frac{x}{2}\right)^x+\left(\frac{2}{x}\right)^x\right)\log_2 x dx$ बराबर है : **[JEE(Main)April,2023]**

(A) $\left(\frac{x}{2}\right)^x \log_2\left(\frac{x}{2}\right)+C$ (B) $\left(\frac{x}{2}\right)^x \log_2\left(\frac{2}{x}\right)+C$

(C) $\left(\frac{x}{2}\right)^x-\left(\frac{2}{x}\right)^x+C$ (D) $\left(\frac{x}{2}\right)^x+\left(\frac{2}{x}\right)^x+C$

Ans. B **Bonus**

Sol. $\int\left(x^x 2^{-x}+2^x x^{-x}\right)\log_2 x dx$

$\int\left(e^{x\ln x}\cdot e^{-x\ln 2}+e^{x\ln 2}\cdot e^{-x\ln x}\right)dx$

$\int\left(e^{x\ln x-x\ln 2}+e^{x\ln 2-x\ln x}\right)\frac{\ln x}{\ln 2}dx$

let $x\ln x-x\ln 2=t$

$(\ln x+1-\ln 2)dx=dt$

28. If $I(x)=\int e^{\sin^2 x}(\cos x\sin 2x-\sin x)dx$ and $I(0)=1$, then $I\left(\frac{\pi}{3}\right)$ is equal to : **[JEE(Main)April,2023]**

यदि $I(x)=\int e^{\sin^2 x}(\cos x\sin 2x-\sin x)dx$ तथा $I(0)=1$ हैं, तो $I\left(\frac{\pi}{3}\right)$ बराबर है : **[JEE(Main)April,2023]**

(A) $e^{\frac{3}{4}}$ (B) $-\frac{1}{2}e^{\frac{3}{4}}$ (C) $-e^{\frac{3}{4}}$ (D) $\frac{1}{2}e^{\frac{3}{4}}$

Ans. D

Sol. $\because I=\int e^{f(x)}\left(g'(x)+f'(x)\cdot g(x)\right)dx=e^{f(x)}\cdot g(x)$

$\therefore I=e^{\sin^2 x}\cdot\cos x+c$

but $I(0)=1\Rightarrow c=0$

$I=e^{\sin^2 x}\cos x$

$I\left(\frac{\pi}{3}\right)=\frac{1}{2}e^{\frac{3}{4}}$

29. For $\alpha, \beta, \gamma, \delta\in\mathbb{N}$, if $\int\left(\left(\frac{x}{e}\right)^{2x}+\left(\frac{e}{x}\right)^{2x}\right)\log_e x\,dx=\frac{1}{\alpha}\left(\frac{x}{e}\right)^{\beta x}-\frac{1}{\gamma}\left(\frac{e}{x}\right)^{\delta x}+C$, where $e=\sum_{n=0}^{\infty}\frac{1}{n!}$ and C is constant of integration, then $\alpha+2\beta+3\gamma-4\delta$ is equal to : **[JEE(Main)April,2023]**

$\alpha, \beta, \gamma, \delta \in \mathbb{N}$ के लिए, यदि $\int\left(\left(\frac{x}{e}\right)^{2x}+\left(\frac{e}{x}\right)^{2x}\right)\log_e x\,dx=\frac{1}{\alpha}\left(\frac{x}{e}\right)^{\beta x}-\frac{1}{\gamma}\left(\frac{e}{x}\right)^{\delta x}+C$ है, जहाँ $e=\sum_{n=0}^{\infty}\frac{1}{n!}$ तथा C समाकलन अचर है, तो $\alpha + 2\beta + 3\gamma - 4\delta$ बराबर है : **[JEE(Main)April,2023]**

(A) –4 (B) 1 (C) 4 (D) –8

Ans. C

Sol. $\text{Let}\left(\frac{x}{e}\right)^{2x}=t$

$2x(\ln x-1)=\ln t$

$\left(2(\ln x-1)+2x\times\frac{1}{x}\right)dx=\frac{1}{t}dt$

$\ln x\,dx=\frac{1}{2t}dt$

$I=\int\left(t+\frac{1}{t}\right)\times\frac{1}{2t}dt=\frac{1}{2}\int\left(1+\frac{1}{t^2}\right)dt$

$=\frac{1}{2}\left(t-\frac{1}{t}\right)+C$

$=\frac{1}{2}\left(\frac{x}{e}\right)^{2x}-\frac{1}{2}\left(\frac{e}{x}\right)^{2x}+C$

$\Rightarrow \alpha=2,\beta=2,\gamma=2,\delta=2$

$\therefore\quad \alpha+2\beta+3\gamma-4\delta=2+4+6-8=4$

30. Let $I(x)=\int\sqrt{\frac{x+7}{x}}dx$ and $I(9)=12+7\log_e 7$. If $I(1)=\alpha+7\log_e(1+2\sqrt{2})$, α^4 is equal to _____.

[JEE(Main)April,2023]

माना $I(x)=\int\sqrt{\frac{x+7}{x}}dx$ तथा $I(9)=12+7\log_e 7$ हैं। यदि $I(1)=\alpha+7\log_e(1+2\sqrt{2})$ है, तो α^4 बराबर है _____

[JEE(Main)April,2023]

Ans. 64

Sol. $I(x)=\int\sqrt{\frac{x+7}{x}}dx$

$=\int\frac{\sqrt{\left(\sqrt{x}\right)^2+\left(\sqrt{7}\right)^2}}{\sqrt{x}}dx$

Let $\sqrt{x} = t \therefore \frac{1}{2\sqrt{x}}dx = dt$

$= 2\int \sqrt{t^2 + (\sqrt{7})^2}\,dt = 2\left(\frac{t}{2}\sqrt{t^2+7} + \frac{7}{2}\ln\left(t + \sqrt{t^2+7}\right)\right) + c$

$I(x) = \sqrt{x}\sqrt{x+7} + 7\ln(\sqrt{x} + \sqrt{x+7}) + c$

$I(9) = 12 + 7\ln 7 = 12 + 7\ln(3+4) + c$

$\Rightarrow c = 0$

$\therefore I(x) = \sqrt{x}\sqrt{x+7} + 7\ln(\sqrt{x} + \sqrt{x+7})$

at $x = 1$

$I(1) = \sqrt{8} + 7\ell n(1+\sqrt{8}) = 7\ell n(1+2\sqrt{2}) + \alpha$

$\alpha = \sqrt{8}$

$\alpha^4 = 64$

31. Let $f(x) = \int \frac{dx}{\left(3+4x^2\right)\sqrt{4-3x^2}}$, $|x| < \frac{2}{\sqrt{3}}$. If $f(0) = 0$ and $f(1) = \frac{1}{\alpha\beta}\tan^{-1}\left(\frac{\alpha}{\beta}\right)$, $\alpha, \beta > 0$, then $\alpha^2 + \beta^2$ is equal to _____. **[JEE(Main)April,2023]**

यदि $f(x) = \int \frac{dx}{\left(3+4x^2\right)\sqrt{4-3x^2}}$, $|x| < \frac{2}{\sqrt{3}}$, $f(0) = 0$ तथा $f(1) = \frac{1}{\alpha\beta}\tan^{-1}\left(\frac{\alpha}{\beta}\right)$, $\alpha, \beta > 0$ हो, तो $\alpha^2 + \beta^2$ का मान gksxk_____ **[JEE(Main)April,2023]**

Ans. 28

Sol. $f(x) = \int \frac{dx}{\left(3+4x^2\right)\sqrt{4-3x^2}}$

Let $x = \frac{1}{t}$

$dx = -\frac{1}{t^2}dt$

$= \int \frac{-\frac{1}{t^2}dt}{\left(3+\frac{4}{t^2}\right)\sqrt{4-\frac{3}{t^2}}}$

$= \int \frac{-tdt}{\left(3t^2+4\right)\sqrt{4t^2-3}}$

Now, let $4t^2 - 3 = p^2$

$4t\,dt = p\,dp$

$$= \int \frac{-\frac{p}{4}dp}{\left(\frac{3p^2+25}{4}\right)p} = -\int \frac{dp}{3p^2+25}$$

$$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3}p}{5}\right) + C$$

$$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3}\cdot\sqrt{4t^2-3}}{5}\right) + C$$

$$= -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3}\cdot\sqrt{\frac{4}{x^2}-3}}{5}\right) + C$$

$$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1}\left(\frac{\sqrt{3}\cdot\sqrt{4-3x^2}}{5|x|}\right) + C$$

$x = 0, y = 0$

$$0 = -\frac{1}{5\sqrt{3}} \tan^{-1}(\infty) + C$$

$$C = \frac{1}{5\sqrt{3}} \times \frac{\pi}{2}$$

$$f(x) = -\frac{1}{5\sqrt{3}} \tan^{-1}\frac{\sqrt{3}\sqrt{4-3x^2}}{5x} + \frac{\pi}{2\left(5\sqrt{3}\right)}$$

$$f(1) = -\frac{1}{5\sqrt{3}} \tan^{-1}\frac{\sqrt{3}}{5} + \frac{\pi}{2\left(5\sqrt{3}\right)}$$

$$= \frac{1}{5\sqrt{3}}\left(\frac{\pi}{2} - \tan^{-1}\frac{\sqrt{3}}{5}\right)$$

$$= \frac{1}{5\sqrt{3}} \cot^{-1}\frac{\sqrt{3}}{5}$$

$$= \frac{1}{5\sqrt{3}} \tan^{-1}\frac{5}{\sqrt{3}}$$

Now, $\alpha = 5,\ \beta = \sqrt{3}$

$\alpha^2 + \beta^2 = 25 + 3 = 28$

32. Let $5f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}+3, x>0$. Then $18\int_1^2 f(x)dx$ is equal to : **[JEE(Main)April,2023]**

माना $5f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}+3, x>0$ है। तब $18\int_1^2 f(x)dx$ का मान है : **[JEE(Main)April,2023]**

(A) $5\log_e 2-3$ (B) $5\log_e 2+3$ (C) $10\log_e 2-6$ (D) $10\log_e 2+6$

Ans. C

Sol. $5f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}+3$

$x\to\frac{1}{x}$

$5f\left(\frac{1}{x}\right)+4f(x)=x+3$

$(1)\times 5-(2)\times 4$

$\Rightarrow f(x)=\frac{5}{9x}-\frac{4}{9}x+\frac{1}{3}$

$\Rightarrow 18\int_1^2 f(x)dx=18\left(\frac{5}{9}\ln 2-\frac{4}{9}\times\frac{3}{2}+\frac{1}{3}\right)$

$=10\ln 2-6$

33. Let f(x) be a function satisfying $f(x)+f(\pi-x)=\pi^2, \forall x\in\mathbb{R}$. Then $\int_0^\pi f(x)\sin x dx$ is equal to : **[JEE(Main)April,2023]**

माना एक फलन f(x) के लिए $f(x)+f(\pi-x)=\pi^2, \forall x\in\mathbb{R}$ है। तो $\int_0^\pi f(x)\sin x dx$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{\pi^2}{2}$ (B) π^2 (C) $\frac{\pi^2}{4}$ (D) $2\pi^2$

Ans. B

Sol. $I=\int_0^\pi f(x)\sin x dx$

$I=\int_0^\pi f(\pi-x)\sin x dx$

$2I=\int_0^\pi \sin x(f(x)+f(\pi-x))dx$

$2I=\pi^2\int_0^\pi \sin x dx$

$2I=2\pi^2\int_0^{\frac{\pi}{2}}\sin x dx$

$I = \pi^2$

34. Let f be a differentiable function such that $x^2 f(x) - x = 4\int_0^x tf(t)dt$, $f(1) = \frac{2}{3}$. Then 18f(3) is equal to :

[JEE(Main)April,2023]

माना एक अवकलनीय फलन f के लिए $x^2 f(x) - x = 4\int_0^x tf(t)dt$, $f(1) = \frac{2}{3}$ हैं। तो 18f(3) बराबर है :

[JEE(Main)April,2023]

(A) 180 (B) 210 (C) 160 (D) 150

Ans. C

Sol. Differentiate the given equation

$\Rightarrow 2xf(x) + x^2 f'(x) - 1 = 4xf(x)$

$\Rightarrow x^2 \frac{dy}{dx} - 2xy = 1$

$\Rightarrow \frac{dy}{dx} + \left(-\frac{2}{x}\right) y = \frac{1}{x^2}$

$I.F. = e^{\int -\frac{2}{x} \ell nx} = \frac{1}{x^2}$

$\therefore y\left(\frac{1}{x^2}\right) = \int \frac{1}{x^4} dx$

$\Rightarrow \frac{y}{x^2} = \frac{-1}{3x^3} + c$

$\Rightarrow y = -\frac{1}{3x^3} + c$

$\Rightarrow y = -\frac{1}{3x} + cx^2$

$\because f(1) = \frac{2}{3} = -\frac{1}{3} + c \Rightarrow c = 1$

$f(x) = -\frac{1}{3x} + x^2$

$18f(3) = 160$

35. Let f be a continuous function satisfying $\int_0^{t^2} \left(f(x) + x^2\right) dx = \frac{4}{3} t^3, \forall t > 0$. Then $f\left(\frac{\pi^2}{4}\right)$ is equal to :

[JEE(Main)April,2023]

माना f एक सतत फलन है तथा $\int_0^{t^2} \left(f(x) + x^2\right) dx = \frac{4}{3} t^3, \forall t > 0$ है। तो $f\left(\frac{\pi^2}{4}\right)$ बराबर है :

(A) $\left(1-\pi\left(1+\frac{\pi^3}{16}\right)\right.$ (B) $-\pi\left(1-\frac{\pi^2}{16}\right)$ (C) $\pi^2\left(1-\frac{\pi^3}{16}\right)$ (D) $\pi\left(1-\frac{\pi^3}{16}\right)$

Ans. D

Sol. $\int_0^{t^2}\left(f(x)+x^2\right)dx=\frac{4}{3}t^3, \forall t>0$

$\left(f\left(t^2\right)+t^4\right)=2t$

$f\left(t^2\right)=2t-t^4$

$t=\frac{\pi}{2}\Rightarrow f\left(\frac{\pi^2}{4}\right)=\frac{2\pi}{2}-\frac{\pi^4}{16}$

$=\pi-\frac{\pi^4}{16}=\pi\left(1-\frac{\pi^3}{16}\right)$

36. The value of the integral $\int_{-\log_e 2}^{\log_e 2} e^x\left(\log_e\left(e^x+\sqrt{1+e^{2x}}\right)\right)dx$ is equal to : **[JEE(Main)April,2023]**

निम्न $\int_{-\log_e 2}^{\log_e 2} e^x\left(\log_e\left(e^x+\sqrt{1+e^{2x}}\right)\right)dx$ का मान ज्ञात कीजिए : **[JEE(Main)April,2023]**

(A) $\log_e\left(\frac{2(2+\sqrt{5})}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$ (B) $\log_e\left(\frac{\sqrt{2}(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$

(C) $\log_e\left(\frac{(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)+\frac{\sqrt{5}}{2}$ (D) $\log_e\left(\frac{\sqrt{2}(3-\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)+\frac{\sqrt{5}}{2}$

Ans. B

Sol. $I=\int_{-\log_e 2}^{\log_e 2} e^x\left[\log_e\left(e^x+\sqrt{1+e^{2x}}\right)\right]dx$

Put $e^x=t$

$e^x dx=dt$

$I=\int_{\frac{1}{2}}^{2} 1\times\log_e\left[t+\sqrt{1+t^2}\right]dt$

$=\left[t\ln\left(\sqrt{t^2+1}+t\right)\right]_{\frac{1}{2}}^{2}-\int_{1/2}^{2}\frac{t}{\sqrt{t^2+1}}dt$

$=\left[t\ln\sqrt{t^2+1}+t-\sqrt{t^2+1}\right]_{\frac{1}{2}}^{2}$

$$=\left[2\ln\sqrt{5}+2-\sqrt{5}\right]-\left[\frac{1}{2}\ln\left(\sqrt{\frac{5}{2}}+\frac{1}{2}\right)-\sqrt{\frac{5}{2}}\right]$$

$$=\ln\left(\frac{\sqrt{2}(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$$

37. If $f:\mathbb{R}\to\mathbb{R}$ be a continuous function satisfying $\int_0^{\frac{\pi}{2}} f(\sin 2x)\sin x\,dx+\alpha\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$, then the value of α is : **[JEE(Main)April,2023]**

यदि $f:\mathbb{R}\to\mathbb{R}$ एक सतत फलन है तथा $\int_0^{\frac{\pi}{2}} f(\sin 2x)\sin x\,dx+\alpha\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$ है, तो α का मान है : **[JEE(Main)April,2023]**

(A) $-\sqrt{3}$ (B) $\sqrt{2}$ (C) $-\sqrt{2}$ (D) $\sqrt{3}$

Ans. C

Sol. $I=\int_0^{\frac{\pi}{4}} f(\sin 2x)\sin x\,dx+\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} f(\sin 2x)\sin x\,dx$

$$+\alpha\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$$

Apply king in first part and put $x-\frac{\pi}{4}=t$ in second part.

$$I=\int_0^{\frac{\pi}{4}} f(\cos 2x)\sin\left(\frac{\pi}{4}-x\right)dx+\int_0^{\frac{\pi}{4}} f(\cos 2t)\sin\left(\frac{\pi}{4}+t\right)dt$$

$$+\alpha\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$$

$$I=\int_0^{\frac{\pi}{4}} f(\cos 2x)\left[2\sin\frac{\pi}{4}\cdot\cos x+\alpha\cos x\right]dx=0$$

$$I=(\alpha+\sqrt{2})\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$$

$$\therefore\alpha=-\sqrt{2}$$

38. Let the function $f:[0,2]\to\mathbb{R}$ be defined as $f(x)=\begin{cases}e^{\min\{x^2,x-[x]\}}, & x\in[0,1)\\ e^{[x-\log_e x]}, & x\in[1,2]\end{cases}$ where [t] denotes the greatest integer less than or equal to t. Then the value of the integral $\int_0^2 xf(x)dx$ is : **[JEE(Main)April,2023]**

माना $f:[0,2]\to\mathbb{R}$, $f(x)=\begin{cases}e^{\min\{x^2,x-[x]\},} & x\in[0,1)\\ e^{[x-\log_e x]}, & x\in[1,2]\end{cases}$ द्वारा परिभाषित है, जहाँ [t] का महत्तम पूर्णांक ≤ t है। तो समाकलन $\int_0^2 xf(x)dx$ का मान है : **[JEE(Main)April,2023]**

(A) $2e-1$ (B) $1+\frac{3e}{2}$ (C) $2e-\frac{1}{2}$ (D) $(e-1)\left(e^2+\frac{1}{2}\right)$

Ans. C

Sol. $f(x)=\begin{cases}e^{x^2} & x\in(0,1)\\ e & x\in[1,2)\end{cases}$

Now $\int_0^2 xf(x)dx=\int_0^1 xe^{x^2}dx+\int_1^2 x\cdot edx=2e-\frac{1}{2}$

39. Among

(S1) : $\lim\limits_{n\to\infty}\frac{1}{n^2}(2+4+6+\ldots\ldots+2n)=1$

(S2) : $\lim\limits_{n\to\infty}\frac{1}{n^{16}}\left(1^{15}+2^{15}+3^{15}+\ldots\ldots+n^{15}\right)=\frac{1}{16}$ **[JEE(Main)April,2023]**

(A) Only (S2) is true (B) Both (S1) and (S2) are true

(C) Only (S1) is true (D) Both (S1) and (S2) are false

(S1) : $\lim\limits_{n\to\infty}\frac{1}{n^2}(2+4+6+\ldots\ldots+2n)=1$

(S2) : $\lim\limits_{n\to\infty}\frac{1}{n^{16}}\left(1^{15}+2^{15}+3^{15}+\ldots\ldots+n^{15}\right)=\frac{1}{16}$ में से : **[JEE(Main)April,2023]**

(A) केवल (S2) सत्य है (B) दोनों (S1) तथा (S2) सत्य हैं

(C) केवल (S1) सत्य है (D) दोनों (S1) तथा (S2) असत्य हैं

Ans. B

Sol. $S_1:\lim\limits_{n\to\infty}\frac{n(n+1)}{n^2}=1\Rightarrow$ True

$S_2:\lim\limits_{n\to\infty}n^{\frac{1}{16}}\left(\sum r^{15}\right)=\lim\limits_{n\to\infty}\frac{1}{n}\sum\left(\frac{r}{n}\right)^{15}$

$=\int_0^1 x^{15}dx=\frac{1}{16}\Rightarrow$ True

40. $\int_0^\infty\frac{6}{e^{3x}+6e^{2x}+11e^x+6}dx=$ **[JEE(Main)April,2023]**

(A) $\log_e\left(\frac{256}{81}\right)$ (B) $\log_e\left(\frac{512}{81}\right)$ (C) $\log_e\left(\frac{32}{27}\right)$ (D) $\log_e\left(\frac{64}{27}\right)$

Ans. C

Sol. Let $e^x = t \Rightarrow e^x dx = dt$

$$I = \int_1^\infty \frac{6}{t(t+1)(t+2)(t+3)}dt$$

$$\Rightarrow I = 6\int_0^\infty \left\{\frac{A}{t} + \frac{B}{t+1} + \frac{C}{t+2} + \frac{D}{t+3}\right\}dt$$

$$\Rightarrow I = 6\int_1^\infty \left(\frac{1}{6t} + \frac{1}{(-2)(t+1)} + \frac{1}{2(t+2)} - \frac{1}{6(t+3)}\right)dt$$

$$\Rightarrow I = \left[\ell nt - 3\ell n(t+1) + 3\ell n(t+2) - \ell n(t+3)\right]_1^\infty$$

$$\Rightarrow I = \left[\ell n \frac{t(t+2)^3}{(t+1)^3(t+3)}\right]_1^\infty$$

$$\Rightarrow I = 0 - \ell n\frac{3^3}{2^3.4} = \ell n\frac{32}{27}$$

41. The value of $\frac{e^{-\frac{\pi}{4}+\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx}}{\int_0^{\frac{\pi}{4}} e^{-x}\left(\tan^{49}x + \tan^{51}x\right)dx}$ is : **[JEE(Main)April,2023]**

$\frac{e^{-\frac{\pi}{4}+\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx}}{\int_0^{\frac{\pi}{4}} e^{-x}\left(\tan^{49}x + \tan^{51}x\right)dx}$ का मान है : **[JEE(Main)April,2023]**

(A) 51 (B) 49 (C) 25 (D) 50

Ans. D

Sol. Let $I_1 = e^{\frac{-\pi}{4}} + \int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx$

$$I_2 = \int_0^{\frac{\pi}{4}} e^{-x}\left(\tan^{49} x + \tan^{51} x\right)dx$$

$$= \int_0^{\frac{\pi}{4}} e^{-x} \tan^{49} x\left(\sec^2 x\right)dx$$

$$= \left|e^{-x}\frac{\tan^{50} x}{50}\right|_0^{\frac{\pi}{4}} + \frac{1}{50}\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50} x dx$$

$$= \frac{e^{-\frac{\pi}{4}}}{50} + \frac{1}{50}\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50} x dx = \frac{I_1}{50}$$

then $\frac{I_1}{I_2} = 50$

42. If $\int_0^1 \frac{1}{\left(5+2x-2x^2\right)\left(1+e^{(2-4x)}\right)}dx = \frac{1}{\alpha}\log_e\left(\frac{\alpha+1}{\beta}\right)$, $\alpha, \beta, > 0$, then $\alpha^4 - \beta^4$ is equal to :

[JEE(Main)April,2023]

यदि $\int_0^1 \frac{1}{\left(5+2x-2x^2\right)\left(1+e^{(2-4x)}\right)}dx = \frac{1}{\alpha}\log_e\left(\frac{\alpha+1}{\beta}\right)$, $\alpha, \beta, > 0$ हो, तो $\alpha^4 - \beta^4$ का मान होगा :

[JEE(Main)April,2023]

(A) 0 (B) 21 (C) –21 (D) 19

Ans. B

Sol. $$I = \int_0^1 \frac{dx}{\left(5+2x-2x^2\right)\left(1+e^{2-4x}\right)} \quad ...(i)$$

$x \to 1 - x$

$$I = \int_0^1 \frac{e^{2-4x}dx}{\left(5+2x-2x^2\right)\left(1+e^{2-4x}\right)} \quad ...(ii)$$

Add (i) and (ii)

$$2I = \int_0^1 \frac{dx}{5+2x-2x^2} = \int_0^1 \frac{dx}{2\left(\frac{11}{4} - \left(x-\frac{1}{2}\right)^2\right)}$$

$$I=\frac{1}{\sqrt{11}}\ln\left(\frac{\sqrt{11}+1}{\sqrt{10}}\right) \quad \begin{matrix}\alpha=\sqrt{11}\\ \beta=\sqrt{10}\end{matrix}$$

$$\alpha^4-\beta^4=121-100=21$$

43. Let $f(x)=\dfrac{x}{\left(1+x^n\right)^{\frac{1}{n}}}, x\in\mathbb{R}-\{-1\}, n\in\mathbb{N}, n>2.$

If $f^n(x)=\left(\text{fofof.....upto n times}\right)(x)$, then $\lim\limits_{n\to\infty}\int_0^1 x^{n-2}\left(f^n(x)\right)dx$ is equal to _____.

[JEE(Main)April,2023]

माना $f(x)=\dfrac{x}{\left(1+x^n\right)^{\frac{1}{n}}}, x\in\mathbb{R}-\{-1\}, n\in\mathbb{N}, n>2$ है।

; fn $f^n(x)=\left(\text{fofof.....upto n times}\right)(x)$ है तो $\lim\limits_{n\to\infty}\int_0^1 x^{n-2}\left(f^n(x)\right)dx$ बराबर है _____

[JEE(Main)April,2023]

Ans. 0

Sol. $\text{Let } f(x)=\dfrac{x}{1+x^n\frac{1}{n}}, x\in\mathbb{R}-\{-1\}, n\in\mathbb{N}, n>2.$

If $f^n(x)=n\ (\text{fofof....upto n times})\ (x)$

then $\lim\limits_{n\to\infty}\int_0^1 x^{n-2}\left(f^n(x)\right)dx$

$$f(f(x))=\frac{x}{\left(1+2x^n\right)^{1/n}}$$

$$f(f(f(x)))=\frac{x}{\left(1+3x^n\right)^{1/n}}$$

$$\text{Similarly } f^n(x)=\frac{x}{\left(1+n\cdot x^n\right)^{1/n}}$$

$$\text{Now } \lim_{n\to\infty}\int\frac{x^{n-2}\cdot x dx}{\left(1+n\cdot x^n\right)^{1/n}}=\lim_{n\to\infty}\int\frac{x^{n-1}\cdot dx}{\left(1+n\cdot x^n\right)^{1/n}}$$

$$\text{Now } 1+nx^n=t$$

$$n^2\cdot x^{n-1}dx=dt$$

$$x^{n-1}dx=\frac{dt}{n^2}$$

$$\Rightarrow \lim_{n\to\infty}\frac{1}{n^2}\int_1^{1+n}\frac{dt}{t^{1/n}}$$

$$\Rightarrow \lim_{n\to\infty}\frac{1}{n^2}\left[\frac{t^{1-\frac{1}{n}}}{1-\frac{1}{n}}\right]_1^{1+n}$$

$$\Rightarrow \lim_{n\to\infty}\frac{1}{n(n-1)}\left((1+n)^{\frac{n-1}{n}}-1\right) \text{ Now let } n=\frac{1}{h}$$

$$\Rightarrow \lim_{h\to 0}\frac{\left(1+\frac{1}{h}\right)^{1-h}-1}{\frac{1}{h}\frac{(1-h)}{h}}$$

Using series expansion

$\Rightarrow 0$

44. Let [t] denote the greatest integer $\le t$. Then $\frac{2}{\pi}\int_{\pi/6}^{5\pi/6}(8[\text{cosec}\,x]-5[\cot x])dx$ is equal to _____.

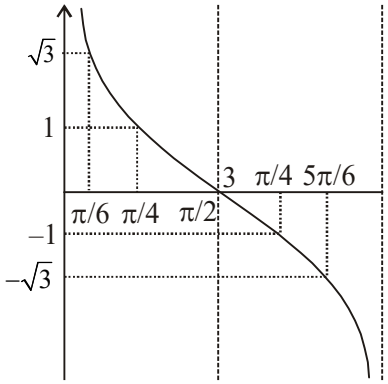
[JEE(Main)April,2023]

माना [t] महत्तम पूर्णांक $\le t$ है। तो $\frac{2}{\pi}\int_{\pi/6}^{5\pi/6}(8[\text{cosec}\,x]-5[\cot x])dx$ बराबर है _____

[JEE(Main)April,2023]

Ans. 14

Sol.



$$I=\frac{2}{\pi}\int_{\pi/6}^{5\pi/6}\left(8[\text{cosec}\,x]-5[\cot x]\right)dx$$

For $x\in(\pi/6,5\pi/6)$

$\text{cosec}\,x\in[1,2)$

$[\text{cosec}\,x]=1$

$$I=\frac{2}{\pi}\left[\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}}8(1)dx-\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}5(1)dx-\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}5(0)dx-\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}}5(-1)dx-\int_{\frac{3\pi}{4}}^{\frac{5\pi}{6}}5(-2)dx\right]=14$$

45. Let [t] denote the greatest integer function. If $\int_0^{2.4}\left[x^2\right]dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}+\delta\sqrt{5}$, then $\alpha+\beta+\gamma+\delta$ is equal to _____. **[JEE(Main)April,2023]**

माना [t] महत्तम पूर्णांक फलन है। यदि $\int_0^{2.4}\left[x^2\right]dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}+\delta\sqrt{5}$ है, तो $\alpha+\beta+\gamma+\delta$ बराबर है : **[JEE(Main)April,2023]**

Ans. 6

Sol. $\int_0^{2.4}\left[x^2\right]dx=\int_0^1 0dx+\int_1^{\sqrt{2}}1dx+\int_{\sqrt{2}}^{\sqrt{3}}2dx$

$+\quad\int_{\sqrt{3}}^{2}3dx+\int_2^{\sqrt{5}}4dx+\int_{\sqrt{5}}^{2.4}5dx$

$=0+(\sqrt{2}-1)+2(\sqrt{3}-\sqrt{2})+3(2-\sqrt{3})+4(\sqrt{5}-2)+5\left(2.4-\sqrt{5}\right)$

$=9-\sqrt{2}-\sqrt{3}-\sqrt{5}$

$\alpha=9,\beta=-1,\gamma=-1,\delta=-1$

46. For m, n > 0, let $\alpha(m,n)=\int_0^2 t^m(1+3t)^n\,dt$. If $11\alpha(10,6)+18\alpha(11,5)=p(14)^6$, then p is equal to _____. **[JEE(Main)April,2023]**

m, n > 0 के लिए माना $\alpha(m,n)=\int_0^2 t^m(1+3t)^n\,dt$ है। यदि $11\alpha(10,6)+18\alpha(11,5)=p(14)^6$ तो p बराबर _____ है। **[JEE(Main)April,2023]**

Ans. 32

Sol. $\alpha(m,n)=\int_0^2 t^m(1+3t)^n dt$

If $11\alpha(10,6)+18\alpha(11,5)=p(14)^6$ then P

$=11\int_0^2\frac{t^{10}}{II}\frac{(1+3t)^6}{I}+10\int_0^2 t^{11}(1+3t)^5dt$

$=11\left[(1+3t)^6\cdot\frac{t^{11}}{11}-\int 6(1+3t)^5\cdot 3\frac{t^{11}}{11}\right]_0^2+18\int_0^2 t^{11}(1+3t)^5dt$

$=\left(t^{11}(1+3t)^6\right)_0^2$

$=2^{11}(7)^6$

$=2^5(14)^6$

$= 32(14)^6$

47. If $\int_{-0.15}^{0.15} \left|100x^2 - 1\right| dx = \frac{k}{3000}$, then k is equal to _____. **[JEE(Main)April,2023]**

यदि $\int_{-0.15}^{0.15} \left|100x^2 - 1\right| dx = \frac{k}{3000}$ है, तो k बराबर है _____ **[JEE(Main)April,2023]**

Ans. 575

Sol. $\int_{-0.15}^{0.15} \left|100x^2 - 1\right| dx = 2\int_{0}^{0.15} \left|100x^2 - 1\right| dx$

$= 2\int_{0}^{0.1} \left(-100x^2 + 1\right) dx + 2\int_{0.1}^{0.15} \left(100x^2 - 1\right) dx$

$= 2 \cdot \left(\frac{-100x^3}{3} + x\right)_0^{0.1} + 2\left(\frac{100x^3}{3} - x\right)_{0.1}^{0.15}$

$= 4\left(-\frac{1}{30} + \frac{1}{10}\right) + 2\left(\frac{9}{80} - \frac{3}{20}\right)$

$= \frac{8}{30} - \frac{3}{40} = \frac{575}{3000} \Rightarrow k = 575$

48. Let for $x \in \mathbb{R}, S_0(x) = x, S_k(x) = C_k x + k\int_0^x S_{k-1}(t)dt$, where $C_0 = 1, C_k = 1 - \int_0^1 S_{k-1}(x)dx, k = 1, 2, 3, \ldots$ Then $S_2(3) + 6C_3$ is equal to _____. **[JEE(Main)April,2023]**

माना $x \in \mathbb{R}, S_0(x) = x, S_k(x) = C_k x + k\int_0^x S_{k-1}(t)dt$ है, जहाँ $C_0 = 1, C_k = 1 - \int_0^1 S_{k-1}(x)dx, k = 1, 2, 3, \ldots$ हैं। तो $S_2(3) + 6C_3$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 18

Sol. $C_0 = 1$

$C_1 = 1 - \int_0^1 S_0(x)dx = 1 - \int_0^1 x dx = 1 - \frac{1}{2} = \frac{1}{2}$

$S_1(x) = C_1 x + \int_0^x t dt$

$= \frac{1}{2}x + \frac{x^2}{2} = \frac{x + x^2}{2}$

$C_2 = 1 - \int_0^1 \left(\frac{x + x^2}{2}\right) dx = 1 - \frac{1}{2}\left(\frac{1}{2} + \frac{1}{3}\right) = 1 - \frac{5}{12} = \frac{7}{12}$

$$S_2(x) = C_2 x + 2\int_0^x \left(\frac{t+t^2}{2}\right)dt = \frac{7}{12}x + \left(\frac{x^2}{2} + \frac{x^3}{3}\right) = \frac{7x+6x^2+4x^3}{12}$$

$$S_2(3) = \frac{21+54+108}{12} = \frac{183}{12} = \frac{61}{4}$$

$$C_3 = 1 - \int_0^1 \left(\frac{7x+6x^2+4x^3}{12}\right)dx$$

$$= 1 - \frac{1}{12}\left[\frac{7}{2} + \frac{6}{3} + \frac{4}{4}\right]$$

$$= 1 - \frac{1}{12}\left(\frac{7}{2} + 3\right) = \frac{24-7-6}{24} = \frac{11}{24}$$

Now, $S_2(3) + 6C_3 = \frac{61}{4} + \frac{11}{4} = \frac{72}{4} = 18$

49. Let $f_n = \int_0^{\frac{\pi}{2}} \left(\sum_{k=1}^{n} \sin^{k-1} x\right)\left(\sum_{k=1}^{n} (2k-1)\sin^{k-1} x\right)\cos x dx, n \in \mathbb{N}$. Then $f_{21} - f_{20}$ is equal to _____.

[JEE(Main)April,2023]

माना $f_n = \int_0^{\frac{\pi}{2}} \left(\sum_{k=1}^{n} \sin^{k-1} x\right)\left(\sum_{k=1}^{n} (2k-1)\sin^{k-1} x\right)\cos x dx, n \in \mathbb{N}$ है। तो $f_{21} - f_{20}$ बराबर है _____

[JEE(Main)April,2023]

Ans. 41

Sol. $f_n(x) = \int_0^{\frac{\pi}{2}} \left(1 + \sin x + \sin^2 x + \sin^3 x + ... + \sin^{n-1}(x)\right)\left(1 + 3\sin x + 5\sin^2 x + .. + (2n-1)\right)\sin^{n-1} x.\cos x dx$

Multiply & divide by $\sqrt{\sin x}$

$$\int_0^{\frac{\pi}{2}} \left((\sin x)^{\frac{1}{2}} + (\sin x)^{\frac{3}{2}} + (\sin x)^{\frac{5}{2}} + (\sin x)^{\frac{7}{2}} + ...(\sin x)^{\frac{2n-1}{2}}\right)$$

$$\left(1 + 3\sin x + 5\sin^2 x + ... + (2n-1)\sin^{n-1}(x)\right)\frac{\cos x}{\sqrt{\sin x}} dx$$

Put $(\sin x)^{1/2} + (\sin x)^{3/2} + (\sin x)^{5/2} + ... (\sin x)^{n-1/2} = t$

$$\frac{1}{2}\frac{\left(1 + 3\sin x + 5\sin^2 x + ...(2n-1)\sin^{n-1} x\right)}{\sqrt{\sin x}}\cos x dx = dt$$

$$f_n = 2\int_0^n t dt$$

$f_n = n^2$

$f_{21} - f_{20} = (21)^2 - (20)^2$

$= 441 - 400$

$= 41$

50. The integral $\int \frac{\left(x^8 - x^2\right)dx}{\left(x^{12} + 3x^6 + 1\right)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$ is equal to : **[JEE(Main)Jan.2024]**

समाकलन $\int \frac{\left(x^8 - x^2\right)dx}{\left(x^{12} + 3x^6 + 1\right)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$ बराबर है : **[JEE(Main)Jan.2024]**

(A) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/3} + C$ (B) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/2} + C$

(C) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right) + C$ (D) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^3 + C$

Ans. A

Sol. $\int \frac{\left(x^8 - x^2\right)dx}{\left(x^{12} + 3x^6 + 1\right)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$

$\tan^{-1}\left(x^3 + \frac{1}{x^3}\right) = t$

$\frac{\left(3x^2 - \frac{3}{x^4}\right)dx}{1 + \left(x^3 + \frac{1}{x^3}\right)^2} = dt$

$\frac{3\left(x^6 - 1\right)dx}{x^4\left(1 + x^6 + \frac{1}{x^6} + 2\right)} = dt$

$\frac{3\left(x^8 - x^2\right)dx}{\left(x^{12} + 3x^6 + 1\right)} = dt$

$= \frac{1}{3}\int \frac{dt}{t}$

$$= \frac{1}{3}\ell n\left|t\right| + c$$

$$\log_e \left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|^{1/3} + c$$

51. For $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, if $y(x) = \int \frac{\cos ec\, x + \sin x}{\cos ecx \sec x + \tan x \sin^2 x} dx$, and $\lim\limits_{x \to \left(\frac{\pi}{2}\right)^-} y(x) = 0$ then $y\left(\frac{\pi}{4}\right)$ is equal to :

[JEE(Main)Jan.2024]

$x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ के लिए यदि $y(x) = \int \frac{\cos ec\, x + \sin x}{\cos ecx \sec x + \tan x \sin^2 x} dx$ है तथा $\lim\limits_{x \to \left(\frac{\pi}{2}\right)^-} y(x) = 0$ है, तो $y\left(\frac{\pi}{4}\right)$ बराबर है :

[JEE(Main)Jan.2024]

(A) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\frac{1}{\sqrt{2}}\tan^{-1}\left(-\frac{1}{2}\right)$ (C) $-\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (D) $\frac{1}{2}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

Ans. B

Sol. $$\int \left(\frac{1 + \sin^2 x}{\frac{1}{\cos x} + \frac{\sin^4 x}{\cos x}}\right) dx$$

$$= \int \frac{\left(1 + \sin^2 x\right)\cos x}{1 + \sin^4 x} dx$$

$\sin x = t, \qquad \cos x\, dx = dt$

$$\int \frac{1 + t^2}{1 + t^4} dt = \int \left(\frac{1 + \frac{1}{t^2}}{t^2 + \frac{1}{t^2}}\right) dt$$

$$= \int \frac{du}{u^2 + 2} \qquad t - \frac{1}{t} = u$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\frac{u}{\sqrt{2}} + c \qquad \left(1 + \frac{1}{t^2}\right)dt = dx$$

$$f(x) = \frac{1}{\sqrt{2}}\tan^{-1}\frac{\left(\sin x - \frac{1}{\sin x}\right)}{\sqrt{2}} + c$$

$$f\left(\frac{\pi}{2}\right)=0\Rightarrow c=0$$

$$f\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\frac{1}{\sqrt{2}}-\sqrt{2}}{\sqrt{2}}\right)$$

$$f\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}\tan^{-1}\left(-\frac{1}{2}\right)$$

52. If $\int\frac{\sin^{\frac{3}{2}}x+\cos^{\frac{3}{2}}x}{\sqrt{\sin^3 x\cos^3 x\sin(x-\theta)}}dx=A\sqrt{\cos\theta\tan x-\sin\theta}+B\sqrt{\cos\theta-\sin\theta\cot x}+C$, where C is the integration constant, then AB is equal to : **[JEE(Main)Jan.2024]**

यदि $\int\frac{\sin^{\frac{3}{2}}x+\cos^{\frac{3}{2}}x}{\sqrt{\sin^3 x\cos^3 x\sin(x-\theta)}}dx=A\sqrt{\cos\theta\tan x-\sin\theta}+B\sqrt{\cos\theta-\sin\theta\cot x}+C$ है, जहां C समाकलन अचर

g$r ksAB बराबर है : **[JEE(Main)Jan.2024]**

(A) 4 sec θ (B) 4 cosec (2θ) (X) 2 σ χ θ (D) 8 cosec (2θ)

Ans. D

Sol. $I=\int\frac{\left(\sin^{\frac{3}{2}}x+\cos^{\frac{3}{2}}x\right)dx}{\sqrt{\sin^3 x\cos^3 x\sin(x-\theta)}}$

$$I=\int\frac{\sec^{\frac{3}{2}}x dx}{\sqrt{\sin(x-\theta)}}+\int\frac{\text{cosec}^{\frac{3}{2}}x dx}{\sqrt{\sin(x-\theta)}}$$

$$I=\int\frac{\sec^2 x dx}{\sqrt{\sec x\sin(x-\theta)}}+\int\frac{\text{cosec}^2 x dx}{\sqrt{\text{cosec}\, x\sin(x-\theta)}}$$

$$I=\int\frac{\sec^2 x dx}{\sqrt{\tan x\cos\theta-\sin\theta}}+\int\frac{\text{cosec}^2 x dx}{\sqrt{\cos\theta-\cot x\sin\theta}}=I_1+I_2$$

$$I_1=\int\frac{\sec^2 x dx}{\sqrt{\tan x\cos\theta-\sin\theta}}$$

Let $\tan x\cos\theta-\sin\theta=t^2$

$$\Rightarrow \sec^2 x dx = \frac{2tdt}{\cos\theta}$$

$$I_1 = \int \frac{\frac{2tdt}{\cos\theta}}{t} = \frac{2t}{\cos\theta} + C_1$$

$$= \frac{2}{\cos\theta}\sqrt{\tan x \cot\theta - \sin\theta} + C_1$$

$$I_2 = \frac{2}{\sin\theta}\sqrt{\cos\theta - \sin\theta \cot x} + C_2$$

$$\Rightarrow I_1 + I_2 \Rightarrow B = \frac{2}{\sin\theta}, A = \frac{2}{\cos\theta}$$

$$\Rightarrow AB = \frac{4}{\sin\theta\cos\theta} = 8\,\text{cosec}\,2\theta$$

53. If $\int_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}}dx = a + b\sqrt{2} + c\sqrt{3}$, where a, b, c are rational numbers, then 2a + 3b – 4c is equal to :

[JEE(Main)Jan.2024]

यदि $\int_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}}dx = a + b\sqrt{2} + c\sqrt{3}$ है, जहाँ a, b, c परिमेय संख्याएँ हैं, तो 2a + 3b – 4c बराबर है :

[JEE(Main)Jan.2024]

(A) 10 (B) 7 (C) 4 (D) 8

Ans. D

Sol. $\int_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}}dx = a + b\sqrt{x} + c\sqrt{3}$

Rationalize

$$\int_0^1 \left(\frac{\sqrt{3+x}-\sqrt{1+x}}{2}\right)dx$$

$$= \frac{1}{2}\left[\frac{2}{3}(3+x)^{\frac{3}{2}} - \frac{2}{3}(1+x)^{3/2}\right]_0^1$$

$$= \frac{1}{3}[(8-2\sqrt{2}) - (3\sqrt{3}-1)]$$

$$= \frac{1}{3}\left[8 - 2\sqrt{2} - 3\sqrt{2} + 1\right]$$

$$= \frac{1}{3}\left[9 - 2\sqrt{2} - 3\sqrt{3}\right]$$

$= 3 - \frac{2}{3}\sqrt{2} - \sqrt{3}$

$= a + b\sqrt{2} + c\sqrt{3}$

$a = 3 \left| b = -\frac{2}{3} \right| c = -1$

$2(3) + 3\left(-\frac{2}{3}\right) - 4(-1)$

$\Rightarrow 6 - 2 + 4$

$\Rightarrow 8$

54. If (a, b) be the orthocentre of the triangle whose vertices are (1,2), (2,3) and (3,1), and $I_1 = \int_a^b x \sin\left(4x - x^2\right) dx$, $I_2 = \int_a^b \sin\left(4x - x^2\right) dx$, then $36\frac{I_1}{I_2}$ is equal to : **[JEE(Main)Jan.2024]**

यदि शीर्षों (1,2), (2,3) तथा (3,1) के त्रिभुज का लंबकेन्द्र (a, b) है, तथा $I_1 = \int_a^b x \sin\left(4x - x^2\right) dx$, $I_2 = \int_a^b \sin\left(4x - x^2\right) dx$ है, तो $36\frac{I_1}{I_2}$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 72 (B) 88 (C) 66 (D) 80

Ans. A

Sol. Slope of AB = 1.

Altitude through C

$y - 1 = -1(x - 3)$

$\Rightarrow x + y = 4.$

Orthocentre (a, b) lies on this

$\Rightarrow a + b = 4$ _____(i)

Now, $I_1 = \int_a^b x.\sin\left(4x - x^2\right) dx$

$= \int_a^b (a + b - x) \sin\left(4(a + b - x) - (a + b - x)^2\right) dx$

$= \int_a^b (4 - x) \sin\left(4x - x^2\right) dx$ {using (i)}

$\Rightarrow I_1 = 4 \cdot I_2 - I_1$

$$\Rightarrow \frac{I_1}{I_2} = 2$$

$$\Rightarrow 36\frac{I_1}{I_2} = 72$$

55. For $0 < a < 1$, the value of the integral $\int_0^{\pi} \frac{dx}{1-2a\cos x + a^2}$ is : **[JEE(Main)Jan.2024]**

$0 < a < 1$ के लिए, समाकलन $\int_0^{\pi} \frac{dx}{1-2a\cos x + a^2}$ का मान है : **[JEE(Main)Jan.2024]**

(A) $\frac{\pi}{1-a^2}$ (B) $\frac{\pi^2}{\pi+a^2}$ (C) $\frac{\pi^2}{\pi-a^2}$ (D) $\frac{\pi}{1+a^2}$

Ans. A

Sol. $I = \int_0^{\pi} \frac{dx}{1-2a\cos x + a^2}; 0 < a < 1$

$$I = \int_0^{\pi} \frac{dx}{1+2a\cos x + a^2}$$

$$2I = 2\int_0^{\pi/2} \frac{2\left(1+a^2\right)}{\left(1+a^2\right)^2 - 4a^2\cos^2 x}dx$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{2\left(1+a^2\right)\cdot\sec^2 x}{\left(1+a^2\right)^2\cdot\sec^2 x - 4a^2}dx$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{2\cdot\left(1+a^2\right)\cdot\sec^2 x}{\left(1+a^2\right)^2\cdot\tan^2 x + \left(1-a^2\right)^2}dx$$

$$\Rightarrow I = \int_0^{\pi/2} \frac{\frac{2\cdot\sec^2 x}{1+a^2}\cdot dx}{\tan^2 x + \left(\frac{1-a^2}{1+a^2}\right)^2}$$

$$\Rightarrow I = \frac{2}{\left(1-a^2\right)}\left[\frac{\pi}{2} - 0\right]$$

$$I = \frac{\pi}{1-a^2}$$

56. If the value of the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2\cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right)dx=\frac{\pi}{4}(\pi+a)-2$, then the value of a is :

[JEE(Main)Jan.2024]

यदि समाकलन $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2\cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right)dx=\frac{\pi}{4}(\pi+a)-2$ है, तो a का मान है : **[JEE(Main)Jan.2024]**

(A) $\frac{3}{2}$ (B) 3 (C) 2 (D) $-\frac{3}{2}$

Ans. B

Sol. By even odd property

$\int_0^{\frac{\pi}{2}}\left(x^2\cos x+\sin^2 x+1\right)dx$

$\int_0^{\frac{\pi}{2}}\left(x^2\cos x+\frac{3}{2}-\frac{\cos 2x}{2}\right)dx$

$\left(x^2\sin x-\int 2x\sin x dx+\frac{3x}{2}-\frac{\sin 2x}{4}\right)_0^{\frac{\pi}{2}}$

$=\left[x^2\sin x-\left[2x(-\cos x)-\int 2(-\cos x)dx\right]+\frac{3x}{2}-\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}}$

$=\left[x^2\sin x+2x\cos x-2\sin x+\frac{3x}{2}-\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}}$

$=\left[\frac{\pi^2}{4}+0-2+\frac{3\pi}{4}-0\right]-[0]=\frac{\pi^2}{4}+\frac{3\pi}{4}-2$

57. $\lim\limits_{x\to\frac{\pi}{2}}\left(\frac{1}{\left(x-\frac{\pi}{2}\right)^2}\int_{x^3}^{\left(\frac{\pi}{2}\right)^3}\cos\left(t^{\frac{t}{3}}\right)dt\right)$ is equal to : **[JEE(Main)Jan.2024]**

$\lim\limits_{x\to\frac{\pi}{2}}\left(\dfrac{1}{\left(x-\frac{\pi}{2}\right)^2}\int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3}\cos\left(t^{\frac{t}{3}}\right)dt\right)$ का मान है : **[JEE(Main)Jan.2024]**

(A) $\dfrac{3\pi}{4}$ (B) $\dfrac{3\pi^2}{8}$ (C) $\dfrac{3\pi^2}{4}$ (D) $\dfrac{3\pi}{8}$

Ans. B

Sol. $\lim\limits_{x\to\frac{\pi}{2}}\dfrac{\int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3}\cos\left(t^{\frac{1}{3}}\right)}{\left(x-\frac{p}{2}\right)^2}=\dfrac{0}{0}$ form

Apply leibniz

$$\lim_{x\to\frac{\pi}{2}}\frac{\cos\left(\frac{\pi}{2}\right)\times 0-\cos(x)\cdot 3x^2}{2\left(x-\frac{\pi}{2}\right)}$$

$x\to\dfrac{\pi}{2}+t$

$\Rightarrow t\to 0$

$$\lim_{t\to 0}\frac{-\cos\left(\frac{\pi}{2}+t\right)3\left(\frac{\pi}{2}\right)^2}{2t}$$

$$\lim_{t\to 0}\frac{3\pi^2\sin t}{8t}=\frac{3\pi^2}{8}$$

58. The value of $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{n^3}{\left(n^2+k^2\right)\left(n^2+3k^2\right)}$ is : **[JEE(Main)Jan.2024]**

$\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{n^3}{\left(n^2+k^2\right)\left(n^2+3k^2\right)}$ का मान है : **[JEE(Main)Jan.2024]**

(A) $\dfrac{\left(2\sqrt{3}+3\right)\pi}{24}$ (B) $\dfrac{\pi}{8\left(2\sqrt{3}+3\right)}$ (C) $\dfrac{13\left(2\sqrt{3}-3\right)\pi}{8}$ (D) $\dfrac{13\pi}{8\left(4\sqrt{3}+3\right)}$

Ans. D

Sol. $$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\frac{n^3}{n^4\left(1+\frac{k^2}{n^2}\right)\left(1+\frac{3k^2}{n^2}\right)}$$

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\frac{n^3}{\left(1+\frac{k^2}{n^2}\right)\left(1+\frac{3k^2}{n^2}\right)}$$

$$=\int_0^1\frac{dx}{3\left(1+x^2\right)\left(\frac{1}{3}+x^2\right)}$$

$$=\frac{1}{3}\times\frac{3}{2}\int_0^1\frac{\left(x^2-1\right)-\left(x^2+\frac{1}{3}\right)}{\left(1+x^2\right)\left(\frac{1}{3}+x^2\right)}$$

$$=\frac{1}{2}\int_0^1\left[\frac{1}{x^2+\left(\frac{1}{\sqrt{3}}\right)^2}-\frac{1}{1+x^2}\right]dx$$

$$=\frac{1}{2}\left[\sqrt{3}\tan^{-1}\left(\sqrt{3}x\right)\right]_0^1-\frac{1}{2}\left(\tan^{-1}x\right)_0^1$$

$$=\frac{\sqrt{3}}{2}\left(\frac{\pi}{3}\right)-\frac{1}{2}\left(\frac{\pi}{4}\right)$$

$$=\frac{\pi}{2\sqrt{3}}-\frac{\pi}{8}$$

$$=\frac{\pi}{2}\left(\frac{1}{\sqrt{3}}-\frac{1}{4}\right)$$

$$=\frac{\pi}{2}\left(\frac{4-\sqrt{3}}{4\sqrt{3}}\right)\left(\frac{4+\sqrt{3}}{4+\sqrt{3}}\right)$$

$$=\frac{\pi}{2}\left(\frac{16-3}{4\sqrt{3}\left(4+\sqrt{3}\right)}\right)$$

$$= \frac{13\pi}{8\left(4\sqrt{3}+3\right)}$$

59. Let $f:\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\to R$ be a differentiable function such that $f(0)=\frac{1}{2}$. If the $\lim\limits_{x\to 0}\frac{x\int_0^x f(t)dt}{e^{x^2}-1}=\alpha$, then $8\alpha^2$ is equal to : **[JEE(Main)Jan.2024]**

माना $f:\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\to R$ एक अवकलनीय फलन है तथा $f(0)=\frac{1}{2}$ है। यदि $\lim\limits_{x\to 0}\frac{x\int_0^x f(t)dt}{e^{x^2}-1}=\alpha$ है, तो $8\alpha^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 16 (B) 2 (C) 4 (D) 1

Ans. B

Sol. $$\lim_{x\to 0}\frac{\int_0^x f(t)\,dt+xf(x)}{e^{x^2}(2x)}=\lim_{x\to 0}\frac{\int_0^x f(t)\,dt}{e^{x^2}(2x)}+\lim_{x\to 0}\frac{f(x)}{e^{x^2}(2)}$$

$$=\lim_{x\to 0}\frac{f(x)}{\left(e^{x^2}+xe^{x^2}(2x)\right)}+\frac{1}{4}=\frac{1}{2}$$

$$\Rightarrow 8\alpha^2=2$$

60. Let $f: R\to R$ be defined as $f(x)=ae^{2x}+be^x+cx$. If $f(0)=-1, f'(\log_e 2)=21$ and $\int_0^{\log_e 4}(f(x)-cx)dx=\frac{39}{2}$, then the value of $|a+b+c|$ equals : **[JEE(Main)Jan.2024]**

माना Let $f: R\to R$, $f(x)=ae^{2x}+be^x+cx$ द्वारा परिभाषित है। यदि $f(0)=-1, f'(\log_e 2)=21$ तथा $\int_0^{\log_e 4}(f(x)-cx)dx=\frac{39}{2}$ हैं, तो $|a+b+c|$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 12 (B) 16 (C) 8 (D) 10

Ans. C

Sol. $f(x)=ae^{2x}+be^x+cx$ $\qquad f(0)=-1$

$a+b=-1$

$f'(x)=2ae^{2x}+be^x+c$ $\qquad f'(\ln 2)=21$

$8a+2b+c=21$

$$\int_0^{\ell n4}\left(ae^{2x}+be^x\right)dx=\frac{39}{2}$$

$$\left[\frac{ae^{2x}}{2}+be^{x}\right]_{0}^{\ell n4}=\frac{39}{2}\Rightarrow 8a+4b-\frac{a}{2}-b=\frac{39}{2}$$

$15a + 6b = 39$

$15a - 6a - 6 = 39$

$9a = 45 \Rightarrow a = 5$

$b = -6$

$c = 21 - 40 + 12 = -7$

$a + b + c - 8$

$|a + b + c| = 8$

61. Let a and b be real constants such that the function f defined by $f(x)=\begin{cases}x^2+3x+a & ,x\le 1\\ bx+2 & ,x>1\end{cases}$ be differentiable on R. Then, the value of $\int_{-2}^{2} f(x)dx$ equal : **[JEE(Main)Jan.2024]**

माना वास्तविक अचर a तथा b इस प्रकार हैं कि फलन $f(x)=\begin{cases}x^2+3x+a & ,x\le 1\\ bx+2 & ,x>1\end{cases}$, R पर अवकलनीय है। तो $\int_{-2}^{2} f(x)dx$ का eku cjkcj gS: **[JEE(Main)Jan.2024]**

(A) 21 (B) $\frac{15}{6}$ (C) 17 (D) $\frac{19}{6}$

Ans. C

Sol. f(x) is continuous at x = 1

$f(1^-) = f(1) = f(1^+)$

$a + 4 = b + 2 \Rightarrow b = a + 2$

f(x) is differential at x = 1

$f'(1^-) = f'(1^+)$

$5 = b \Rightarrow a = 3$

Now $\int_{-2}^{2} f(x)dx = \int_{-2}^{1}\left(x^2+3x+3\right)dx+\int_{1}^{2}(5x+2)dx$

$$=\left(\frac{x^3}{3}+\frac{3x^2}{2}+3x\right)_{-2}^{1}+\left(\frac{5x^2}{2}+2x\right)_{1}^{2}=\left(\frac{1}{3}(1+8)+\frac{3}{2}(1-4)+9\right)+\frac{5}{2}(4-1)+2$$

$$=\frac{1}{3}+\frac{3}{2}+3-\left(\frac{-8}{3}+6-6\right)+\left(10+4-\frac{5}{2}-2\right)$$

$$=6+\frac{3}{2}+12-\frac{5}{2}=18-1=17$$

62. Let y = f(x) be a thrice differentiable function in (–5, 5). Let the tangents to the curve y = f(x) at (1, f(1)) and (3, f(3)) make angles $\frac{\pi}{6}$ and $\frac{\pi}{4}$, respectively with positive x-axis. If $27\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)dt=\alpha+\beta\sqrt{3}$ where α, β are integers, then the value of $\alpha+\beta$ equals : **[JEE(Main)Jan.2024]**

माना फलन y = f(x) अंतराल (–5, 5) तीन बार अवकलनीय है। माना वक्र y = f(x) के बिन्दुओं (1, f(1)) तथा (3, f(3)) पर स्पर्श रेखाऐं धनात्मक x-अक्ष से $\frac{\pi}{6}$ तथा $\frac{\pi}{4}$ के कोण बनाती हैं। यदि $27\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)dt=\alpha+\beta\sqrt{3}$ हैं, जहाँ α, β पूर्णांक हैं, तो $\alpha+\beta$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 26 (B) –16 (C) –14 (D) 36

Ans. A

Sol. $y=f(x)\Rightarrow\frac{dy}{dx}=f'(x)$

$$\left.\frac{dy}{dx}\right)_{(1,f(1))}=f'(1)=\tan\frac{\pi}{6}=\frac{1}{\sqrt{3}}\Rightarrow f'(1)=\frac{1}{\sqrt{3}}$$

$$\left.\frac{dy}{dx}\right)_{(3,f(3))}=f'(3)=\tan\frac{\pi}{4}=1\Rightarrow f'(3)=1$$

$$27\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)dt=\alpha+\beta\sqrt{3}$$

$$I=\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)dt$$

$f'(t)=z\Rightarrow f''(t)\,dt=dz$

$z=f'(3)=1$

$$z=f'(1)=\frac{1}{\sqrt{3}}$$

$$I=\int_{\frac{1}{\sqrt{3}}}^{1}\left(z^2+1\right)dz=\left(\frac{z^3}{3}+z\right)_{\frac{1}{\sqrt{3}}}^{1}$$

$$=\left(\frac{1}{3}+1\right)-\left(\frac{1}{3}\cdot\frac{1}{3\sqrt{3}}+\frac{1}{\sqrt{3}}\right)$$

$$=\frac{4}{3}-\frac{10}{9\sqrt{3}}=\frac{4}{3}-\frac{10}{27}\sqrt{3}$$

$$\alpha+\beta\sqrt{3}=27\left(\frac{4}{3}-\frac{10}{27}\sqrt{3}\right)=36-10\sqrt{3}$$

$\alpha = 36, \beta = -10$

$\alpha + \beta = 36 - 10 = 26$

63. Let $f: R \to R$ be a function defined by $f(x) = \frac{x}{\left(1+x^4\right)^{1/4}}$ and $g(x) = f\left(f\left(f\left(f(x)\right)\right)\right)$. Then, $18\int_0^{\sqrt{2\sqrt{5}}} x^2 g(x)dx$ equal to : **[JEE(Main)Jan.2024]**

माना फलन $f: R \to R$, $f(x) = \frac{x}{\left(1+x^4\right)^{1/4}}$ द्वारा परिभाषित है तथा $g(x) = f\left(f\left(f\left(f(x)\right)\right)\right)$ है। तो $18\int_0^{\sqrt{2\sqrt{5}}} x^2 g(x)dx$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 42 (B) 33 (C) 36 (D) 39

Ans. D

Sol. $f(x) = \frac{x}{\left(1+x^4\right)^{\frac{1}{4}}}$

$$fof(x) = \frac{f(x)}{\left(1+f(x)^4\right)^{\frac{1}{4}}} = \frac{\frac{x}{\left(1+x^4\right)^{\frac{1}{4}}}}{\left(1+\frac{x^4}{1+x^4}\right)^{\frac{1}{4}}} = \frac{x}{\left(1+2x^4\right)^{\frac{1}{4}}}$$

$$f\left(f\left(f\left(f(x)\right)\right)\right) = \frac{x}{\left(1+4x^4\right)^{\frac{1}{4}}}$$

$$18\int_0^{\sqrt{2\sqrt{5}}} \frac{x^3}{\left(1+4x^4\right)^{\frac{1}{4}}}dx$$

Let $1 + 4x^4 = t^4$

$16x^3\,dx = 4t^3\,dt$

$$\frac{18}{4}\int_1^3 \frac{t^3 dt}{t}$$

$$= \frac{9}{2}\left(\frac{t^3}{3}\right)_1^3$$

$$= \frac{3}{2}[26] = 39$$

64. Let a be the sum of all coefficients in the expansion of $(1 - 2x + 2x^2)^{2023}(3 - 4x^2 + 2x^3)^{2024}$ and

$b = \lim\limits_{x\to 0}\left(\dfrac{\int_0^x \frac{\log(1+t)}{t^{2024}+1}dt}{x^2}\right)$. If the equtions $cx^2 + dx + e = 0$ and $2bx^2 + ax + 4 = 0$ have a common root, where c, d, e $\in$ R, then d : c : e equals : **[JEE(Main)Jan.2024]**

माना $(1 - 2x + 2x^2)^{2023}(3 - 4x^2 + 2x^3)^{2024}$ के प्रसार में सभी पदों के गुणांक का योग a है तथा $b = \lim\limits_{x\to 0}\left(\dfrac{\int_0^x \frac{\log(1+t)}{t^{2024}+1}dt}{x^2}\right)$ है।

यदि समीकरणों $cx^2 + dx + e = 0$ तथा $2bx^2 + ax + 4 = 0$ का एक उभयनिष्ठ मूल हैं, जहाँ c, d, e $\in$ R हैं, तो d : c : e बराबर gS **[JEE(Main)Jan.2024]**

(A) 1 : 1 : 4 (B) 4 : 1 : 4 (C) 2 : 1 : 4 (D) 1 : 2 : 4

Ans. A

Sol. By putting x = 1

$a = (1 - 2 + 2)^{2023}(3 - 4 + 2)^{2024} = (1)(1) = 1$

$$b = \lim_{x\to 0}\frac{\int_0^x \frac{\ell n(1+t)}{t^{2024}+1}dt}{x^2}\ \left(\frac{0}{0}\text{ form}\right)$$

By DL hospital Rule

$$\lim_{x\to 0}\frac{\frac{\ell n(1+x)}{x^{2024}+1}}{2x}$$

$$= \frac{1}{2}\cdot\frac{1}{(0+1)} = \frac{1}{2}$$

Now $cx^2 + dx + e = 0$ and $x^2 + x + 4 = 0$ have a common root.
But D < 0 for second equation. So both roots are common.

$$\therefore \frac{c}{1} = \frac{d}{1} = \frac{e}{1}$$

d : c : e = 1 : 1 : 4

65. Let $f, g : (0,\infty) \to \mathbb{R}$ be two functions defined by $f(x) = \int_{-x}^{x}\left(|t| - t^2\right)e^{-t^2}dt$ and $g(x) = \int_0^{x^2} t^{1/2}e^{-t}dt$. Then the value of $9\left(f\left(\sqrt{\log_e 9}\right) + g\left(\sqrt{\log_e 9}\right)\right)$ is equal to : **[JEE(Main)Jan.2024]**

माना $f(x)=\int_{-x}^{x}\left(|t|-t^2\right)e^{-t^2}dt$ तथा $g(x)=\int_0^{x^2}t^{1/2}e^{-t}dt$ द्वारा परिभाषित दो फलन $f,g:(0,\infty)\to\mathbb{R}$ हैं। तो $9\left(f\left(\sqrt{\log_e 9}\right)+g\left(\sqrt{\log_e 9}\right)\right)$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 9 (B) 6 (C) 10 (D) 8

Ans. D

Sol. $f(x)=\int_{-x}^{x}\left(|t|-t^2\right)e^{-t^2}dt$

$\Rightarrow f'(x)=2\cdot\left(|x|-x^2\right)e^{-x^2}$ .....(1)

$g(x)=\int_0^{x^2}t^{\frac{1}{2}}e^{-t}dt$

$g'(x)=x^{-x^2}(2x)-0$

$f'(x)+g'(x)=2x^{-x^2}-2x^2e^{-x^2}+2x^2e^{-x^2}$

Integrating both sides w.r.t.x

$f(x)+g(x)=\int_0^{\alpha}2xe^{-x^2}dx$

$x^2=t$

$\Rightarrow\int_0^{\sqrt{\alpha}}e^{-t}dt=\left[-e^{-t}\right]_0^{\sqrt{\alpha}}$

$=-e^{\left(\log_e(9)^{-1}\right)+1}$

$\Rightarrow 9(f(x)+g(x))=\left(1-\frac{1}{9}\right)9=8$

66. The value of the integral $\int_0^{\frac{\pi}{4}}\frac{xdx}{\sin^4(2x)+\cos^4(2x)}$ equals : **[JEE(Main)Jan.2024]**

समाकलन $\int_0^{\frac{\pi}{4}}\frac{xdx}{\sin^4(2x)+\cos^4(2x)}$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) $\frac{\sqrt{2}\pi^2}{32}$ (B) $\frac{\sqrt{2}\pi^2}{8}$ (C) $\frac{\sqrt{2}\pi^2}{16}$ (D) $\frac{\sqrt{2}\pi^2}{64}$

Ans. A

Sol. By property P–6

$2I=\int_0^{\frac{\pi}{4}}\left(\frac{x}{\cos^4 2x+\sin^4 2x}+\frac{\frac{\pi}{4}-x}{\sin^4 2x+\cos^4 2x}\right)dx \Rightarrow \frac{\pi}{4}\frac{1}{2}\int_0^{\frac{\pi}{4}}\frac{d\theta}{\sin^4\theta+\cos^4\theta}=\frac{\pi}{8}\int_0^1\frac{1+t^2}{t^4+1}dt,$

$t = \tan\theta$

$$= \frac{\pi}{8}\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{t-\frac{1}{t}}{\sqrt{2}}\right)_0^1 = \frac{\pi^2}{16\sqrt{2}}$$

67. The value of $\int_0^1\left(2x^3-3x^2-x+1\right)^{\frac{1}{3}}dx$ is equal to : **[JEE(Main)Jan.2024]**

$\int_0^1\left(2x^3-3x^2-x+1\right)^{\frac{1}{3}}dx$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 1 (B) 2 (C) 0 (D) –1

Ans. C

Sol. $2x^3-3x^2-x+1=\frac{1}{4}(2x-1)\left((2x-1)^2-5\right)$

$$I=\int_0^1\left(\left(\frac{(2x-1)\left((2x-1)^2-5\right)}{4}\right)\right)^{\frac{1}{3}}dx$$

put $2x-1=t$

$$I=\frac{1}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}}\left(\frac{t\left(t^2-5\right)}{4}\right)^{\frac{1}{3}}dt$$

$= 0$ ($\because$ odd function in t)

68. If $\int_0^{\frac{\pi}{3}}\cos^4x\,dx = a\pi + b\sqrt{3}$, where a and b are rational numbers, then 9a + 8b is equal to : **[JEE(Main)Jan.2024]**

यदि $\int_0^{\frac{\pi}{3}}\cos^4x\,dx = a\pi + b\sqrt{3}$, जहाँ a तथा b परिमेय संख्याएँ है, तो 9a + 8b बराबर होगा : **[JEE(Main)Jan.2024]**

(A) $\frac{3}{2}$ (B) 2 (C) 1 (D) 3

Ans. B

Sol. $\int_0^{\pi/3}\cos^4x dx$

$$=\int_0^{\pi/3}\left(\frac{1+\cos 2x}{2}\right)^2dx$$

$$=\frac{1}{4}\int_0^{\pi/3}\left(1+2\cos 2x+\cos^2 2x\right)dx$$

$$=\frac{1}{4}\left[\int_0^{\pi/3}dx+2\int_0^{\pi/3}\cos 2x dx+\int_0^{\pi/3}\frac{1+\cos 4x}{2}dx\right]$$

$$=\frac{1}{4}\left[\frac{\pi}{3}+(\sin 2x)_0^{\pi/3}+\frac{1}{2}\left(\frac{\pi}{3}\right)+\frac{1}{8}(\sin 4x)_0^{\pi/3}\right]$$

$$=\frac{1}{4}\left[\frac{\pi}{3}+(\sin 2x)_0^{\pi/3}+\frac{1}{2}\left(\frac{\pi}{3}\right)+\frac{1}{8}(\sin 4x)_0^{\pi/3}\right]$$

$$=\frac{1}{4}\left[\frac{\pi}{2}+\frac{\sqrt{3}}{2}+\frac{1}{8}\times\left(-\frac{\sqrt{3}}{2}\right)\right]$$

$$=\frac{\pi}{2}+\frac{7\sqrt{3}}{64}$$

$$\therefore a=\frac{1}{8};b=\frac{7}{64}$$

$$\therefore 9a+8b=\frac{9}{8}+\frac{7}{8}=2$$

69. Let $f(x)=\int_0^x g(t)\log_e\left(\frac{1-t}{1+t}\right)dt$, where g is a continuous odd function.

If $\int_{-\pi/2}^{\pi/2}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)dx=\left(\frac{\pi}{\alpha}\right)^2-\alpha$, then α is equal to _____. **[JEE(Main)Jan.2024]**

माना $f(x)=\int_0^x g(t)\log_e\left(\frac{1-t}{1+t}\right)dt$ है, जहाँ g एक संतत विषम फलन है। यदि $\int_{-\pi/2}^{\pi/2}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)dx=\left(\frac{\pi}{\alpha}\right)^2-\alpha$ है,

तो α बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. B

Sol. $f(x)=\int_0^x g(t)\ell n\left(\frac{1-t}{1+t}\right)dt$

$$f(-x)=\int_0^{-x} g(t)\ell n\left(\frac{1-t}{1+t}\right)dt$$

$$f(-x)=-\int_0^x g(-y)\ell n\left(\frac{1+y}{1-y}\right)dy$$

$$=-\int_0^x g(y)\ell n\left(\frac{1-y}{1+y}\right)dy \text{ (g is odd)}$$

$f(-x) = -f(x) \Rightarrow f$ is also odd

Now,

$$I = \int_{-\pi/2}^{\pi/2} \left(f(x) + \frac{x^2 \cos x}{1+e^x} \right) dx$$

$$I = \int_{-\pi/2}^{\pi/2} \left(f(-x) + \frac{x^2 e^x \cos x}{1+e^x} \right) dx$$

$$2I = \int_{-\pi/2}^{\pi/2} x^2 \cos x dx = 2\int_0^{\pi/2} x^2 \cos x dx$$

$$I = \left(x^2 \sin x \right)_0^{\pi/2} - \int_0^{\pi/2} 2x \sin x dx$$

$$= \frac{\pi^2}{4} - 2\left(-x\cos x + \int \cos x dx \right)_0^{\pi/2}$$

$$= \frac{\pi^2}{4} - 2(0+1) = \frac{\pi^2}{4} - 2 \Rightarrow \left(\frac{\pi}{2} \right)^2 - 2$$

$\therefore$ a = 2

70. Let the slope of the line 45x + 5y + 3 = 0 be $27r_1 + \frac{9r_2}{2}$ for some $r_1, r_2 \in R$. Then

$$\lim_{x\to 3} \left(\int_3^x \frac{8t^2}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x} dt \right)$$ is equal to _____. **[JEE(Main)Jan.2024]**

माना किसी $r_1, r_2 \in R$ के लिए रेखा 45x + 5y + 3 = 0 की प्रवणता $27r_1 + \frac{9r_2}{2}$ है। तो $\lim_{x\to 3} \left(\int_3^x \frac{8t^2}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x} dt \right)$

बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 12

Sol. Slope of 45x + 5y + 3 = 0 is m = – 9

$$\therefore 27r_1 + \frac{9r_2}{2} = -9$$

$$\lim_{x\to 3} \left(\int_3^x \frac{8t^2}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x} dt \right)$$

$$\lim_{x\to 3}\frac{\int_3^x 8t^2dt}{\frac{3r_2x}{2}-r_2x^2-r_1x^3-3x}$$

$$= \lim_{x\to 3}\frac{8x^2}{\frac{3r_2}{2}-2r_2x-3r_1x^2-3} \text{ by L.H. rule}$$

$$=\frac{8\times 9}{\frac{3r_2}{2}-6r_2-27r_1-3}$$

$$=\frac{8\times 9}{-\frac{9r_2}{2}-27r_1-3}$$

$$=\frac{8\times 9}{6}=12$$

71. If $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\sin 2x}dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}$, where α, β and γ are rational numbers, then $3\alpha + 4\beta - \gamma$ is equal to _____. **[JEE(Main)Jan.2024]**

यदि $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\sin 2x}dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}$, जहाँ α, β तथा γ परिमेय संख्याऐं हैं, तो $3\alpha + 4\beta - \gamma$ बराबर है_____. **[JEE(Main)Jan.2024]**

Ans. 6

Sol. $=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\sin 2x}\,dx$

$$=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}|\sin x-\cos x|\,dx$$

$$=\int_{\frac{\pi}{6}}^{\frac{\pi}{4}}(\cos x-\sin x)dx+\int_{\frac{\pi}{4}}^{\frac{\pi}{3}}(\sin x-\cos x)dx$$

$$=-1+2\sqrt{2}-\sqrt{3}$$

$$=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}$$

$\alpha = -1$, $\beta = 2$, $\gamma = -1$

$3\alpha + 4\beta - \gamma = 6$

72. The value of $9\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]dx$, where [t] denotes the greatest integer less than or equal to t, is _____.

[JEE(Main)Jan.2024]

$9\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]dx$, जहाँ [t] महत्तम पूर्णांक $\le t$ है, का मान है _____। **[JEE(Main)Jan.2024]**

Ans. 155

Sol. $9\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]dx$

Let $I=\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]dx$

$\frac{10x}{x+1}=1\Rightarrow x=\frac{1}{9}$

$\frac{10x}{x+1}=4 \qquad \Rightarrow x=\frac{2}{3}$

$\frac{10x}{x+1}=9 \qquad \Rightarrow x=9$

$I=\int_0^{\frac{1}{9}}0dx+\int_{\frac{1}{9}}^{\frac{2}{3}}1dx+\int_{\frac{2}{3}}^{9}2dx$

$=0+\left[x\right]_{\frac{1}{9}}^{\frac{2}{3}}+\left[2x\right]_{\frac{2}{3}}^{9}$

$I=\frac{2}{3}-\frac{1}{9}+18-\frac{4}{3}$

$\Rightarrow I=\frac{155}{9}$

$\Rightarrow 9I=155$

73. Let $f:R\to R$ be a function defined by $f(x)=\frac{4^x}{4^x+2}$ and $M=\int_{f(a)}^{f(1-a)}x\sin^4\left(x(1-x)\right)dx$,

$N=\int_{f(a)}^{f(1-a)}\sin^4\left(x(1-x)\right)dx$; $a\neq\frac{1}{2}$. If $\alpha M=\beta N$, $\alpha,\beta\in N$, then the least value of $\alpha^2+\beta^2$ is equal to _____.

[JEE(Main)Jan.2024]

माना फलन $f: R \to R$, $f(x)=\frac{4^x}{4^x+2}$ द्वारा परिभाषित है तथा $M=\int\limits_{f(a)}^{f(1-a)} x\sin^4\left(x(1-x)\right)dx$,

$N=\int\limits_{f(a)}^{f(1-a)} \sin^4\left(x(1-x)\right)dx$; $a \neq \frac{1}{2}$ हैं। यदि $\alpha M = \beta N$, $\alpha, \beta \in N$ हैं, तो $\alpha^2+\beta^2$ का न्यूनतम मान है _____.

[JEE(Main)Jan.2024]

Ans. 5

Sol. $f(x)+f(1-x)=\frac{4^x}{4^x+2}+\frac{4^{1-x}}{4^{1-x}}$

$=\frac{4^x}{4^x+2}+\frac{4}{4+2\cdot 4^x}$

$=\frac{4^x+2}{4^x+2}=1$

$\Rightarrow M=\int\limits_{f(a)}^{f(1-a)} x\sin^4\left(x(1-x)\right)dx$

$x \to f(a)+f(1-a)-x=1-x$

$M=\int\limits_{f(a)}^{f(1-a)} (1-x)\sin^4\left((1-x)x\right)dx$

$M=N-M \Rightarrow \frac{M}{N}=\frac{1}{2}=\frac{\beta}{\alpha}$

$\Rightarrow \left(\alpha^2+\beta^2\right)_{least}=1+4=5$

74. If the integral $525\int\limits_0^{\frac{\pi}{2}} \sin 2x\cos^{\frac{11}{2}}x\left(1+\cos^{\frac{5}{2}}x\right)^{\frac{1}{2}}dx$ is equal to $\left(n\sqrt{2}-64\right)$, then n is equal to _____.

[JEE(Main)Jan.2024]

यदि समाकलन $525\int\limits_0^{\frac{\pi}{2}} \sin 2x\cos^{\frac{11}{2}}x\left(1+\cos^{\frac{5}{2}}x\right)^{\frac{1}{2}}dx=\left(n\sqrt{2}-64\right)$ है, तो n बराबर है_____.

[JEE(Main)Jan.2024]

Ans. 176

Sol. $I = 525\int \sin 2x \cos^{\frac{11}{2}} x\left(1+\cos^{\frac{5}{2}} x\right)^{\frac{1}{2}} dx$

put $\sqrt{\cos x} = t \Rightarrow \cos x = t^2$

$-\sin x\, dx = 2t dt$

$\sin x\, dx = -2t\, dt$

$$I = 525\int_0^1 2(2t)t^2 t^{11}\sqrt{1+t^5}\,dt$$

$$= 525\times 4\int_0^1 \sqrt{1+t^5}\cdot t^{14}dt$$

$$= 2100\int_0^1 \sqrt{1+t^5}\cdot t^{14}dt$$

again put $\sqrt{1+t^5} = u \quad \Rightarrow 1+t^5 = u^2$

$5t^4dt = 2udu$

$$I = 2100\int_1^{\sqrt{2}} u\cdot\frac{2u}{5}\cdot\left(u^2-1\right)^2 du$$

$$= 420\times 2\left(\int_1^{\sqrt{2}} u^2\left(u^4-2u^2+1\right)du\right)$$

$$= 840\int_1^{\sqrt{2}}\left(u^6-2u^4+u^2\right)du$$

$$= 840\left[\frac{u^7}{7}-\frac{2u^5}{5}+\frac{u^3}{3}\right]_1^{\sqrt{2}}$$

$$= 840\left[\frac{8\sqrt{2}}{7}-\frac{8\sqrt{2}}{5}+\frac{2\sqrt{2}}{3}-\frac{1}{7}+\frac{2}{5}-\frac{1}{3}\right]$$

$= 176\sqrt{2}-64$

Compare with $n\sqrt{2}-64$

$n = 176$

75. $\left|\frac{120}{\pi^3}\int_0^{\pi}\frac{x^2\sin x\cos x}{\sin^4 x+\cos^4 x}dx\right|$ is equal to _____. **[JEE(Main)Jan.2024]**

$\left|\frac{120}{\pi^3}\int_0^{\pi}\frac{x^2\sin x\cos x}{\sin^4 x+\cos^4 x}dx\right|$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 15

Sol. $I=\int_0^{\pi}\frac{x^2\sin x\cos x}{\sin^4 x+\cos^4 x}dx$

$2I=\int_0^{\pi}\frac{\sin x\cos x}{\sin^4 x+\cos^4 x}\left(x^2-(\pi-x)^2\right)dx$

$2I=\int_0^{\pi}\frac{\pi(2x-\pi)\sin x\cos x}{\sin^4 x+\cos^4 x}dx$

$\Rightarrow \quad I=\pi\int_0^{\pi/2}\frac{(2x-\pi)\sin x\cos x}{\sin^4 x+\cos^4 x}dx$

$\Rightarrow \quad I=\pi\int_0^{\pi/2}\frac{-2x\sin x\cos x}{\sin^4 x+\cos^4 x}dx$

$2I=-\pi^2\int_0^{\pi/2}\frac{\sin x\cos x}{\sin^4 x+\cos^4 x}dx$

$I=\frac{-\pi}{2\times 2}\int_0^{\pi/2}\frac{\sin 2x}{\sin^4 x+\cos^4 x}dx$

$I=\frac{-\pi^2}{4}\int_0^{\pi/2}\frac{2\tan x\sec^2 x}{\tan^4 x+1}dx$

$I=\frac{-\pi^2}{4}\tan^{-1}\left(\tan^2 x\right)_0^{\pi/2}$

$\Rightarrow \quad I=\frac{-\pi^3}{8}$

$\therefore \quad \left|\frac{120}{\pi^3}I\right|=15$

76. If $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x dx}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)}=\alpha\pi+\beta\log_e\left(3+2\sqrt{2}\right)$, where α, β are integers, then $\alpha^2+\beta^2$ equals _____.

[JEE(Main)Jan.2024]

यदि $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x dx}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)}=\alpha\pi+\beta\log_e\left(3+2\sqrt{2}\right)$ है, जहाँ α, β पूर्णांक हैं, तो $\alpha^2+\beta^2$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 8

Sol. $$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)}dx \qquad ...(1)$$

Apply king

$$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x\left(e^{\sin x}\right)}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)}dx \qquad ...(2)$$

adding (1) & (2)

$$2I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x}{1+\sin^4 x}dx$$

$$I=\int_{0}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x}{1+\sin^4 x}dx$$

$\sin x = t$

$$I=\int_{0}^{1}\frac{8\sqrt{2}}{1+t^4}dx$$

$$I=4\sqrt{2}\int_{0}^{1}\left(\frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}}-\frac{1-\frac{1}{t^2}}{t^2+\frac{1}{t^2}}\right)dt$$

$$I=4\sqrt{2}\int_{0}^{1}\frac{\left(1+\frac{1}{t^2}\right)}{\left(t-\frac{1}{t}\right)^2+2}-\frac{\left(1-\frac{1}{t^2}\right)}{\left(t+\frac{1}{t}\right)^2-2}dt$$

Let $t-\frac{1}{t}=z$ & $t+\frac{1}{t}=k$

$$=4\sqrt{2}\left[\int_{-\infty}^{0}\frac{dz}{z^2+2}-\int_{\infty}^{2}\frac{dk}{k^2-2}\right]$$

$$=4\sqrt{2}\left[\frac{1}{\sqrt{2}}\tan^{-1}\frac{z}{\sqrt{2}}\right]_{-\infty}^{0}-\left[\frac{1}{2\sqrt{2}}\ell n\left(\frac{k-\sqrt{2}}{k+\sqrt{2}}\right)\right]_{\infty}^{2}$$

$$= 4\sqrt{2}\left[\frac{\pi}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}\left[\ell n\frac{2-\sqrt{2}}{2+\sqrt{2}}\right]\right]$$

$$= 2\pi + 2\ell n\left(3+2\sqrt{2}\right)$$

$\alpha = 2$

$\beta = 2$

77. Let $f: (0, \infty) \to R$ and $F(x) = \int_0^x t\,f(t)\,dt$. If $F(x^2) = x^4 + x^5$, then $\sum_{r=1}^{12} f\left(r^2\right)$ is equal to _____.

[JEE(Main)Jan.2024]

माना $f: (0, \infty) \to R$ तथा $F(x) = \int_0^x t\,f(t)\,dt$ हैं। यदि $F(x^2) = x^4 + x^5$ है, तो $\sum_{r=1}^{12} f\left(r^2\right)$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 219

Sol. $F(x) = \int_0^x tf(t)dt$ and $F(x)^2 = x^4 + x^5$

$F'(x) = xf(x)$ and $2xF'\left(x^2\right) = 4x^3 + 5x^4$

$2F'\left(x^2\right) = 4x^2 + 5x^3$

$2F'(x) = 4x + 5x^{\frac{3}{2}}$

$\Rightarrow 2x + \frac{5}{2}x^{\frac{3}{2}} = xf(x)$

$\Rightarrow f(x) = 2 + \frac{5}{2}x^{\frac{1}{2}}$

$\Rightarrow f\left(x^2\right) = 2 + \frac{5}{2}x$

$$\sum_{r=1}^{12} f\left(r^2\right) = \sum_{r=1}^{12}\left(2+\frac{5}{2}r\right)$$

$$= 24 + \frac{5}{2}\sum_{r=1}^{12} r$$

$$= 24 + \frac{5}{2}\times\frac{12\times13}{2}$$

$24 + 195 = 219$

78. If $\int \frac{1}{a^2\sin^2 x + b^2\cos^2 x}dx = \frac{1}{12}\tan^{-1}(3\tan x) +$ constant, then the maximum value of $a\sin x + b\cos x$, is :

यदि $\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} dx = \frac{1}{12} \tan^{-1}(3 \tan x) +$ अचर है, तो $\sin x + b \cos x$ का उच्चतम मान है :

[JEE(Main)April,2024]

(A) $\sqrt{40}$ (B) $\sqrt{42}$ (C) $\sqrt{39}$ (D) $\sqrt{41}$

Ans. A

Sol. $\int \frac{\sec^2 x dx}{a^2 \tan^2 x + b^2}$

let $\tan x = t$

$\sec^2 dx = dt$

$\int \frac{dt}{a^2 t^2 + b^2}$

$\frac{1}{a^2} \int \frac{dt}{t^2 + \left(\frac{b}{a}\right)^2}$

$\frac{1}{a^2} \frac{1}{\frac{b}{a}} \tan^{-1}\left(\frac{t}{b} a\right) + c$

$\frac{1}{ab} \tan^{-1}\left(\frac{\alpha}{b} \tan x\right) + c$

on compairing $\frac{a}{b} = 3$

$ab = 12$

$a = 6, b = 2$

maximum value of

$6 \sin x + 2 \cos x$ is $\sqrt{40}$

79. Let $I(x) = \int \frac{6}{\sin^2 x (1 - \cot x)^2} dx$. If $I(0) = 3$, then $I\left(\frac{\pi}{12}\right)$ is equal to : [JEE(Main)April,2024]

माना $I(x) = \int \frac{6}{\sin^2 x (1 - \cot x)^2} dx$ है। यदि $I(0) = 3$, तब $I\left(\frac{\pi}{12}\right)$ बराबर होगा : [JEE(Main)April,2024]

(A) $2\sqrt{3}$ (B) $\sqrt{3}$ (C) $3\sqrt{3}$ (D) $6\sqrt{3}$

Ans. C

Sol. $I(x)=\int \frac{6dx}{\sin^2(1-\cot x)^2}$ $\qquad I(0)=3$

$I\left(\frac{\pi}{12}\right)=?$

$$I(x)=\int \frac{6dx}{\sin^2 x\frac{(\sin x-\cos x)^2}{\sin^2 x}}=\int \frac{6dx}{\sin^2 x+\cos^2 x-2\sin\cos x}$$

$$=\int \frac{6dx}{1-2\sin x\cos x}=\int \frac{6\sec^2 xdx}{1+\tan^2 x-2\tan x}$$

$$=\int \frac{6\sec^2 xdx}{(\tan x-1)^2}$$

$= \tan x = t$

$= \sec^2 x\, dx = dt$

$$=\int \frac{6dt}{(t-1)^2}=6\int (t-1)^{-2}\,dt$$

$$I(x)=\frac{-6}{(t-1)}+C$$

$$I(x)=-\frac{6}{\tan x-1}+C$$

$I(0) = 3$

$3 = 6 + C$

$C = -3$

$$I(x)=-\frac{6}{\tan x-1}-3$$

$$I\left(\frac{\pi}{12}\right)=-\frac{6}{2-\sqrt{3}-1}-3$$

$$=-\frac{6}{1-\sqrt{3}}-3=-\left(\frac{6+3-3\sqrt{3}}{1-\sqrt{3}}\right)$$

$$=-\left(\frac{9-3\sqrt{3}}{1-\sqrt{3}}\times\frac{1+\sqrt{3}}{1+\sqrt{3}}\right)$$

$$=-\left(\frac{9+9\sqrt{3}-3\sqrt{3}-9}{1-3}\right)=\frac{6\sqrt{3}}{2}=3\sqrt{3}$$

80. Let $\int \frac{2-\tan x}{3+\tan x}dx = \frac{1}{2}\left(\alpha x + \log_e|\beta \sin x + \gamma \cos x|\right) + C$, where C is the constant of integration. Then $\alpha + \frac{\gamma}{\beta}$ is equal to : **[JEE(Main)April,2024]**

माना $\int \frac{2-\tan x}{3+\tan x}dx = \frac{1}{2}\left(\alpha x + \log_e|\beta \sin x + \gamma \cos x|\right) + C$ है, जहाँ C समाकलन अचर है। तो $\alpha + \frac{\gamma}{\beta}$ बराबर है : **[JEE(Main)April,2024]**

(A) 3 (B) 1 (C) 7 (D) 7

Ans. D

Sol. $\int \frac{2-\tan x}{3+\tan x}dx = \int \frac{2\cos x - \sin x}{3\cos x + \sin x}dx$

Put, $2\cos x - \sin x = a(-3\sin x + \cos x) + b(3\cos x + \sin x)$

$a + 3b = 2$ .....(i)

$-3a + b = -1$ .....(ii)

From equation (i) and (ii) $a = b = \frac{1}{2}$

$$I = \frac{1}{2}\int \frac{-3\sin x + \cos x}{3\cos x + \sin x}dx + \frac{1}{2}\int \frac{3\cos x + \sin x}{3\cos x + \sin x}dx$$

$$I = \frac{1}{2}\ell n|3\cos x + \sin x| + \frac{1}{2}x = \frac{1}{2}\left(\alpha x + \log|\beta \sin x + \gamma \cos x|\right)$$

On comparing, we get

$\alpha = 1, \beta = 1, \gamma = 3$

$$\alpha + \frac{\gamma}{\beta} = 1 + \frac{3}{1} = 4$$

81. If $\int \text{cosec}^5 x dx = \alpha \cot x \,\text{cosec}\, x\left(\text{cosec}^2 x + \frac{3}{2}\right) + \beta \log_e\left|\tan\frac{x}{2}\right| + C$ where $\alpha, \beta \in R$ and C is the constant of integration, then the value of $8(\alpha + \beta)$ equals _____. **[JEE(Main)April,2024]**

यदि $\int \text{cosec}^5 x dx = \alpha \cot x \,\text{cosec}\, x\left(\text{cosec}^2 x + \frac{3}{2}\right) + \beta \log_e\left|\tan\frac{x}{2}\right| + C$ है, जहाँ $\alpha, \beta \in R$ तथा C समाकल अचर है, तो $8(\alpha + \beta)$ का मान बराबर है _____ **[JEE(Main)April,2024]**

Ans. 1

Sol. $\int \text{cosec}^3 x \cdot \text{cosec}^2 x dx = I$

By applying integration by parts

$$I = -\cot x\,\text{cosec}^3 x + \int \cot x\left(-3\text{cosec}^2 x \cot x \,\text{cosec}\, x\right)dx$$

$I = -\cot x \operatorname{cosec}^3 x - 3I - 3\int \operatorname{cosec}^3 x\left(\operatorname{cosec}^2 x - 1\right)dx$

$I = -\cot x \operatorname{cosec}^3 x - 3I + 3\int \operatorname{cosec}^3 x dx$

Let

$I_1 = \int \operatorname{cosec}^3 x dx = -\operatorname{cosec} x \cot x - \int \cot^2 x \operatorname{cosec} x dx$

$I_1 = -\operatorname{cosec} x \cot x - \int\left(\operatorname{cosec}^2 x - 1\right)\operatorname{cosec} x dx$

$2I_1 = -\operatorname{cosec} x \cot x + \ell n\left|\tan\frac{x}{2}\right|$

$I_1 = -\frac{1}{2}\operatorname{cosec} x \cot x + \frac{1}{2}\ell n\left|\tan\frac{x}{2}\right|$

$4I = -\cot x \operatorname{cosec}^3 x - \frac{3}{2}\operatorname{cosec} x \cot x + \frac{3}{2}\ell n\left|\tan\frac{x}{2}\right| + 4c$

$I = -\frac{1}{4}\operatorname{cosec} x \cot x\left(\operatorname{cosec}^2 x + \frac{3}{2}\right) + \frac{3}{2}\ell n\left|\tan\frac{x}{2}\right| + c$

$\therefore \alpha = \frac{-1}{4},\ \beta = \frac{3}{8} \rightarrow 8(\alpha + \beta) = 1$

82. If $\int \frac{1}{\sqrt[5]{(x-1)^4(x+3)^6}}dx = A\left(\frac{\alpha x - 1}{\beta x + 3}\right)^B + C$, where C is the contant of integration, then the value of $\alpha + \beta +$ 20AB is _____. **[JEE(Main)April,2024]**

यदि $\int \frac{1}{\sqrt[5]{(x-1)^4(x+3)^6}}dx = A\left(\frac{\alpha x - 1}{\beta x + 3}\right)^B + C$ है, जहाँ C समाकलन अचर है, तो $\alpha + \beta + 20AB$ का मान है_____.

[JEE(Main)April,2024]

Ans. 7

Sol. $\int \frac{1}{\sqrt[5]{(x-1)^4(x+3)^6}}dx = A\left(\frac{\alpha x - 1}{\beta x + 3}\right)^B + C$

$I = \int \frac{1}{(x-1)^{\frac{4}{5}}(x+3)^{\frac{6}{5}}}dx$

$$I=\int \frac{1}{\left(\frac{x-1}{x+3}\right)^{\frac{4}{5}}(x+3)^2}dx$$

$\left(\frac{x-1}{x+3}\right)=t$ $\qquad$ $\frac{4}{(x+3)^2}dx=dt$

$$I=\frac{1}{4}\int \frac{1}{t^{\frac{4}{5}}}dt=\frac{1}{4}\left(\frac{t^{\frac{1}{5}}}{\frac{1}{5}}\right)+c$$

$$I=\frac{5}{4}\left(\frac{x-1}{x+3}\right)^{\frac{1}{5}}+C$$

$A=\frac{5}{4}$ $\qquad$ $\alpha=\beta=1$ $\qquad$ $B=\frac{1}{5}$

$$\alpha+\beta+20AB=2+20\times\frac{5}{4}\times\frac{1}{5}=7$$

83. Let $f(x)=\begin{cases}-2, & -2\le x\le 0\\ x-2, & 0<x\le 2\end{cases}$ and $h(x)=f(|x|)+|f(x)|$. Then $\int_{-2}^{2} h(x)dx$ is equal to :

[JEE(Main)April,2024]

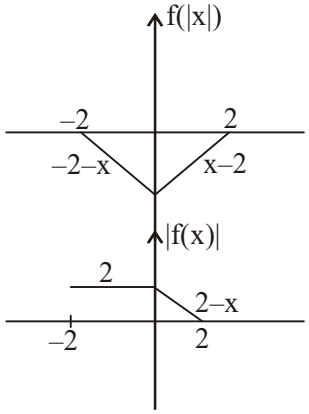
माना $f(x)=\begin{cases}-2, & -2\le x\le 0\\ x-2, & 0<x\le 2\end{cases}$ तथा $h(x)=f(|x|)+|f(x)|$ हैं। तो $\int_{-2}^{2} h(x)dx$ बराबर है :

[JEE(Main)April,2024]

(A) 2 $\qquad$ (B) 6 $\qquad$ (C) 4 $\qquad$ (D) 1

Ans. A

Sol.



$$\therefore h(x)=\begin{cases}-x & x\in[-2,0]\\ 0 & x\in[0,2]\end{cases}$$

$$\int_{-2}^{2} h(x)dx=\int_{-2}^{0}(-x)dx$$

$$=\left(-\frac{x^2}{22}\right)_{-2}^{0}$$

$$=-(-2)=2$$

84. If the value of the integral $\int_{-1}^{1}\frac{\cos\alpha x}{1+3^x}dx$ is $\frac{2}{\pi}$. Then, a value of α is : **[JEE(Main)April,2024]**

यदि समाकलन $\int_{-1}^{1}\frac{\cos\alpha x}{1+3^x}dx$ का मान $\frac{2}{\pi}$ है। तो α का एक मान है : **[JEE(Main)April,2024]**

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

Ans. C

Sol. Let $I=\int_{-1}^{+1}\frac{\cos\alpha x}{1+3^x}dx$ _____(1)

$$I=\int_{-1}^{+1}\frac{\cos\alpha x}{1+3^{-x}}dx$$

(using $\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$) _____(2)

Add (1) and (2)

$$2I = \int_{-1}^{+1} \cos(\alpha x)dx = 2\int_{0}^{1} \cos(\alpha x)dx$$

$$I = \frac{\sin\alpha}{\alpha} = \frac{2}{\pi} \text{ (given)}$$

$$\therefore \alpha = \frac{\pi}{2}$$

85. Let $f(x) = \int_0^x \left(t + \sin\left(1 - e^t\right)\right)dt$, $x \in R$. Then, $\lim\limits_{x\to 0} \frac{f(x)}{x^3}$ is equal to : **[JEE(Main)April,2024]**

माना $f(x) = \int_0^x \left(t + \sin\left(1 - e^t\right)\right)dt$, $x \in R$ है। तो $\lim\limits_{x\to 0} \frac{f(x)}{x^3}$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{1}{6}$ (B) $-\frac{1}{6}$ (C) $-\frac{2}{3}$ (D) $\frac{2}{3}$

Ans. B

Sol. $\lim\limits_{x\to 0} \frac{f(x)}{x^3}$

Using L Hopital Rule.

$$\lim_{x\to 0} \frac{f'(x)}{3x^2} = \lim_{x\to 0} \frac{x + \sin\left(1 - e^x\right)}{3x^2} \text{ (Again L hospital)}$$

Using L Hopital Rule

$$\lim_{x\to 0} \frac{-\left[\sin\left(1 - e^x\right)\left(-e^x\right)\cdot e^x + \cos\left(1 - e^x\right)\cdot e^x\right]}{6}$$

$$= -\frac{1}{6}$$

86. The integral $\int_0^{\frac{\pi}{4}} \frac{136\sin x}{3\sin x + 5\cos x}dx$ is equal to : **[JEE(Main)April,2024]**

समाकलन $\int_0^{\frac{\pi}{4}} \frac{136\sin x}{3\sin x + 5\cos x}dx$ बराबर है : **[JEE(Main)April,2024]**

(A) $3\pi - 30\log_e 2 + 20\log_e 5$ (B) $3\pi - 50\log_e 2 + 20\log_e 5$

(C) $3\pi - 10\log_e\left(2\sqrt{2}\right) + 10\log_e 5$ (D) $3\pi - 25\log_e 2 + 10\log_e 5$

Ans. B

Sol. $(136)\int_0^{\pi/4}\frac{\sin x}{3\sin x+5\cos x}dx$

$136\sin x=\lambda(3\sin x+5\cos x)+\mu(3\cos x-5\sin x)$

Compare coffi of sinx, cosx

$136=3\lambda-5\mu$ ...(1)

$0=5\lambda+3\mu$ ...(2)

$408=34\lambda\Rightarrow\lambda=12$

$\mu=-20$

$(136)\int_0^{\pi/4}\frac{\sin x}{3\sin x+5\cos x}dx$

$=12(x)_0^{\pi/4}-20\left[\ell n(3\sin x+5\cos x)\right]_0^{\pi/4}$

$=12\frac{\pi}{4}-20\left[\ell n4\sqrt{2}-\ell n5\right]$

$3\pi-20\ell n\left(4\sqrt{2}\right)+20\ell n5$

$=3\pi-50\ell n2+20\ell n5$

87. The value of $\int_{-\pi}^{\pi}\frac{2y(1+\sin y)}{1+\cos^2 y}dy$ is : **[JEE(Main)April,2024]**

$\int_{-\pi}^{\pi}\frac{2y(1+\sin y)}{1+\cos^2 y}dy$ का मान बराबर है : **[JEE(Main)April,2024]**

(A) π^2 (B) $\frac{\pi}{2}$ (C) $\frac{\pi^2}{2}$ (D) $2\pi^2$

Ans. A

Sol. $=\int_{-\pi}^{\pi}\frac{2y(1+\sin y)}{1+\cos^2 y}dy$

$=\int_0^{\pi}\left(\frac{2y(1+\sin y)}{1+\cos^2 y}+\frac{-2y(1-\sin y)}{1+\cos^2 y}\right)dy$

$=\int_0^{\pi}\left(\frac{2y+2y\sin y-2y+2y\sin y}{1+\cos^2 y}\right)dy$

$I=4\int_0^{\pi}\left(\frac{y\sin y}{1+\cos^2 y}\right)dy$ .....(1)

$$I = 4\int_0^{\pi}\left(\frac{(\pi-y)\sin(\pi-y)}{1+\cos^2(\pi-y)}\right)dy$$

$$I = 4\left[\int_0^{\pi}\left(\frac{\pi\sin y}{1+\cos^2 y}\right)dy - \int_0^{\pi}\frac{y\sin y}{1+\cos^2 y}dy\right] \quad .....(2)$$

Adding equation (1) and (2)

$$2I = 4\int_0^{\pi}\left(\frac{\pi\sin y}{1+\cos^2 y}\right)dy$$

$$I = 2\pi\int_0^{\pi}\left(\frac{\sin y}{1+\cos^2 y}\right)dy$$

$$= 2\pi\times\frac{\pi}{2}$$

$$= \pi^2$$

88. Let $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$, m, n > 0 . If $\int_0^1 (1-x^{10})^{20}dx = a\times\beta(b,c)$, then 100 (a + b + c) equals :

[JEE(Main)April,2024]

माना $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$, m, n > 0 है। यदि $\int_0^1 (1-x^{10})^{20}dx = a\times\beta(b,c)$ है, तो 100 (a + b + c) बराबर है :

[JEE(Main)April,2024]

(A) 1021 (B) 2120 (C) 2012 (D) 1120

Ans. B

Sol. $I = \int_0^1 (1-x^{10})^{20}\,dx$

$x^{10} = t \Rightarrow x = (t)^{1/10}$

$$I = \frac{1}{10}\int_0^1 (t)^{-9/10}(1-t)^{20}\,dt$$

$$a = \frac{1}{10},\ b = \frac{1}{10},\ c = 21$$

$$100\left(\frac{1}{10}+\frac{1}{10}+21\right)$$

$\Rightarrow$ 10 + 10 + 2100 = 2120

89. $\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{\left(\cos^3 x + \sin^3 x\right)^2} dx$ is equal to : **[JEE(Main)April,2024]**

$\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{\left(\cos^3 x + \sin^3 x\right)^2} dx$ बराबर है : **[JEE(Main)April,2024]**

(A) 1/9 (B) 1/6 (C) 1/3 (D) 1/12

Ans. B

Sol. $\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{\left(\cos^3 x + \sin^3 x\right)^2} dx$

$\Rightarrow \int_0^{\pi/4} \frac{\tan^2 x \cdot \sec^2 x}{\left(1+\tan^3 x\right)^2} \cdot dx$

Let $1 + \tan^3 x = t$

$3 \cdot \tan^2 x \cdot \sec^2 x \cdot dx = dt$

$\Rightarrow \frac{1}{3}\int_1^2 \frac{dt}{\left(t^2\right)^2}$

$\Rightarrow \frac{1}{3}\left[-\frac{1}{t}\right]_1^2$

$\Rightarrow -\frac{1}{3}\left[+\frac{1}{2}-1\right] \Rightarrow \frac{1}{6}$

90. The value of $k \in N$ for which the integral $I_n = \int_0^1 \left(1-x^k\right)^n dx$, $n \in N$, satisfies $147\, I_{20} = 148\, I_{21}$ is :

[JEE(Main)April,2024]

$k \in N$ का मान जिसके लिए समाकलन $I_n = \int_0^1 \left(1-x^k\right)^n dx$, $n \in N$ एवं $147\, I_{20} = 148\, I_{21}$ को संतुष्ट करता है :

[JEE(Main)April,2024]

(A) 8 (B) 7 (C) 10 (D) 14

Ans. B

Sol. $I_n = \int_0^1 \left(1-x^k\right)^n dx$

$$I_{21} = \int_0^1 \left(1 - x^k\right)^{21} dx$$

$$\Rightarrow \int_0^1 \left(1 - x^k\right)\left(1 - x^k\right)^{20} dx$$

$$\Rightarrow \int_0^1 \left(1 - x^k\right)^{20} dx - \int_0^1 x^k \left(1 - x^k\right)^{20} dx$$

$$I_{21} = I_{20} - \int_0^1 x^k \left(1 - x^k\right)^{20} dx$$

$$\int_0^1 xx^{(k-1)} \left(1 - x^k\right)^{20} dx$$

$1 - x^k = t$

$-K(x^{K-1}\, dx) = dt$

Integration by part

$$x\left[-\frac{\left(1 - x^K\right)^{21}}{21K}\right]_0^1 - \frac{1}{21K}\int_0^1 \left(1 - x^k\right)^{21} dx$$

$$I_{21} = I_{20} + 0 - \frac{I_{21}}{21K}$$

$$= \left(\frac{21K + 1}{21K}\right) I_{21} = I_{20}$$

$= (21\ K + 1)\ I_{21} = 21KI_{20}$

$K = 7$

91. Let $\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$. Then e^{α} and $e^{-\alpha}$ are the roots of the equation : **[JEE(Main)April,2024]**

माना $\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$ है। तो e^{α} तथा $e^{-\alpha}$ निम्न में से किस समीकरण के मूल हैं : **[JEE(Main)April,2024]**

(A) $2x^2 - 5x - 2 = 0$ (B) $x^2 - 2x - 8 = 0$ (C) $2x^2 - 5x + 2 = 0$ (D) $x^2 + 2x - 8 = 0$

Ans. C

Sol. $\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$

Let $e^x - 1 = t^2$

$e^x\, dx = 2t\,dt$

$$\int \frac{2t\,dt}{e^x \cdot t} = \int \frac{2dt}{t^2 + 1}$$

$= 2\tan^{-1} t$

$$= 2\tan^{-1}\left(\sqrt{e^x - 1}\right)_{\alpha}^{\log_e 4}$$

$$= 2\left[\tan^{-1}\sqrt{3} - \tan^{-1}\sqrt{e^{\alpha} - 1}\right] = \frac{\pi}{6}$$

$$= \frac{\pi}{3} - \tan^{-1}\sqrt{e^{\alpha} - 1} = \frac{\pi}{12}$$

$$= \tan^{-1}\sqrt{e^{\alpha} - 1} = \frac{\pi}{4}$$

$e^{\alpha} = 2, \qquad e^{-\alpha} = \frac{1}{2}$

$$x^2 - \left(2 + \frac{1}{2}\right)x + 1 = 0$$

$2x^2 - 5x + 2 = 0$

92. $\lim\limits_{x \to \frac{\pi}{2}} \left(\dfrac{\int_{x^3}^{\left(\frac{\pi}{2}\right)^3} \left(\sin\left(2t^{\frac{1}{3}}\right) + \cos\left(t^{\frac{1}{3}}\right) \right) dt}{\left(x - \frac{\pi}{2}\right)^2} \right)$ is equal to : **[JEE(Main)April,2024]**

$\lim\limits_{x\to\frac{\pi}{2}}\left(\dfrac{\int\limits_{x^3}^{\left(\frac{\pi}{2}\right)^3}\left(\sin\left(2t^{\frac{1}{3}}\right)+\cos\left(t^{\frac{1}{3}}\right)\right)dt}{\left(x-\frac{\pi}{2}\right)^2}\right)$ बराबर है : **[JEE(Main)April,2024]**

(A) $\dfrac{11\pi^2}{10}$ (B) $\dfrac{3\pi^2}{2}$ (C) $\dfrac{5\pi^2}{9}$ (D) $\dfrac{9\pi^2}{8}$

Ans. D

Sol. L. hopital's rule

$$\Rightarrow \lim_{x\to\frac{\pi}{2}}\frac{-3x^2\cdot(\sin 2x+\cos x)}{2\left(x-\frac{\pi}{2}\right)}=\lim_{x\to\frac{\pi}{2}}\frac{-3x^2\cos x(2\sin x+1)}{2\left(x-\frac{\pi}{2}\right)}$$

$$=\lim_{x\to\frac{\pi}{2}}\frac{-9\times\frac{\pi^2}{4}}{2}\times\frac{\cos x}{x-\frac{\pi}{2}}=\lim_{x\to\frac{\pi}{2}}\frac{-9\pi^2}{8}\times\frac{-\sin x}{1}$$

$$=\frac{9\pi^2}{8}$$

93. The integral $\int\limits_{\frac{1}{4}}^{\frac{3}{4}}\cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$ is equal to : **[JEE(Main)April,2024]**

समाकलन $\int\limits_{\frac{1}{4}}^{\frac{3}{4}}\cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$ बराबर है : **[JEE(Main)April,2024]**

(A) $\dfrac{1}{4}$ (B) $-\dfrac{1}{2}$ (C) $-\dfrac{1}{4}$ (D) $\dfrac{1}{2}$

Ans. C

Sol. $\int_{\frac{1}{4}}^{\frac{3}{4}}\cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$

$x = \cos 2\theta$

$\therefore \quad dx = (-2 \sin 2\theta \, d\theta)$

Tokε λμ ιτ οασα and β

$$-2\int_{\alpha}^{\beta} \cos 2\theta \cdot \sin 2\theta d\theta$$

$$= \int_{\alpha}^{\beta} \sin 4\theta d\theta$$

$$= \frac{-\cos 4\theta}{4}\Bigg|_{\alpha}^{\beta}$$

$$= -\frac{1}{4}\left(2\cdot\left(x^2\right)-1\right)\Bigg|_{1/4}^{3/4}$$

$$= -\frac{1}{4}\left(2x^2-1\right)\Bigg|_{1/4}^{3/4}$$

$$= -\frac{1}{4}\left(\frac{18}{16}-1-\frac{2}{16}+1\right)$$

$$= -\frac{1}{4}$$

94. The value of the integral $\int_{-1}^{2} \log_e\left(x+\sqrt{x^2+1}\right)dx$ is : **[JEE(Main)April,2024]**

समाकलन $\int_{-1}^{2} \log_e\left(x+\sqrt{x^2+1}\right)dx$ का मान है : **[JEE(Main)April,2024]**

(A) $\sqrt{2}-\sqrt{5}+\log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$ (B) $\sqrt{5}-\sqrt{2}+\log_e\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$

(C) $\sqrt{2}-\sqrt{5}+\log_e\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$ (D) $\sqrt{5}-\sqrt{2}+\log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$

Ans. C

Sol. $\int_{-1}^{2} \log_e\left(x+\sqrt{x^2+1}\right)dx$

$$= \left[x\log_e\left(x+\sqrt{x^2+1}\right)\right]_{-1}^{2} - \int_{-1}^{2} \frac{x}{\left(x+\sqrt{x^2+1}\right)}\left(1+\frac{x}{\sqrt{x^2+1}}\right)dx$$

$$= 2\log_e(2+\sqrt{5}) + \log_e(\sqrt{2}-1) - \int_{-1}^{2} \frac{x}{\sqrt{x^2+1}}dx$$

$$= \log_e\left[(2+\sqrt{5})^2(\sqrt{2}-1)\right] - \left[\sqrt{x^2+1}\right]_{-1}^{2}$$

$$= \log_e\left[\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right] - \sqrt{5} + \sqrt{2}$$

$$= \sqrt{2} - \sqrt{5} + \log_e\left[\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right]$$

95. If the shortest distance between the lines $\frac{x+2}{2} = \frac{y+3}{3} = \frac{z-5}{4}$ and $\frac{x-3}{1} = \frac{y-2}{-3} = \frac{z+4}{2}$ is $\frac{38}{3\sqrt{5}}k$, and $\int_0^k \left[x^2\right]dx = \alpha - \sqrt{\alpha}$, where [x] denotes the greatest integer function, then $6\alpha^3$ is equal to _____.

[JEE(Main)April,2024]

यदि रेखाओं $\frac{x+2}{2} = \frac{y+3}{3} = \frac{z-5}{4}$ तथा $\frac{x-3}{1} = \frac{y-2}{-3} = \frac{z+4}{2}$ के बीच न्यूनतम $\frac{38}{3\sqrt{5}}k$ है तथा $\int_0^k \left[x^2\right]dx = \alpha - \sqrt{\alpha}$ है, जहाँ [x] महत्तम पूर्णांक फलन है, तो $6\alpha^3$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 48

Sol. $L_1 \equiv \frac{x+2}{2} = \frac{y+3}{3} = \frac{z-5}{4}$

$L_2 \equiv \frac{x-3}{1} = \frac{y-2}{-3} = \frac{z+4}{2}$

Shortest distance between L_1 and L_2 is $= \frac{\begin{vmatrix} 5 & 5 & -9 \\ 2 & 3 & 4 \\ 1 & -3 & 2 \end{vmatrix}}{\sqrt{(18)^2 + (0)^2 + (-9)^2}}$

$$\frac{38}{3\sqrt{5}}k = \frac{90+0+81}{9\sqrt{5}} = \frac{19}{\sqrt{5}}$$

$$\Rightarrow k = \frac{3}{2}$$

Now $\int_0^k \left[x^2\right]dx = \int_0^{\frac{3}{2}} \left[x^2\right]dx$

$$= \int_0^1 \left[x^2\right] dx + \int_1^{\sqrt{2}} \left[x^2\right] dx + \int_{\sqrt{2}}^{\frac{3}{2}} \left[x^2\right] dx$$

$$= 0 + \left(\sqrt{2} - 1\right) + 2\left(\frac{3}{2} - \sqrt{2}\right)$$

$$= 2 - \sqrt{2}$$

$\Rightarrow \alpha = 2$

$\Rightarrow 6\alpha^3 = 48$

96. If $\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1 + \sin x \cos x} dx = \frac{1}{a} \log_e \left(\frac{a}{3}\right) + \frac{\pi}{b\sqrt{3}}$, where $a, b \in N$, then $a + b$ is equal to _____.

[JEE(Main)April,2024]

यदि $\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1 + \sin x \cos x} dx = \frac{1}{a} \log_e \left(\frac{a}{3}\right) + \frac{\pi}{b\sqrt{3}}$, जहाँ $a, b \in N$ है, तो $a + b$ बराबर है _____

[JEE(Main)April,2024]

Ans. 8

Sol. $\ell = \int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1 + \sin x \cos x} dx = \int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{\sin^2 x + \cos^2 x + \sin x \cos x} dx$

$$\ell = \int_0^{\frac{\pi}{4}} \frac{\tan^2 x}{1 + \tan x + \tan^2 x} dx = \int_0^{\frac{\pi}{4}} \frac{\tan^2 x \cdot \sec^2 x dx}{\left(1 + \tan^2 x\right)\left(1 + \tan x + \tan^2 x\right)}$$

Let $\tan x = t$

$$\ell = \int_0^1 \frac{t^2}{\left(1 + t^2\right)\left(1 + t + t^2\right)} dt$$

$$= \int_0^1 \left(\frac{x}{1 + x^2} - \frac{x}{1 + x + x^2}\right) dx$$

$$= \frac{1}{2} \int_0^1 \frac{2x}{1 + x^2} dx - \int_0^1 \frac{\frac{1}{2}(2x + 1) - \frac{1}{2}}{1 + x + x^2} dx$$

$$= \frac{1}{2} \ell n 2 - \frac{1}{2} \ln 3 + \frac{1}{2} \int_0^1 \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}}$$

$$= \frac{1}{2}\ell n\frac{2}{3} + \frac{1}{2}\cdot\frac{2}{\sqrt{3}}\left[\tan^{-1}\frac{2x+1}{\sqrt{3}}\right]_0^1$$

$$= \frac{1}{2}\ell n\frac{2}{3} + \frac{1}{\sqrt{3}}\left(\frac{\pi}{3}-\frac{\pi}{6}\right)$$

$$= \frac{1}{2}\ell n\frac{2}{3} + \frac{1}{\sqrt{3}}\cdot\frac{\pi}{6}$$

$\therefore$ a = 2, b = 6

$\therefore$ a + b = 8

97. If $f(t) = \int_0^{\pi} \frac{2x\,dx}{1-\cos^2 t\sin^2 x}$, $0 < t < \pi$, then the value of $\int_0^{\pi/2} \frac{\pi^2 dt}{f(t)}$ equals _____.

[JEE(Main)April,2024]

यदि $f(t) = \int_0^{\pi} \frac{2x\,dx}{1-\cos^2 t\sin^2 x}$, $0 < t < \pi$ है, तो $\int_0^{\pi/2} \frac{\pi^2 dt}{f(t)}$ का मान है _____ **[JEE(Main)April,2024]**

Ans. 1

Sol. $f(t) = \int_0^{\pi} \frac{2xdx}{1-\cos^2 t\sin^2 x}$

$x \to \pi - x$

$$f(t) = \int_0^{\pi} \frac{2(\pi-x)dx}{1-\cos^2 t\sin^2 x} = 2\pi\int_0^{\pi} \frac{dx}{1-\cos^2 t\sin^2 x} - f(t)$$

$$\Rightarrow f(t) = \pi\int_0^{\pi} \frac{dx}{1-\cos^2 t\sin^2 x}$$

$$= 2\pi\int_0^{\frac{\pi}{2}} \frac{dx}{1-\cos^2 t\sin^2 x}$$

$$f(t) = 2\pi\int_0^{\frac{\pi}{2}} \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t\tan^2 x}$$

$$I_1 = \int \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t\tan^2 x}$$

Put cost tanx = $\lambda \Rightarrow$ cost $\sec^2 x dx = d\lambda$

$$I_1 = \int \frac{d\lambda}{\cos t.\left(1+\lambda^2 \sec^2 t - \lambda^2\right)} = \int \frac{d\lambda}{\cos t\left(1+\lambda^2 \tan^2 t\right)}$$

$$= \frac{1}{\cos t.\tan^2 t}.\int \frac{d\lambda}{\lambda^2 + \cos^2 t} = \frac{1}{\cos t \tan^2 t} \times \frac{1}{\cos t} \tan^{-1}\left(\lambda \tan t\right)$$

$$= \frac{1}{\sin t} \tan^{-1}\left(\sin t \tan x\right)$$

$$\Rightarrow f(t) = \frac{2\pi}{\sin t} \tan^{-1}\left(\sin t \tan x\right)\Big]_0^{\frac{\pi}{2}} = \frac{\pi^2}{\sin t}$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} \frac{\pi^2 dt}{f(t)} = \int_0^{\frac{\pi}{2}} \sin t dt = 1$$

98. Let $r_k = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}, k \in \mathbb{N}$. Then the value of $\sum_{k=1}^{10} \frac{1}{7\left(r_k - 1\right)}$ is equal to _____.

[JEE(Main)April,2024]

माना $r_k = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}, k \in \mathbb{N}$ है। तो $\sum_{k=1}^{10} \frac{1}{7\left(r_k - 1\right)}$ का मान बराबर है _____

[JEE(Main)April,2024]

Ans. 65

Sol. $r_k = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}$

$$\int_0^1 \left(1-x^7\right)^{k+1} dx = \int_0^1 \left(1-x^7\right)^k \cdot \left(1-x^7\right) dx$$

$$= \int_0^1 \left(1-x^7\right)^k dx - \int_0^1 x^7 \left(1-x^7\right)^k dx$$

$$= \int_0^1 \left(1-x^7\right)^k - \int_0^1 x \cdot x^6 \left(1-x^7\right)^k dx$$

$$= \int_0^1 \left(1-x^7\right)^k - x\int_0^1 x^6 \left(1-x^7\right)^k dx + \int_0^1 \left[1 \cdot \int x^6 \cdot \left(1-x^7\right)^k dx\right] dx$$

$$= \int_0^1 \left(1-x^7\right)^k - x\frac{\left(1-x^7\right)^{k+1}}{-7(k+1)}\Bigg|_0^1 + \int_0^1 \frac{\left(1-x^7\right)^{k+1}}{-7(k+1)} dx$$

$$\Rightarrow \int_0^1 \left(1-x^7\right)^{k+1} dx \left\{1+\frac{1}{7(k+1)}\right\} = \int_0^1 \left(1-x^7\right)^k dx - 0$$

$$\frac{7k+8}{7(k+1)} = \frac{\int_0^1 \left(1-x^7\right)^k dx}{\int_0^1 \left(1-x^7\right)^{k+1} dx}$$

$$\sum_{k=1}^{10} \frac{1}{7\left(r_k-1\right)}$$

$$\Rightarrow \sum_{k=1}^{10} \frac{1}{7\left(1+\frac{1}{7k+7}-1\right)}$$

$$\Rightarrow \sum_{k=1}^{10} (K+1) \qquad \Rightarrow \frac{10\times 11}{2}+10 \Rightarrow 55+10=65$$

99. Let [t] denote the largest integer less than or equal to t. If

$\int_0^3 \left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx = a+b\sqrt{2}-\sqrt{3}-\sqrt{5}+c\sqrt{6}-\sqrt{7}$, where a, b, c $\in$ Z, then a + b + c is equal to _____.

[JEE(Main)April,2024]

माना [t], t से कम या t के बराबर महत्तम पूर्णांक को दर्शाता है। यदि

$\int_0^3 \left(\left[x^2\right]+\left[\frac{x^2}{2}\right]\right)dx = a+b\sqrt{2}-\sqrt{3}-\sqrt{5}+c\sqrt{6}-\sqrt{7}$, जहाँ a, b, c $\in$ Z तो, a + b + c बराबर है _____

[JEE(Main)April,2024]

Ans. 23

Sol. $\int_0^3 \left(\left[x^2\right]+\int_0^3 \left[\frac{x^2}{2}\right]\right)dx$

$$= \int_0^1 0dx + \int_1^{12} 1dx + \int_{\sqrt{2}}^{\sqrt{3}} 2dx$$

$$+ \int_{\sqrt{3}}^{2} 3dx + \int_2^{\sqrt{5}} 4dx + \int_{\sqrt{5}}^{\sqrt{6}} 5dx + \int_{\sqrt{6}}^{\sqrt{7}} 6dx + \int_{\sqrt{7}}^{\sqrt{8}} 7dx + \int_{\sqrt{8}}^{3} 8dx + \int_0^{\sqrt{2}} 0dx + \int_{\sqrt{2}}^{2} 1dx$$

$$+ \int_2^{\sqrt{6}} 2dx + \int_{\sqrt{6}}^{\sqrt{8}} 3dx + \int_{\sqrt{8}}^{3} 4dx = 31-6\sqrt{2}-\sqrt{3}-\sqrt{5}-2\sqrt{5}\sqrt{6}-\sqrt{7}$$

a = 31 b = –6, c = –2

a + b + c = 31 – 6 – 2 = 23

100. Let $\lim\limits_{n\to\infty}\left(\frac{n}{\sqrt{n^4+1}}-\frac{2n}{\left(n^2+1\right)\sqrt{n^4+1}}+\frac{n}{\sqrt{n^4+16}}-\frac{8}{\left(n^2+4\right)\sqrt{n^4+16}}+....+\frac{n}{\sqrt{n^4+n^4}}-\frac{2n\cdot n^2}{\left(n^2+n^2\right)\sqrt{n^4+n^4}}\right)$ be $\frac{\pi}{k}$, using only the principal values of the inverse trigonometric functions. Then k^2 is equal to _____.

[JEE(Main)April,2024]

माना $\lim\limits_{n\to\infty}\left(\frac{n}{\sqrt{n^4+1}}-\frac{2n}{\left(n^2+1\right)\sqrt{n^4+1}}+\frac{n}{\sqrt{n^4+16}}-\frac{8}{\left(n^2+4\right)\sqrt{n^4+16}}+....+\frac{n}{\sqrt{n^4+n^4}}-\frac{2n\cdot n^2}{\left(n^2+n^2\right)\sqrt{n^4+n^4}}\right)=\frac{\pi}{k}$ है, जब प्रतिलोम त्रिकोणमितीय फलनों के केवल मुख्य मान लिए गए है। तो k^2 बराबर है _____

[JEE(Main)April,2024]

Ans. 32

Sol. $\sum\limits_{r=1}^{\infty}\frac{n}{\sqrt{n^4+r^4}}-\frac{2nr^2}{\left(n^2+r^2\right)\sqrt{n^4+r^4}}$

$\sum\limits_{r=1}^{\infty}\frac{\frac{1}{n}}{\sqrt{1+\left(\frac{r}{n}\right)^4}}-\frac{2\left(\frac{1}{n}\right)\left(\frac{r}{n}\right)^2}{\left(1+\left(\frac{r}{n}\right)^2\right)\sqrt{1+\left(\frac{r}{n}\right)^4}}$

$\Rightarrow\int\limits_0^1\frac{dx}{\sqrt{1+x^4}}-\frac{2x^2dx}{\left(1+x^2\right)\sqrt{1+x^4}}$

$\Rightarrow\int\limits_0^1\frac{1-x^2}{\left(1+x^2\right)\sqrt{1+x^4}}dx$

$$\Rightarrow \int_0^1 \frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{x^2+\frac{1}{x^2}}}dx$$

$$\Rightarrow -\int_0^1 \frac{\frac{1}{x^2}-1}{\left(x+\frac{1}{x}\right)\sqrt{\left(x+\frac{1}{x}\right)^2-2}}dx$$

$$x+\frac{1}{x}=t \Rightarrow 1-\frac{1}{x^2}dx=dt$$

$$\Rightarrow -\int_\infty^2 \frac{dt}{t\sqrt{t^2-2}}$$

$$\Rightarrow -\int_\infty^2 \frac{tdt}{t^2\sqrt{t^2-2}}$$

take $t^2 - 2 = \alpha^2$

$tdt = ad\alpha$

$$\Rightarrow -\int_\infty^{\sqrt{2}} \frac{\alpha d\alpha}{\left(\alpha^2+2\right)\alpha}$$

$$\Rightarrow -\int_\infty^{\sqrt{2}} \frac{d\alpha}{\alpha^2+2}$$

$$\Rightarrow \frac{-1}{\sqrt{2}}\tan^{-1}\frac{\alpha}{\sqrt{2}}\Big]_\infty^{\sqrt{2}}$$

$$\Rightarrow \frac{-1}{\sqrt{2}}\{\tan^{-1}\}+\frac{1}{\sqrt{2}}\tan^{-1}\infty$$

$$\Rightarrow \frac{1}{\sqrt{2}}\left\{\frac{\pi}{2}-\frac{\pi}{4}\right\}$$

$$\Rightarrow \frac{\pi}{\sqrt{2}}=\frac{\pi}{K}$$

So, $K = 4\sqrt{2}$

$K^2 = 32$

Exercise J-A

1. The integral $\int \frac{\sec^2 x}{(\sec x+\tan x)^{9/2}}dx$ equals (for some arbitrary constant K) **[JEE-2012]**

समाकलन $\int \frac{\sec^2 x}{(\sec x+\tan x)^{9/2}}dx$ बराबर होगा (किसी स्वेच्छ नियतांक K के लिए) **[JEE-2012]**

(A) $-\frac{1}{(\sec x+\tan x)^{11/2}}\left\{\frac{1}{11}-\frac{1}{7}(\sec x+\tan x)^2\right\}+K$ (B) $\frac{1}{(\sec x+\tan x)^{11/2}}\left\{\frac{1}{11}-\frac{1}{7}(\sec x+\tan x)^2\right\}+K$

(C) $-\frac{1}{(\sec x+\tan x)^{11/2}}\left\{\frac{1}{11}+\frac{1}{7}(\sec x+\tan x)^2\right\}+K$ (D) $\frac{1}{(\sec x+\tan x)^{11/2}}\left\{\frac{1}{11}+\frac{1}{7}(\sec x+\tan x)^2\right\}+K$

Ans. C

Sol. Let $I=\int\frac{\sec^2 x}{(\sec x+\tan x)^{9/2}}dx$

$=\int\frac{\sec x(\sec x+\tan x)\sec x}{(\sec x+\tan x)^{11/2}}dx$

Put $\sec x+\tan x=t$

$\Rightarrow$ $(\sec x\tan x+\sec^2 x)\,dx=dt$

Also $\because$ $\sec^2 x-\tan^2 x=1$

$\Rightarrow$ $\sec x-\tan x=\frac{1}{t}$

$\therefore$ $\sec x=\frac{1}{2}\left(t+\frac{1}{t}\right)$

$\therefore$ $I=\frac{1}{2}\int\frac{\left(1+\frac{1}{t}\right)dt}{t^{11/2}}=\frac{1}{2}\cdot\int\left(t^{-9/2}+t^{-13/2}\right)dt$

$=\frac{1}{2}\left(-\frac{2t^{-7/2}}{7}-\frac{2t^{-11/2}}{11}\right)+K$

$=-\frac{1}{t^{11/2}}\left(\frac{1}{11}+\frac{t^2}{7}\right)+K$

$=-\frac{1}{(\sec x+\tan x)^{11/2}}\left\{\frac{1}{11}+\frac{1}{7}(\sec x+\tan x)^2\right\}+K$

2. The value of the integral $\int_{-\pi/2}^{\pi/2}\left(x^2+\ln\frac{\pi+x}{\pi-x}\right)\cos x\,dx$ is : **[JEE-2012]**

$\int_{-\pi/2}^{\pi/2}\left(x^2+\ln\frac{\pi+x}{\pi-x}\right)\cos x\,dx$ का मान होगा : **[JEE-2012]**

(A) 0 (B) $\frac{\pi^2}{2}-4$ (C) $\frac{\pi^2}{2}+4$ (D) $\frac{\pi^2}{2}$

Ans. B

Sol. $\int\limits_{-\pi/2}^{\pi/2} x^2\cos x\,dx+\int\limits_{-\pi/2}^{\pi/2}\ell n\left(\frac{\pi+x}{\pi-x}\right)\cos x\,dx$

$=\int\limits_{-\pi/2}^{\pi/2} x^2\cos x\,dx=2\int\limits_{0}^{\pi/2}\underset{I}{x^2}\underset{II}{\cos x}\,dx$

$=2\left((x^2\sin x)_0^{\pi/2}-2\int\limits_0^{\pi/2}x\sin x\,dx\right)$

$=2\left(\frac{\pi^2}{4}-2\left(-(x\cos x)_0^{\pi/2}+\int\limits_0^{\pi/2}\cos x\,dx\right)\right)$

$=2\left(\frac{\pi^2}{4}-2\int\limits_0^{\pi/2}\cos x\,dx\right)$

$=2\left(\frac{\pi^2}{4}-2\right)=\frac{\pi^2}{2}-4$

3. For $a\in R$ (the set of all real numbers), $a\neq -1$, $\lim\limits_{n\to\infty}\frac{(1^a+2^a+...+n^a)}{(n+1)^{a-1}[(na+1)+(na+2)+...+(na+n)]}=\frac{1}{60}$. Then a = **[JEE(Adv.)2013]**

$a\in R$ (सभी वास्तविक संख्याओं का समुच्चय), $a\neq -1$, के लिए $\lim\limits_{n\to\infty}\frac{(1^a+2^a+...+n^a)}{(n+1)^{a-1}[(na+1)+(na+2)+...+(na+n)]}=\frac{1}{60}$

तब a = **[JEE(Adv.)2013]**

(A) 5 (B) 7 (C) $\frac{-15}{2}$ (D) $\frac{-17}{2}$

Ans. B

Sol. $$L=\lim_{n\to\infty}\frac{1^a+2^a+.....+n^a}{(n+1)^{a-1}\left[\underbrace{na+na+na+...+na}_{n\text{ times}}+1+2+3+...+n\right]}=\lim_{n\to\infty}\frac{\sum_{r=1}^{n}r^a}{(n+1)^{a-1}\left[n^2a+\frac{n(n+1)}{2}\right]}$$

$$=\lim_{n\to\infty}\frac{\left(\frac{1}{n}\sum_{r=1}^{n}\frac{r^a}{n^a}\right)n^{a+1}}{(n+1)^{a-1}\left[n^2a+\frac{n(n+1)}{2}\right]}=\lim_{n\to\infty}\frac{\left(\frac{1}{n}\sum_{r=1}^{n}\frac{r^a}{n^a}\right)}{\left(\frac{n+1}{n}\right)^{a-1}\left[\frac{n^2a+\frac{n(n+1)}{2}}{n^2}\right]}$$

$$=\frac{\int_0^1 x^a dx}{\left(a+\frac{1}{2}\right)}=\frac{1}{60}\quad\Rightarrow\frac{2}{(a+1)(2a+1)}=\frac{1}{60}$$

$\Rightarrow 2a^2+3a-119=0 \quad\Rightarrow\quad a=7\ \&\ -\frac{17}{2}$

$a=-\frac{17}{2}$ will be rejected as $\int_0^1 x^{-\frac{17}{2}}dx$ is not defined.

$\Rightarrow a=7$

4. Let $f:[a,b]\to[1,\infty)$ be a continuous function and let $g:R\to R$ be defined as $g(x)=\begin{cases}0 & \text{if } x<a,\\ \int_a^x f(t)\,dt & \text{if } a\le x\le b,\\ \int_a^b f(t)\,dt & \text{if } x>b\end{cases}$

Then: **[JEE(Adv.)2014]**

(A) g(x) is continuous but not differentiable at a

(B) g(x) is differentiable on R

(C) g(x) is continuous but not differentiable at b

(D) g(x) is continuous and differentiable at either a or b but not both

माना कि $f:[a,b]\to[1,\infty)$ एक सतत फलन है तथा $g:R\to R$ निम्नानुसार $g(x)=\begin{cases}0 & if\ x<a,\\ \int_a^x f(t)\,dt & if\ a\le x\le b,\\ \int_a^b f(t)\,dt & if\ x>b\end{cases}$ परिभाषित है तब:

[JEE(Adv.)2014]

(A) a पर g(x) सतत (continuous) है परन्तु अवकलनीय नहीं है

(B) R पर g(x) अवकलनीय है

(C) b पर g(x) संतत है परन्तु अवकलनीय नहीं है।

(D) a या b पर g(x) संतत एवं अवकलनीय है परन्तु दोनों पर नहीं

Ans. AC

Sol. $\lim\limits_{x\to a^-} g(x)=0$

$$\lim_{x\to a^+} g(x)=\lim_{x\to a^+}\int_a^x f(t)dt=\int_a^a f(t)dt=0$$

$$g(a)=\int_a^a f(t)dt=0$$

Similarly at x = b we will be get continuous

So g (x) is continuous $\forall\ x\in R$

$$g'(x)=\begin{cases}0 & x<a\\ f(x) & a\le x\le b\\ 0 & x>b\end{cases}$$

$g'(a^-)=0 \qquad g'(b^-)=f(b)$

$g'(a^+)=f(a) \qquad g'(b^+)=0$

Since f(x) co-domain is [1, ∞) f(a) and f(b) can never be zero.

Hence it is non derivable at x = a and x = b

5. The value of $\int_0^1 4x^3\left\{\frac{d^2}{dx^2}(1-x^2)^5\right\}dx$ is : **[JEE(Adv.)2014]**

निम्न $\int_0^1 4x^3\left\{\frac{d^2}{dx^2}(1-x^2)^5\right\}dx$ का मान है : **[JEE(Adv.)2014]**

Ans. 2

Sol. $\left(4x^3.\frac{d}{dx}\left(1-x^2\right)^5\right)_0^1 - 12\int_0^1 x^2.\frac{d}{dx}\left(1-x^2\right)^5 dx$

$= \left[4x^3 \cdot 5\left(1-x^2\right)^4 \cdot -2x\right]_0^1 - 12\left[x^2\left(1-x^2\right)^5\right]_0^1 + 24\int_0^1 x\left(1-x^2\right)^5 dx = 2$

Alternative :

$\int_0^1 4x^3\left\{\frac{d^2}{dx^2}\left(1-x^2\right)^5\right\}dx$

$\frac{d}{dx}\left(\frac{d\left(1-x^2\right)^5}{dx}\right) = \frac{d}{dx}\left(5\left(1-x^2\right)^4\left(-13x\right)\right)$

$= -10\frac{d}{dx}\left(x\left(1-x^2\right)^4\right)$

$= -10\left[(1-x^2)^4 + x^4(1-x^2)^3(-2x)\right]$

$= \left[-10(1-x^2)^3\right]\left[1-x^2-8x^2\right]$

Hence Integral

$= -40\int_0^1 x^3\left(1-x^2\right)^3\left(1-9x^2\right)dx$ Put $x = \sin\theta$

$= -40\int_0^{\pi/2} \sin^3\theta\cos^7\theta d\theta + 360\int_0^1 \sin^5\theta\cos^7\theta d\theta$

$= -40.1.\frac{2.6.4.2}{10.8.6.2} + 360.\frac{4.2.6.4.2}{12.10.8.6.4.2} = -1 + 3 = 2$

6. The following integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2\,\text{cosec}\ x)^{17}\,dx$ is equal to : **[JEE(Adv.)2014]**

निम्न समाकल $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2\,\text{cosec}\ x)^{17}\,dx$ नीचे दिये गये विकल्पो में से किसके समान है : **[JEE(Adv.)2014]**

(A) $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16}\,du$ (B) $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17}\,du$

(C) $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17}\,du$ (D) $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16}\,du$

Ans. A

Sol. $I = \int_{\pi/4}^{\pi/2} \left(2\,\mathrm{cosec}\,x\right)^{17} dx$

Put $\ell n\ \tan\frac{x}{2} = t \qquad \Rightarrow \tan\frac{x}{2} = e^t$

$$\Rightarrow \sin x = \frac{2e^t}{1+e^{2t}}$$

$$\mathrm{cosec}\,x = \frac{e^t + e^{-t}}{2}$$

$$I = 2\int_{\ell n\left(\sqrt{2}-1\right)}^{0} \left(e^t + e^{-t}\right)^{16}.dt$$

$$= 2\int_{-\ell n\left(\sqrt{2}+1\right)}^{0} \left(e^t + e^{-t}\right)^{16}.dt$$

since $(e^t + e^{-t})^{16}$ is an even function

Hence $I = \int_{0}^{\ell n\left(\sqrt{2}+1\right)} 2\left(e^t + e^{-t}\right)^{16} dt$

7. Let f:[0,2] $\to$ R be a function which is continuous on [0,2] and is differentiable on (0,2) with f(0) = 1. Let $F(x) = \int_0^{x^2} f(\sqrt{t})\,dt$ for x $\in$ [0,2]. If F' (x) = f'(x) for all x $\in$ (0, 2), then F(2) equals : **[JEE(Adv.)2014]**

माना कि f:[0,2] $\to$ R एक ऐसा फलन है जो [0,2] पर संतत है एवं (0,2) पर अवकलनीय है तथा f(0) = 1 है। माना कि सभी x $\in$ [0,2] के लिये माना $F(x) = \int_0^{x^2} f(\sqrt{t})\,dt$ है। यदि सभी x $\in$ (0,2) के लिये F' (x) = f'(x) है, तब F(2) का मान है :

[JEE(Adv.)2014]

(A) $e^2 - 1$ (B) $e^4 - 1$ (C) e – 1 (D) e^4

Ans. B

Sol. $F'(x) = f(x).2x$

$$\Rightarrow \frac{F'(x)}{f(x)} = 2x$$

$$\Rightarrow \int \frac{f'(x)}{f(x)}dx = \int 2x\,dx \left\{\because f'(x) = F'(x)\right\}$$

$$\Rightarrow F(x) = e^{x^2} + C$$

$\therefore F(0)=0 \Rightarrow C=-1$

$\Rightarrow F(x)=e^{x^2}-1$

$\Rightarrow F(2)=e^4-1$

Paragraph For Question Nos. 8 to 9

Given that for each a $\in (0,1)$, $\lim\limits_{h\to 0^+}\int_h^{1-h} t^{-a}(1-t)^{a-1}dt$ exists. Let this limit be g(a). In addition, it is given that the function g(a) is differentiable on (0, 1).

दिया गया है कि प्रत्येक a $\in (0,1)$ के लिये सीमा $\lim\limits_{h\to 0^+}\int_h^{1-h} t^{-a}(1-t)^{a-1}dt$ अस्तित्व में है। माना कि यह सीमा g(a) है। इसके अतिरिक्त यह भी दिया गया है कि अन्तराल (0, 1) पर फलन g(a) अवकलनीय है।

8. The value of $g\left(\frac{1}{2}\right)$ is : **[JEE(Adv.)2014]**

$g\left(\frac{1}{2}\right)$ का मान है : **[JEE(Adv.)2014]**

(A) π (B) 2π (C) $\pi/2$ (D) $\pi/4$

Ans. A

Sol. Uisng leibniz rule

$$g'(a)=\int_h^{1-h}\frac{\delta}{\delta a}t^{-a}(1-t)^{a-1}dt$$

$$=-\int_h^{1-h}t^{-a}(1-t)^{a-1}dt+t^{-a}(1-t)^{a-1}dt=0$$

$g(a)=\text{constant} \quad \Rightarrow g(a)=\lambda$

$$=-\int_h^{1-h}t^{-a}(1-t)^{a-1}dt+t^{-a}(1-t)^{a-1}dt=0$$

$$=\int_h^{1-h}\frac{dt}{\sqrt{-\left(t-\frac{1}{2}\right)^2+\frac{1}{4}}}=\left(\sin^{-1}\frac{t-\frac{1}{2}}{\frac{1}{2}}\right)_h^{1-h}=\sin^{-1}(2t-1)\Big|_h^{1-h}=\sin^{-1}(1-2h)-\sin^{-1}(2h-1)=\pi$$

9. The value of $g'\left(\frac{1}{2}\right)$ is : **[JEE(Adv.)2014]**

$g'\left(\frac{1}{2}\right)$ का मान है : **[JEE(Adv.)2014]**

(A) $\pi/2$ (B) π (C) $-\pi/2$ (D) 0

Ans. D

Sol. $g(a) = \lim\limits_{h\to 0^+} \int\limits_h^{1-h} t^{-a}(1-t)^{a-1}\,dt$

$$g(1-a) = \lim_{h\to 0^+} \int_h^{1-h} t^{-(1-a)}(1-t)^{(1-a)-1}\,dt = \lim_{h\to 0^+} \int_h^{1-h} t^{a-1}(1-t)^{-a}\,dt$$

$$= \lim_{h\to 0^+} \int_h^{1-h} (1-t)^{a-1}\left(1-(1-t)\right)^{-a}dt \quad \left\{\text{by}\int_a^b f(x)dx = \int_a^b f(a+b-x)\,dx\right\}$$

$$= \lim_{h\to 0^+} \int_h^{1-h} (1-t)^{a-1}t^{-a}dt$$

$g(1-a) = g(a)$

$-\,g'(1-a) = g'(a)$

at $a = \frac{1}{2}$

$$-g'\left(\frac{1}{2}\right) = g'\left(\frac{1}{2}\right) \quad \Rightarrow g'\left(\frac{1}{2}\right) = 0$$

10. Match the following column : **[JEE(Adv.)2014]**

Column-I	**Column-II**
P. The number of polynomials f(x) with non-negative integer coefficients of degree $\le$ 2, satisfying $f(0) = 0$ and $\int_0^1 f(x)\,dx = 1$, is	**1.** 8
Q. The number of points in the interval $\left[-\sqrt{13}, \sqrt{13}\right]$ at which $f(x) = \sin(x^2)+\cos(x^2)$ attains its maximum value, is	**2.** 2
R. $\int_{-2}^{2} \frac{3x^2}{(1+e^x)}dx$ equals	**3.** 4
S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right)dx\right)}{\left(\int_{0}^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right)dx\right)}$ equals	**4.** 0

	P	Q	R	S		P	Q	R	S
(A)	3	2	4	1	(B)	2	3	4	1
(C)	3	2	1	4	(D)	2	3	1	4

निम्न कॉलम को सुमेलित कीजिए : **[JEE(Adv.)2014]**

Column-I		**Column-II**	
P. अऋणात्मक पूर्णांक गुणांकों वाले बहुपदों f(x), जिनकी घात ≤ 2 है तथा जो $f(0)=0$ एवं $\int_0^1 f(x)\,dx=1$ को सन्तुष्ट करती है, की संख्या है		**1.**	8
Q. अंतराल $\left[-\sqrt{13},\sqrt{13}\right]$ में स्थित उन बिन्दुओं की संख्या जिन पर $f(x)=\sin(x^2)+\cos(x^2)$ का मान अधिकतम है,		**2.**	2
R. $\int_{-2}^{2}\frac{3x^2}{(1+e^x)}dx$ का मान है		**3.**	4
S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}}\cos 2x\log\left(\frac{1+x}{1-x}\right)dx\right)}{\left(\int_{0}^{\frac{1}{2}}\cos 2x\log\left(\frac{1+x}{1-x}\right)dx\right)}$ का मान है		**4.**	0

	P	Q	R	S		P	Q	R	S
(A)	3	2	4	1	(B)	2	3	4	1
(C)	3	2	1	4	(D)	2	3	1	4

Ans. D

Sol. (P) Let $f(x) = ax^2 + bx$, $a, b \in W$ (as $f(0) = 0$)

$$\int_1^0 \left(ax^2+bx\right)dx = \frac{a}{3}+\frac{b}{2}=1 \quad \Rightarrow\ 2a+3b=6$$

$(a,b) \equiv (3, 0), (0, 2)$

$\Rightarrow$ Number of such polynomials = 2

(Q) $f(x)=\sqrt{2}\sin\left(x^2+\frac{\pi}{4}\right)$

if f(x) is maximum

$$x^2+\frac{\pi}{4}=2n\pi+\frac{\pi}{2}$$

$$x^2=2n\pi+\frac{\pi}{4}$$

$$\left\{\therefore x\in\left[-\sqrt{13},\sqrt{13}\right]\right\}$$

$\Rightarrow n = 0 \text{ or } 1 \text{ or } -1 \text{ or } -2$

(R) $\int_{-2}^{2} \frac{3x^2}{1+e^x}dx = \int_{0}^{2} 3x^2\left(\frac{1}{1+e^x}+\frac{1}{1+e^{-x}}\right)dx \left\{\int_{-a}^{a} f(x)dx = \int_{0}^{a}\left(f(x)+f(-x)\right)dx\right\}$

$= \int_{300}^{2} 3x^2\left(\frac{1}{1+e^x}+\frac{e^x}{1+e^x}\right)dx \int_{0}^{2} 3x^2dx = x^3\Big|_0^2 = 8$

(S) $\int_{-1/2}^{1/2} \cos 2x \ln\left(\frac{1+x}{1-x}\right)dx = 0$ (as it is an odd function)

Hence P $\rightarrow$ 2, Q $\rightarrow$ 3, R $\rightarrow$ 1, S $\rightarrow$ 4

11. Let f: $(0, \infty) \rightarrow R$ be given by $f(x) = \int_{\frac{1}{x}}^{x} e^{-\left(t+\frac{1}{t}\right)}\frac{dt}{t}$. then: **[JEE(Adv.)2014]**

(A) f(x) is monotonically increasing on $[1, \infty)$ (B) f(x) is monotonically decreasing on (0, 1)

(C) $f(x) + f\left(\frac{1}{x}\right) = 0$, for all $x \in (0, \infty)$ (D) $f(2^x)$ is an odd function of x on R

माना कि f: $(0, \infty) \rightarrow R$ निम्न के द्वारा $f(x) = \int_{\frac{1}{x}}^{x} e^{-\left(t+\frac{1}{t}\right)}\frac{dt}{t}$ परिभाषित है। तब : **[JEE(Adv.)2014]**

(A) $[1, \infty)$ पर f(x) एकदिष्ट वर्धमान है (B) (0, 1) पर f(x) एकदिष्ट ह्रासमान है

(C) सभी $x \in (0, \infty)$ के लिये, $f(x) + f\left(\frac{1}{x}\right) = 0$ (D) R पर $f(2^x)$, x का एक विषम फलन है

Ans. ACD

Sol. $f(x) = \int_{\frac{1}{x}}^{x} e\frac{-\left(t+\frac{1}{t}\right)}{t}dt$

$\Rightarrow f'(x) = \frac{e^{-\left(x+\frac{1}{x}\right)}}{x} + \frac{1}{x^2}e^{-\left(-x+\frac{1}{x}\right)}.x$

$= e^{-\left(x+\frac{1}{x}\right)}.\frac{2}{x}$

$\Rightarrow f'(x) > 0 \forall x \in (0,\infty)$

$\Rightarrow f(x)$ is increasing in $(0, \infty)$

$\Rightarrow f(x)$ is increasing iin $(0, \infty)$

Now $f(x)+f\left(\frac{1}{x}\right)=\int_{\frac{1}{x}}^{x}\frac{e^{-\left(t+\frac{1}{t}\right)}}{t}dt+\int_{x}^{\frac{1}{x}}\frac{e^{-\left(t+\frac{1}{t}\right)}}{t}dt$

$=\int_{\frac{1}{x}}^{\frac{1}{x}}\frac{e^{-\left(t+\frac{1}{t}\right)}}{t}dt=0$

Similarly

$f\left(2^{x}\right)+f\left(\frac{1}{2^{x}}\right)=0$

$\Rightarrow f\left(2^{-x}\right)=-f\left(2^{x}\right)$

12. Let $f: R \to R$ be a function defined by $f(x)=\begin{cases}[x], & x \le 2\\ 0, & x>2\end{cases}$ where [x] is the greatest integer less than or equal to x. If $I=\int_{-1}^{2}\frac{x f\left(x^{2}\right)}{2+f(x+1)}dx$, then the value of $(4I-1)$ is : **[JEE(Adv.)2015]**

माना कि फलन $f: R \to R$, $f(x)=\begin{cases}[x], & x \le 2\\ 0, & x>2\end{cases}$ से परिभाषित है, जहाँ [x], x से कम या x के बराबर के महतम पूर्णांक को दर्शाता gSA यदि $I=\int_{-1}^{2}\frac{x f\left(x^{2}\right)}{2+f(x+1)}dx$, तब $(4I-1)$ का मान है : **[JEE(Adv.)2015]**

Ans. 0

Sol. $I=\int_{-1}^{2}\frac{x\left[x^{2}\right]}{2+[x+1]}dx=\int_{-1}^{2}\frac{x\left[x^{2}\right]}{2+[x+1]}dx=\int_{-1}^{0}\frac{0}{2+0}dx+\int_{0}^{1}\frac{0}{2+0}dx+\int_{1}^{\sqrt{2}}\frac{x.1}{2+2}dx$

$=\frac{1}{4}$

$\therefore 4I-1=0$

13. If $\alpha = \int_0^1 \left(e^{9x+3\tan^{-1}x}\right)\left(\frac{12+9x^2}{1+x^2}\right) dx$ where $\tan^{-1}x$ takes only principal values, then the value of $\left(\log_e|1+\alpha| - \frac{3\pi}{4}\right)$ is : **[JEE(Adv.)2015]**

यदि $\alpha = \int_0^1 \left(e^{9x+3\tan^{-1}x}\right)\left(\frac{12+9x^2}{1+x^2}\right) dx$ जहाँ $\tan^{-1}x$ केवल मुख्य मानों (principal values) को लेता है, तब $\left(\log_e|1+\alpha| - \frac{3\pi}{4}\right)$ का मान है : **[JEE(Adv.)2015]**

Ans. 9

Sol. $\alpha = \int_0^1 e^{9x+3\tan^{-1}x}.\left(\frac{12+9x^2}{1+x^2}\right)dx$

$\Rightarrow \quad \alpha = \left(e^{9x+3\tan^{-1}x}\right)_0^1$

$\Rightarrow \quad \alpha = e^{9+\frac{3\pi}{4}} - 1$

$\Rightarrow \quad \ln(1+\alpha) = 9 + \frac{3\pi}{4}$

Alter :

$\alpha = \int_0^1 e^{\left(9x+3\tan^{-1}x\right)}\left(\frac{12+9x^2}{1+x^2}\right)dx$

Let $9x + 3\tan^{-1}x = t$

$\Rightarrow \left(9+\frac{3}{1+x^2}\right)dx - dt \Rightarrow \left(\frac{12+9x^2}{1+x^2}\right)dx = dt$

$\Rightarrow \alpha = \int_0^{9+3\pi/4} e^t dt = \left(e^t\right)_0^{9+3\pi/4} = e^{9+3\pi/4} - 1$

Now $\log_e|1+\alpha| - 3\pi/4 = \log_e e^{(9+3\pi/4)} - 3\pi/4 = 9$

14. Let $f : R \to R$ be a continuous odd function, which vanishes exactly at one point and $f(1) = \frac{1}{2}$. Suppose that $F(x) = \int_{-1}^{x} f(t)\,dt$ for all $x \in [-1, 2]$ and $G(x) = \int_{-1}^{x} t\left|f\left(f(t)\right)\right| dt$ for all $x \in [-1, 2]$. If $\lim_{x\to 1}\frac{F(x)}{G(x)} = \frac{1}{14}$, then the value of $f\left(\frac{1}{2}\right)$ is : **[JEE(Adv.)2015]**

माना कि $f: R \to R$ एक संतत विषम फलन है जिसका मान केवल एक बिन्दु पर ही शून्य होता है तथा $f(1) = \frac{1}{2}$ है। माना कि सभी x $\in [-1, 2]$ के लिए $F(x) = \int_{-1}^{x} f(t)dt$ एवं सभी $x \in [-1, 2]$ के लिए $G(x) = \int_{-1}^{x} t\left|f(f(t))\right| dt$ हैं। यदि $\lim_{x\to 1} \frac{F(x)}{G(x)} = \frac{1}{14}$ है,

r c $f\left(\frac{1}{2}\right)$ का मान है : **[JEE(Adv.)2015]**

Ans. 7

Sol. $F(x) = \int_{-11}^{x} f(t)dt = \int_{1}^{x} f(t)dt$

$G(x) = \int_{-1}^{x} t\,|f(f(t))|dt = \int_{-1}^{x} t\,|f(f(t))|\,dt$

$\lim_{x\to 1} \frac{F(x)}{G(x)} = \frac{1}{14}$

L'hospitals

$\lim_{x\to 1} \frac{f(x)}{x\,|f(f(x))|} = \frac{1}{14}$

$\frac{\frac{1}{2}}{1\left|f\left(\frac{1}{2}\right)\right|} = \frac{1}{14}$

$f\left(\frac{1}{2}\right) = 7$

15. The option (s) with the values of a and L that satisfy the following equation is (are) $\frac{\int_{0}^{4\pi} e^t\left(\sin^6 at + \cos^4 at\right)dt}{\int_{0}^{\pi} e^t\left(\sin^6 at + \cos^4 at\right)dt} = L?$

[JEE(Adv.)2015]

निम्नलिखित में से a तथा L के कौन सा (से) मान समीकरण $\frac{\int_{0}^{4\pi} e^t\left(\sin^6 at + \cos^4 at\right)dt}{\int_{0}^{\pi} e^t\left(\sin^6 at + \cos^4 at\right)dt} = L?$ **[JEE(Adv.)2015]**

(A) $a=2,\ L=\dfrac{e^{4\pi}-1}{e^{\pi}-1}$ (B) $a=2,\ L=\dfrac{e^{4\pi}+1}{e^{\pi}+1}$ (C) $a=4,\ L=\dfrac{e^{4\pi}-1}{e^{\pi}-1}$ (D) $a=4,\ L=\dfrac{e^{4\pi}+1}{e^{\pi}+1}$

Ans. AC

Sol. $I_1=\int_0^{\pi} e^t\left(\sin^6 at+\cos^4 at\right)dt+\int_{\pi}^{2\pi} e^t\left(\sin^6 at+\cos^4 at\right)dt+\int_{2\pi}^{3\pi} e^t\left(\sin^6 at+\cos^4 at\right)dt$

$+\int_{2\pi}^{4\pi} e^t\left(\sin^6 at+\cos^4 at\right)dt$

$=\left(1+e^{\pi}+e^{2\pi}+e^{3\pi}\right)\int_{\pi}^{2\pi} e^t\left(\sin^6 at+\cos^4 at\right)dt$

$=\left(1+e^{\pi}+e^{2\pi}+e^{3\pi}\right)\int_{\pi}^{2\pi} e^t\left(\sin^6 at+\cos^4 at\right)dt$

$\Rightarrow \dfrac{I_1}{I_2}=1+e^{\pi}+e^{2\pi}+e^{3\pi}=\dfrac{e^{4\pi}-1}{e^{\pi}-1}$

16. Let $f(x)=7\tan^8 x+7\tan^6 x-3\tan^4 x-3\tan^2 x$ for all $x\in\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$. Then the correct expression (s) is (are) :

[JEE(Adv.)2015]

माना कि सभी $x\in\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$ के लिए, $f(x)=7\tan^8 x+7\tan^6 x-3\tan^4 x-3\tan^2 x$ है, तब सही कथन है (हैं) :

[JEE(Adv.)2015]

(A) $\int_0^{\pi/4} x f(x)dx=\dfrac{1}{12}$ (B) $\int_0^{\pi/4} f(x)dx=0$ (C) $\int_0^{\pi/4} xf(x)dx=\dfrac{1}{6}$ (D) $\int_0^{\pi/4} f(x)dx=1$

Ans. AB

Sol. $f(x)=(7\tan^6 x-3\tan^2 x).\sec^2 x$

$\therefore \int_0^{\pi/4} f(x)dx=\int_0^1\left(7t^6-3t^2\right)dt=\left(t^7-t^3\right)_0^1=0$

Now $\int_0^{\pi/4} f(x)dx=\int_0^1 \underbrace{\left(7t^6-3t^2\right)}_{II}\underbrace{\tan^{-1}t}_{I}dt$

$=\left(\tan^{-1}t.\left(t^7-t^3\right)\right)_0^1-\int_0^1\left(t^7-t^3\right).\dfrac{1}{1+t^2}dt$

$$= \int_0^1 \frac{t^3\left(1-t^4\right)}{1+t^2}dt = \int_0^1 t^3\left(1-t^2\right)dt$$

$$= \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$$

17. Let $f'(x) = \frac{192x^3}{2+\sin^4 \pi x}$ for all $x \in R$ with $f\left(\frac{1}{2}\right) = 0$. If $m \le \int_{1/2}^{1} f(x)dx \le M$, then the possible values of m and M are : **[JEE(Adv.)2015]**

माना कि सभी $x \in R$ के लिए, $f'(x) = \frac{192x^3}{2+\sin^4 \pi x}$ एवं $f\left(\frac{1}{2}\right) = 0$ है। यदि $m \le \int_{1/2}^{1} f(x)dx \le M$, तब m और M के सही l alHo eku gS(gSa) : **[JEE(Adv.)2015]**

(A) m = 13, M = 24 (B) $m = \frac{1}{4}, M = \frac{1}{2}$ (C) m = –11, M = 0 (D) m = 1, M = 12

Ans. D

Sol. $f'(x) = \frac{192x^3}{2+\sin^4(\pi x)} \quad \forall\, x \in R$

If $x \in \left[\frac{1}{2}, 1\right]$ then $f''(x) > 0$

$\Rightarrow f'\left(\frac{1}{2}\right) \le f'(x) \le f'(1)$

$\Rightarrow \frac{24}{3} \le f'(x) \le \frac{192}{2}$

$8 \le f'(x) \le 96$

$\int_{1/2}^{x} 8dx \le \int_{1/2}^{x} f'(x)\,dx \le \int_{1/2}^{x} 96$

$8\left(x - \frac{1}{2}\right) \le f(x) - f\left(\frac{1}{2}\right) \le 96\left(x - \frac{1}{2}\right)$

$8x - 4 \le f(x) \le 96x - 48$

$\int_{1/2}^{1} (8x-4)dx \le \int_{1/2}^{1} f(x)\,dx \le \int_{1/2}^{1} (96x-48)dx$

$\left[4x^2 - 4x\right]_{1/2}^{1} \le \int_{1/2}^{1} f(x)dx \le \left[48x^2 - 48x\right]_{1/2}^{1}$

$1 \le \int_{1/2}^{1} f(x)dx \le 12$

$\Rightarrow m = 1$ & $m = 12$

Paragraph for Question Nos. 18 to 19

Let $F : R \to R$ be a thrice differentiable function. Suppose that $F(1) = 0$, $F(3) = -4$ and $F'(x) < 0$ for all $x \in (1/2, 3)$. Let $f(x) = xF(x)$ for all $x \in R$.

माना कि $F : R \to R$ एक फलन है जो तीन बार अवकलनीय (thrice differentiable) है। माना कि $F(1) = 0$, $F(3) = -4$ और सभी $x \in (1/2, 3)$ के लिए, $F'(x) < 0$ है। माना कि सभी $x \in R$ के लिए, $f(x) = xF(x)$ है।

18. The correct statement (s) is (are) : **[JEE(Adv.)2015]**

(A) $f'(1) < 0$ (B) $f(2) < 0$

(C) $f'(x) \neq 0$ for any $x \in (1, 3)$ (D) $f'(x) = 0$ for some $x \in (1, 3)$

निम्नलिखित में से सही कथन है (हैं) : **[JEE(Adv.)2015]**

(A) $f'(1) < 0$ (B) $f(2) < 0$

(C) किसी भी $x \in (1, 3)$ के लिए $f'(x) \neq 0$ (D) कुछ $x \in (1, 3)$ के लिए $f'(x) = 0$

Ans. ABC

Sol. $f'(x) = x\,F'(x) + F(x)$

$\Rightarrow f'(1) = F'(1) + F(1) = F'(1) < 0 \Rightarrow$ (A)

$f(2) = 2F(2) < 0 \Rightarrow$ (B)

for $x \in (1, 3)$ $f'(x) = xF'(x) + F(x) < 0 \Rightarrow$ (C)

19. If $\int_1^3 x^2F'(x)\,dx = -12$ and $\int_1^3 x^3F''(x)\,dx = 40$, then the correct expression (s) is (are) :

[JEE(Adv.)2015]

यदि $\int_1^3 x^2F'(x)\,dx = -12$ और $\int_1^3 x^3F''(x)\,dx = 40$ है, तब सही कथन है (हैं) : **[JEE(Adv.)2015]**

(A) $9f'(3) + f'(1) - 32 = 0$ (B) $\int_1^3 f(x)\,dx = 12$

(C) $9f'(3) - f'(1) + 32 = 0$ (D) $\int_1^3 f(x)\,dx = -12$

Ans. CD

Sol. $\int_1^3 x^2F'(x)\,dx = -12$

$\Rightarrow \left[x^2F(x)\right]_1^3 - 2\int_1^3 xF(x)\,dx = -12$

$$\Rightarrow \left[x f(x)\right]_1^3 - 2\int_1^3 f(x)\,dx = -12$$

$$\Rightarrow \int_1^3 f(x)\,dx = -12$$

Now $\int_1^3 x^3 F''(x)\,dx = \left[x^3 F'(x)\right]_1^3 - 3\int_1^3 x^2 F'(x)\,dx = 40$

$$\Rightarrow 27F'(3) - F'(1) + 36 = 40$$

$$9f'(3) - f'(1) + 32 = 0$$

20. The total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^2}{1+t^4}\,dt = 2x - 1$ is. **[JEE(Adv.)2016]**

ऐसे सभी भिन्न $x \in [0, 1]$, जिनके लिए $\int_0^x \frac{t^2}{1+t^4}\,dt = 2x - 1$ है, की कुल संख्या है। **[JEE(Adv.)2016]**

Ans. 1

Sol. Let $f(x) = \int_0^x \frac{t^2}{1+t^4}\,dt$

Differentiating both sides, we get

$$f'(x) = \frac{x^2}{1+x^4}$$

Observing $f'(x)$, we can say that, $f'(x) > 0$

$\Rightarrow$ f(x) is an increasing function

Now at x = 0, f(0) = 0 and at x = 1, f(1) $= \int_0^1 \frac{t^2}{1+t^4}\,dt$

Because, $0 < \frac{x^2}{1+x^4} < 1$

$$\Rightarrow 0 < \int_0^1 \frac{t^2}{1+t^4}\,dt < \int_0^1 1\,dt$$

$$\Rightarrow 0 < \int_0^1 \frac{t^2}{1+t^2}\,dt < 1$$

Plotting graph of y = f(x) and y = 2x – 1, we find that they intersect at only one point

21. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{1+e^x} dx$ is equal to : **[JEE(Adv.)2016]**

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{1+e^x} dx$ का मान है : **[JEE(Adv.)2016]**

(A) $\frac{\pi^2}{4}-2$ (B) $\frac{\pi^2}{4}+2$ (C) $\pi^2-e^{\frac{\pi}{2}}$ (D) $\pi^2+e^{\frac{\pi}{2}}$

Ans. A

Sol. Apply $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

Let $I = \int_{-\pi/2}^{\pi/2} \frac{x^2 \cos x}{1+e^x} dx$ $\Rightarrow 2I = \int_{-\pi/2}^{\pi/2} \frac{x^2 \cos x}{1+e^x} dx + \int_{-\pi/2}^{\pi/2} \frac{x^2 \cos x}{1+e^{-x}} dx$

$I = \int_0^{\pi/2} x^2 \cos x\, dx = \left[x^2 \sin x - 2x(-\cos x) + 2(-\sin x)\right]_0^{\pi/2} = \frac{\pi^2}{4}-2$

22. Let $f(x) = \lim_{n\to\infty} \left(\frac{n^n (x+n)(x+\frac{n}{2}).....(x+\frac{n}{n})}{n!(x^2+n^2)(x^2+\frac{n^2}{4}).....(x^2+\frac{n^2}{n^2})} \right)^{\frac{x}{n}}$, for all $x > 0$. Then : **[JEE(Adv.)2016]**

माना कि सभी $x > 0$ के लिए $f(x) = \lim_{n\to\infty} \left(\frac{n^n (x+n)(x+\frac{n}{2}).....(x+\frac{n}{n})}{n!(x^2+n^2)(x^2+\frac{n^2}{4}).....(x^2+\frac{n^2}{n^2})} \right)^{\frac{x}{n}}$ है। तब : **[JEE(Adv.)2016]**

(A) $f\left(\frac{1}{2}\right) \geq f(1)$ (B) $f\left(\frac{1}{3}\right) \leq f\left(\frac{2}{3}\right)$ (C) $f'(2) \leq 0$ (D) $\frac{f'(3)}{f(3)} \geq \frac{f'(2)}{f(2)}$

Ans. BC

Sol. $$f(x)=\lim_{n\to\infty}\left(\frac{\prod_{r=1}^{n}\left(x+\frac{n}{r}\right)}{\prod_{r=1}^{n}\left(\frac{r}{n}\right)\prod_{r=1}^{n}\left(x^2+\frac{n^2}{r^2}\right)}\right)^{\frac{x}{n}}$$

$$\Rightarrow \ln f(x)=x\lim_{n\to\infty}\left[\left\{\sum_{r=1}^{n}\log\left(x+\frac{1}{r/n}\right).\frac{1}{n}\right\}-\left\{\sum_{r=1}^{n}\log\left(\frac{r}{n}\right).\frac{1}{n}\right\}-\left\{\sum_{r=1}^{n}\log\left(x^2+\frac{1}{(r/n)^2}\right).\frac{1}{n}\right\}\right]$$

$$\Rightarrow \ln f(x)=x\int_0^1\log\left(x+\frac{1}{t}\right)dt-\int_0^1\log(t)\,dt-\int_0^1\log\left(x^2+\frac{1}{t^2}\right)dt$$

$$\Rightarrow \ln f(x)=x\int_0^1\log\left(\frac{tx+1}{t^2x^2+1}\right)dt$$

$$\text{put } tx=u$$

$$\Rightarrow \ln f(x)=\int_0^x\log\left(\frac{u+1}{u^2+1}\right)du$$

$$\Rightarrow f(x)=\int_0^x\log\left(\frac{u+1}{u^2+1}\right)du$$

23. If $I=\sum_{k=1}^{98}\int_k^{k+1}\frac{k+1}{x(x+1)}dx$, then : **[JEE(Adv.)2017]**

यदि $I=\sum_{k=1}^{98}\int_k^{k+1}\frac{k+1}{x(x+1)}dx$, तब : **[JEE(Adv.)2017]**

(A) $I>\frac{49}{50}$ (B) $I<\frac{49}{50}$ (C) $I<\log_e 99$ (D) $I>\log_e 99$

Ans. AC

Sol. $$I=\sum_{k=1}^{98}\left(\int_k^{k+1}\frac{k+1}{x(x+1)}dx\right)$$

For $\int_k^{k+1}\frac{k+1}{x(x+1)}dx$

$$\frac{k+1}{x(x+1)}<\frac{k+1}{x(k+1)}\ (\because \text{ Least value of } x+1 \text{ is } k+1)$$

$$\Rightarrow \frac{k+1}{x(x+1)} < \frac{1}{x}$$

$x < k + 1$

$$\frac{k+1}{x} > 1$$

$$\Rightarrow \frac{k+1}{x(x+1)} > \frac{1}{x+1}$$

$$\Rightarrow \frac{1}{x+1} < \frac{k+1}{x(x+1)} < \frac{1}{x}$$

$$\Rightarrow \int_k^{k+1} \frac{1}{x+1}\,dx < \int_k^{k+1} \frac{k+1}{x(x+1)}\,dx < \int_k^{k+1} \frac{1}{x}\,dx$$

$$\Rightarrow \left[\ln(x+1)\right]_k^{k+1} < \int_k^{k+1} \frac{k+1}{x(x+1)}\,dx < (\ln x)_k^{k+1}$$

$$\sum_{k=1}^{98} \ln\left(\frac{k+2}{k+1}\right) < I < \sum_{k=1}^{98} \ln\left(\frac{k+1}{k}\right)$$

$$\sum_{k=1}^{98} \ln\left(\frac{k+2}{k+1}\right) = (\ln 3 - \ln 2 + \ln 4 - \ln 3 + \ldots\ldots + \ln 100 - \ln 99)$$

$$= \ln\left(\frac{100}{2}\right) = \ln 50$$

$$\sum_{k=1}^{98} \ln\left(\frac{k+1}{k+1}\right) = \sum_{k=1}^{98} \left(\ln(k+1) - \ln k\right)$$

$= \ln 2 - \ln 1 + \ln 3 - \ln 2 + \ldots\ldots + \ln 99 - \ln 98$

$= \ln(99)$

$\Rightarrow \ln(50) < I < \ln(99)$

as $\ln(50) > \frac{49}{50}$

$$\Rightarrow I > \frac{49}{50} \text{ \& } I < \ln(99)$$

24. If $g(x) = \int_{\sin x}^{\sin(2x)} \sin^{-1}(t)dt$, then : **[JEE(Adv.)2017]**

यदि $g(x) = \int_{\sin x}^{\sin(2x)} \sin^{-1}(t)dt$ है, तब : **[JEE(Adv.)2017]**

(A) $g'\left(-\frac{\pi}{2}\right)=-2\pi$ (B) $g'\left(-\frac{\pi}{2}\right)=2\pi$ (C) $g'\left(\frac{\pi}{2}\right)=2\pi$ (D) $g'\left(\frac{\pi}{2}\right)=-2\pi$

Ans. BONUS

Sol. $g(x)=\int_{\sin x}^{\sin 2x}\sin^{-1}(t)dt$

$g'(x)=\sin^{-1}(\sin 2x)\cdot\cos 2x\cdot 2-\sin^{-1}(\sin x)\cdot\cos x$

$=2\cos 2x\cdot\sin^{-1}(\sin 2x)-\cos x\cdot\sin^{-1}(\sin x)$

$g'\left(-\frac{\pi}{2}\right)=\cos(-\pi)\sin^{-1}(\sin(-\pi))-\cos\left(-\frac{\pi}{2}\right).\sin^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right)=0$

$g'\left(-\frac{\pi}{2}\right)=2\cos(\pi)\sin^{-1}(\sin(\pi))-\cos\left(\frac{\pi}{2}\right).\sin^{-1}\left(\sin\left(\frac{\pi}{2}\right)\right)=0$

25. Let $f: R\to R$ be a differentiable function such that $f(0)=0$, $f\left(\frac{\pi}{2}\right)=3$ and $f'(0)=1$.

If $g(x)=\int_{x}^{\frac{\pi}{2}}[f'(t)\operatorname{cosec} t-\cot t\operatorname{cosec} t\ f(t)]dt$ for $x\in(0,\frac{\pi}{2}]$, then $\lim_{x\to 0} g(x)=$ **[JEE(Adv.)2017]**

माना $f: R\to R$ इस प्रकार का अवकलनीय फलन (differentiable function) है, कि $f(0)=0$, $f\left(\frac{\pi}{2}\right)=3$ तथा $f'(0)=1$ है।

; fin $x\in(0,\frac{\pi}{2}]$ के लिये $g(x)=\int_{x}^{\frac{\pi}{2}}[f'(t)\operatorname{cosec} t-\cot t\operatorname{cosec} t\ f(t)]dt$ है, तब $\lim_{x\to 0} g(x)=$ **[JEE(Adv.)2017]**

Ans. 2

Sol. $g(x)=\left[f(t)\operatorname{cosec} t\right]_0^{\pi/2}$

$=f(\pi/2)-f(x)\operatorname{cosec} x$

$g(x)=3-\frac{f(x)}{\sin x}$

$\lim_{x\to 0} g(x)=3-\lim_{x\to 0}\frac{f(x)}{\sin x}$

$=3-\lim_{x\to 0}\frac{f'(x)}{\cos x}$ (Use L.H. rule)

$=3-1=2$

26. Let $f: \mathbb{R}\to(0,1)$ be a continuous function. Then, which of the following function(s) has(have) the value zero at some point in the interval (0, 1)? **[JEE(Adv.)2017]**

माना कि $f: \mathbb{R} \to (0, 1)$ एक सतत फलन है। तब निम्न फलनों में से कौनसे फलन/फलनों का/के मान अन्तराल (0, 1) के किसी बिन्दु ij 'kwU; gksxk\? : **[JEE(Adv.)2017]**

(A) $x^9 - f(x)$ (B) $f(x) + \int_0^{\frac{\pi}{2}} f(t)\sin t\, dt$ (C) $x - \int_0^{\frac{\pi}{2}-x} f(t)\cos t\, dt$ (D) $e^x - \int_0^x f(t)\sin t\, dt$

Ans. AC

Sol. Let $g(x) = x^9 - f(x)$

$\Rightarrow g(0) = -f(0) < 0$ and $g(1) = 1 - f(1) > 0$

$\Rightarrow g(x) = 0$ for some $x \in (0, 1)$

Let $h(x) = x - \int_0^{\frac{\pi}{2}-x} f(t)\cos t\, dt$

$h(0) = 0 - \int_0^{\frac{\pi}{2}-0} f(t)\cos t\ dt < 0$

$h(1) = 1 - \int_0^{\frac{\pi}{2}-1} f(t)\cos t\ dt > 0$

$\Rightarrow h(x) = 0$ for some $x \in (0, 1)$

$e^x - \int_0^x f(t)\sin t\, dt$

$\because x \in (0,1) \Rightarrow e^x \in (1, e)$

and $0 < f(t) < 1$ and $0 < \sin t < 1$

$e^x - \int_0^x f(t)\sin t\, dt \neq 0$

$f(x) + \int_0^{\frac{\pi}{2}} f(t)\sin t\, dt > 0$

27. The value of the integral $\int_0^{\frac{1}{2}} \frac{1+\sqrt{3}}{\left((x+1)^2(1-x)^6\right)^{1/4}} dx$ is : **[JEE(Adv.)2018]**

समाकल $\int_0^{\frac{1}{2}} \frac{1+\sqrt{3}}{\left((x+1)^2(1-x)^6\right)^{1/4}} dx$ का मान है : **[JEE(Adv.)2018]**

Ans. 2

Sol. $\frac{1+\sqrt{3}}{\left((x+1)^2(1-x)^6\right)^{1/4}}$

$$\int_0^{1/2} \frac{\left(1+\sqrt{3}\right)dx}{\left[\left(1+x\right)^2\left(1-x\right)^6\right]^{1/4}}$$

Put $\frac{1-x}{1+x}=t \quad \Rightarrow \frac{-2dx}{\left(1+x\right)^2}=dt$

$$I=\int_0^{1/3}\frac{\left(1+\sqrt{3}\right)dt}{-2t^{6/4}}=\frac{-\left(1+\sqrt{3}\right)}{2}\times\left|\frac{-2}{\sqrt{t}}\right|_1^{1/3}=\left(1+\sqrt{3}\right)\left(\sqrt{3}-1\right)=2$$

28. For each positive integer n, let $y_n=\frac{1}{n}\left((n+1)(n+2)...(n+n)\right)^{\frac{1}{n}}$. For $x\in R$, let [x] be the greatest integer less than or equal to x. If $\lim_{n\to\infty} y_n = L$, then the value of [L] is_____. **[JEE(Adv.)2018]**

प्रत्येक धनात्मक पूर्णांक n के लिये, माना कि $y_n=\frac{1}{n}\left((n+1)(n+2)...(n+n)\right)^{\frac{1}{n}}$.

$x\in R$ के लिये माना कि [x], x से छोआ या x के बराबर महत्तम पूर्णांक है। यदि $\lim_{n\to\infty} y_n = L$, तब [L] का मान है _____

[JEE(Adv.)2018]

Ans. 1

Sol. $y_n=\left(\frac{n+1}{n}\frac{n+2}{n}.....\frac{n+n}{n}\right)^{1/n}$

$$\log L=\lim_{x\to\infty}\frac{1}{n}\sum_{r=1}^{n}\log\left(1+\frac{r}{n}\right)$$

$$=\int_0^1\log\left(1+x\right)dx=\int_1^2\log x dx=|\,x\log x-x\,|_1^2=2\log 2=\log\frac{4}{3}$$

$$\Rightarrow \quad L=\frac{4}{e} \quad \Rightarrow [L]=1$$

29. For $a\in R$, $|a|>1$, Let $\lim_{n\to\infty}\left(\frac{1+\sqrt[3]{2}+....+\sqrt[3]{n}}{n^{7/3}\left(\frac{1}{(an+1)^2}+\frac{1}{(an+2)^2}+....+\frac{1}{(an+n)^2}\right)}\right)=54$. Then the possible value (s) of a is/are : **[JEE(Adv.)2018]**

माना कि $a \in R, |a| > 1$ के लिए $\lim\limits_{n\to\infty}\left(\dfrac{1+\sqrt[3]{2}+....+\sqrt[3]{n}}{n^{7/3}\left(\dfrac{1}{(an+1)^2}+\dfrac{1}{(an+2)^2}+....+\dfrac{1}{(an+n)^2}\right)}\right) = 54$ तब a का (के) सम्भावित मान

g\$हैं) **[JEE(Adv.)2018]**

(A) –6 (B) 8 (C) 7 (D) –9

Ans. BD

Sol. $\lim\limits_{x\to\infty}\left(\dfrac{1+(2)^{1/3}+(3)^{1/3}....(n)^{1/3}}{n^{1/3}.\dfrac{n^2}{n^2}\left(\dfrac{1}{\left(a+\dfrac{1}{n}\right)^2}+\dfrac{1}{\left(a+\dfrac{2}{n}\right)^2}....\dfrac{1}{\left(a+\dfrac{n}{n}\right)^2}\right)}\right)$

$\lim\limits_{x\to\infty}\left(\dfrac{\dfrac{1}{n}\sum\limits_{r=1}^{n}\left(\dfrac{r}{n}\right)^{1/3}}{\dfrac{1}{n}\sum\limits_{r=1}^{n}\dfrac{1}{\left(a+\dfrac{r}{n}\right)^2}}\right)$

$\dfrac{r}{n}=x, \dfrac{1}{x}=dx$

$\dfrac{\int\limits_0^1 (x)^{1/3}\,dx}{\int\limits_0^1 \dfrac{1}{(a+x)^2}dx}=54$

$\dfrac{\left(\dfrac{x^{4/3}}{4/3}\right)_0^1}{-\left(\dfrac{1}{a+x}\right)_0^1}=54$

$$\frac{\frac{3}{4}}{-\left[\frac{1}{a+1}-\frac{1}{a}\right]}=54$$

$$\frac{a(a+1)}{-(a-a-1)}=72$$

$$a^2+a-72=0$$

$$a^2+9a-8a-72=0$$

$$a=-9,\ a=8$$

30. Ler $f:R\to R$ be given by $f(x)=(x-1)(x-2)(x-5)$ define $F(x)=\int_0^x f(t)dt, x>0$. Then which of the following options is/are correct : **[JEE(Adv.)2019]**

(A) $F(x)\neq 0$ for all $x\in(0,5)$ (B) F has a local maximum at $x=2$

(C) F has a local minimum at $x=1$ (D) F has two local maxima and one local minimum in $(0,\infty)$

माना कि $f:R\to R, f(x)=(x-1)(x-2)(x-5)$ द्वारा दिया गया है। परिभाषित कीजिए। $F(x)=\int_0^x f(t)dt, x>0$ तब निम्न में l sd lkS l k(से) विकल्प सही है (हैं)? **[JEE(Adv.)2019]**

(A) सभी $x\in(0,5)$ के लिए $F(x)\neq 0$

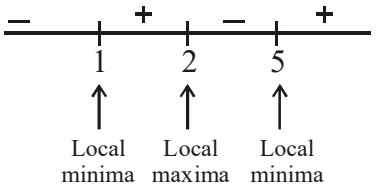
; ठ द्धश्य का एक स्थानीय उच्चतम (local maximum) $x=2$ है

(C) F का एक स्थानीय निम्नतम (local minimum) $x=1$ पर है

(D) F के दो स्थानीय उच्चतम और एक स्थानीय निम्नतम $(0,\infty)$ में हैं

Ans. ABC

Sol. $F'(x)=f(x)=(x-1)(x-1)(x-5)$



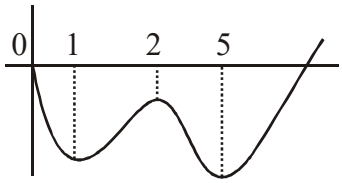
$$F(x)=\int_0^x\left(x^3-8x^2+17x-10\right)dx$$

$$F(x)=\frac{x^4}{4}-\frac{8x^3}{3}+\frac{17}{2}x^2-10x$$

$$F(1)=\frac{1}{4}-\frac{8}{3}+\frac{17}{2}-10<0$$

$$F(2)=\frac{16}{4}-\frac{64}{3}+\frac{68}{2}-20<0$$

$$F(5)=\frac{625}{4}-\frac{1000}{3}+\frac{17\times25}{2}-50<0$$



31. If $I=\frac{2}{\pi}\int\limits_{-\pi/4}^{\pi/4}\frac{dx}{\left(1+e^{\sin x}\right)\left(2-\cos 2x\right)}$ then $27I^2$ equals. **[JEE(Adv.)2019]**

यदि $I=\frac{2}{\pi}\int\limits_{-\pi/4}^{\pi/4}\frac{dx}{\left(1+e^{\sin x}\right)\left(2-\cos 2x\right)}$ तब $27I^2$ बराबर। **[JEE(Adv.)2019]**

Ans. 4

Sol. Apply king property and add

$$2I=\frac{2}{\pi}\int\limits_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\frac{dx}{\left(2-\cos 2x\right)}$$

$$\Rightarrow I=\frac{2}{\pi}\int\limits_{0}^{\frac{\pi}{4}}\frac{dx}{2-\left(\frac{1-\tan^2 x}{1+\tan^2 x}\right)}$$

let $\tan x = t$

$\sec^2 x\, dx = dt$

$$I=\frac{2}{\pi}\int\limits_{0}^{1}\frac{dt}{2+2t^2-1+t^2}$$

$$I=\frac{2}{\pi}\int\limits_{0}^{1}\frac{dt}{1+3t^2}=\frac{2}{3\pi}\int\limits_{0}^{1}\frac{dt}{t^2+\left(\frac{1}{\sqrt{3}}\right)^2}$$

$$=\frac{2}{3\pi}\cdot\frac{1}{\frac{1}{\sqrt{3}}}\left(\tan^{-1}\left(\sqrt{3}t\right)\right)_0^1$$

$$I = \frac{2}{\pi\sqrt{3}} \cdot \frac{\pi}{3} = \frac{2}{3\sqrt{3}}$$

$$I^2 = \frac{4}{27}$$

32. The value of the integral $\int_0^{\pi/2} \frac{3\sqrt{\cos\theta}}{\left(\sqrt{\cos\theta}+\sqrt{\sin\theta}\right)^5} d\theta$ equals_____. **[JEE(Adv.)2019]**

समाकल (integral) $\int_0^{\pi/2} \frac{3\sqrt{\cos\theta}}{\left(\sqrt{\cos\theta}+\sqrt{\sin\theta}\right)^5} d\theta$ का मान बराबर _____ **[JEE(Adv.)2019]**

Ans. 0.5

Sol. $\int_0^{\pi/2} \frac{3\sqrt{\cos\theta}}{\left(\sqrt{\cos\theta}+\sqrt{\sin\theta}\right)^5} d\theta$

$\int_0^{\pi/2} \frac{3\sec^2\theta}{\left(1+\sqrt{\tan\theta}\right)^5} d\theta$ Let $\tan\theta = t^2$

$\int_0^{\infty} \frac{6t\,dt}{(1+t)^5}$ $\sec^2\theta\, d\theta = 2tdt$

$$= 6\left[\int_0^{\infty}\left(\frac{1}{(1+t)^4} - \frac{1}{(1+t)^5}\right)dt\right.$$

$$= 6\left(-\frac{1}{3}\frac{1}{(1+t)^3} + \frac{1}{4(1+t)^4}\right)_0^{\infty}$$

$$= 6\left(\frac{1}{3} - \frac{1}{4}\right)$$

$$= \frac{6}{12} = 0.5$$

33. Which of the following inequalities is/are TRUE : **[JEE(Adv.)2020]**

निम्न में से कौनसी असमिका(यें) सत्य है/हैं : **[JEE(Adv.)2020]**

(A) $\int_0^1 x\cos x\ dx \geq \frac{3}{8}$ (B) $\int_0^1 x\sin x\ dx \geq \frac{3}{10}$ (C) $\int_0^1 x^2\cos x\ dx \geq \frac{1}{2}$ (D) $\int_0^1 x^2\sin x\ dx \geq \frac{2}{9}$

Ans. ABD

Sol. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + ...$

$\Rightarrow \cos x \geq 1 - \frac{x^2}{2}$

(A) $\int_0^1 x \cos\, dx \geq \int_0^1 x\left(1 - \frac{x^2}{2}\right)dx$

$\geq \left(\frac{x^2}{2} - \frac{x^4}{8}\right)_0^1$

$\geq \frac{3}{8}$

(B) $\sin x \geq x - \frac{x^3}{3!}$

$\int_0^1 x \sin x\, dx \geq \int_0^1 \left(x^2 - \frac{x^4}{6}\right)dx$

$\geq \left(\frac{x^3}{3} - \frac{x^5}{30}\right)_0^1$

$\geq \frac{3}{10}$

(C) $\cos x \leq 1$

$\int_0^1 x^2 \cos x\, dx \leq \int_0^1 x^2\, dx$

$\leq \left(\frac{x^3}{3}\right)_0^1$

$\leq \frac{1}{3}$

(D) $\sin x \geq x - \frac{x^3}{3!}$

$\int_0^1 x^2 \sin x\, dx \geq \int_0^1 \left(x^3 - \frac{x^5}{6}\right)dx$

$$\geq \left(\frac{x^4}{4}-\frac{x^6}{36}\right)_0^1$$

$$\geq \frac{2}{9}$$

34. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that its derivative f' is continuous and $f(\pi) = -6$. If $F: [0, \pi] \to \mathbb{R}$ is defined by $F(x) = \int_0^x f(t)dt$, and if $\int_0^{\pi} (f'(x) + F(x)) \cos x \, dx = 2$, then the value of $f(0)$ is _____ **[JEE(Adv.)2020]**

माना कि $f: \mathbb{R} \to \mathbb{R}$ एक ऐसा अवकलनीय फलन है जिसका अवकलज f' संतत है, एवं $f(\pi) = -6$ है। यदि $F: [0, \pi] \to \mathbb{R}$, $F(x) = \int_0^x f(t)dt$ द्वारा परभाषित है, तथा यदि $\int_0^{\pi} (f'(x) + F(x)) \cos x \, dx = 2$, तब $f(0)$ का मान है _____

[JEE(Adv.)2020]

Ans. 4

Sol. $F(x) = \int_0^x f(t)\,dt$

differentiating wrt x

$F'(x) = f(x)$ .....(1)

$\int_0^{\pi} \left(f'(x)\cos x + F(x)\cos x\right) dx = 2$

$\Rightarrow \int_0^{\pi} f'(x)\cos x \, dx + \int_0^{\pi} F(x)\cos x \, dx = 2$

Using intgeration by parts

$\left[f(x)\cos x\right]_0^{\pi} + \int_0^{\pi} \sin x . f(x)\,dx + \left[F(x).\sin x\right]_0^{\pi} - \int_0^{\pi} \sin x \, F'(x)\,dx = 2$

Using (1)

$\left[f(x)\cos x\right]_0^{\pi} + \int_0^{\pi} \sin x . f(x)\,dx + \left[F(x).\sin x\right]_0^{\pi} - \int_0^{\pi} \sin x \, f(x)\,dx = 2$

$\Rightarrow -f(\pi) - f(0) + F(\pi).0 - F(0).0 = 2$

$\Rightarrow -f(\pi) - f(0) = 2$

$\Rightarrow 6 - f(0) = 2$

$\Rightarrow f(0) = 4$ Ans.

35. Let $f:\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\to\mathbb{R}$ be a continuous function such that $f(0)=1$ and $\int_0^{\frac{\pi}{3}} f(t)dt = 0$. The which of the following statements is (are) **TRUE?** **[JEE(Adv.)2021]**

(A) The equation $f(x) - 3\cos 3x = 0$ has at least one solution in $\left(0,\frac{\pi}{3}\right)$

(B) The equation $f(x) - 3\sin 3x = -\frac{6}{\pi}$ has at least one solution in $\left(0,\frac{\pi}{3}\right)$

(C) $\lim\limits_{x\to 0}\frac{\int_0^x f(t)dt}{1-e^{x^2}} = 1$

(D) $\lim\limits_{x\to 0}\frac{\sin x\int_0^x f(t)dt}{x^2} = -1$

माना $f:\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\to\mathbb{R}$ एक सतत् फलन है जबकि $f(0)=1$ तथा $\int_0^{\frac{\pi}{3}} f(t)dt = 0$ तब निम्न में से कौनसा/कौनसे कथन सही है/हैं? **[JEE(Adv.)2021]**

(A) $\left(0,\frac{\pi}{3}\right)$ में समीकरण $f(x) - 3\cos 3x = 0$ का कम से कम एक हल है

(B) $\left(0,\frac{\pi}{3}\right)$ में समीकरण $f(x) - 3\sin 3x = -\frac{6}{\pi}$ का कम से कम एक हल है

(C) $\lim\limits_{x\to 0}\frac{\int_0^x f(t)dt}{1-e^{x^2}} = -1$

(D) $\lim\limits_{x\to 0}\frac{\sin x\int_0^x f(t)dt}{x^2} = -1$

Ans. ABC

Sol. (A) $\int_0^{\frac{\pi}{3}}\left(f(x)-3\cos 3x\right)dx = \int_0^{\frac{\pi}{3}} f(x)dx - \int_0^{\frac{\pi}{3}} 3\cos 3x\,dx = 0 - \sin 3x\big|_0^{\pi/3} = 0$

$\Rightarrow$ $f(x) - 3\cos 3x = 0$ has atleast one solution in $\left(0,\frac{\pi}{3}\right)$

(B) $\int_0^{\frac{\pi}{3}}\left(f(x)-3\sin 3x+\frac{6}{\pi}\right)dx = \int_0^{\frac{\pi}{3}} f(x)\,dx - 3\int_0^{\frac{\pi}{3}}\sin 3x\,dx + \frac{6}{\pi}\int_0^{\frac{\pi}{3}} dx = 2-2=0$

$\Rightarrow f(x)-3\sin 3x+\frac{6}{\pi}=0$ has atleast one solution in $\left(0,\frac{\pi}{3}\right)$.

(C) $\lim\limits_{x\to\infty}\frac{\int_0^x f(t)dt}{1-e^{x^2}}\left(\frac{0}{0}\right)$

Use L'hospital Rule

$\lim\limits_{x\to 0}\frac{x.f(x)+\int_0^x f(t)\,dt}{-2xe^{x^2}}\left(\frac{0}{0}\right)$

Use L'Hospital Rule again

$\lim\limits_{x\to 0}\frac{x.f'(x)+2f(x)}{-2e^{x^2}-4x^2.e^{x^2}}=\frac{2.f(0)}{-2}=-1$

(D) $\lim\limits_{x\to 0}\left(\frac{\sin x}{x}\right).\frac{\int_0^x f(t)\,dt}{x}$

$\lim\limits_{x\to 0}\frac{\int_0^x f(t)\,dt}{x}\left(\frac{0}{0}\right)$

Use L'Hospital Rule

$\lim\limits_{x\to 0}\frac{f(x)}{1}=f(0)=1$

As ABC

Question Stem for Question Nos. 36 and 37

Let $g_i:\left[\frac{\pi}{8},\frac{3\pi}{8}\right]\to R, i=1,2,$ and $f:\left[\frac{\pi}{8},\frac{3\pi}{8}\right]\to R$ be functions such that

$g_1(x)=1, g_2(x)=|4x-\pi|$ and $f(x)=\sin^2 x,$ for all $x\in\left[\frac{\pi}{8},\frac{3\pi}{8}\right]$

Define $S_i = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} f(x) \cdot g_i(x) dx, \quad i = 1, 2$

माना $g_i : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \to R, i = 1, 2,$ तथा $f : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \to R$ फलन है जो इस प्रकार हैं कि सभी $x \in \left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$ के लिए

$g_1(x) = 1, g_2(x) = |4x - \pi|$ तथा $f(x) = \sin^2 x$ है। $S_i = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} f(x) \cdot g_i(x) dx, \quad i = 1, 2$ को परिभाषित कीजिए।

36. The value of $\frac{16S_1}{\pi}$ is____. **[JEE(Adv.)2021]**

$\frac{16S_1}{\pi}$ का मान ____है। **[JEE(Adv.)2021]**

Ans. 2

Sol.

37. The value of $\frac{48S_2}{\pi^2}$ is_____. **[JEE(Adv.)2021]**

$\frac{48S_2}{\pi^2}$ का मान _____है। **[JEE(Adv.)2021]**

Ans. 1.5

Sol $S_1 = \int_{\pi/8}^{3\pi/8} f(x).1dx = \int_{\pi/8}^{3\pi/8} \sin^2 x dx$ ....(1)

apply $\int_a^b f(a+b-x)dx = \int_a^b f(x)dx$

$S_1 = \int_{\pi/8}^{3\pi/8} \cos^2 x dx$ .... (2)

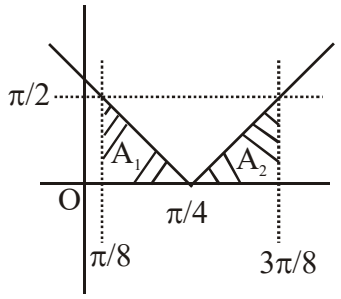
$(1) + (2) \Rightarrow 2S_1 = \int_{\pi/8}^{3\pi/8} dx = \frac{\pi}{4} \Rightarrow S_1 = \frac{\pi}{8}$

$\frac{16S_1}{\pi} = 2$

$$S_2 = \int_{\pi/8}^{3\pi/8} \sin^2 x\,|4x-\pi|\,dx \qquad .....(3)$$

apply $\int_a^b f(a+b-x)\,dx = \int_a^b f(x)\,dx$

$$S_2 = \int_{\pi/8}^{3\pi/8} \cos^2 x\,|4x-\pi|\,dx \qquad ....(4)$$



$$(3)+(4) \Rightarrow 2S_2 = \int_{\pi/8}^{3\pi/8} |4x-\pi|\,dx = \frac{\pi^2}{16}$$

$$A_1 + A_2 = \frac{1}{2}\times\frac{\pi}{4}\times\frac{\pi}{2} = \frac{\pi^2}{16}$$

$$\Rightarrow S_2 = \frac{\pi^2}{32}$$

$$\Rightarrow \frac{48}{\pi^2}S_2 = 1.5$$

Paragraph for Question Nos. 38 to 39

Let $\psi_1:[0,\infty)\to R,\ \psi_2:[0,\infty)\to R,\ f:[0,\infty)\to R$ and $g:[0,\infty)\to R$ be functions such that $f(0)=g(0)=0,$

$\psi_1(x)=e^{-x}+x, \quad x\ge 0,$

$\psi_2(x)=x^2-2x-2e^{-x}+2, \quad x\ge 0,$

$f(x)=\int_{-x}^{x}(|t|-t^2)e^{-t^2}dt, \quad x>0$

and $g(x)=\int_0^{x^2}\sqrt{t}e^{-t}dt, \quad x>0.$

माना $\psi_1:[0,\infty)\to R,\ \psi_2:[0,\infty)\to R,\ f:[0,\infty)\to R$ तथा $g:[0,\infty)\to R$ फलन है जो इस प्रकार हैं कि $f(0)=g(0)=0,$

$\psi_1(x)=e^{-x}+x, \quad x\ge 0,$

$\psi_2(x)=x^2-2x-2e^{-x}+2, \quad x\ge 0,$

$f(x)=\int_{-x}^{x}(|t|-t^2)e^{-t^2}dt, \quad x>0$

तथा $g(x)=\int_0^{x^2}\sqrt{t}e^{-t}dt, \quad x>0.$

38. Which of the following statements is TRUE? **[JEE(Adv.)2021]**

(A) $f(\sqrt{\ln 3})+g(\sqrt{\ln 3})=\frac{1}{3}$

(B) For every x > 1, there exists an $\alpha \in (1,x)$ such that $\psi_1(x)=1+\alpha x$

(C) For every x > 0, there exists a $\beta \in (0,x)$ such that $\psi_2(x)=2x(\psi_1(\beta)-1)$

(D) f is an increasing function on the interval $\left[0,\frac{3}{2}\right]$

निम्न में से कौनसा कथन सत्य है? **[JEE(Adv.)2021]**

(A) $f(\sqrt{\ln 3})+g(\sqrt{\ln 3})=\frac{1}{3}$

(B) प्रत्येक x > 1 के लिए, एक $\alpha \in (1,x)$ इस प्रकार विद्यमान है कि $\psi_1(x)=1+\alpha x$

(C) प्रत्येक x > 0 के लिए एक, $\beta \in (0,x)$ इस प्रकार विद्यमान है कि $\psi_2(x)=2x(\psi_1(\beta)-1)$

(D) अन्तराल $\left[0,\frac{3}{2}\right]$ में f एक वर्धमान फलन है

Ans. C

Sol. $\phi(\xi)=2\int_0^x\left(t-t^2\right)e^{-t^2}dt$

$f'(x)=2\left(x-x^2\right)e^{-x^2}$

$g'(x)=2x^2e^{-x^2}$

Option (A)

Let consider

$f'(x)+g'(x)=2\left(x-x^2\right)e^{-x^2}+2x^2e^{-x^2}$

Integrating both side

$f(x)+g(x)=-e^{-x^2}+c$

put x = 0

then c = 1

$f(x)+g(x)=1-e^{-x^2}$

$f(\sqrt{\ln 3})+g(\sqrt{\ln 3})=1-e^{-\ln 3}=1-\frac{1}{3}=\frac{2}{3}$

hence option (A) is wrong.

Option (B)

Let consider

$H(x) = \psi_1(x) - 1 - \alpha x$

$H(1) = \psi_1(1) - 1 - \alpha$

$H(1) = e^{-1} + 1 - 1 - \alpha$

$H(1) = e^{-1} - \alpha < 0$

$H'(x) = \psi_1'(x) - \alpha$

$H'(x) = -e^{-x} + 1 - \alpha < 0$

hence H(x) is decresing.

So H(x) is always negative and $H(x) \neq 0$

Hence option (B) is wrong.

Option (C)

$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2$

Applying LMVT in (0, x)

$$\psi'_2(\beta) = \frac{\psi_2(x) - \psi_2(0)}{x - 0}$$

$$\psi'_2(\beta) = \frac{\psi_2(x) - 0}{x} \quad .....(1)$$

$\psi'_2(x) = 2x - 2 + 2e^{-x}$

$\psi'_2(\beta) = 2\beta - 2 + 2e^{-\beta}$

$\psi'_2(\beta) = 2\left(\beta + e^{-\beta}\right) - 2$

$\psi'_2(\beta) = 2\psi_1(\beta) - 2$

Now by (1)

$$2(\psi_1(\beta) - 1) = \frac{\psi_2(x)}{x}$$

$\psi_2(x) = 2x(\psi_1(\beta) - 1)$

hence option (C) is correct.

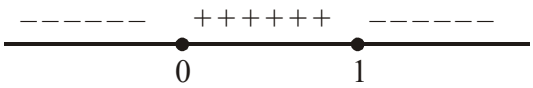
Option (D)

$$f(x) = \int_{-x}^{x} (|t|) - t^2)e^{-t^2}\, dt$$

$$f(x) = 2\int_{0}^{x} (t - t^2)e^{-t^2} dt$$

$f'(x) = 2(x - x^2)e^{-x^2}$

$f'(x) = 2x(1 - x)e^{-x^2}$



So option (D) is wrong.

39. Which of the following statements is TRUE ? **[JEE(Adv.)2021]**

(A) $\psi_1(x) \le 1,$ for all $x > 0$

(B) $\psi_2(x) \le 0,$ for all $x > 0$

(C) $f(x) \ge 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5,$ for all $x \in \left(0, \frac{1}{2}\right)$

(D) $g(x) \le \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7,$ for all $x \in \left(0, \frac{1}{2}\right)$

निम्न में से कौनसा कथन सत्य है? **[JEE(Adv.)2021]**

(A) $\psi_1(x) \le 1,$ सभी $x > 0$ के लिए

(B) $\psi_2(x) \le 0,$ सभी $x > 0$ के लिए

(C) $f(x) \ge 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5,$ for all $x \in \left(0, \frac{1}{2}\right)$

(D) $g(x) \le \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7,$ सभी $x \in \left(0, \frac{1}{2}\right)$ के लिए

Ans. D

Sol. Option (A)

$\psi_1(x) = e^{-x} + x$

$\psi'_1(x) = -e^{-x} + 1$

$\psi'_1(x) > 0$

hence $\psi_1(x)$ is increasing

then

$\psi_1(x) \ge \psi_1(0)$

$\psi_1(x) \ge 1$

So option (A) is wrong.

Option (B)

$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2$

$\psi_2'(x) = 2x - 2 + 2e^{-x}$

$\psi_2'(x) > 0$

hence $\psi_2'(x)$ is incresing.

then

$\psi_2(x) \geq \psi_2(0)$

$\psi_2(x) \geq 0$

hence option (B) is wrong.

Option (C)

$f(x) \geq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$

Consider

$H(x) = f(x) - 1 + e^{-x^2} + \frac{2}{3}x^3 - \frac{2}{5}x^5$

$H'(x) = f'(x) - 2xe^{-x^2} + 2x^2 - 2x^4$

but $f'(x) = 2(x - x^2)e^{-x^2}$

Now

$H'(x) = 2x^2(1 - x^2 - e^{-x^2})$

$H'(x) = 2x^2(1 - x^2 - 1 + \frac{x^2}{1} - \frac{x^4}{2} + ...) < 0$

then H(x) is decreasing

$H(x) \leq H(0)$

$H(x) \leq f(0)$

$H(x) \leq 0$

hence

$f(x) \leq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$

hence option (C) is wrong.

Option (D)

$g(x) \leq \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7$

Let consider

$H(x) = g(x) - \frac{2}{3}x^3 + \frac{2}{5}x^5 - \frac{1}{7}x^7$

$H'(x) = g'(x) - 2x^2 + 2x^4 - x^6$

but $g'(x) = 2x^2e^{-x^2}$

$H'(x) = 2x^2e^{-x^2} - 2x^2 + 2x^4 - x^6$

$$H'(x) = 2x^2\left(1-\frac{x^2}{1}+\frac{x^4}{2}-\frac{x^6}{3}.....\right)-2x^2+2x^4-x^6<0$$

H(x) is decreasing.

So

$H(x) \le H(0)$

$$g(x) -\frac{2}{3}x^3+\frac{2}{5}x^5-\frac{1}{7}x^7 \le 0$$

$$g(x) \le \frac{2}{3}x^3-\frac{2}{5}x^5+\frac{1}{7}x^7$$

hence option (D) is correct.

40. For any real number x, let [x] denote the largest integer less than or equal to x. If $I=\int_0^{10}\left[\sqrt{\frac{10x}{x+1}}\right]dx,$ then the value of 9I is_____. **[JEE(Adv.)2021]**

किसी वास्तविक संख्या x के लिए, माना [x] के बराबर या इससे कम महत्तम पूर्णांक को निरूपित करता है। यदि $I=\int_0^{10}\left[\sqrt{\frac{10x}{x+1}}\right]dx,$ तब 9I का मान_____है। **[JEE(Adv.)2021]**

Ans. 182

Sol. $f(x)=\frac{10x}{x+1}$

$$f'(x)=\frac{10}{(x+1)^2}>0$$

hence f'(x) is increasing

Now

$$I=\int_0^{\frac{1}{9}}0dx+\int_{\frac{1}{9}}^{\frac{2}{3}}1dx+\int_{\frac{2}{3}}^{9}2dx+\int_9^{10}3dx=\frac{182}{9}$$

Now 9I = 182

Ans = 182

41. Consider the equation $\int_1^e \frac{(\log_e x)^{1/2}}{x\left(a-(\log_e x)^{3/2}\right)^2}dx=1, a\in(-\infty,0)\cup(1,\infty)$. Which of the following statements is/are

TRUE? **[JEE(Adv.)2022]**

(A) No a satisfies the above equation

(B) An integer a satisfies the above equation

(C) An irrational number a satisfies the above equation

(D) More than one a safisfy the above equation

समीकरण $\int_1^e \frac{\left(\log_e x\right)^{1/2}}{x\left(a-(\log_e x)^{3/2}\right)^2} dx = 1, a \in (-\infty, 0) \cup (1, \infty)$ पर विचार कीजिए। निम्न कथनों में से कौन सा सत्य हैं?

[JEE(Adv.)2022]

(A) कोई भी a उपरोक्त समीकरण को संतुष्ट नहीं करता है

(B) एक पूर्णांक (integer) a उपरोक्त समीकरण को संतुष्ट करता है

(C) एक अपरिमेय संख्या (irrational number) a उपरोक्त समीकरण को संतुष्ट करता है

(D) एक से ज्यादा a उपरोक्त समीकरण को संतुष्ट करते है

Ans. CD

Sol. Let $I = \int_1^e \frac{(\ln x)^{1/2} dx}{x\left(a-(\ln x)^{3/2}\right)^2}$

put $a-(\ln x)^{3/2} = t$

$\Rightarrow \quad -\frac{3}{2}(\ln x)^{1/2} \cdot \frac{1}{x} dx = dt$

$\therefore \quad I = \int_a^{a-1} \frac{\left(-\frac{2}{3}\right) dt}{t^2}$

$= \left(-\frac{2}{3}\right) \frac{t^{-2+1}}{-2+1}\Big|_a^{a-1}$

$= \frac{2}{3t}\Big|_a^{a-1} = \frac{2}{3}\left(\frac{1}{a-1} - \frac{1}{a}\right)$

$\therefore \quad I = \left(\frac{2}{3}\right) \frac{1}{a(a-1)} = 1$

$\Rightarrow 2 = 3a^2 - 3a$

$\Rightarrow 3a^2 - 3a - 2 = 0$

$\Rightarrow \quad a = \frac{3 \pm \sqrt{9-4(3)(-2)}}{6}$

$a = \frac{3+\sqrt{33}}{6}, \frac{3-\sqrt{33}}{6}$

42. The greatest integer less than or equal to $\int_1^2 \log_2\left(x^3+1\right)dx+\int_1^{\log_2 9}\left(2^x-1\right)^{\frac{1}{3}}dx$ is _____.

[JEE(Adv.)2022]

वह महत्तम पूर्णांक, जो $\int_1^2 \log_2\left(x^3+1\right)dx+\int_1^{\log_2 9}\left(2^x-1\right)^{\frac{1}{3}}dx$ से कम या बराबर हो _____ है।

[JEE(Adv.)2022]

Ans. 5

Sol. $y=\int_1^2 \log_2\left(x^3+1\right)dx+\int_1^{2\log_2 3}\left(2^x-1\right)^{\frac{1}{3}}dx$

$=4\log_2 3-1$

we know $\int_a^b f(x)dx+\int_{f(a)}^{f(b)}\left(f^{-1}(x)\right)dx=bf(b)-af(a)$

So, [y] = 5

43. Let $f:(0,1)\to R$ be the function defined as $f(x)=\sqrt{n}$ if $x\in\left[\frac{1}{n+1},\frac{1}{n}\right)$ where $n\in N$. Let $g:(0,1)\to R$ be a function such that $\int_{x^2}^{x}\sqrt{\frac{1-t}{t}}dt<g(x)<2\sqrt{x}$ for all $x\in(0,1)$. Then $\lim\limits_{x\to 0} f(x)g(x)$: **[JEE(Adv.)2023]**

(A) does **NOT** exist (B) is equal to 1

(C) is equal to 2 (D) is equal to 3

माना कि फलन $f:(0,1)\to R$ इस तरह से परिभाषित है कि $f(x)=\sqrt{n}$ यदि $x\in\left[\frac{1}{n+1},\frac{1}{n}\right)$ जहाँ $n\in N$ है। माना कि फलन $g:(0,1)\to R$ इस प्रकार है कि सभी $x\in(0,1)$ के लिए $\int_{x^2}^{x}\sqrt{\frac{1-t}{t}}dt<g(x)<2\sqrt{x}$ है। तब $\lim\limits_{x\to 0} f(x)g(x)$:

[JEE(Adv.)2023]

(A) का अस्तित्व **नहीं** है (B) 1 के बराबर है

(C) 2 के बराबर है (D) 3 के बराबर है

Ans. C

Sol. $f(x)=\sqrt{x}$

$\frac{1}{n+1}\le x<\frac{1}{n}$

$n<\frac{1}{x}\le n+1$

$$n < \frac{1}{x} \quad \Big| \quad n \geq \frac{1}{x} - 1$$
$$n \geq \frac{1-x}{x}$$

$$\frac{1-x}{x} \leq n < \frac{1}{x}$$

$$\sqrt{\frac{1-x}{x}} \leq \sqrt{n} < \frac{1}{\sqrt{x}}$$

$$\sqrt{\frac{1-x}{x}} \leq f(x) < \frac{1}{\sqrt{x}}$$

$$\int_{x^2}^{x} \sqrt{\frac{1-t}{t}}dt \times \sqrt{\frac{1-x}{x}} \leq f(x)g(x) < \frac{1}{\sqrt{x}} \times 2\sqrt{x}$$

$$\lim_{x\to 0} \int_{x^2}^{x} \sqrt{\frac{1-t}{t}}dt \sqrt{\frac{1-x}{x}} \leq \lim_{x\to 0} f(x)g(x) < \lim_{x\to 0} 2$$

$$\lim_{x\to 0} \frac{\int_{x^2}^{x} \left(\sqrt{\frac{1-t}{t}}\right)dt}{\sqrt{\frac{x}{1-x}}} \left(\frac{0}{0}\right)$$

L.H. Rule

$$\lim_{x\to 0} \frac{\sqrt{\frac{1-x}{x}} - \sqrt{\frac{1-x^2}{x^2}} \times (2x)}{\frac{1}{2\sqrt{\frac{x}{1-x}}} \times \left(\frac{1-x+x}{(1-x)^2}\right)} = \lim_{x\to 0} \frac{2\sqrt{\frac{1-x}{x}}\left(1 - \sqrt{\frac{1+x}{x}} \times 2x\right)}{\sqrt{\frac{1-x}{x}} \times \frac{1}{(1-x)^2}} = 2$$

44. Let $f: [1, \infty) \to R$ be a differentiable function such that $f(1) = \frac{1}{3}$ and $3\int_1^x f(t)dt = xf(x) - \frac{x^3}{3}$, $x \in [1, \infty)$. Let e denote the base of the natural logarithm. Then the value of f(e) is : **[JEE(Adv.)2023]**

माना कि $f: [1, \infty) \to R$ एक अवकलनीय (differentiable) फलन इस प्रकार है कि $f(1) = \frac{1}{3}$ है एवं $x \in [1, \infty)$ के लिए $3\int_1^x f(t)dt = xf(x) - \frac{x^3}{3}$ है। माना कि e प्राकृतिक लघुगणक के आधार (base of the natural logarithm) को निरूपित d j r kgSar c f(e) का मान है : **[JEE(Adv.)2023]**

(A) $\frac{e^2+4}{3}$ (B) $\frac{\log_e 4+e}{3}$ (C) $\frac{4e^2}{3}$ (D) $\frac{e^2-4}{3}$

Ans. C

Sol. Diff. w.r.t 'x'

$3f(x) = f(x) + xf'(x) - x^2$

$$\frac{dy}{dx}-\left(\frac{2}{x}\right)y = x$$

$$IF = e^{-2\ell nx} = \frac{1}{x^2}$$

$$y\left(\frac{1}{x^2}\right) = \int x.\frac{1}{x^2}dx$$

$$y = x^2\ell nx + cx^2$$

$$\therefore y(1) = \frac{1}{3} \Rightarrow c = \frac{1}{3}$$

$$y(e) = \frac{4e^2}{3}$$

45. For $x \in R$, let $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the minimum value of the function $f: R \to R$ defined by $f(x) = \int_0^{x\tan^{-1}x} \frac{e^{(t-\cos t)}}{1+t^{2023}}dt$ is _____. **[JEE(Adv.)2023]**

माना कि $x \in R$ के लिए $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ है। तब फलन $f: R \to R$ जो $f(x) = \int_0^{x\tan^{-1}x} \frac{e^{(t-\cos t)}}{1+t^{2023}}dt$ से परिभाषित है का U;wure eku (Minimum value) है _____ **[JEE(Adv.)2023]**

Ans. 0

Sol. $$f'(x) = \frac{\left(e^{x\tan^{-1}x-\cos\left(x\tan^{-1}x\right)}\right)}{1+\left(x\tan^{-1}x\right)^{2023}}\left(\tan^{-1}x+\frac{x}{1+x^2}\right)$$

$f'(x) = 0$

$\Rightarrow \quad \tan^{-1} x + \frac{x}{1+x^2} = 0 \quad \Rightarrow \boxed{x = 0}$

and $\quad f'(0^+) > 0$

$f'(0^-) < 0$

So $\quad$ min at $x = 0$

$\boxed{f(0) = 0}$ $\quad$ Ans. = 0

46. Let the function $f: [1, \infty) \to R$ be defined by

$$f(t) = \begin{cases} (-1)^{n+1} 2, & \text{if } t = 2n-1, n \in N \\ \frac{(2n+1-t)}{2} f(2n-1) + \frac{(t-(2n-1))}{2} f(2n+1), & \text{if } 2n-1 < t < 2n+1, n \in N \end{cases}$$

Define $g(x) = \int_1^x f(t)dt, x \in (1, \infty)$. Let α denote the number of solutions of the equation $g(x) = 0$ in the interval $(1, 8]$ and $\beta = \lim\limits_{x \to 1+} \frac{g(x)}{x-1}$. Then the value of $\alpha + \beta$ is equal to _____. **[JEE(Adv.)2024]**

माना कि फलन $f: [1, \infty) \to R$

$$f(t) = \begin{cases} (-1)^{n+1} 2, & \text{यदि } t = 2n-1, n \in N \\ \frac{(2n+1-t)}{2} f(2n-1) + \frac{(t-(2n-1))}{2} f(2n+1), & \text{यदि } 2n-1 < t < 2n+1, n \in N \end{cases}$$

द्वारा परिभाषित है। $g(x) = \int_1^x f(t)dt, x \in (1, \infty)$ से परिभाषित कीजिए। माना कि α, अंतराल $(1, 8]$ में समीकरण $g(x) = 0$ के हलों (solutions) की संख्या को दर्शाता है और $\beta = \lim\limits_{x \to 1+} \frac{g(x)}{x-1}$ है। तब $\alpha + \beta$ का मान _____ है। **[JEE(Adv.)2024]**

Ans. 5

Sol. If $t = 2n-1, n \in N$

$f(t) = (-1)^{n+1} .2$

$\Rightarrow f(1) = 2, f(3) = -2, f(15) = 2, f(7) = -2, f(9) = 2, ...$

If $t \in (2n-1, 2n+1)$

$$f(t) = \frac{(2n+1-t)}{2} f(2n-1) + \frac{(t-(2n-1))}{2} f(2n+1)$$

Now n = 1

$1 < t < 3$

$$f(t) = \frac{(3-t)}{2}(2) + \frac{(t-1)}{2}(-2)$$

$\Rightarrow 3 - t - t + 1$

$\Rightarrow 4 - 2t$

now n = 2

$3 < t < 5$

$$f(t) = \frac{(5-t)}{2}(-2) + \left(\frac{t-3}{2}\right)(2)$$

$= -5 + t + t - 3$

$\Rightarrow -8 + 2t$

n = 3

$5 < t < 7$

$$f(t) = \left(\frac{7-t}{2}\right)(2) + \left(\frac{t-5}{2}\right)(-2)$$

$= 7 - t - t + 5$

$\Rightarrow 12t - 2t$

n = 4

$7 < t < 9$

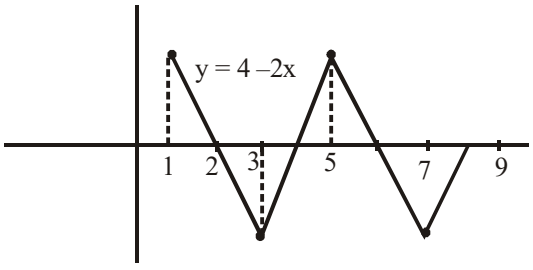
$$f(t) = \left(\frac{9-t}{2}\right)(-2) + \left(\frac{t-7}{2}\right)(2)$$

$= t - 9 + t - 7$

$\Rightarrow 2t - 16$

Plot f(x)

No. οφσολυτιον οφγ$(\xi) = 3 = \alpha$



$$\beta = \lim_{x \to 1^+} \frac{g(x)}{x-1}$$

$$= \lim_{x \to 1^+} g'(x)$$

$\Rightarrow f(x) \Rightarrow f(1^+) \Rightarrow 2$

$\alpha + \beta$

$\Rightarrow 3 + 2$

$\Rightarrow 5$

PARAGRAPH

Paragraph for Q. No 47 to 48

Let $f:\left[0,\frac{\pi}{2}\right] \to [0,1]$ be the function defined by $f(x) = \sin^2 x$ and let $g:\left[0,\frac{\pi}{2}\right] \to [0,\infty)$ be the function defined by $g(x) = \sqrt{\frac{\pi x}{2} - x^2}$. **[JEE(Adv.)2024]**

(There are two questions based on PARAGRAPH)

माना कि $f:\left[0,\frac{\pi}{2}\right] \to [0,1]$ वह फलन (function) है जो $f(x) = \sin^2 x$ द्वारा परिभाषित है, और माना कि $g:\left[0,\frac{\pi}{2}\right] \to [0,\infty)$ वह फलन है जो $g(x) = \sqrt{\frac{\pi x}{2} - x^2}$ द्वारा परिभाषित है। **[JEE(Adv.)2024]**

(PARAGRAPH पर आधारित दो प्रश्न हैं, नीचे दिया गया प्रश्न उनमें से एक है)

47. The value of $2\int_0^{\frac{\pi}{2}} f(x)g(x)dx - \int_0^{\frac{\pi}{2}} g(x)dx$ is _____.

$2\int_0^{\frac{\pi}{2}} f(x)g(x)dx - \int_0^{\frac{\pi}{2}} g(x)dx$ का मान _____ है।

Ans. 0

Sol. Let $I_1 = 2\int_0^{\frac{\pi}{2}} \sin^2 x\sqrt{\frac{\pi x}{2} - x^2}dx$ _____(1)

King

$I_1 = 2\int_0^{\frac{\pi}{2}} \cos^2 x\sqrt{\frac{\pi x}{2} - x^2}dx$ _____(2)

(1) + (2)

$2I_1 = 2\int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi x}{2} - x^2}dx$

$$I_1 = \int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi x}{2} - x^2}\, dx$$

Let $I_2 = \int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi x}{2} - x^2}\, dx$

Ans. $= I_1 - I_2$

$= 0$

48. The value of $\frac{16}{\pi^3}\int_0^{\frac{\pi}{2}} f(x)g(x)dx$ is _____. **[JEE(Adv.)2025]**

$\frac{16}{\pi^3}\int_0^{\frac{\pi}{2}} f(x)g(x)dx$ का मान _____ है। **[JEE(Adv.)2025]**

Ans. 0.25

Sol. $I = \frac{16}{\pi^3}\int_0^{\frac{\pi}{2}} \sin^2 x \sqrt{\frac{\pi x}{2} - x^2}\, dx$ _____(1)

Apply king property

$I = \frac{16}{\pi^3}\int_0^{\frac{\pi}{2}} \cos^2 x \sqrt{\frac{\pi x}{2} - x^2}\, dx$ _____(2)

(1) + (2)

$2I = \frac{16}{\pi^3}\int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi x}{2} - x^2}\, dx$

$I = \frac{8}{\pi^3}\int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi^2}{16} - \left(x - \frac{\pi}{4}\right)^2}\, dx$

Put $x - \frac{\pi}{4} = t$

$dx = dt$

$$I=\frac{8}{\pi^3}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\sqrt{\frac{\pi^2}{16}-t^2}\,dt$$

Even function

$$I=\frac{16}{\pi^3}\int_{0}^{\frac{\pi}{4}}\sqrt{\frac{\pi^2}{16}-t^2}\,dt$$

$$=\frac{16}{\pi^3}\left[t\sqrt{\frac{\pi^2}{16}-t^2}+\frac{\pi^2}{32}\sin^{-1}\left(\frac{t}{\frac{\pi}{4}}\right)\right]_0^{\frac{\pi}{4}}$$

$$=\frac{16}{\pi^3}\times\frac{\pi^2}{32}\times\frac{\pi}{2}=\frac{1}{4}=0.25$$

49. Let α and β be the real numbers such that

$$\lim_{x\to 0}\frac{1}{x^3}\left(\frac{\alpha}{2}\int_0^x\frac{1}{1-t^2}\,dt+\beta x\cos x\right)=2.$$

Then the value $\alpha+\beta$ is _____. **[JEE(Adv.)2025]**

मान लीजिए कि α और β वह वास्तविक संख्याएँ हैं कि

$$\lim_{x\to 0}\frac{1}{x^3}\left(\frac{\alpha}{2}\int_0^x\frac{1}{1-t^2}\,dt+\beta x\cos x\right)=2.$$

है। तब $\alpha+\beta$ का मान है _____ **[JEE(Adv.)2025]**

Ans. 2.4

Sol. $$\lim_{x\to 0}\frac{1}{x^3}\left[\frac{\alpha}{2}\int_0^x\frac{1}{1-t^2}dt+\beta x\cos x\right]=2$$

Apply L–H Rule

$$\lim_{x\to 0}\frac{\left[\frac{\alpha}{2\left(1-x^2\right)}+\beta(\cos x-x\sin x)\right]}{3x^2}=2$$

$$\lim_{x\to 0}\left[\frac{\alpha+\beta\times 2\left(1-x^2\right)(\cos x - x\sin x)}{2\left(1-x^2\right)\cdot 3x^2}\right]=2$$

$$\frac{1}{6}\lim_{x\to 0}\left[\frac{\alpha+\left(2\beta-2\beta x^2\right)(\cos x - x\sin x)}{x^2}\right]=2$$

$$\frac{1}{6}\lim_{x\to 0}\frac{\left[\alpha+2\beta\cos x-2\beta x\sin x-2\beta x^2\cos x+2\beta x^3\sin x\right]}{x^2}=2$$

$$\frac{1}{6}\lim_{x\to 0}\frac{\left[\alpha+2\beta\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}\right)\right]-2\beta x\left(x-\frac{x^3}{3!}\right)-2\beta x^2\left(1-\frac{x^2}{2!}\right)+2\beta x^4\right]}{x^2}=2$$

$$\frac{1}{6}\lim_{x\to 0}\frac{\left[(\alpha+2\beta)+x^2[-\beta-2\beta-2\beta]+x^4\left(\frac{2\beta}{4!}+\frac{2\beta}{3!}+\frac{2\beta}{2!}+2\beta\right)\right]}{x^2}=2$$

$\alpha + 2\beta = 0$ and $-\frac{5\beta}{6}=2$

$$\beta=-\frac{12}{5}$$

$\alpha = -2\beta$

$$=-2\left(\frac{-12}{5}\right)$$

$$=\frac{24}{5}$$

$$\alpha+\beta=\frac{24}{5}-\frac{12}{5}$$

$$=\frac{12}{5}=2.4$$

50. If $\alpha=\int_{\frac{1}{2}}^{2}\frac{\tan^{-1}x}{2x^2-3x+2}dx$, then the value of $\sqrt{7}\tan\left(\frac{2\alpha\sqrt{7}}{\pi}\right)$ is _____.

(Here, the inverse trigonometric function $\tan^{-1} x$ assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.) **[JEE(Adv.)2025]**

. यदि $\alpha = \int_{\frac{1}{2}}^{2} \frac{\tan^{-1} x}{2x^2 - 3x + 2} dx$ है, तब $\sqrt{7} \tan\left(\frac{2\alpha\sqrt{7}}{\pi}\right)$ का मान _____ है।

(यहाँ, प्रतिलोम त्रिकोणमितीय फलन (inverse trigonometric function) $\tan^{-1} x$ के मान $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ में हैं।)

[JEE(Adv.)2025]

Ans. 21

Sol. $\alpha = \int_{1/2}^{2} \frac{\tan^{-1} x}{2x^2 - 3x + 2} dx \quad$ put $x = \frac{1}{y}$

$$\alpha = \int_{2}^{1/2} \frac{\tan^{-1}\left(\frac{1}{y}\right)\left(\frac{-1}{y^2}\right) dy}{\frac{2}{y^2} - \frac{3}{y} + 2} = -\int_{2}^{1/2} \frac{\cot^{-1}(y) dy}{2y^2 - 3y + 2}$$

$$\Rightarrow 2\alpha = \int_{1/2}^{2} \frac{(\pi/2) dx}{2x^2 - 3x + 2} \Rightarrow \alpha = \frac{\pi}{4} \cdot \frac{1}{2} \int_{1/2}^{2} \frac{dx}{x^2 - \frac{3x}{2} + 1}$$

$$\Rightarrow \alpha = \frac{\pi}{8} \int_{1/2}^{2} \frac{dx}{\left(x - \frac{3}{4}\right)^2 + \left(\frac{\sqrt{7}}{4}\right)^2}$$

$$\Rightarrow \alpha = \frac{\pi}{8} \cdot \frac{4}{\sqrt{7}} \tan^{-1}\left(\frac{x - 3/4}{\sqrt{7}/4}\right)\Bigg|_{1/2}^{2}$$

$$\alpha = \frac{4\pi}{8\sqrt{7}} \tan^{-1}\left(\frac{4x - 3}{\sqrt{7}}\right)_{1/2}^{2}$$

$$\alpha = \frac{\pi}{2\sqrt{7}}\left(\tan^{-1}\frac{5}{\sqrt{7}} - \tan^{-1}\left(-\frac{1}{\sqrt{7}}\right)\right)$$

$$\alpha = \frac{\pi}{2\sqrt{7}}\left(\tan^{-1}\left(\frac{5}{\sqrt{7}}\right)+\tan^{-1}\left(\frac{1}{\sqrt{7}}\right)\right)$$

$$\frac{2\alpha\sqrt{7}}{\pi} = \tan^{-1}\left(\frac{5}{\sqrt{7}}\right)+\tan^{-1}\left(\frac{1}{\sqrt{7}}\right)$$

$$\tan\left(\frac{2\alpha\sqrt{7}}{\pi}\right) = \frac{6/\sqrt{7}}{1-5/7} = \frac{6/\sqrt{7}}{2/7} = 3\sqrt{7}$$

Ans. 21

51. The value of the definite integral $\int_0^2 \frac{1}{3^x+3}dx$ is : **[JEE(Adv.)2026]**

निश्चित समाकल $\int_0^2 \frac{1}{3^x+3}dx$ का मान होगा : **[JEE(Adv.)2026]**

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{\log_e 3}{3}$ (D) $\frac{\log_e 3}{2}$

Ans. B

Sol. $I = \int_0^2 \frac{1}{3x+3}dx$

$3^x = t \Rightarrow 3^x \log_e 3\, dx = dt$

$dx = \frac{1}{\log_e 3 \cdot t}dt$

$$I = \int_1^9 \frac{1}{t+3}\cdot\frac{dt}{t.\log_e 3} \Rightarrow \quad I = \frac{1}{\log_e 3}\int_1^9\left(\frac{-\frac{1}{3}}{t+3}+\frac{\frac{1}{3}}{t}\right)dt$$

$$I = \frac{1}{\log_e 3}\cdot\frac{1}{3}\log_e\left|\frac{t}{t+3}\right|_1^9 \Rightarrow \qquad I = \frac{1}{3\log_e 3}\left\{\log_e\left(\frac{9}{12}\right)-\log_e\left(\frac{1}{4}\right)\right\}$$

$$I = \frac{1}{3\log_e 3}\cdot \ell n_e 3 = \frac{1}{3}$$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. If $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx = l n\left(f(x)\right) + g(x) + C$ where C is the constant of integration and f(x) is positive then f(x) + g (x) has the value equal to :

यदि $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx = l n\left(f(x)\right) + g(x) + C$ जहां C समाकलन अचर है तथा f(x) धनात्मक तब f(x) + g (x) का eku cj kcj :

(A) $e^x + \sin x + 2x$ (B) $e^x + \sin x$ (C) $e^x - \sin x$ (D) $e^x + \sin x + x$

Ans. B

Sol. $I = \int \frac{(e^x + \cos x + 1) - (e^x + \sin x + x)}{e^x + \sin x + x} dx$

$= l n(e^x + \sin x + x) - x + C$

$\therefore \quad f(x) = e^x + \sin x + x \quad$ and $\quad g(x) = -x$

$f(x) + g(x) = e^x + \sin x$

2. $\int \left(\sin(101x) \cdot \sin^{99} x\right) dx$ equals :

$\int \left(\sin(101x) \cdot \sin^{99} x\right) dx$ बराबर :

(A) $\frac{\sin(100x)(\sin x)^{100}}{100} + C$ (B) $\frac{\cos(100x)(\sin x)^{100}}{100} + C$

(C) $\frac{\cos(100x)(\cos x)^{100}}{100} + C$ (D) $\frac{\sin(100x)(\sin x)^{101}}{101} + C$

Ans. A

Sol. $I = \int \left(\sin(100x + x) \cdot (\sin x)^{99} dx = \int [(\sin(100x)\cos x + \cos 100x \cdot \sin x)](\sin x)^{99} dx\right.$

$= \int \underbrace{\sin(100x)}_{I} \underbrace{\cos x \cdot (\sin x)^{99}}_{II} dx + \int \cos(100x) \cdot (\sin x)^{100} dx$

$= \frac{\sin(100x)(\sin x)^{100}}{100} - \frac{100}{100} \int \cos(100x)(\sin x)^{100} dx + \int \cos(100x)(\sin x)^{100} dx$

$= \frac{\sin(100x)(\sin x)^{100}}{100} + C$

3. $\int a^x \left(l n x + l n a . l n \left(\frac{x}{e}\right)^x \right) dx =$

(A) $a^x l\,n\left(\frac{e}{x}\right)^{2x}+C$ (B) $a^x l\,n\left(\frac{x}{e}\right)^{x}+C$

(C) $a^x + l\,n\left(\frac{x}{e}\right)^{x}+C$ (D) None of these/इनमें से कोई नहीं

Ans. B

Sol. $\int a^x\left(\ell nx+\ell na\cdot\ell n\left(\frac{x}{e}\right)^x\right)dx$

$=\int\frac{d}{dx}\left(a^x\ell n\left(\frac{x}{e}\right)^x\right)dx$

$a^x\ell n\left(\frac{x}{e}\right)^x+C$

4. $\int\frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}}dx$ is equal to :

$\int\frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}}dx$ बराबर होगा :

(A) $\frac{\sqrt{2x^4-2x^2+1}}{x^2}+C$ (B) $\frac{\sqrt{2x^4-2x^2+1}}{x^3}+C$

(C) $\frac{\sqrt{2x^4-2x^2+1}}{x}+C$ (D) $\frac{\sqrt{2x^4-2x^2+1}}{2x^2}+C$

Ans. D

Sol. $\int\frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}}dx=\int\frac{\left(\frac{1}{x^3}-\frac{1}{x^5}\right)dx}{\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}}$

Let $2-\frac{2}{x^2}+\frac{1}{X^4}=t \quad\Rightarrow\left(\frac{4}{x^3}-\frac{4}{x^5}\right)dx=dt$

$=\frac{1}{4}\int\frac{dt}{\sqrt{t}}=\frac{1}{2}\sqrt{t}+C=\frac{1}{2x^2}\sqrt{2x^4-2x^2+1+C}$

5. $\int \frac{8x^{43}+13x^{38}}{\left(x^{13}+x^5+1\right)^4}dx =$

(A) $\frac{x^{39}}{3\left(x^{13}+x^5+1\right)^3}+C$ (B) $\frac{x^{39}}{\left(x^{13}+x^5+1\right)^3}+C$

(C) $\frac{x^{39}}{5\left(x^{13}+x^5+1\right)^5}+C$ (D) None of these/इनमें से कोई नहीं

Ans. A

Sol. $\int \frac{8x^{43}+13x^{38}}{\left(x^{13}+x^5+1\right)^4}dx$

$= \int \frac{8x^{-9}+13x^{-14}}{\left(1+x^{-8}+x^{-13}\right)^4}dx$

Let $1+x^{-8}+x^{-13}=t$

$\Rightarrow -\left(8x^{-9}+13x^{-14}\right)dx = dt$

$= -\int \frac{1}{t^4}dt = \frac{1}{t^3}+C$

$= \frac{1}{3\left(1+x^{-8}+x^{-13}\right)^3}+C$ $= \frac{x^{39}}{3\left(x^{13}+x^5+1\right)^3}+C$

6. $\int_0^1 \frac{\tan^{-1}x}{x}dx =$

(A) $\int_0^{\pi/4} \frac{\sin x}{x}dx$ (B) $\int_0^{\pi/2} \frac{x}{\sin x}dx$ (C) $\frac{1}{2}\int_0^{\pi/2} \frac{x}{\sin x}dx$ (D) $\frac{1}{2}\int_0^{\pi/4} \frac{x}{\sin x}dx$

Ans. C

Sol. $I = \int_0^1 \frac{\tan^{-1}x}{x}\,dx$ $x = \tan\theta$; $dx = \sec^2\theta\, d\theta$

$I = \int_0^{\pi/4} \frac{\theta.\sec^2\theta}{\tan\theta}d\theta = \int_0^{\pi/4} \frac{\theta}{\sin\theta\cos\theta}d\theta = \int_0^{\pi/4} \frac{2\theta}{\sin 2\theta}d\theta$ $2\theta = y$

$$= \frac{1}{2}\int_0^{\pi/2} \frac{y}{\sin y}\,dy = \frac{1}{2}\int_0^{\pi/2} \frac{x}{\sin x}\,dx \Rightarrow (C)$$

7. The value of the definite integral $\int_0^{\pi/2} (\sin x + \cos x)\cdot\sqrt{\frac{e^x}{\sin x}}\,dx$ equals :

निश्चित समाकलन $\int_0^{\pi/2} (\sin x + \cos x)\cdot\sqrt{\frac{e^x}{\sin x}}\,dx$ का मान है :

(A) $2\sqrt{e^{\pi/2}}$ (B) $\sqrt{e^{\pi/2}}$ (C) $2\sqrt{e^{\pi/2}}\cdot\cos 1$ (D) $\frac{1}{2}e^{\pi/4}$

Ans. A

Sol. $I = \int_0^{\pi/2} \frac{e^x(\sin x + \cos x)}{\sqrt{e^x \sin x}}dx$

put $e^x \sin x = t^2 \Rightarrow e^x(\sin x + \cos x)dx = 2t\,dt$

$\therefore \quad I = 2\int_0^{e^{\sqrt{\pi/2}}} dt \;\; 2t\Big]_0^{e^{\sqrt{\pi/2}}} = 2e^{\sqrt{\pi/2}} = 2\sqrt{e^{\pi/2}}$

8. The absolute value of $\frac{\int_0^{\pi/2}(x\cos x+1)e^{\sin x}\,dx}{\int_0^{\pi/2}(x\sin x-1)e^{\cos x}\,dx}$ is equal to :

$\frac{\int_0^{\pi/2}(x\cos x+1)e^{\sin x}\,dx}{\int_0^{\pi/2}(x\sin x-1)e^{\cos x}\,dx}$ का निरपेक्ष मान होगा :

(A) e (B) πe (C) e/2 (D) π/e

Ans. A

Sol. $\frac{\int_0^{\pi/2}(x\cos x+1)e^{\sin x}dx}{\int_0^{\pi/2}(x\sin x-1)e^{\cos x}dx} = \frac{\left[e^{\sin x}\cdot x\right]_0^{\frac{\pi}{2}}}{\left[-e^{\cos x}\cdot x\right]_0^{\frac{\pi}{2}}}$

$$= \frac{e \cdot \frac{\pi}{2}}{-\frac{\pi}{2}} = -e$$

$\therefore$ absolute value is e.

9. If $\int_0^1 \frac{e^t\,dt}{1+t} = A$ then the value of $\int_0^1 \frac{e^t\,dt}{(1+t)^2}$ is :

यदि $\int_0^1 \frac{e^t\,dt}{1+t} = A$ तो $\int_0^1 \frac{e^t\,dt}{(1+t)^2}$ का मान है :

(A) $A + \frac{e}{2} - 1$ (B) $A - \frac{e}{2} + 1$ (C) $A - \frac{e}{2} - 1$ (D) $A + \frac{e}{2} + 1$

Ans. B

Sol. $I = \int_0^1 \frac{e^t\,(t+1-t)}{(1+t)^2}\,dt = \int_0^1 \frac{e^t}{1+t}\,dt - \int_0^1 e^t \left(\frac{1}{1+t} - \frac{1}{(1+t)^2}\right) dt$

$= A - \frac{e^t}{1+t}\Big]_0^1 = A - \frac{e}{2} + 1$; Alternatively I. B. P. directly

10. If $\beta + 2\int_0^1 x^2\, e^{-x^2}\,dx = \int_0^1 e^{-x^2}\,dx$ then the value of β is :

(A) e^{-1} (B) e (C) 1/2e (D) can not be determined

यदि $\beta + 2\int_0^1 x^2\, e^{-x^2}\,dx = \int_0^1 e^{-x^2}\,dx$ तो β का मान होगा :

(A) e^{-1} (B) e (C) 1/2e ;क ज्ञात नहीं किया जा सकता है

Ans. A

Sol. $\beta + \int_0^1 \underset{I}{x}\,\underbrace{2xe^{-x^2}}_{II}\,dx = \int_0^1 e^{-x^2}\,dx$

$\beta + \left[-xe^{-x^2}\right]_0^1 - \int_0^1 -e^{-x^2}\,dx = \int_0^1 e^{-x^2}\,dx \quad \beta = \frac{1}{e}$

11. The value of the definite integral $\int_0^{\pi/2} \sin\left|2x - \alpha\right| dx$ where $\alpha \in [0, \pi]$:

निश्चित समाकलन $\int_0^{\pi/2} \sin|2x-\alpha|\,dx$ का मान ज्ञात कीजिए जहाँ $\alpha \in [0, \pi]$:

(A) 1 (B) $\chi o \sigma \alpha$ (C) $\frac{1+\cos\alpha}{2}$ (D) $(1 + \sin\alpha)$

Ans. A

Sol. $I = \int_{-\alpha}^{(\pi-\alpha)} \sin|t|\,dt$ where $2x - \alpha = t \quad \Rightarrow \quad dx = \frac{dt}{2}$

$$= \frac{1}{2}\int_{-\alpha}^{0} -\sin t\,dt + \frac{1}{2}\int_{0}^{\pi-\alpha} \sin t\,dt = \frac{1}{2}\cos t\Big]_{-\alpha}^{0} - \frac{1}{2}\cos t\Big]_{0}^{\pi-\alpha} = \frac{1}{2}[1 - \cos\alpha] - \frac{1}{2}[-\cos\alpha - 1]$$

$$= \frac{1}{2}(1 - \cos\alpha) + \frac{1}{2}(1 + \cos\alpha) = 1$$

12. The value of the definite integral $\int_2^4 \Big(x(3-x)(4+x)(6-x)(10-x) + \sin x\Big)dx$ equals :

$\int_2^4 \Big(x(3-x)(4+x)(6-x)(10-x) + \sin x\Big)dx$ का मान है :

(A) cos 2 + cos 4 (B) cos 2 – cos 4 (C) sin 2 + sin 4 (D) sin 2 – sin 4

Ans. B

Sol. We have $I = \int_2^4 \Big(\underbrace{x(3-x)(4+x)(6-x)(10-x)}_{E} + \sin x \Big)dx$ ....(1)

Now $I = \int_2^4 \Big((6-x)(3-(6-x))(4+(6-x))(6-(6-x))(10-(6-x)) + \sin(6-x) \Big)dx$

(Applying King property)

$$= \int_2^4 \Big(\underbrace{(6-x)(x-3)(10-x)x(4+x)}_{E} + \sin(6-x) \Big)dx \quad ....(2)$$

$\therefore$ On adding (1) and (2), we get $\Rightarrow I = \cos 2 - \cos 4$

13. $\int_{-2}^{0} \Big(x^3 + 3x^2 + 3x + 3 + (x+1)\cos(x+1)\Big)dx$ is equal to :

$\int_{-2}^{0}\left(x^3+3x^2+3x+3+(x+1)\cos(x+1)\right)dx$ बराबर होगा :

(A) -4 (B) 0 (C) 4 (D) 6

Ans. C

Sol. $I=\int_{-2}^{0}[x^3+3x^2+3x+3+(x+1)\cos(x+1)]dx$

$$=\left[\frac{x^4}{4}+x^3+\frac{3x^2}{2}+3x+(x+1)\sin(x+1)+\cos(x+1)\right]_{-2}^{0}$$

$=4$

14. Value of the definite integral $\int_{-1/2}^{1/2}(\sin^{-1}(3x-4x^3)-\cos^{-1}(4x^3-3x))\,dx$:

$\int_{-1/2}^{1/2}(\sin^{-1}(3x-4x^3)-\cos^{-1}(4x^3-3x))\,dx$ का मान होगा :

(A) 0 (B) $-\frac{\pi}{2}$ (C) $\frac{7\pi}{2}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. $I=\int_{-1/2}^{1/2}(\sin^{-1}(3x-4x^3)-\cos^{-1}(4x^3-3x))\,dx$

Apply king property

$I=\int_{-1/2}^{1/2}(\sin^{-1}(-3x+4x^3)-\cos^{-1}(-4x^3+3x))\,dx$

$I=\int_{-1/2}^{1/2}-\sin^{-1}\left(3x-4x^3\right)-\pi+\cos^{-1}\left(4x^3-3x\right)dx$

$\Rightarrow 2I=-\pi$

15. The value of the definite integral $\int_0^{\pi/2}\sqrt{\tan x}\,dx$, is :

$\int_0^{\pi/2}\sqrt{\tan x}\,dx$ का मान होगा :

(A) $\sqrt{2}\,\pi$ (B) $\frac{\pi}{\sqrt{2}}$ (C) $2\sqrt{2}\,\pi$ (D) $\frac{\pi}{2\sqrt{2}}$

Ans. B

Sol. $I = \int_0^{\pi/2} \sqrt{\tan x}\,dx$ ....(1) ; $I = \int_0^{\pi/2} \sqrt{\cot x}\,dx$ ....(2)

adding (1) and (2), we get

$$2I = \int_0^{\pi/2} \left(\sqrt{\tan x} + \sqrt{\cot x}\right) dx = \sqrt{2}\int_0^{\pi/2} \frac{\sin x + \cos x}{\sqrt{\sin 2x}}\,dx = \sqrt{2}\int_0^{\pi/2} \frac{\sin x + \cos x}{\sqrt{1-(\sin x - \cos x)^2}}\,dx$$

$$= \sqrt{2}\int_{-1}^{1} \frac{dt}{\sqrt{1-t^2}} = 2\sqrt{2}\int_0^1 \frac{dt}{\sqrt{1-t^2}} = \sqrt{2}\,\pi \quad (\text{where } \sin x - \cos x = t)$$

$\therefore \quad I = \frac{\pi}{\sqrt{2}}$

16. If $y(t) = \int_{-ln\,t}^{ln\,t} \cos(te^x)dx$, then $y'(1)$ equals :

यदि $y(t) = \int_{-ln\,t}^{ln\,t} \cos(te^x)dx$, तो $y'(1)$ का मान है :

(A) 2 cos 1 (B) – 2 cos 1 (C) – cos 1 (D) cos 1

Ans. A

Sol. Let $I = \int_{-ln\,t}^{ln\,t} \cos(t\,e^x)\,dx$; Put $te^x = y \Rightarrow t\,e^x\,dx = dy$

$$\therefore \quad I = \int_1^{t^2} \frac{\cos(y)\,dy}{y}$$

$$\therefore \quad y'(t) = \frac{dI}{dt} = \frac{(\cos t^2)2t}{t^2} - 0 = \frac{2\cos t^2}{t}$$

So, $y'(1) = 2\cos 1$

17. $\lim_{x\to\infty}\left(x^3 \int_{-1/x}^{1/x} \frac{ln(1+t^2)}{1+e^t}dt\right)$ equals :

$\lim\limits_{x\to\infty}\left(x^3\int\limits_{-1/x}^{1/x}\frac{ln(1+t^2)}{1+e^t}dt\right)$ का मान है :

(A) 1/3 (B) 2/3 (C) 1 (D) 0

Ans. A

Sol. Consider $I=\int\limits_{-1/x}^{1/x}\frac{ln(1+t^2)}{1+e^t}dt$ ....(1)

$=\int\limits_{-1/x}^{1/x}\frac{ln(1+t^2)}{1+e^{-t}}dt$ (Using King)

$I=\int\limits_{-1/x}^{1/x}\frac{ln(1+t^2)e^t}{1+e^t}dt$ ....(2)

(1) + (2)

$2I=\int\limits_{-1/x}^{1/x}ln(1+t^2)dt=2\int\limits_{0}^{1/x}ln(1+t^2)dt \quad\Rightarrow\quad I=\int\limits_{0}^{1/x}ln(1+t^2)dt$

hence $l=\lim\limits_{x\to\infty}x^3\int\limits_{0}^{1/x}ln(1+t^2)dt=\lim\limits_{x\to\infty}\frac{\int\limits_{0}^{1/x}ln(1+t^2)dt}{x^{-3}}$ $\left(\frac{0}{0}\text{ form}\right)$

Using L'Hospital's Rule

$$l=\lim_{x\to\infty}\frac{x^4\,ln\left(1+\frac{1}{x^2}\right)\cdot\left(-\frac{1}{x^2}\right)}{-3}=\frac{1}{3}\lim_{x\to\infty}x^2\,ln\left(1+\frac{1}{x^2}\right)=\frac{1}{3}\lim_{x\to\infty}ln\left(1+\frac{1}{x^2}\right)^{x^2}\quad(1^\infty\text{ form})$$

$$=\lim_{x\to\infty}\frac{1}{3}x^2\left(1+\frac{1}{x^2}-1\right)=\frac{1}{3}$$

18. Let $C_n=\int\limits_{\frac{1}{n+1}}^{\frac{1}{n}}\frac{\tan^{-1}(nx)}{\sin^{-1}(nx)}dx$ then $\lim\limits_{n\to\infty}n^2\cdot C_n$ equals :

माना $C_n=\int\limits_{\frac{1}{n+1}}^{\frac{1}{n}}\frac{\tan^{-1}(nx)}{\sin^{-1}(nx)}dx$, तो $\lim\limits_{n\to\infty}n^2\cdot C_n$ है :

(A) 1 (B) 0 (C) – 1 (D) $\frac{1}{2}$

Ans. D

Sol. $C_n = \int\limits_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1}(nx)}{\sin^{-1}(nx)}dx$ (put $nx = t$) $\Rightarrow$ $C_n = \frac{1}{n}\int\limits_{\frac{n}{n+1}}^{1} \frac{\tan^{-1}(t)}{\sin^{-1}(t)}dt$

$L = \lim\limits_{n\to\infty} n^2 \cdot C_n = \lim\limits_{n\to\infty} n \cdot \int\limits_{\frac{n}{n+1}}^{1} \frac{\tan^{-1} t}{\sin^{-1} t}dt$ $(\infty \times 0)$; $L = = \frac{\int\limits_{\frac{n}{n+1}}^{1} \frac{\tan^{-1} t}{\sin^{-1} t}dt}{\frac{1}{n}}$ $\left(\frac{0}{0}\right)$

applying Leibnitz rule

$$L = \lim_{n\to\infty} \frac{0 - \frac{\tan^{-1}\frac{n}{n+1}}{\sin^{-1}\frac{n}{n+1}}\left(\frac{1}{(n+1)^2}\right)}{-\frac{1}{n^2}} = \frac{\pi}{4}\cdot\frac{2}{\pi} = \frac{1}{2}$$

19. $\lim\limits_{\lambda\to 0}\left(\int\limits_0^1 (1+x)^{\lambda}dx\right)^{1/\lambda}$ is equal to :

$\lim\limits_{\lambda\to 0}\left(\int\limits_0^1 (1+x)^{\lambda}dx\right)^{1/\lambda}$ का मान होगा :

(A) 2 *l*n 2 (B) $\frac{4}{e}$ (C) *l*n $\frac{4}{e}$ (D) 4

Ans. B

Sol. $\lim\limits_{\lambda\to 0}\left(\int\limits_0^1 (1+x)^{\lambda}dx\right)^{1/\lambda} = \lim\limits_{\lambda\to 0}\left(\frac{(1+x)^{\lambda+1}}{\lambda+1}\Big|_0^1\right)^{1/\lambda} = \lim\limits_{\lambda\to 0}\left(\frac{2^{\lambda+1}-1}{\lambda+1}\right)^{1/\lambda}$ (1^{∞} form)

$$= e^{\lim\limits_{\lambda\to 0}\frac{1}{\lambda}\left(\frac{2^{\lambda+1}-1-\lambda-1}{\lambda+1}\right)} = e^{\lim\limits_{\lambda\to 0}\left(\frac{2^{\lambda+1}-2-\lambda}{\lambda(\lambda+1)}\right)} = e^{\lim\limits_{\lambda\to 0}\left(\frac{2(2^{\lambda}-1)}{\lambda}-1\right)} = e^{2\,l\text{n}\,2\,-\,1} = e^{l\text{n}\left(\frac{4}{e}\right)} = \frac{4}{e}$$

20. For positive integers k = 1, 2, 3, ..., n, let S_k denotes the area of ΔAOB_k (where 'O' is origin) such that $\angle AOB_k = \frac{k\pi}{2n}$, OA = 1 and $OB_k = k$. The value of the $\lim\limits_{n\to\infty} \frac{1}{n^2}\sum\limits_{k=1}^{n} S_k$ is :

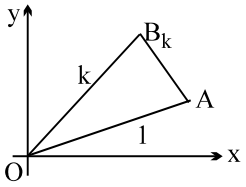
k = 1, 2, 3, ..., n, के लिए माना S_k, ΔAOB_k का क्षेत्रफल (जहाँ 'O' मूल बिन्दु है) इस प्रकार है कि $\angle AOB_k = \frac{k\pi}{2n}$, OA = 1 तथा $OB_k = k$ तो $\lim\limits_{n\to\infty} \frac{1}{n^2}\sum\limits_{k=1}^{n} S_k$ का मान होगा :

(A) $\frac{2}{\pi^2}$ (B) $\frac{4}{\pi^2}$ (C) $\frac{8}{\pi^2}$ (D) $\frac{1}{2\pi^2}$

Ans. A

Sol. $OB_k = k$

$\angle AOB_k = \frac{k\pi}{2n}$



$S_k = \frac{1}{2}k\sin\frac{k\pi}{2n}$ $\quad (\text{using } \Delta = \frac{1}{2}ab\sin\theta)$

$\therefore \quad L = \frac{k}{2n^2}\sum\limits_{n=1}^{\infty}\sin\frac{k\pi}{2n} = \frac{1}{2n}\sum\limits_{n=1}^{\infty}\frac{k}{n}\sin\frac{k\pi}{2n} = \frac{1}{2}\int\limits_0^1 x\cdot\sin\frac{\pi x}{2}dx$

$$= \frac{1}{2}\left[\underbrace{\frac{-2}{\pi}x\cos\frac{\pi x}{2}\Big|_0^1}_{\text{zero}} + \frac{2}{\pi}\int\limits_0^1\cos\frac{\pi x}{2}dx\right] = \frac{1}{2}\left[0+\frac{2}{\pi}\cdot\frac{2}{\pi}\left(\sin\frac{\pi x}{2}\right)_0^1\right] = \frac{2}{\pi^2}$$

21. The interval [0, 4] is divided into *n* equal sub-intervals by the points x_0, x_1, x_2, ..., x_{n-1}, x_n where $0 = x_0 < x_1 < x_2 < x_3 ..... < x_n = 4$. If $\delta x = x_i - x_{i-1}$ for i = 1, 2, 3, ... n then $\lim\limits_{\delta x\to 0}\sum\limits_{i=1}^{n} x_i\delta x$ is equal to:

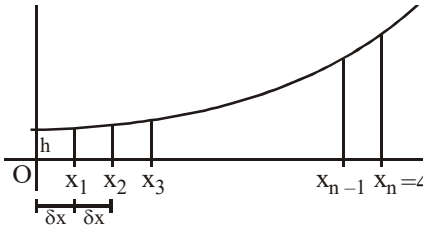
माना अन्तराल [0, 4], *n* समान उपअन्तरालों में बिन्दुओं x_0, x_1, x_2, ..., x_{n-1}, x_n द्वारा बांटा जाता है, जहाँ $0 = x_0 < x_1 < x_2 < x_3 ..... < x_n = 4$. यदि $\delta x = x_i - x_{i-1}$, i = 1, 2, 3, ... n के लिए तो $\lim\limits_{\delta x\to 0}\sum\limits_{i=1}^{n} x_i\delta x$ का मान होगा :

(A) 4 (B) 8 (C) $\frac{32}{3}$ (D) 16

Ans. B

Sol. $L = \lim\limits_{\delta x\to 0}\delta x(x_1 + x_2 + x_3 + ... + x_n)$

$$= \lim_{n\to\infty} \frac{4}{n}\left[\frac{4}{n}+\frac{8}{n}+\frac{12}{n}+.....+4\cdot\frac{n}{n}\right] \quad (\delta x = \frac{4}{n})$$



$$= \lim_{n\to\infty} \frac{16}{n^2}(1+2+3+.....+n) = \lim_{n\to\infty} \frac{16}{n^2}\cdot\frac{n(n+1)}{2} = 8$$

22. If $f(\theta) = \frac{4}{3}\left(1-\cos^6\theta-\sin^6\theta\right)$, then $\lim_{n\to\infty}\frac{1}{n}\left[\sqrt{f\left(\frac{1}{n}\right)}+\sqrt{f\left(\frac{2}{n}\right)}+\sqrt{f\left(\frac{3}{n}\right)}+.....+\sqrt{f\left(\frac{n}{n}\right)}\right] =$

यदि $f(\theta) = \frac{4}{3}\left(1-\cos^6\theta-\sin^6\theta\right)$ है, तो $\lim_{n\to\infty}\frac{1}{n}\left[\sqrt{f\left(\frac{1}{n}\right)}+\sqrt{f\left(\frac{2}{n}\right)}+\sqrt{f\left(\frac{3}{n}\right)}+.....+\sqrt{f\left(\frac{n}{n}\right)}\right] =$

(A) $\frac{1-\cos1}{2}$ (B) $1-\cos2$ (C) $\frac{\sin2}{2}$ (D) $\frac{1-\cos2}{2}$

Ans. D

Sol. $f(\theta) = \frac{4}{3}\left(1-\cos^6\theta-\sin^6\theta\right)$

$$\lim_{n\to\infty}\frac{1}{n}\left[\sqrt{f\left(\frac{1}{n}\right)}+\sqrt{f\left(\frac{2}{n}\right)}+\sqrt{f\left(\frac{3}{n}\right)}+.....+\sqrt{f\left(\frac{n}{n}\right)}\right] = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\sqrt{f\left(\frac{r}{n}\right)}$$

$$= \int_0^1 \sqrt{f(x)}dx = \sqrt{\frac{4}{3}}\int_0^1 \sqrt{1-\cos^6 x-\sin^6 x}\,dx$$

$$= 2\int_0^1 \sin x\cos x\,dx = \left[\frac{\cos 2x}{-2}\right]_0^1$$

$$= \frac{1-\cos2}{2}$$

23. Let f(x) be a function satisfying $f'(x)$ = f(x) with f(0) = 1 and g(x) be a function that satisfies $f(x)+g(x)=x^2$. Then the value of the integral $\int_0^1 f(x)\,g(x)\,dx$, is :

यदि f(x) फलन इस प्रकार है कि जो $f'(x) = f(x)$ तथा $f(0) = 1$ को संतुष्ट करता है एवं g(x) इस प्रकार है कि जो $f(x) + g(x) = x^2$ को संतुष्ट करता है तो समाकलन $\int_0^1 f(x)\, g(x)\, dx$ का मान होगा :

(A) $e + \frac{e^2}{2} + \frac{5}{2}$ (B) $e - \frac{e^2}{2} - \frac{5}{2}$ (C) $e + \frac{e^2}{2} - \frac{3}{2}$ (D) $e - \frac{e^2}{2} - \frac{3}{2}$

Ans. D

Sol. $f'(x) = f(x) \text{ and } f(0) = 1 \quad \Rightarrow f(x) = e^x$

$$I = \int_0^1 f(x) g(x) dx = \int_0^1 e^x \left(x^2 - e^x\right) dx$$

$$= e - \frac{e^2}{2} - \frac{3}{2}$$

24. The integral $\int \sqrt{\cot x}\, e^{\sqrt{\sin x}} \sqrt{\cos x}\, dx$ equals :

$\int \sqrt{\cot x}\, e^{\sqrt{\sin x}} \sqrt{\cos x}\, dx$ बराबर :

(A) $\frac{\sqrt{\tan x}\, e^{\sqrt{\sin x}}}{\sqrt{\cos x}} + C$ (B) $2e^{\sqrt{\sin x}} + C$ (C) $-\frac{1}{2} e^{\sqrt{\sin x}} + C$ (D) $\frac{\sqrt{\cot x}\, e^{\sqrt{\sin x}}}{2\sqrt{\cos x}} + C$

Ans. B

Sol. $e^{\sqrt{\sin x}} = t$

$$\frac{e^{\sqrt{\sin x}}}{2\sqrt{\sin x}} \cos x ; \quad e^{\sqrt{\sin x}} \sqrt{\cot x} \sqrt{\cos x}\, dx = 2\, dt$$

Hence $\int 2\, dt = 2t + C \Rightarrow \quad 2e^{\sqrt{\sin x}} + C$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

25. $\int_0^\infty \frac{x}{(1+x)(1+x^2)} dx =$

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$

(C) is same as $\int_0^\infty \frac{dx}{(1+x)(1+x^2)}$ (D) cannot be evaluated

$$\int_0^{\infty} \frac{x}{(1+x)(1+x^2)}dx =$$

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$

(C) $\int_0^{\infty} \frac{dx}{(1+x)(1+x^2)}$ के समान (D) ज्ञात नही किया जा सकता

Ans. AC

Sol. $I = \int_0^{\infty} \frac{x}{(1+x)(1+x^2)}dx$ _____(1)

let $x = \frac{1}{t}$

$$I = \int_{\infty}^{0} \frac{1}{t\left(\frac{t+1}{t}\right)\left(\frac{t^2+1}{t^2}\right)} \times \frac{-1}{t^2}dt$$

$$\Rightarrow I = \int_0^{\infty} \frac{dt}{(t+1)(1+t^2)}$$

$$\Rightarrow I = \int_0^{\infty} \frac{1}{(1+x)(1+x^2)}dx$$ _____(2)

By (1) + (2)

$$2I = \int_0^{\infty} \frac{dx}{1+x^2} = \left[\tan^{-1} x\right]_0^{\infty} = \frac{\pi}{2}$$

$$\Rightarrow I = \frac{\pi}{4}$$

26. Let $f(x) = \int_0^x e^t \cdot \sin(x-t)dt$ and $g(x) = f(x) + f''(x)$. Which of the following statements are correct?

(A) g (x) is positive $\forall$ $x \in R$. (B) g (x) is a constant function

(C) g (1) = e (D) g ' (x) = g (x)

माना $f(x) = \int_0^x e^t \cdot \sin(x-t)dt$ तथा $g(x) = f(x) + f''(x)$. निम्न में से कौनसे कथन सत्य है?

(A)g (x) धनात्मक है $\forall$ x $\in$ R. ; ठ द्धह ; गद्धएक अचर फलन है

(C)g (1) = e (D)g ' (x) = g (x)

Ans. ACD

Sol. $f(x) = \int_0^x e^t (\sin x \cos t - \cos x \sin t)\, dt$

$$f(x) = \sin x \int_0^x e^t \cos t\, dt - \cos x \int_0^x e^t \sin t\, dt \quad ....(1)$$

$$f'(x) = \sin x \cdot e^x \cos x + \left(\int_0^x e^t \cos t\, dt\right)\cos x - \left[\cos x \cdot e^x \sin x - \left(\int_0^x e^t \sin t\, dt\right)\sin x\right]$$

$$f'(x) = \cos x \int_0^x e^t \cos t\, dt + \sin x \int_0^x e^t \sin t\, dt$$

$$f''(x) = \cos x \cdot e^x \cdot \cos x - \sin x \int_0^x e^t \cos t\, dt + \sin x \cdot e^x \cdot \sin x + \left(\int_0^x e^t \sin t\, dt\right)\cos x$$

$$f''(x) = e^x + \cos x \int_0^x e^t \sin t\, dt - \sin x \int_0^x e^t \cos t\, dt \quad ....(2)$$

(1) + (2)

$$f(x) + f''(x) = e^x = g(x) \quad \Rightarrow \quad A, C, D$$

27. Let $I_n = \int_0^{\sqrt{3}} \frac{dx}{1+x^n}$ $(n = 1, 2, 3 .....)$ and $\lim_{n\to\infty} I_n = I_0$ (say), then which of the following statement(s)is/are correct?

(Given : e = 2.71828)

माना $I_n = \int_0^{\sqrt{3}} \frac{dx}{1+x^n}$ $(n = 1, 2, 3 .....)$ तथा $\lim_{n\to\infty} I_n = I_0$ (say) तो निम्न में से कौनसे कथन सत्य है?

(दिया गया है : e = 2.71828)

(A) $I_1 > I_0$ (B) $I_2 < I_0$ (C) $I_0 + I_1 + I_2 > 3$ (D) $I_0 + I_1 > 2$

Ans. ACD

Sol. We have $I_1 = \ln(1 + \sqrt{3})$; $I_2 = \int_0^{\sqrt{3}} \frac{dx}{1+x^2}$

$$I_2 = \frac{\pi}{3}$$

$$I_0 = \lim_{n\to\infty} I_n = \lim_{n\to\infty}\left(\int_0^1 \frac{dx}{1+x^n} + \underbrace{\int_1^{\sqrt{3}} \frac{dx}{1+x^n}}_{\text{zero}} \right) = \int_0^1 dx = 1$$

Hence $I_0 = 1$. Now verify all alternatives

28. Let $I_n = \int_0^{\pi/4} (\tan x)^n \, dx$ and let $J_n = (-1)^n I_{2n}$ for $n = 0, 1, 2$ then which of the following hold(s) good?

(A) $I_n + I_{n+2} = \frac{1}{n+1}$ (B) $J_n - J_{n-1} = \frac{(-1)^n}{2n-1}$ for $n \geq 1$

(C) If $u = \tan x$ then $I_n = \int_0^1 \frac{u^n}{1+u^2} du$ (D) $\lim_{n\to\infty} J_n \to 0$

माना $I_n = \int_0^{\pi/4} (\tan x)^n \, dx$ and let $J_n = (-1)^n I_{2n}$, $n = 0, 1, 2$ के लिए, तो निम्न में से कौनसे कथन सत्य है?

(A)$I_n + I_{n+2} = \frac{1}{n+1}$ (B)$J_n - J_{n-1} = \frac{(-1)^n}{2n-1}$, $n \geq 1$ के लिये

(C)यदि $u = \tan x$ तो $I_n = \int_0^1 \frac{u^n}{1+u^2} du$ (D)$\lim_{n\to\infty} J_n \to 0$

Ans. ABCD

Sol. (A) $I_n + I_{n+2} = \int_0^{\pi/4} \tan^n x \sec^2 x \, dx = \int_0^1 t^n \, dt = \frac{1}{n+1}$ ....(1) $\Rightarrow$ (A) is correct

(B) $J_n - J_{n-1} = (-1)^n I_{2n} - (-1)^{n-1} \cdot I_{2n-2}$

$= (-1)^n [I_{2n} + I_{2n-2}] = (-1)^n \frac{1}{2n-2+1}$ (putting $n \to 2n-2$ in (1))

$= \frac{(-1)^n}{2n-1}$ $\Rightarrow$ (B) is correct

(C) If $u = \tan x \Rightarrow$ $du = \sec^2 x = dx$ $\Rightarrow$ $dx = \frac{du}{1+u^2}$

$\therefore$ $I_n = \int_0^1 \frac{u^n}{1+u^2} du$ $\Rightarrow$ (C) is also correct

(D) obviously correct

29. Which of the following statement(s) is/are TRUE?

(A) If $f(x)$ is continuous function such that $f(c) \neq 0$ then $f(x)\, f(c) > 0 \ \forall\, x \in (c-h, c+h)$ where h is sufficiently small positive quantity.

(B) $\underset{n\to\infty}{\text{Limit}} \frac{1}{n} \ln\left(\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right).....\left(1+\frac{n}{n}\right)\right) = 1 + 2 \ln 2.$

(C) Let f be a continuous and non-negative function defined on [a. b].

If $\int_a^b f(x)dx = 0$ then $f(x) = 0 \ \ \forall\, x \in [a,b]$.

(D) Let f be a continuous function defined on [a, b] such that $\int_a^b f(x)dx = 0$, then there exists atleast one $c \in (a, b)$ for which $f(c) = 0$

निम्न में से कौनसे कथन सत्य हैं :

; । द्धयदि फलन $y = f(x)$, $x = c$ पर सतत् है तथा $f(c) \neq 0$ हो तो $f(x)\, f(c) > 0 \ \forall\, x \in (c-h, c+h)$ जहाँ h नगण्य धनात्मक संख्या है।

(B) $\underset{n\to\infty}{\text{Limit}} \frac{1}{n} \ln\left(\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right).....\left(1+\frac{n}{n}\right)\right) = 1 + 2 \ln 2.$

(C) माना f अन्तराल [a. b] में सतत् तथा अऋणात्मक फलन है। यदि $\int_a^b f(x)dx = 0$ हो तो $f(x) = 0 \ \ \forall\, x \in [a,b]$.

; क द्धमाना फलन f अन्तराल [a, b] में इस प्रकार परिभाषित है कि $\int_a^b f(x)dx = 0$, तो कम से कम एक $c \in (a, b)$ हो जिसके लिए $f(c) = 0$ है।

Ans. ACD

Sol. (A) The expression $f(x)\, f(c) \ \forall\, x \in (c-h, c+h)$ where $h \to 0^+$ is equivalent to $\underset{x\to c}{\text{Lim}}\ f(x)\, f(c)$ which equals to $(f(c))^2$ because $f(x)$ is continuous.

$\therefore\ f(x)\, f(c) > 0 \ \ \forall\, x \in (c-h, c+h)$ where $h \to 0^+$.

(B) We have $I = \underset{n\to\infty}{\text{Lim}} \frac{1}{n} \ln\left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right).....\left(1+\frac{n}{n}\right)\right] = \underset{n\to\infty}{\text{Lim}} \frac{1}{n} \ln \prod_{k=1}^{n}\left(1+\frac{k}{n}\right)$

$= \underset{n\to\infty}{\text{Lim}} \frac{1}{n} \sum_{k=1}^{n} \ln\left(1+\frac{k}{n}\right) = \int_1^2 \ln x\, dx = \left[x(\ln x - 1)\right]_{x=1}^{x=2} = -1 + 2\ln 2 \approx -0.4.$

(C) Given $f(x) \geq 0 \ \Rightarrow \int_a^b f(x)dx \geq 0$.

But given $\int_a^b f(x)dx = 0$, so this can be true only when $f(x) = 0$.

(D) $\int_a^b f(x)\,dx = 0 \Rightarrow$ $y = f(x)$ cuts x axis at least once.

So there exists at least one $c \in (a, b)$ for which $f(c) = 0$

30. Let $f(x) = \int_0^x e^{t-[t]}\ dt\ (x > 0)$, where [x] denotes greatest integer less than or equal to x, is :

(A) continuous and differentiable $\forall\ x \in (0,3]$ (B) continuous but not differentiable $\forall\ x \in (0, 3]$

(C) $f(1) = e$ (D) $f(2) = 2(e - 1)$

माना $f(x) = \int_0^x e^{t-[t]}\ dt\ (x > 0)$, जहाँ [x] महत्तम पूर्णांक फलन है :

; । द्धग $\in (0, 3]$ में सतत् तथा अवकलनीय है (B) $x \in (0, 3]$ में सतत् परन्तु अवकलनीय नही है

(C) $f(1) = e$ (D)$f(2) = 2(e - 1)$

Ans. BD

Sol. We have $f(x) = \int_0^x e^{t-[t]}dt = \int_0^x e^{\{t\}}dt$, so

$$f(x) = \begin{cases} \int_0^x e^t\,dt & \text{if } x \in [0,1) \\ \int_0^1 e^t\,dt + \int_1^x e^{t-1}\,dt & \text{if } x \in [1,2) \\ \int_0^1 e^t\,dt + \int_1^2 e^{t-1}\,dt + \int_2^x e^{t-2}\,dt & \text{if } x \in [2,3) \end{cases} \Rightarrow f(x) = \begin{cases} e^x - 1 & \text{if } x \in [0, 1) \\ (e-1) + (e^{x-1} - 1) & \text{if } x \in [1,2) \\ 2(e-1) + (e^{x-2} - 1) & \text{if } x \in [2,3) \end{cases}$$

Clearly $f(x)$ is continuous $\forall\ x > 0$ but not differentiable $\forall\ x \in N \Rightarrow$ (B)

Also $f(2) = 2(e - 1) = 0 = 2(e - 1) \Rightarrow$ (D)

31. If $a, b, c \in R^+$ then $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{n}{(k + an)(k + bn)}$ is equal to :

(A) $\frac{1}{a-b} ln \frac{b(b+1)}{a(a+1)}$ if $a \neq b$ (B) $\frac{1}{a-b} ln \frac{a(b+1)}{b(a+1)}$ if $a \neq b$

(C) non existent if $a = b$ (D) $\frac{1}{a(1+a)}$ if $a = b$

यदि a, b, c $\in R^+$ हो, तो $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{n}{(k+an)(k+bn)}$ का मान है :

(A) $\dfrac{1}{a-b}\ell n\dfrac{b(b+1)}{a(a+1)}$ यदि $a\neq b$ (B) $\dfrac{1}{a-b}\ell n\dfrac{a(b+1)}{b(a+1)}$ यदि $a\neq b$

;ं द्धयदि a = b तो अस्तित्व में नहीं (D) $\dfrac{1}{a(1+a)}$ यदि a = b

Ans. BD

Sol. We have $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{n}{n^2\left(a+\dfrac{k}{n}\right)\left(b+\dfrac{k}{n}\right)}=\lim\limits_{n\to\infty}\dfrac{1}{n}\sum\limits_{k=1}^{n}\dfrac{1}{\left(a+\dfrac{k}{n}\right)\left(b+\dfrac{k}{n}\right)}$

$$=\int_0^1\frac{1}{(a+x)(b+x)}dx=\frac{1}{a-b}\int_0^1\frac{(a+x)-(b+x)}{(a+x)(b+x)}dx=\frac{1}{a-b}\int_0^1\left(\frac{1}{b+x}-\frac{1}{a+x}\right)dx$$

$$=\frac{1}{a-b}\left[\ln(b+x)-\ln(a+x)\right]_0^1=\frac{1}{a-b}\left[\ln\frac{(b+x)}{(a+x)}\right]_0^1=\frac{1}{a-b}\ell n\frac{(b+1)a}{(a+1)b}\Rightarrow\text{(B)}$$

Now if a = b then $\ell=\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{1}{n}\dfrac{1}{\left(a+\dfrac{k}{n}\right)^2}=\int_0^1\frac{dx}{(a+x)^2}=\left.\frac{1}{a+x}\right]_1^0=\frac{1}{a}-\frac{1}{a+1}=\frac{1}{a(a+1)}\Rightarrow$ (D)

32. If $\int_{-\alpha}^{\alpha}\left(e^x+\cos x\ell n\left(x+\sqrt{1+x^2}\right)\right)dx>\dfrac{3}{2}$, then possible value of α can be :

यदि $\int_{-\alpha}^{\alpha}\left(e^x+\cos x\ell n\left(x+\sqrt{1+x^2}\right)\right)dx>\dfrac{3}{2}$ है, तो α के संभावित मान होगें :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. ABCD

Sol. $I=\int_{-\alpha}^{\alpha}e^x+\cos x\,\ell n\left(x+\sqrt{1+x^2}\right)dx$

Apply king prop.

$$I=\int_{-\alpha}^{\alpha}\left(e^{-x}-\cos x\,\ell n\left(\sqrt{x^2+1}+x\right)\right)dx$$

$$\Rightarrow 2I=\int_{-\alpha}^{\alpha}\left(e^x+e^{-x}\right)dx$$

$$= 2\int_0^{\alpha}\left(e^{x}+e^{-x}\right)dx$$

$$\Rightarrow I=\left[e^{\alpha}-1\right]-\left[e^{-\alpha}-1\right]$$

$$\Rightarrow I=\frac{e^{2\alpha}-1}{e^{\alpha}}$$

Given $I>\frac{3}{2}$

$$\Rightarrow 2e^{2\alpha}-2>3e^{\alpha}$$

let $e^{\alpha}=t$

$$2t^{2}-3t-2>0$$

$$\Rightarrow 2t^{2}-3t-2>0$$

$$\Rightarrow e^{\alpha}<-\frac{1}{2} \text{ or } e^{\alpha}>2$$

$\Rightarrow \alpha$ can be 1, 2, 3 or 4

CHALLENGER EXERCISES

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. $\int \frac{\left(e^{\sqrt{x}}-e^{-\sqrt{x}}\right)\cos\left(e^{\sqrt{x}}+e^{-\sqrt{x}}+\frac{\pi}{4}\right)+\left(e^{\sqrt{x}}+e^{-\sqrt{x}}\right)\cos\left(e^{\sqrt{x}}-e^{-\sqrt{x}}+\frac{\pi}{4}\right)}{\sqrt{x}}dx$

Ans. $2\sqrt{2}\left(\cos\left(e^{-\sqrt{x}}\right)\right)\left(\sin\left(e^{\sqrt{x}}\right)+\cos\left(e^{\sqrt{x}}\right)\right)+C$

Sol. $I=\int \frac{\left(e^{\sqrt{x}}-e^{-\sqrt{x}}\right)\cos\left(e^{\sqrt{x}}+e^{-\sqrt{x}}+\frac{\pi}{4}\right)+\left(e^{\sqrt{x}}+e^{-\sqrt{x}}\right)\cos\left(e^{\sqrt{x}}-e^{-\sqrt{x}}+\frac{\pi}{4}\right)}{\sqrt{x}}dx$

$=2\int\left(\cos\left(e^{\sqrt{x}}+e^{-\sqrt{x}}+\frac{\pi}{4}\right)\frac{d}{dx}\left(e^{\sqrt{x}}+e^{-\sqrt{x}}+\frac{\pi}{4}\right)\right)dx \;+2\int\left(\cos\left(e^{\sqrt{x}}-e^{-\sqrt{x}}+\frac{\pi}{4}\right)\frac{d}{dx}\left(e^{\sqrt{x}}-e^{-\sqrt{x}}+\frac{\pi}{4}\right)\right)dx$

$=2\sin\left(e^{\sqrt{x}}+e^{-\sqrt{x}}+\frac{\pi}{4}\right)+2\sin\left(e^{\sqrt{x}}-e^{-\sqrt{x}}+\frac{\pi}{4}\right)+C$

$=4\sin\left(\frac{\pi}{4}+e^{\sqrt{x}}\right)\cos\left(e^{-\sqrt{x}}\right)+C$

$=2\sqrt{2}\left(\cos(e^{-\sqrt{x}})\right)\left(\sin(e^{\sqrt{x}})+\cos(e^{\sqrt{x}})\right)+C$

2. If the value $\int \frac{1-(\cot x)^{2008}}{\tan x+(\cot x)^{2009}}dx=\frac{1}{k}ln\,|\sin^k x+\cos^k x|+C$, then find k.

यदि $\int \frac{1-(\cot x)^{2008}}{\tan x+(\cot x)^{2009}}dx=\frac{1}{k}ln\,|\sin^k x+\cos^k x|+C$, तो k का मान बताओं।

Ans. 2010

Sol. $I=\int \frac{1-(\cot x)^{2008}}{\tan x+(\cot x)^{2009}}dx$

$=\int \frac{\left(\sin^{2008}x-\cos^{2008}x\right)\sin x\cos x}{\sin^{2010}x+\cos^{2010}x}dx$

Let $\sin^{2010}x+\cos^{2010}x=t$

$=\frac{1}{2010}\int\frac{dt}{t}$

$$= \frac{1}{2010}\ell n\left|\sin^{2010} x + \cos^{2010} x\right| + C$$

$$\Rightarrow k = 2010$$

3. $\int \cos 2\theta \, . \, ln \frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta} d\theta$

Ans. $\frac{1}{2}(\sin 2\theta) ln\left(\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right) - \frac{1}{2} ln(\sec 2\theta) + C$

Sol. $\int \cos 2\theta \ell n\left(\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right) d\theta$

$$= \ell n\left(\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right)\frac{\sin 2\theta}{2} - \int \frac{\sin 2\theta}{2} . \frac{2}{\cos 2\theta} d\theta$$

$$= \frac{1}{2}\ell n\left(\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right)\sin 2\theta - \frac{1}{2}\ell n(\sec 2\theta) + C$$

4. $\int \frac{x}{(7x - 10 - x^2)^{3/2}} dx$

Ans. $\frac{2(7x-20)}{9\sqrt{7x-10-x^2}} + C$

Sol. $I = \int \frac{x}{(7x - 10 - x^2)^{3/2}} dx$

$$= \int \frac{x dx}{\left(\frac{9}{4} - \left(x - \frac{7}{2}\right)^2\right)^{\frac{3}{2}}}$$

$$= -\frac{1}{2}\left[\int \frac{(-2x + 7) dx}{\left(\frac{9}{4} - \left(x - \frac{7}{2}\right)^2\right)^{\frac{3}{2}}} - 7\int \frac{dx}{\left(\frac{9}{4} - \left(x - \frac{7}{2}\right)^2\right)^{\frac{3}{2}}}\right]$$

$\downarrow$ $\downarrow$

Let $\frac{9}{4} - \left(x - \frac{7}{2}\right)^2 = t^2$ $\left(x - \frac{7}{2}\right) = \frac{3}{2}\sin\theta$

$$= -\frac{1}{2}\left[\int \frac{2}{t^2}dt - \frac{28}{9}\int \sec^2\theta d\theta\right]$$

$$= \frac{1}{t} + \frac{14}{9}\tan\theta + C$$

$$I = \frac{2(7x-20)}{9\sqrt{7x-10-x^2}} + C$$

5. $$\int\left[\frac{\sqrt{x^2+1}\left[\ell n\left(x^2+1\right)-2\ell nx\right]}{x^4}\right]dx$$

Ans. $$\frac{\left(x^2+1\right)\sqrt{x^2+1}}{9x^3}.\left[2-3\ln\left(1+\frac{1}{x^2}\right)\right]$$

Sol. $$I = \int\left[\frac{\sqrt{x^2+1}\left[\ell n\left(x^2+1\right)-2\ell nx\right]}{x^4}\right]dx$$

$$= \int \frac{\sqrt{x^2+1}}{x^4}\ell n\left(1+\frac{1}{x^2}\right)dx$$

Let $1+\frac{1}{x^2} = t^2$

$$\Rightarrow I = -2\int t^2 \ell ntdt$$

$$= -2\left(\frac{t^3}{3}\ell nt - \frac{t^3}{9}\right) + C$$

$$= -\frac{2}{3}\left(t^3\right)\left(\ell nt - \frac{1}{3}\right) + C$$

$$I = \frac{\left(x^2+1\right)\sqrt{x^2+1}}{9x^3}.\left[2-3\ln\left(1+\frac{1}{x^2}\right)\right] + C$$

6. $\int \sqrt{\frac{\sin(x-a)}{\sin(x+a)}}dx$

Ans. $\cos a \cdot \arccos\left(\frac{\cos x}{\cos a}\right) - \sin a \cdot \ln\left(\sin x + \sqrt{\sin^2 x - \sin^2 a}\right) + C$

Sol. $I = \int \sqrt{\frac{\sin(x-a)}{\sin(x+a)}}dx$

$$= \int \frac{\sin(x-a)}{\sqrt{\sin^2 x - \sin^2 a}}dx$$

$$= \int \frac{\sin x \cos a}{\sqrt{1-\cos^2 x - \sin^2 a}}dx - \int \frac{\cos x \sin a}{\sqrt{\sin^2 x - \sin^2 a}}dx$$

$$= \int \frac{\sin x \cos a}{\sqrt{\cos^2 a - \cos^2 x}}dx - \int \frac{\cos x \sin a}{\sqrt{\sin^2 x - \sin^2 a}}dx$$

$\cos x = t$ $\qquad\qquad$ $\sin x = v$

$$= -\cos a \int \frac{dt}{\sqrt{\cos^2 a - t^2}} - \int \frac{\sin a dv}{\sqrt{v^2 - \sin^2 a}}$$

$$= \cos a \cos^{-1}\left(\frac{t}{\cos a}\right) + C - \sin a \ell n\left(v + \sqrt{v^2 - \sin^2 a}\right)$$

$$= \cos a \cos^{-1}\left(\frac{\cos x}{\cos a}\right) - \sin a \ell n\left(\sin x + \sqrt{\sin^2 x - \sin^2 a}\right) + C$$

$$= \cos a \cdot \arccos\left(\frac{\cos x}{\cos a}\right) - \sin a \cdot \ln\left(\sin x + \sqrt{\sin^2 x - \sin^2 a}\right) + C$$

7. $\int \frac{\cot x\, dx}{(1-\sin x)(\sec x + 1)}$

Ans. $\frac{1}{2} \ln\left|\tan\frac{x}{2}\right| + \frac{1}{4} \sec^2 \frac{x}{2} + \tan \frac{x}{2} + C$

Sol. $\int \frac{\cot x dx}{(1-\sin x)(\sec x + 1)} = \int \frac{\cos^2 x\ dx}{\sin x(1-\sin x)(1+\cos x)}$

$$= \int \frac{(1+\sin x)}{\sin x(1+\cos x)}dx = \int \frac{dx}{\sin x(1+\cos x)} + \int \frac{1}{1+\cos x}dx$$

$$= \int \frac{\sin x\ dx}{(1-\cos^2 x)(1+\cos x)} + \tan\frac{x}{2} + C$$

$$= \int \frac{-dt}{(1+t)(1-t^2)} + \tan\frac{x}{2} + C$$

$$= \frac{1}{2}\ell n\left|\tan\frac{x}{2}\right| + \frac{1}{4}\sec^2\frac{x}{2} + \tan\frac{x}{2} + C$$

8. $\int \frac{\sin(x+\alpha)}{\cos^3 x}\sqrt{\frac{\text{cosec}\, x + \sec x}{\text{cosec}\, x - \sec x}}\, dx$

Ans. $-(\cos\alpha + \sin\alpha)\sin 2\theta - 2\theta\sin\alpha - \theta\cos\alpha - \cos\alpha\frac{\sin 4\theta}{4} + C$

Sol. $I = \int \frac{\sin(x+\alpha)}{\cos^3 x}\sqrt{\frac{\cos ecx + \sec x}{\cos ecx - \sec x}}dx$

$$= \int\left(\frac{\sin x\cos\alpha + \cos x\sin\alpha}{\cos^3 x}\right)\sqrt{\frac{\cos x + \sin x}{\cos x - \sin x}}dx$$

$$= \int(\tan x\cos\alpha + \sin\alpha)\sec^2 x\sqrt{\frac{1+\tan x}{1-\tan x}}dx$$

let $\tan x = t$

$$\Rightarrow I = \int(t\cos\alpha + \sin\alpha)\sqrt{\frac{1+t}{1-t}}dt$$

let $t = \cos 2\theta$

$$\Rightarrow I = -\int(\sin\alpha + \cos\alpha\cos 2\theta)\cot\theta\sin 2\theta\cdot 2d\theta$$

$$\Rightarrow I = -4\int(\sin\alpha + \cos\alpha\cos 2\theta)\cos^2\theta d\theta$$

$$\Rightarrow I = -4\int(\sin\alpha + \cos\alpha\cos 2\theta)\cos^2\theta d\theta$$

$$= -4\left[\int\sin\alpha\cos^2\theta d\theta + \int\cos\alpha\cos 2\theta\cos^2\theta d\theta\right]$$

$$= -4\left[\frac{\sin\alpha}{2}\left(\theta+\frac{\sin 2\theta}{2}\right)+\frac{\cos\alpha}{2}\int\left(\cos 2\theta+\cos^2 2\theta\right)d\theta\right]$$

$$= -4\left[\sin\alpha\frac{(2\theta+\sin 2\theta)}{4}+\frac{\cos\alpha}{2}\left[\frac{\sin 2\theta}{2}+\frac{\theta}{2}+\frac{\sin 4\theta}{8}\right]\right]+C$$

$$= -\left[\sin\alpha(2\theta+\sin 2\theta)+\frac{4\cos\alpha}{2}\left[\frac{\theta}{2}+\frac{\sin 2\theta}{2}+\frac{\sin 4\theta}{8}\right]\right]+C$$

$$= -\sin\alpha(2\theta+\sin 2\theta)-\cos\alpha\left(\theta+\sin 2\theta+\frac{\sin 4\theta}{4}\right)+C$$

$$= -(\cos\alpha+\sin\alpha)\sin 2\theta-2\theta\sin\alpha-\theta\cos\alpha-\cos\alpha\frac{\sin 4\theta}{4}+C$$

9. $\int\frac{dx}{\sqrt{\sin^3 x\sin(x+\alpha)}}$

Ans. $C-\frac{2}{\sin\alpha}\sqrt{\frac{\sin(x+\alpha)}{\sin x}}$

Sol. $I=\int\frac{dx}{\sqrt{\sin^3 x\sin(x+\alpha)}}$

Let $\sqrt{\frac{\sin(x+\alpha)}{\sin x}}=t$

$$\Rightarrow -\frac{\sin\alpha}{2\sin^2 x\sqrt{\frac{\sin(x+\alpha)}{\sin x}}}dx=dt$$

$$\Rightarrow I=-\frac{2}{\sin\alpha}\int dt=-\frac{2t}{\sin\alpha}+C$$

$$I=C-\frac{2}{\sin\alpha}\sqrt{\frac{\sin(x+\alpha)}{\sin x}}$$

10. $\int\frac{dx}{\cos^3 x-\sin^3 x}$

Ans. $\frac{2}{3}\tan^{-1}(\sin x+\cos x)+\frac{1}{3\sqrt{2}}\,ln\left|\frac{\sqrt{2}+\sin x+\cos x}{\sqrt{2}-\sin x-\cos x}\right|+C$

Sol. $I = \int \frac{dx}{\cos^3 x - \sin^3 x}$

$$= \int \frac{dx}{(\cos x - \sin x)(1 + \sin x \cos x)}$$

let $\cos x - \sin x = t$

$\Rightarrow 2 \sin x \cos x = 1 - t^2$

$$= -2\int \frac{\sqrt{2 - t^2}}{t\left(3 - t^2\right)} dt$$

let $2 - t^2 = p^2$

$\Rightarrow -2tdt = 2pdp$

$$I = \int \frac{2p^2 dp}{\left(2 - p^2\right)\left(1 + p^2\right)}$$

$$= 2\int \frac{\left(1 + p^2 - 1\right)}{\left(2 - p^2\right)\left(1 + p^2\right)} dp$$

$$= \int \frac{2dp}{2 - p^2} - \int \frac{2dp}{\left(2 - p^2\right)\left(1 + p^2\right)}$$

$$= 2\int \frac{dp}{2 - p^2} - \frac{2}{3}\int \frac{\left(1 + p^2\right) + \left(2 - p^2\right)}{\left(1 + p^2\right)\left(2 - p^2\right)} dp$$

$$= 2\int \frac{dp}{2 - p^2} - \frac{2}{3}\int \frac{dp}{2 - p^2} - \frac{2}{3}\int \frac{dp}{1 + p^2}$$

$$= \frac{4}{3}\int \frac{dp}{2 - p^2} - \frac{2}{3}\tan^{-1}(p)$$

$$= \frac{4}{3}\frac{1}{2\sqrt{2}}\ell n\left(\frac{\sqrt{2} + p}{\sqrt{2} - p}\right) - \frac{2}{3}\tan^{-1} p + C$$

$$= \frac{2}{3\sqrt{2}}\ell n\left(\frac{\sqrt{2} + \sin x + \cos x}{\sqrt{2} - \sin x - \cos x}\right) - \frac{2}{3}\tan^{-1}(\sin x + \cos x) + C$$

11. $\int \frac{\sqrt{\sin^4 x + \cos^4 x}}{\sin^3 x \cos x} dx$, $x \in \left(0, \frac{\pi}{2}\right)$

Ans. $C - \frac{\sqrt{1+t^2}}{2} - \frac{1}{4} \ell n \frac{\sqrt{t^2+1}-1}{\sqrt{t^2+1}+1}$, where $t = \cot^2 x$

Sol. $I = \int \frac{\sqrt{\sin^4 x + \cos^4 x}}{\sin^3 x \cos x} dx; x \in \left(0, \frac{\pi}{2}\right)$

$$I = \int \frac{\sqrt{1+\cot^4 x}}{\sin x \cos x} dx$$

Let $\cot^2 x = t$

$$\Rightarrow 2 \cot x \left(-\cos ec^2 x\right) dx = dt$$

$$\Rightarrow \frac{-2\cos x}{\sin^3 x} dx = dt \Rightarrow \frac{-2\cot^2 x}{\sin x \cos x} dx = dt$$

$$I = \int \frac{\sqrt{1+t^2}}{-2t} dt$$

$$I = -\frac{1}{2}\int \frac{\sqrt{1+t^2}}{t} dt = -\frac{1}{2}\int \sqrt{1+\frac{1}{t^2}} dt$$

Let $t = \cot \theta$

$$\Rightarrow I = -\frac{1}{2}\int \sqrt{1+\tan^2 \theta}(\left(-\cos ec^2 \theta\right) d\theta$$

$$\Rightarrow I = \frac{1}{2}\int \sec \theta \cos ec^2 \theta d\theta$$

$$\Rightarrow I = \frac{1}{2}\int \frac{d\theta}{\cos \theta \sin^2 \theta} = \frac{1}{2}\int \sec \theta d\theta + \frac{1}{2}\int \cot \theta \cos ec\theta d\theta$$

$$I = \frac{1}{2} \ell n \left|\sec \theta + \tan \theta\right| - \frac{1}{2} \cos ec\theta + C$$

$$\Rightarrow I = \frac{1}{2} \ell n \left|\sqrt{1+\frac{1}{t^2}} + \frac{1}{t}\right| - \frac{1}{2}\sqrt{1+t^2} + C$$

$$\Rightarrow I = \frac{1}{2}\ell n\left|\frac{1+\sqrt{1+t^2}}{t}\right| - \frac{1}{2}\sqrt{1+t^2} + C$$

12. $\int \frac{x^2}{(x\cos x - \sin x)(x\sin x + \cos x)}dx$

Ans. $ln\left|\frac{x\sin x + \cos x}{x\cos x - \sin x}\right|$

Sol. $I = \int \frac{x^2}{(x\cos x - \sin x)(x\sin x + \cos x)}dx$

$$= \int \frac{x^2(x\cos x - \sin x)}{(x\cos x - \sin x)^2(x\sin x + \cos x)}dx$$

let $\frac{x\sin x + \cos x}{x\cos x - \sin x} = t$

$$\Rightarrow \left(\frac{(x\cos x - \sin x)x\cos x + (x\sin x + \cos x)x\sin x}{(x\cos x - \sin x)^2}\right)dx = dt$$

$$\Rightarrow \frac{x^2(1-0)}{(x\cos x - \sin x)^2}dx = dt$$

$$\Rightarrow I = \int \frac{dt}{t} = \ell n|t| + C$$

$$= \ell n\left|\frac{x\sin x + \cos x}{x\cos x - \sin x}\right| + C$$

13. Suppose $\int \frac{1-7\cos^2 x}{\sin^7 x\ \cos^2 x}dx = \frac{g(x)}{\sin^7 x} + C$, where C is arbitrary constant of integration. Then find the value of

$g'(0) + g''\left(\frac{\pi}{4}\right)$

दिया है $\int \frac{1-7\cos^2 x}{\sin^7 x\ \cos^2 x}dx = \frac{g(x)}{\sin^7 x} + C$, जहाँ C स्वेच्छ समाकलन स्थिरांक है तब $g'(0) + g''\left(\frac{\pi}{4}\right)$ का मान ज्ञात कीजिए।

Ans. 5

Sol. $\int \frac{1-7\cos^2 x}{\sin^7 x \cos^2 x}dx = \frac{g(x)}{\sin^7 x} + C$

$\int (\cos ec^7 x)(\sec^2 x)dx - 7\int \cos ec^7 x dx = \frac{g(x)}{\sin^7 x} + C$

$(\cos ec^7 x)\tan x + \int 7\cos ec^7 x dx - 7\int \cos ec^7 x dx + C = \frac{g(x)}{\sin^7(x)} + C$

$\cos ec^7 x \tan x + C = \frac{g(x)}{\sin^7 x} + C$

$\frac{\tan x}{\sin^7 x} + C = \frac{g(x)}{\sin^7 x} + C$

$\Rightarrow g(x) = \tan x$

$g'(x) = \sec^2 x$

$g''(x) = 2\sec^2 x \tan x$

$g'(0) + g''\left(\frac{\pi}{4}\right) = 1 + 2 \times 2 = 5$

14. $\int_1^2 \frac{(x^2-1)dx}{x^3 \cdot \sqrt{2x^4 - 2x^2 + 1}} = \frac{u}{v}$ where u and v are in their lowest form. Find the value of $\frac{(1000)u}{v}$.

$\int_1^2 \frac{(x^2-1)dx}{x^3 \cdot \sqrt{2x^4 - 2x^2 + 1}} = \frac{u}{v}$ जहाँ u एवं v सापेक्षतः अभाज्य है; $\frac{(1000)u}{v}$ का मान बताओं।

Ans. 125

Sol. $I = \int_1^2 \frac{(x^2-1)dx}{x^3\sqrt{2x^4 - 2x^2 + 1}}$

$= \int_1^2 \frac{\left(1 - \frac{1}{x^2}\right)}{x^3\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}}}dx$

let $\frac{1}{x^2} = t$

$\Rightarrow$ At x = 1, t = 1

$\Rightarrow$ At $x = 2,\ t = \frac{1}{4}$

$$I = \int_{1}^{\frac{1}{4}} \frac{1-t}{\sqrt{2-2t+t^2}}\left(\frac{1}{2}\right)dt$$

$$= \frac{1}{4}\int_{1}^{\frac{1}{4}} \frac{2(t-1)}{\sqrt{t^2-2t+2}}dt$$

$$= \frac{1}{4}\left[\frac{\sqrt{t^2-2t+2}}{\frac{1}{2}}\right]_{1}^{\frac{1}{4}}$$

$$= \frac{1}{4}\left[\sqrt{\frac{1}{16}-\frac{2}{4}+2}-1\right]$$

$$I = \frac{1}{2}\left[\frac{5}{4}-1\right] = \frac{1}{8}$$

$$I = \frac{u}{v} = \frac{1}{8}$$

$$\frac{(1000)u}{v} = \frac{1000}{8} = 125$$

15. Let $h(x) = (fog)(x) + K$ where K is any constant. If $\frac{d}{dx}(h(x)) = -\frac{\sin x}{\cos^2(\cos x)}$ then compute the value of $j(0)$ where $j(x) = \int_{g(x)}^{f(x)} \frac{f(t)}{g(t)}dt$, where f and g are trigonometric functions.

माना $h(x) = (fog)(x) + K$ जहाँ K कोई अचर है यदि $\frac{d}{dx}(h(x)) = -\frac{\sin x}{\cos^2(\cos x)}$ तो $j(0)$ का मान बताओं जहाँ $j(x) = \int_{g(x)}^{f(x)} \frac{f(t)}{g(t)}dt$, जहाँ f एवं g त्रिकोणमितीय फलन है।

Ans. $1 - \sec(1)$

Sol. $h(x) = (fog)(x) + K$

$$h'(x) = f'(g(x))g'(x) = -\frac{\sin x}{\cos^2(\cos x)}$$

Given f(x) & g(x) are trigonometric function.

$\Rightarrow$ f(x) = tan x & g(x) = cos x

$$j(x) = \int_{g(x)}^{f(x)} \frac{f(t)}{g(t)}dt$$

$$\Rightarrow j(0) = \int_{g(0)}^{f(0)} \frac{f(t)}{g(t)}dt = \int_{1}^{0} \frac{\tan t}{\cos t}dt = 1 - \sec(1)$$

16. $$\int_{1}^{\frac{1+\sqrt{5}}{2}} \frac{x^2+1}{x^4-x^2+1} ln\left(1+x-\frac{1}{x}\right)dx$$

Ans. $\frac{\pi}{8} ln\, 2$

Sol. $$I = \int_{1}^{\frac{1+\sqrt{5}}{2}} \frac{x^2+1}{x^4-x^2+1} \ell n\left(1+x-\frac{1}{x}\right)dx$$

Let $1+x-\frac{1}{x} = t$

$$I = \int_{1}^{2} \frac{\ell nt}{(t-1)^2+1}dt$$

$$= \left(\tan^{-1}(t-1)\,\ell nt\right)_1^2 - \int_{1}^{2} \frac{\tan^{-1}(t-1)}{t}dt$$

$$= \frac{\pi}{4}\ell n2 - \int_{1}^{2} \frac{\tan^{-1}(t-1)}{t}dt$$

$$I_1 = \int_{1}^{2} \frac{\tan^{-1}(t-1)}{t}dt$$

let t – 1 = tan θ

$$I_1 = \int_{0}^{\frac{\pi}{4}} \underbrace{\theta}_{I} \underbrace{\left(\frac{\sec^2\theta}{1+\tan\theta}\right)}_{II} d\theta$$

$$= \left(\theta \ell n\left(1+\tan\theta\right)\right)_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \ell n\left(1+\tan\theta\right) d\theta$$

Now $I = \int_0^{\frac{\pi}{4}} \ell n\left(1+\tan\theta\right) d\theta$

Apply King property

$$I = \int_0^{\frac{\pi}{4}} \ell n\left(\frac{2}{1+\tan\theta}\right) d\theta$$

$$I = \frac{\pi}{8} \ell n 2$$

17. $\int_0^{\sqrt{3}} \sin^{-1} \frac{2x}{1+x^2}\, dx$

Ans. $\frac{\pi\sqrt{3}}{3}$

Sol. $I = \int_0^{\sqrt{3}} \sin^{-1} \frac{2x}{1+x^2} dx$

Let $x = \tan\theta$

$$I = \int_0^{\frac{\pi}{3}} \sin^{-1}\left(\sin 2\theta\right) \sec^2\theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} 2\theta \sec^2\theta d\theta + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \left(\pi - 2\theta\right) \sec^2\theta d\theta$$

$$= 2\left(\frac{\pi}{4} - \int_0^{\frac{\pi}{4}} \tan\theta d\theta\right) + \pi\left(\tan\theta\right)_{\frac{\pi}{4}}^{\frac{\pi}{3}} - 2\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \theta \sec^2\theta d\theta$$

$$= \sqrt{3}\pi - \frac{2\pi}{\sqrt{3}} = \frac{\pi}{\sqrt{3}}$$

18. $\int_1^{16} \tan^{-1}\sqrt{\sqrt{x}-1}dx$

Ans. $\frac{16\pi}{3}-2\sqrt{3}$

Sol. $I=\int_1^{16} \tan^{-1}\sqrt{\sqrt{x}-1}dx$

Let $\sqrt{x}=\sec^2\theta$

$$I=4\int_0^{\frac{\pi}{3}} \tan^{-1}(\tan\theta)\tan\theta\sec^4\theta d\theta$$

$$=4\int_0^{\frac{\pi}{3}} \underset{I}{\theta}\underbrace{\left(1+\tan^2\theta\right)\sec^2\theta\tan\theta}_{II}d\theta$$

$$I=\frac{4\pi}{3}\left(\frac{\tan^2\frac{\pi}{3}}{2}+\frac{\tan^4\frac{\pi}{3}}{4}\right)-4\int_0^{\frac{\pi}{3}}\left(\frac{\tan^2\theta}{2}+\frac{\tan^4\theta}{4}\right)d\theta$$

$$=\frac{4\pi}{3}\left(\frac{3}{2}+\frac{9}{4}\right)-\frac{4}{2}\left(\tan\theta-\theta\right)_0^{\frac{\pi}{3}}-\int_0^{\frac{\pi}{3}}\tan^4\theta d\theta$$

$$=\frac{16\pi}{3}-2\sqrt{3}$$

19. Let α, β be the distinct positive roots of the equation $\tan x = 2x$ then evaluate $\int_0^1 (\sin\alpha x\cdot\sin\beta x)\,dx$, independent of α and β.

यदि α, β समीकरण $\tan x = 2x$ के भिन्न धनात्मक मूल है तो $\int_0^1 (\sin\alpha x\cdot\sin\beta x)\,dx$, का मान ज्ञात करों।

Ans. 0

Sol. $\int_0^1 (\sin\alpha x\cdot\sin\beta x)\,dx=\frac{1}{2}\int_0^1\left(\cos(\alpha x-\beta x)-\cos(\alpha x+\beta x)\right)dx$

$$= \frac{1}{2\left(\alpha^2 - \beta^2\right)}\left((\alpha+\beta)\sin(\alpha-\beta) - (\alpha-\beta)\sin(\alpha+\beta)\right)$$

$$= \frac{1}{2\left(\alpha^2 - \beta^2\right)}\left(\beta \sin\alpha\cos\beta - \alpha\cos\alpha\sin\beta\right)$$

$$= \frac{\cos\alpha\cos\beta}{\left(\alpha^2 - \beta^2\right)}\left(\beta\tan\alpha - \alpha\tan\beta\right)$$

$\therefore$ α and β are root of $\tan x - 2x = 0$.

$$\Rightarrow \int_0^1 \left(\sin\alpha x \cdot \sin\beta x\right)dx = 0$$

20. Determine a pair of number *a* and *b* for which $\int_0^1 \frac{ax+b}{(x^2+3x+2)^2}dx = \frac{5}{2}$.

संख्या युग्म *a* एवं *b* ज्ञात करो जहाँ $\int_0^1 \frac{ax+b}{(x^2+3x+2)^2}dx = \frac{5}{2}$.

Ans. $a = 15,\ b = \frac{45}{2}$

Sol. $$\frac{ax+b}{(x+1)^2(x+2)^2} = \frac{A}{(x+1)} + \frac{B}{(x+2)} + \frac{C}{(x+1)^2} + \frac{D}{(x+2)^2}$$

$$\Rightarrow ax+b = (A+B)x^3 + (2A+B+3A+3B+C+D)x^2 +$$

$$(6A+3B+2A+2B+2D+4C)x + (4C+D+4A+2B)$$

$\Rightarrow A = 3a - 2b,\ B = 2b - 3a,\ C = b - a$ & $D = b - 2a$

$$\int_0^1 \frac{ax+b}{(x^2+3x+2)^2}dx = \frac{5}{2}$$

$$= (3a-2b)\ell n2 + (2b-3a)\ell n\frac{3}{2} + \frac{(b-a)}{2} + \frac{(b-2a)}{6}$$

$$\Rightarrow a\ell n\frac{8}{27} + b\ell n\frac{9}{4} + \frac{2b}{3} - \frac{5a}{6} = \frac{5}{2}$$

$$(3a-2b)\ell n\frac{2}{3} + \frac{(4b-5a)}{6} = \frac{5}{2}$$

$\Rightarrow a = 15$ & $b = \frac{45}{2}$

21. Comment upon the nature of roots of the quadratic equation $x^2 + 2x = k + \int_0^1 |t + k|\,dt$ depending on the value of $k \in R$.

$k \in R$ के मानों के लिये द्विघातीय समीकरण $x^2 + 2x = k + \int_0^1 |t + k|\,dt$ के मूलो की प्रकृति की विवेचना कीजिए।

Ans. real & distinct $\forall\ k \in R$

Sol. $x^2 + 2x + 1 = k + 1 + \int_0^1 |t + k|\,dt$

$(x + 1)^2 = (k + 1) + \int_0^1 |t + k|\,dt$

If $k \geq -1$ R.H.S. ≥ 0

so there will be two real and distinct roots for $k \geq -1$

If $k < -1$

$(x + 1)^2 = k + 1 - \int_0^1 (t + k)dt$

$(x + 1)^2 = 1/2$

so there will have two real and distinct roots for $k < -1$

$\Rightarrow$ The equation will have two real and distinct roots for $k \in R$,

22. Evaluate: $\int_0^{\pi} e^{|\cos x|}\left(2\sin\left(\frac{1}{2}\cos x\right) + 3\cos\left(\frac{1}{2}\cos x\right)\right)\sin x\,dx$.

ज्ञात करो: $\int_0^{\pi} e^{|\cos x|}\left(2\sin\left(\frac{1}{2}\cos x\right) + 3\cos\left(\frac{1}{2}\cos x\right)\right)\sin x\,dx$.

Ans. $\frac{24}{5}\left(e\cos\left(\frac{1}{2}\right) + \frac{e}{2}\sin\left(\frac{1}{2}\right) - 1\right)$

Sol. Let $I = \int_0^{\pi} e^{|\cos x|}\left[2\sin\left(\frac{1}{2}\cos x\right) + 3\cos\left(\frac{1}{2}\cos x\right)\right]\sin x\,dx$

$$= \int_0^{\pi} e^{|\cos x|} 2\sin\left(\frac{1}{2}\cos x\right)\sin x dx$$

$$+\int_0^{\pi} e^{|\cos x|} 3\cos\left(\frac{1}{2}\cos x\right)\sin x\, dx$$

$$= I_1 + I_2$$

Now using the property that

$$\int_0^{2a} f(x)dx = 0 \qquad \text{if } f(2a-x) = -f(x)$$

$$= 2\int_0^{a} f(x)dx \qquad \text{if } f(2a-x) = f(x)$$

We get, $I_1 = 0$

and $I_2 = 2\int_0^{\pi/2} e^{|\cos x|} 3\cos\left(\frac{1}{2}\cos x\right)\sin x\, dx$

$$= 6\int_0^{\pi/2} e^{\cos x}\cos\left(\frac{1}{2}\cos x\right)\sin x\, dx$$

Put $\cos x = t \Rightarrow -\sin x\, dx = dt$, we get

or $I_2 = 6\int_0^1 e^t \cos\frac{t}{2}dt$

$$I_2 = 6[(e^t \cos\frac{t}{2})_0^1 + \frac{1}{2}\int_0^1 e^t \sin\frac{t}{2}dt]$$

$$= 6\left[e\cos(1/2) - 1 + \frac{1}{2}\left\{(e^t \sin t/2)_0^1 - \frac{1}{2}\int_0^1 e^t \cos t/2 dt\right\}\right]$$

$$I_2 = 6\left[e\cos\left(\frac{1}{2}\right) - 1 + \frac{1}{2}\left\{e\sin(1/2) - \frac{1}{2}.\frac{1}{6}I_2\right\}\right]$$

$$I_2 + \frac{1}{4}I_2 = 6\left[e\cos(1/2) + \frac{1}{2}e\sin(1/2) - 1\right]$$

$$\Rightarrow I_2 = \frac{24}{5}\left[e\cos(1/2) + \frac{1}{2}e\sin\left(\frac{1}{2}\right) - 1\right]$$

23. If the value of the definite integral $\int_{-1}^{1} \cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\cdot\left(\cot^{-1}\frac{x}{\sqrt{1-(x^2)^{|x|}}}\right)dx = \frac{\pi^2\left(\sqrt{a}-\sqrt{b}\right)}{\sqrt{c}}$ where $a, b, c \in N$ in their lowest form, then find the value of $(a + b + c)$.

यदि $\int_{-1}^{1}\cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\cdot\left(\cot^{-1}\frac{x}{\sqrt{1-(x^2)^{|x|}}}\right)dx=\frac{\pi^2\left(\sqrt{a}-\sqrt{b}\right)}{\sqrt{c}}$ जहाँ a, b, c ∈ N अपने न्यूनतम रूप में लिखे गये है तो (a + b + c) का मान बताओं।

Ans. 7

Sol. $I=\int_{-1}^{1}\cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\cot^{-1}\left(\frac{x}{\sqrt{1-\left(x^2\right)^{|x|}}}\right)dx$

Apply King property

$$I=\int_{-1}^{1}\cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\cot^{-1}\left(\frac{-x}{\sqrt{1-\left(x^2\right)^{|x|}}}\right)dx$$

$$\Rightarrow 2I=\int_{-1}^{1}\cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\pi dx$$

$$\Rightarrow 2I=\pi\int_{-1}^{1}\cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)dx$$

$$\Rightarrow I=\pi\int_{0}^{1}\cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)dx$$

$$I=\pi\left[\left[x\cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right]_0^1-\int_0^1 x\left(\frac{-1}{1+\frac{1}{1-x^2}}\right)\frac{xdx}{\left(1-x^2\right)^{\frac{3}{2}}}\right.\right.$$

$$I=\pi\int_{0}^{1}\frac{x^2\left(1-x^2\right)}{2-x^2}\frac{1}{\left(1-x^2\right)^{\frac{3}{2}}}dx$$

$$I=\pi\int_{0}^{1}\frac{x^2}{\sqrt{1-x^2}\left(2-x^2\right)}dx$$

$$I=\pi\int_{0}^{1}\frac{x^2}{\sqrt{1-x^2}\left(1+1-x^2\right)}dx$$

let $x = \sin\theta$

$$\Rightarrow I = \pi\int_0^{\frac{\pi}{2}} \frac{\sin^2\theta\cos\theta d\theta}{\cos\theta\left(1+\cos^2\theta\right)}$$

$$\Rightarrow I = \pi\int_0^{\frac{\pi}{2}} \frac{\sin^2\theta d\theta}{1+\cos^2\theta}$$

$$= \pi\int_0^{\frac{\pi}{2}} \frac{1-\cos 2\theta}{3+\cos 2\theta} d\theta$$

$$\Rightarrow I = \pi\int_0^{\frac{\pi}{2}} \frac{4}{3+\cos 2\theta} - \frac{\pi^2}{2} = \pi\int_0^{\frac{\pi}{2}} \frac{4\sec^2 x dx}{4+2\tan^2 x} - \frac{\pi^2}{2}$$

$$I = \frac{\pi^2}{\sqrt{2}} - \frac{\pi^2}{2}$$

$$= \frac{\pi^2}{\sqrt{2}}\left[1-\frac{1}{\sqrt{2}}\right]$$

$$= \frac{\pi^2}{2}\left[\sqrt{2}-1\right]$$

$\Rightarrow a = 2, b = 1, c = 4$

$\Rightarrow a + b + c = 7$

24. Let $y = f(x)$ be a quadratic function with $f'(2) = 1$. Find the value of the integral $\int_{2-\pi}^{2+\pi} f(x)\cdot\sin\left(\frac{x-2}{2}\right)dx$.

यदि $y = f(x)$ एक द्विघातीय फलन है जहाँ $f'(2) = 1$ तो समाकलन $\int_{2-\pi}^{2+\pi} f(x)\cdot\sin\left(\frac{x-2}{2}\right)dx$ का मान बताओं।

Ans. $I = 8$ as $\int_0^{\pi/2} y\sin y\,dy = 1$

Sol. Given $f(x) = ax^2 + bx + c$

$f'(x) = 2ax + b$

$\Rightarrow 4a + b = 1$

$\Rightarrow f(x) = ax^2 + (1 - 4a)x + c = a(x - 2)^2 + x + c - 4a$

$$I=\int_{2-\pi}^{2+\pi} f(x)\cdot\sin\left(\frac{x-2}{2}\right)dx$$

Let $\frac{x-2}{2}=y$

$$\Rightarrow I=2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(2y+2)\sin y\,dy$$

$$=2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(4ay^2+2y+2+c-40\right)\sin y\,dy$$

$$=4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} y\sin y\,dy$$

$$=8\int_{0}^{\frac{\pi}{2}} y\sin y\,dy=8\ \{\because\ \int_{0}^{\pi/2} y\sin y\,dy\ =1\}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 25 to 27

Let $f(x)$ be a polynomial function of degree 2 satisfying

$$\int\frac{f(x)}{x^3-1}dx=\ln\left|\frac{x^2+x+1}{x-1}\right|+\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C,$$

where C is indefinite integration constant.

माना एक बहुपद फलन $f(x)$, की कोटि 2 है और

$$\int\frac{f(x)}{x^3-1}dx=\ln\left|\frac{x^2+x+1}{x-1}\right|+\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C,$$ को सन्तुष्ट करता है

जहां C समाकलन अचर है।

25. The value of $f(1)$ is equal to :

$f(1)$ का मान बराबर होगा:

(A) 1 (B) 2 (C) – 1 (D) – 3

Ans. D

Sol. $\int \frac{f(x)}{x^3-1}dx = \ln\left|\frac{x^2+x+1}{x-1}\right| + \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C$

Differentiating both sides, we get

$$\frac{f(x)}{x^3-1} = \frac{x-1}{x^2+x+1}\cdot\frac{(x-1)(2x+1)-(x^2+x+1).1}{(x-1)^2} + \frac{2}{\sqrt{3}}\cdot\frac{1}{1+\left(\frac{2x+1}{\sqrt{3}}\right)^2}\cdot\frac{2}{\sqrt{3}}$$

$$= \frac{2x^2-x-1-x^2-x-1}{x^3-1} + \frac{4}{3}\cdot\frac{3}{3+(4x^2+4x+1)} = \frac{x^2-2x-2}{x^3-1} + \frac{1}{x^2+x+1}$$

$f(x) = (x^2-2x-2)+(x-1) = x^2-x-3 \Rightarrow f(1) = -3$

26. Let $\int \frac{1-6\,\text{cosec}\,x}{6+f(\sin x)}d(\sin x) = g(x)+K$, where $g(x)$ contains no constant term. Then $\lim\limits_{t\to\frac{\pi}{2}} g(t)$ is equal to :

(where K is indefinite integration constant)

माना $\int \frac{1-6\,\text{cosec}\,x}{6+f(\sin x)}d(\sin x) = g(x)+K$, जहाँ $g(x)$ में कोई अचर पद नहीं है। तब $\lim\limits_{t\to\frac{\pi}{2}} g(t)$ बराबर होगी : (जहाँ K समाकलन अचर है।)

(A) $\ln 1$ (B) $\ln 2$ (C) $\ln 3$ (D) $\ln 4$

Ans. C

Sol. $I = \int \frac{1-6\,\text{cosec}\,x}{6+f(\sin x)}d(\sin x) = \int \frac{1-\frac{6}{\sin x}}{6+\sin^2 x-\sin x-3}d(\sin x)$

$$\int \frac{1-\frac{6}{\sin x}}{\sin^2 x-\sin x+3}d(\sin x) \quad (\text{Put } \sin x = t)$$

$$= \int \frac{1-\frac{6}{t}}{t^2-t+3}dt = \int \frac{\frac{1}{t^2}-\frac{6}{t^3}}{1-\frac{1}{t}+\frac{3}{t^2}}dt = \ln\left(1-\frac{1}{t}+\frac{3}{t^2}\right)+K = \ln\left(1-\frac{1}{\sin x}+\frac{3}{\sin^2 x}\right)+K$$

$$\Rightarrow \quad g(x) = \ln\left(1-\frac{1}{\sin x}+\frac{3}{\sin^2 x}\right)$$

$$\therefore \quad g(t) = \ln\left(1-\frac{1}{\sin t}+\frac{3}{\sin^2 t}\right)$$

Now, $\lim_{t\to\frac{\pi}{2}} g(t) = l\text{n}3$

27. Let $\int \frac{5+f(\sin x)+f(\cos x)}{\sin x+\cos x}\,dx = h(x)+\lambda$, where $h(1)=-1$. The value of $\tan^{-1}\left(h(2)\right)+\tan^{-1}\left(h(3)\right)$ is equal to : (where λ is indefinite integration constant)

माना $\int \frac{5+f(\sin x)+f(\cos x)}{\sin x+\cos x}\,dx = h(x)+\lambda$, जहाँ $h(1)=-1$ है। $\tan^{-1}\left(h(2)\right)+\tan^{-1}\left(h(3)\right)$ का मान बराबर होगा (जहाँ λ समाकलन अचर है।)

(A) $\frac{\pi}{4}$ (B) $\frac{-\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\frac{-3\pi}{4}$

Ans. D

Sol. $I = \int \frac{5+f(\sin x)+f(\cos x)}{\sin x+\cos x}\,dx = \int \frac{5+\sin^2 x-\sin x-3+\cos^2 x-\cos x-3}{\sin x+\cos x}\,dx$

$= \int -dx = -x+\lambda$

$\therefore\ h(x) = -x \quad (\text{since } h(1)=-1)$

Hence $\tan^{-1}\left(h(2)\right)+\tan^{-1}\left(h(3)\right) = \tan^{-1}(-2)+\tan^{-1}(-3) = \frac{-3\pi}{4}$

Paragraph for Question Nos. 28 to 30

Consider $f(n) = \int_0^{\pi/2} \frac{1-\cos 2nx}{1-\cos 2x}\,dx$ where $n \in N$ then :

माना $f(n) = \int_0^{\pi/2} \frac{1-\cos 2nx}{1-\cos 2x}\,dx$ जहाँ $n \in N$ हो तो :

28. $\frac{f(n+1)+f(n-1)}{f(n)}$ has the value equal to :

$\frac{f(n+1)+f(n-1)}{f(n)}$ का मान है :

(A) 2 (B) 1 (C) 1/2 ;क द्धछवदमः कोई नहीं

Ans. A

Sol. $f(n) = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x} dx$

Consider $f(n+1) + f(n-1) = \int_0^{\pi/2} \frac{\sin^2(n+1)x + \sin^2(n-1)x}{\sin^2 x} dx$

$= \int_0^{\pi/2} \frac{1 - \left(\cos^2(n+1)x - \sin^2(n-1)x\right)dx}{\sin^2 x} = \int_0^{\pi/2} \frac{1 - \cos 2nx \cdot \cos 2x \, dx}{\sin^2 x}$

$= \int_0^{\pi/2} \frac{1 - \cos 2nx(1 - 2\sin^2 x)\, dx}{\sin^2 x} = 2\int_0^{\pi/2} \frac{1 - \cos 2nx \, dx}{1 - \cos 2x} + \underbrace{2\int_0^{\pi/2} \cos 2nx \, dx}_{\text{zero}} = 2 f(n)$

hence $\frac{f(n+1) + f(n-1)}{f(n)} = 2$ Ans. $\Rightarrow$ $f(n-1), f(n), f(n+1)$ are in A.P.

29. If the sum $\sum_{r=1}^{100} f(r)$ equals $(5050)\lambda$ then λ equals :

यदि $\sum_{r=1}^{100} f(r)$ का मान $(5050)\lambda$ हो, तो λ का मान होगा :

(A) π (B) $\pi/2$ (C) $\pi/4$ (D) $\pi/8$

Ans. B

Sol. Now, $f(1), f(2), f(3), \ldots\ldots f(n)$ are in A.P.

$f(1) = \frac{\pi}{2}$; $f(2) = \pi$ etc. $\Rightarrow$ $d = \frac{\pi}{2}$

$\therefore$ $\sum_{r=1}^{n} f(r) = \frac{n}{2}\left[\pi + (n-1)\frac{\pi}{2}\right] = \frac{n}{2}\left[(n+1)\frac{\pi}{2}\right] = n(n+1)\frac{\pi}{4}$ $\Rightarrow$ $1 = \frac{\pi}{4}$

If $n = 100$ then the sum $= (5050)\frac{\pi}{2} \Rightarrow \lambda = \frac{\pi}{2}$

30. The value of the determinant $\begin{vmatrix} f(1) & f(2) & f(3) \\ f(4) & f(5) & f(6) \\ f(7) & f(8) & f(9) \end{vmatrix}$ is equal to :

सारणिक $\begin{vmatrix} f(1) & f(2) & f(3) \\ f(4) & f(5) & f(6) \\ f(7) & f(8) & f(9) \end{vmatrix}$ का मान होगा :

(A) $\pi/2$ (B) π (C) 2π (D) 0

Ans. D

Sol. Using $C_3 \to C_3 - C_2$ and $C_2 \to C_2 - C_1$

$$\Delta = \begin{vmatrix} \pi/2 & d & d \\ f(4) & d & d \\ f(7) & d & d \end{vmatrix} \quad \Rightarrow \quad D = 0$$

Paragraph for Question Nos. 31 to 33

Let f and g be two real-valued differentiable functions on R satisfying $\int_0^x g(t)\,dt = 3x + \int_x^0 \cos^2 t\, g(t)\,dt$ and

$$f(x) = \lim_{\alpha \to 0} \frac{1}{\alpha^4} \int_0^\alpha \frac{\left(e^{x+t} - e^x\right) ln^2(1+t)}{2t^3+3}\,dt.$$

माना f तथा g दो वास्तविक मान के R में अवकलनीय फलन है जो $\int_0^x g(t)\,dt = 3x + \int_x^0 \cos^2 t\, g(t)\,dt$ तथा

$f(x) = \lim_{\alpha \to 0} \frac{1}{\alpha^4} \int_0^\alpha \frac{\left(e^{x+t} - e^x\right) ln^2(1+t)}{2t^3+3}\,dt$ को संतुष्ट करते है।

31. $f(ln\,2)$ is :

$f(ln\,2)$ का मान है :

(A) 0 (B) $\frac{1}{6}$ (C) $\frac{3}{20}$ (D) $\frac{4}{25}$

Ans. B

Sol. We have $f(x) = \lim_{\alpha \to 0} \frac{1}{\alpha^4} \int_0^\alpha \frac{\left(e^{x+t} - e^x\right) ln^2(1+t)}{2t^3+3}\,dt$

$$f(x) = \lim_{\alpha \to 0} \frac{e^x \int_0^\alpha \frac{(e^t - 1) ln^2(1+t)}{(2t^3+3)}\,dt}{\alpha^4} = \lim_{\alpha \to 0} e^x \frac{(e^\alpha - 1) ln^2(1+\alpha)}{(2\alpha^3+3)4\alpha^3}$$

$$= e^x \lim_{\alpha \to 0} \frac{(e^\alpha - 1)}{\alpha} \cdot \frac{ln^2(1+\alpha)}{\alpha^2} \cdot \frac{1}{4(2\alpha^3+3)} = \frac{e^x}{12}$$

Clearly $f(ln\,2) = \frac{e^{ln 2}}{12} = \frac{2}{12} = \frac{1}{6}$

32. Range of g(x) is equal to :

g(x) का परिसर है :

(A) $\left[\frac{3}{2}, 3\right]$ (B) $\left[\frac{1}{2}, 3\right]$ (C) $\left[\frac{3}{2}, \infty\right)$ (D) [– 3, 3]

Ans. A

Sol. We have $f(x) = \underset{\alpha \to 0}{\text{Lim}} \frac{1}{\alpha^4} \int_0^{\alpha} \frac{\left(e^{x+t} - e^x\right) ln^2(1+t)}{2t^3+3} dt$

$$f(x) = \underset{\alpha \to 0}{\text{Lim}} \frac{e^x \int_0^{\alpha} \frac{(e^t - 1) ln^2(1+t)}{(2t^3+3)} dt}{\alpha^4} = \underset{\alpha \to 0}{\text{Lim}}\, e^x \frac{(e^{\alpha} - 1) ln^2(1+\alpha)}{(2\alpha^3+3) 4\alpha^3}$$

$$= e^x \underset{\alpha \to 0}{\text{Lim}} \frac{(e^{\alpha} - 1)}{\alpha} \cdot \frac{ln^2(1+\alpha)}{\alpha^2} \cdot \frac{1}{4(2\alpha^3+3)} = \frac{e^x}{12}$$

We have $\int_0^x g(t)\, dt = 3x + \int_x^0 \cos^2 t\, g(t)\, dt$

Now, on differentiating both the sides with respect to x, we get

$$g(x) = 3 - \cos^2 x\, g(x) \Rightarrow \quad g(x) = \frac{3}{1+\cos^2 x}$$

Clearly $\frac{3}{2} \le \frac{3}{1+\cos^2 x} \le 3.$

Hence range of $g(x) = \left[\frac{3}{2}, 3\right]$

33. The value of definite integral $\int_0^{\frac{\pi}{2}} g(x)\, dx$ lies in the interval :

नश्चित समाकलन $\int_0^{\frac{\pi}{2}} g(x)\, dx$ का मान अन्तराल में स्थित है :

(A) $\left(\frac{2\pi}{3}, \frac{4\pi}{5}\right)$ (B) $\left(\frac{\pi}{2}, \frac{5\pi}{6}\right)$ (C) $\left(\frac{3\pi}{4}, \frac{6\pi}{5}\right)$ (D) $\left(\pi, \frac{3\pi}{2}\right)$

Ans. CD

Sol. We have $f(x) = \lim_{\alpha \to 0} \frac{1}{\alpha^4} \int_0^{\alpha} \frac{\left(e^{x+t} - e^x\right) ln^2(1+t)}{2t^3+3} dt$

$$f(x) = \lim_{\alpha \to 0} \frac{e^x \int_0^{\alpha} \frac{(e^t - 1) ln^2(1+t)}{(2t^3+3)} dt}{\alpha^4} = \lim_{\alpha \to 0} e^x \frac{(e^{\alpha} - 1) ln^2(1+\alpha)}{(2\alpha^3+3) 4\alpha^3}$$

$$= e^x \lim_{\alpha \to 0} \frac{(e^{\alpha} - 1)}{\alpha} \cdot \frac{ln^2(1+\alpha)}{\alpha^2} \cdot \frac{1}{4(2\alpha^3+3)} = \frac{e^x}{12}$$

Let $I = \int_0^{\frac{\pi}{2}} g(x)\,dx = \int_0^{\frac{\pi}{2}} \frac{3}{1+\cos^2 x} dx = \int_0^{\frac{\pi}{2}} \frac{3\sec^2 x}{\sec^2 x + 1} dx = \int_0^{\frac{\pi}{2}} \frac{3\sec^2 x}{\tan^2 x + 2} dx$

Put $\tan x = t \Rightarrow \sec^2 x\, dx = dt$,

so $I = \int_0^{\infty} \frac{3dt}{t^2+2} = \frac{3}{\sqrt{2}} \left(\tan^{-1} \frac{t}{\sqrt{2}} \right)_0^{\infty} = \frac{3}{\sqrt{2}} \left(\frac{\pi}{2} - 0 \right) = \frac{3\pi}{2\sqrt{2}}$

Paragraph for Questions 34 to 36

Let f(x) be function defined on [0, 1] such that $f(1) = 0$ and for any $a \in (0, 1]$, $\int_0^a f(x)\,dx - \int_a^1 f(x)\,dx = 2f(a) + 3a + b$ where b is constant.

माना [0, 1] में फलन f(x) इस प्रकार परिभाषित है कि $f(1) = 0$ तथा $a \in (0, 1]$ के लिए, $\int_0^a f(x)\,dx - \int_a^1 f(x)\,dx = 2f(a) + 3a + b$ जहाँ b एक नियतांक है।

34. $b =$

(A) $\frac{3}{2e} - 3$ (B) $\frac{3}{2e} - \frac{3}{2}$ (C) $\frac{3}{2e} + 3$ (D) $\frac{3}{2e} + \frac{3}{2}$

Ans. A

35. The length of the subtangent of the curve $y = f(x)$ at $x = 1/2$ is :

वक्र $y = f(x)$ के बिन्दु $x = 1/2$ पर अधोस्पर्शी की लम्बाई होगी :

(A) $\sqrt{e} - 1$ (B) $\frac{\sqrt{e} - 1}{2}$ (C) $\sqrt{e} + 1$ (D) $\frac{\sqrt{e} + 1}{2}$

Ans. A

36. $\int_0^1 f(x)\,dx =$

(A) $\frac{1}{e}$ (B) $\frac{1}{2e}$ (C) $\frac{3}{2e}$ (D) $\frac{2}{e}$

Ans. C

Sol. $\int_0^a f(x)\,dx - \int_a^1 f(x)\,dx = 2f(a) + 3a + b$ _____(i)

diff. w.r.t. a

$\Rightarrow 2f(a) = 2f'(a) + 3$

$\Rightarrow f'(a) - f(a) = \frac{-3}{2}$

I.F. $= e^{-a}$

$\Rightarrow f(a)e^{-a} = \frac{3}{2}e^{-a} + C$

$\Rightarrow f(x) = \frac{3}{2} + Ce^{x}$

$\because f(1) = 0 \Rightarrow c = \frac{-3}{2e}$

$\Rightarrow f(x) = \frac{3}{2}\left(1 - e^{x-1}\right)$

Plugging the value a = 1 in (i)

$\Rightarrow \int_0^1 f(x)\,dx = 2f(1) + 3 + b$

$\Rightarrow \frac{3}{2} - \frac{3}{2e}(e-1) = 3 + b \Rightarrow b = \frac{3}{2e} - 3$

$f(x) = \frac{3}{2} - \frac{3}{2e}e^{x} = y$

length of subtangent $= \left|\left(\frac{y}{\frac{dy}{dx}}\right)\right|_{\text{at } x=\frac{1}{2}}$

$$= \left| \left[\begin{matrix} \frac{3}{2} - \frac{3}{2e} e^x \\ -\frac{3}{2} e \end{matrix} \right]_{x=\frac{1}{2}} \right|$$

$$= \left| \frac{\left(1 - \frac{\sqrt{e}}{e}\right)}{-\frac{1}{e}\sqrt{e}} \right| = \left| \frac{\left(1 - \frac{1}{\sqrt{e}}\right)}{-\frac{1}{\sqrt{e}}} \right|$$

$$= \left|1 - \sqrt{e}\right| = \sqrt{e-1}$$

$$\int_0^1 f(x)\,dx = \int_0^1 \left(\frac{3}{2} - \frac{3}{2} e^{x-1}\right) dx = \frac{3}{2} - \frac{3}{2e}(e-1) = \frac{3}{2e}$$

MATCH THE COLUMN TYPE QUESTION(S)

37. Match the following column :

Column-I		**Column-II**	
(A)	Let $f(x) = \int x^{\sin x}(1 + x\cos x \cdot \ln x + \sin x)dx$ and $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$ then the value of $f(\pi)$ is	(P)	rational
(B)	Let $g(x) = \int \frac{1+2\cos x}{(\cos x + 2)^2} dx$ and $g(0) = 0$ then the value of $g\left(\frac{\pi}{2}\right)$ is	(Q)	irrational
		(R)	integral
(C)	If real numbers x and y satisfy $(x + 5)^2 + (y - 12)^2 = (14)^2$ then the minimum value of $\sqrt{(x^2 + y^2)}$ is	(S)	prime

(D) Let $k(x) = \int \frac{(x^2+1)\,dx}{\sqrt[3]{x^3 + 3x + 6}}$ and $k(-1) = \frac{1}{\sqrt[3]{2}}$ then the value of $k(-2)$ is

निम्न कॉलम को सुमेलित कीजिए :

Column-I **Column-II**

(A) यदि $f(x) = \int x^{\sin x}\left(1 + x\cos x \cdot \ell n\, x + \sin x\right)dx$ और $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$ (P) परिमेय

r c $f(\pi)$ का मान है

(B) यदि $g(x) = \int \frac{1+2\cos x}{(\cos x+2)^2}dx$ और $g(0) = 0$ तब $g\left(\frac{\pi}{2}\right)$ का मान है (Q) अपरिमेय

(C) यदि वास्तविक संख्याऐं x और y सम्बन्ध $(x + 5)^2 + (y - 12)^2 = (14)^2$ को सन्तुष्ट (R) पूर्णांक

करते हैं तब $\sqrt{(x^2+y^2)}$ का न्यूनतम मान है (S) अभाज्य

(D) यदि $k(x) = \int \frac{(x^2+1)\,dx}{\sqrt[3]{x^3+3x+6}}$ और $k(-1) = \frac{1}{\sqrt[3]{2}}$ तब $k(-2)$ का मान है।

Ans. (A) Q; (B) P; (C) P, R; (D) P, R, S

Sol. (A) $f(x) = \int x^{\sin x}\left(1 + x\cos x \cdot \ell n\, x + \sin x\right)dx$

if $F(x) = x^{\sin x} = e^{\sin x\, \ell n\, x}$

$\therefore \quad x\,F'(x) = x^{\sin x}(x\cos x\, \ell n\, x + \sin x)$

$\therefore \quad f(x) = \int\left(F(x) + x\,F'(x)\right) = x\,F(x) + C$

$f(x) = x \cdot x^{\sin x} + C$

$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2}\cdot\frac{\pi}{2} + C \quad \Rightarrow \quad C = 0$

$\therefore \quad f(x) = x\,(x)^{\sin x}$; $f(\pi) = \pi(\pi)^0 = \pi$ (irrational) $\Rightarrow$ **(Q)**

(B) $g(x) = \int \frac{\cos x(\cos x+2) + \sin^2 x}{(\cos x+2)^2}dx$

$= \int \underbrace{\cos x}_{II}\cdot\underbrace{\frac{1}{(\cos x+2)}}_{I}dx + \int \frac{\sin^2 x}{\cos x+2}dx$

$= \frac{1}{\cos x+2}\cdot\sin x - \int \frac{\sin^2 x}{(\cos x+2)^2}dx + \int \frac{\sin^2 x}{(\cos x+2)^2}dx$

$g(x) = \frac{\sin x}{\cos x+2} + C$

Alternatively: Consider $\frac{d}{dx}\left(\frac{\sin x}{\cos x+2}\right) = \frac{(\cos x+2)\cos x + \sin^2 x}{(\cos x+2)^2} = \frac{2\cos x+1}{(\cos x+2)^2}$

$$\therefore \quad \int \frac{2\cos x + 1}{(\cos x + 2)^2} dx = \frac{\sin x}{\cos x + 2} + C$$

$$\therefore \quad g(x) = \frac{\sin x}{\cos x + 2} + C$$

$$g(0) = 0 \quad \Rightarrow \quad C = 0$$

$$\therefore \quad g(x) = \frac{\sin x}{\cos x + 2} \quad \Rightarrow \quad g\left(\frac{\pi}{2}\right) = \frac{1}{2} \text{ (rational) } \textbf{Ans.} \quad \Rightarrow \quad \textbf{(P)}$$

(C) Let $\xi + 5 = 14\ \chi o \sigma \theta$

$$y - 12 = 14 \sin\theta$$

$$\therefore \quad x^2 + y^2 = (14\cos\theta - 5)^2 + (14\sin\theta + 12)^2$$

$$= 196 + 25 + 144 + 28(12\sin\theta - 5\cos\theta)$$

$$= 365 + 28(12\sin\theta - 5\cos\theta)$$

$$\therefore \quad \sqrt{(x^2 + y^2)}\Big|_{min} = \sqrt{365 - 28\times 13} = \sqrt{365 - 364} = 1 \text{ Ans.} \quad \Rightarrow \quad \textbf{P, R}$$

(D) $$k(x) = \int \frac{(x^2 + 1)\,dx}{(x^3 + 3x + 6)^{1/3}}$$

put $x^3 + 3x + 6 = t^3 \Rightarrow \quad 3(x^2 + 1)dx = 3t^2\,dt$

$$k(x) = \int \frac{t^2\,dt}{t} = \frac{t^2}{2} + C$$

$$k(x) = \frac{1}{2}(x^3 + 3x + 6)^{2/3} + C$$

$$k(-1) = \frac{1}{2}(2)^{2/3} + C \quad \Rightarrow \quad C = 0$$

$$\therefore \quad k(x) = \frac{1}{2}(x^3 + 3x + 6)^{2/3} \ ; \quad f(-2) = \frac{1}{2}(-8)^{2/3} = \frac{1}{2}[(-2)^3]^{2/3} = 2 \ \textbf{Ans.} \Rightarrow \textbf{P, R, S}$$

38. Match the following column :

Column-I		**Column-II**	
(A)	If $f(x) = \int_0^{g(x)} \frac{dt}{\sqrt{1+t^3}}$ where $g(x) = \int_0^{\cos x} (1+\sin t^2)dt$ then the value of $f'(\pi/2)$	(P)	3
(B)	If $f(x)$ is a non zero differentiable function such that $\int_0^x f(t)dt = (f(x))^2$ for all x, then $f(2)$ equals	(Q)	2
		(R)	1

(C) If $\int_a^b (2+x-x^2)\,dx$ $(b>a)$ is maximum then $(a+b)$ is equal to (S) – 1

(D) If $\lim_{x\to 0}\left(\frac{\sin 2x}{x^3}+a+\frac{b}{x^2}\right)=0$ then $(3a+b)$ has the the value equal to

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Column-I		**Column-II**	
(A)	यदि $f(x)=\int_0^{g(x)}\frac{dt}{\sqrt{1+t^3}}$ जहाँ $g(x)=\int_0^{\cos x}(1+\sin t^2)dt$ तो $f'(\pi/2)$ का मान होगा।	(P)	3
(B)	यदि $f(x)$ अशून्य अवकलनीय फलन इस प्रकार है कि x के सभी मानों के लिए $\int_0^x f(t)dt=(f(x))^2$, तो $f(2)$ का मान होगा।	(Q)	2
		(R)	1
(C)	यदि $\int_a^b (2+x-x^2)\,dx$ $(b>a)$ अधिकतम हो, तो $(a+b)$ का मान होगा।	(S)	– 1
(D)	यदि $\lim_{x\to 0}\left(\frac{\sin 2x}{x^3}+a+\frac{b}{x^2}\right)=0$ तो $(3a+b)$ का मान होगा।		

Ans. (A) S; (B) R; (C) R; (D) Q

Sol. (A) $f'(x)=\frac{g'(x)}{\sqrt{1+g^3(x)}}$ and $g'(x)=[1+\sin(\cos^2 x)](-\sin x)$

hence $f'(x)=\frac{[1+\sin(\cos^2 x)](-\sin x)}{\sqrt{1+g^3(x)}}$

$f'\left(\frac{\pi}{2}\right)=\frac{1+0}{\sqrt{1+g^3(\pi/2)}}=\frac{-1}{1+0}=-1$ as $g\left(\frac{\pi}{2}\right)=0$

$\therefore \quad f'\left(\frac{\pi}{2}\right)=-1$ Ans.

(C) Maximum when $a=-1;\ b=2$

$\Rightarrow \quad a+b=1$

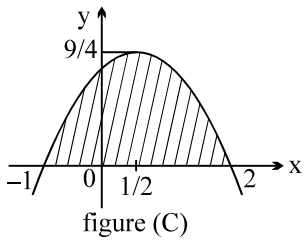


figure (C)

(D) If $\lim_{x\to 0}\frac{\sin 2x}{x^3}+a+\frac{b}{x^2}=0 \quad \Rightarrow \quad \lim_{x\to 0}\frac{\sin 2x+ax^3+bx}{x^3}=0$

for limit to exist $\quad 2 + b = 0 \quad \Rightarrow \quad \boxed{b = -2}$

$\therefore \quad \lim\limits_{x\to 0} \dfrac{\sin 2x + ax^3 - 2x}{x^3} = 0$

apply LHS rule, $\quad \lim\limits_{x\to 0} \dfrac{2\cos 2x + 3ax^2 - 2}{3x^2} = 0$

$\therefore \quad a = \lim\limits_{x\to 0} \dfrac{2(1-\cos 2x)}{3x^2} \quad \Rightarrow \quad \boxed{a = \dfrac{4\sin^2 x}{3x^2} = \dfrac{4}{3}}$

$\therefore \quad 3a + b = 3 \cdot \dfrac{4}{3} - 2 = 2$

39. Match the following column :

Column-I		**Column-II**	
(A)	$\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} \dfrac{3}{n} \cdot \sin\left(2\pi + \dfrac{3\pi i}{n}\right)$ is equal to	(P)	$\dfrac{1}{e}$
(B)	$\int\limits_0^{\infty} [x] e^{-x}\, dx$ equals (where [ ] denotes the greatest integer function)	(Q)	$\dfrac{2}{e}$
(C)	If $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} \left(\dfrac{\ln 2i - \ln n}{n}\right) = \ln k$, then 'k' equals	(R)	$\dfrac{1}{\pi}$
(D)	$\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} \dfrac{i}{n^2} \sin\left(\dfrac{i^2\pi}{n^2}\right)$ is equal to	(S)	$\dfrac{2}{\pi}$
		(T)	$\dfrac{1}{e-1}$

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Column-I		**Column-II**	
(A)	$\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} \dfrac{3}{n} \cdot \sin\left(2\pi + \dfrac{3\pi i}{n}\right)$ का मान है	(P)	$\dfrac{1}{e}$
(B)	$\int\limits_0^{\infty} [x] e^{-x}\, dx$ का मान है। (जहाँ [ ] महत्तम पूर्णांक फलन है।)	(Q)	$\dfrac{2}{e}$
(C)	यदि $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} \left(\dfrac{\ln 2i - \ln n}{n}\right) = \ln k$, तो 'k' का मान है	(R)	$\dfrac{1}{\pi}$

(D) $\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}\frac{i}{n^2}\sin\left(\frac{i^2\pi}{n^2}\right)$ का मान है। (S) $\frac{2}{\pi}$

(T) $\frac{1}{e-1}$

Ans. (A) S; (B) T; (C) Q; (D) R

Sol. (A) $3\int_0^1 \sin(2\pi+3\pi x)\,dx = 3\int_0^1 \sin 3\pi x\,dx$

$= -\frac{3}{3\pi}[\cos 3\pi x]_0^1 = -\frac{1}{\pi}[(-1)-(1)] = \frac{2}{\pi}$

(B) $I = \int_0^\infty [x]e^{-x}\,dx$

$= \int_0^1 [x]e^{-x}\,dx + \int_1^2 [x]e^{-x}\,dx + \int_2^3 [x]e^{-x}\,dx + ...$

$= 0 + 1\int_1^2 e^{-x}\,dx + 2\int_2^3 e^{-x}\,dx + 3\int_3^4 e^{-x}\,dx + ...$

$= (e^{-1}-e^{-2}) + 2(e^{-2}-e^{-3}) + 3(e^{-3}-e^{-4}) + ... \infty$

$= e^{-1} + e^{-2} + e^{-3} + e^{-4} + ... \infty$

$= \frac{e^{-1}}{1-e^{-1}} = \frac{1}{e-1}$

(C) $\frac{1}{n}\sum_{i=1}^{n} ln\left(\frac{2i}{n}\right)$

$\sum_{i=1}^{n}\frac{ln\,2 + ln\,(i/n)}{n} = \int_0^1 (ln\,2 + ln\,x)\,dx$

$= [x\,ln\,2 + x\,ln\,x - x]_0^1 = (ln\,2 - 1) - 0 \qquad \left(\text{as } \lim_{x\to 0} x\,ln\,x = 0\right)$

$= ln\,\frac{2}{e} \Rightarrow \qquad k = \frac{2}{e}$

(D) $l = \lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{i=1}^{n}\frac{i}{n}\sin\left(\frac{i^2}{n^2}\right)\pi = \int_0^1 x\sin(\pi x^2)dx = \frac{1}{\pi}$

40. Match the following column :

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Column-I		**Column-II**	
(A)	$\int_0^{\pi} x\left(\sin^2(\sin x)+\cos^2(\cos x)\right)dx$	(P)	π^2
(B)	$\int_0^{\pi} \frac{x\,dx}{1+\sin^2 x}$	(Q)	$\frac{\pi^2}{2}$
(C)	$\int_0^{\pi^2/4}\left(2\sin\sqrt{x}+\sqrt{x}\cos\sqrt{x}\right)dx =$	(R)	$\frac{\pi^2}{4}$
		(S)	$\frac{\pi^2}{2\sqrt{2}}$

Ans. (A) Q; (B) S; (C) Q

Sol. (A) $I = \int_0^{\pi} x\left(\sin^2(\sin x)+\cos^2(\cos x)\right)dx$; $I = \int_0^{\pi}(\pi-x)\left(\sin^2(\sin x)+\cos^2(\cos x)\right)dx$

$2I = \pi\int_0^{\pi}\left(\sin^2(\sin x)+\cos^2(\cos x)\right)dx$; $2I = 2\pi\int_0^{\pi/2}\left(\sin^2(\sin x)+\cos^2(\cos x)\right)dx$ (Using Queen)

also $I = \pi\int_0^{\pi/2}\left(\sin^2(\cos x)+\cos^2(\sin x)\right)dx$; use king and add,

$2I = \pi\int_0^{\pi/2} 2\,dx \Rightarrow I = \pi\int_0^{\pi/2} dx = \frac{\pi^2}{2}$ Ans. $\Rightarrow$ (Q)

(B) $I = \int_0^{\pi}\frac{\pi-x}{1+\sin^2 x}dx$ (Using King); $\therefore$ $2I = \pi\int_0^{\pi}\frac{dx}{1+\sin^2 x}$

$2I = 2\pi\int_0^{\pi/2}\frac{dx}{1+\sin^2 x} \Rightarrow I = \pi\int_0^{\pi/2}\frac{\sec^2 x\,dx}{1+2\tan^2 x}$

$= \frac{\pi}{2}\int_0^{\pi/2}\frac{\sec^2 x\,dx}{\tan^2 x+(1/\sqrt{2})^2} = \frac{\pi\sqrt{2}}{2}\cdot\frac{\pi}{2} = \frac{\pi^2}{2\sqrt{2}}$ **Ans.** $\Rightarrow$ **(S)**

(C) $f(x) = 2\sin\sqrt{x}$

$f'(x) = \frac{2\cos\sqrt{x}}{2\sqrt{x}} = \frac{\cos\sqrt{x}}{\sqrt{x}}$

$x\,f'(x) = \sqrt{x}\cos\sqrt{x}$

$$\therefore \quad I = \int_0^{\pi^2/4} \big(f(x) + xf'(x)\big)dx = \; x\,f(x)\Big]_0^{\pi^2/2} = \; 2x\sin\sqrt{x}\Big]_0^{\pi^2/2} = \frac{2\pi^2}{4} = \frac{\pi^2}{2}$$ **Ans.** $\Rightarrow$ **(Q)**

41. Match the following column :

Column-I		**Column-II**	
(A)	$\lim\limits_{n\to\infty} 4\left[\frac{e^{\frac{1}{n}}}{n^2} + \frac{2}{n^2}e^{\frac{2}{n}} + \frac{3}{n^2}e^{\frac{3}{n}} + ... + \frac{1}{n}e\right] =$	(P)	0
(B)	$\int_0^1 \ln\left(\frac{1}{x} - 1\right) dx =$	(Q)	1
(C)	$\int_0^{10\pi}\left(\lim\limits_{x\to y}\left(\frac{\sin x - \sin y}{x - y}\right)\right) dy =$	(R)	2
(D)	$\int_0^{\infty} \frac{\ln\left(x + \frac{1}{x}\right)dx}{\left(1 + x^2\right)} = \frac{\pi}{2}\ln a$, then a =	(S)	4
		(T)	5

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Column-I		**Column-II**	
(A)	$\lim\limits_{n\to\infty} 4\left[\frac{e^{\frac{1}{n}}}{n^2} + \frac{2}{n^2}e^{\frac{2}{n}} + \frac{3}{n^2}e^{\frac{3}{n}} + ... + \frac{1}{n}e\right] =$	(P)	0
(B)	$\int_0^1 \ln\left(\frac{1}{x} - 1\right) dx =$	(Q)	1
(C)	$\int_0^{10\pi}\left(\lim\limits_{x\to y}\left(\frac{\sin x - \sin y}{x - y}\right)\right) dy =$	(R)	2
(D)	$\int_0^{\infty} \frac{\ln\left(x + \frac{1}{x}\right)dx}{\left(1 + x^2\right)} = \frac{\pi}{2}\ln a$, तब a =	(S)	4
		(T)	5

Ans. (A)-S, (B)-P, (C)-P, (D)-S

Sol. (A) $\lim\limits_{n\to\infty} 4\left[\dfrac{e^{\frac{1}{n}}}{n^2}+\dfrac{2}{n^2}e^{\frac{2}{n}}+\dfrac{3}{n^2}e^{\frac{3}{n}}+.....+\dfrac{1}{n}e\right]=\lim\limits_{n\to\infty} 4\left[\sum\limits_{r=1}^{n}\dfrac{r}{n^2}e^{\frac{r}{n}}\right]$

$=4\int\limits_0^1 xe^x dx=4$

(B) $\int\limits_0^1 \ln\left(\dfrac{1}{x}-1\right)dx=\int\limits_0^1\left(\ell n(1-x)-\ell nx\right)dx$

$=\int\limits_0^1 \ell n(1-x)\,dx-\int\limits_0^1 \ell nx\,dx$

$=-\int\limits_1^0 \ell n(t)\,dt-\int\limits_0^1 \ell nx\,dx=0$

(C) $\int\limits_0^{10\pi}\left(\lim\limits_{x\to y}\left(\dfrac{\sin x-\sin y}{x-y}\right)\right)dy=\int\limits_0^{10\pi}\cos y\,dy=0$

(D) $\int\limits_0^{\infty}\dfrac{\ln\left(x+\frac{1}{x}\right)dx}{\left(1+x^2\right)}=\int\limits_0^{\infty}\dfrac{\ell n\left(x+\frac{1}{x}\right)}{x\left(x+\frac{1}{x}\right)}dx$

Let $x=\tan\theta$

$=\int\limits_0^{\frac{\pi}{2}}\dfrac{\ell n(\tan\theta+\cot\theta)}{\sec^2\theta}\sec^2\theta\,d\theta$

$=-\int\limits_0^{\frac{\pi}{2}}\left(\ell n\sin\theta+\ell n\cos\theta\right)d\theta$

$=-\left(-\dfrac{\pi}{2}\ell n2+\dfrac{-\pi}{2}\ell n2\right)=\dfrac{\pi}{2}\ell n4$

$\Rightarrow a=4$